

# Mobility dividend and cultural divergence in science

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## Abstract

Scientists cross national borders far more often than a generation ago, and science has drawn a dividend from it, while the moves they now make also span greater cultural distances — a transformation that should, in theory, have widened the space of combinations from which science can draw, since new ideas arise from recombining existing ones and movement brings into contact what was apart. Over the same half century, science has become less likely to break with the past, and a given rate of advance has come to demand ever larger research inputs. Whether the transformation of mobility has helped or hindered science is therefore unknown, and answering it requires knowing not only who moves but how far. Here we introduce a measure of early-career cultural distance and apply it to 584,712 scientists whose first publications appeared between 1956 and 1996, matching those who moved to those who did not and following productivity, impact and disruptiveness over career years 6–25. A move recombines both the collaborators a scientist works with and the material they work on, which carries productivity and disruptiveness, and it opens elite ties and the team standing that carry impact; within the mobile group these divide along cultural distance, shorter moves securing the second and longer moves the first, at the cost of the standing a nearer setting affords. Neither those who move nor those who move further reach the extreme of the disruptiveness distribution more often, since the breadth that lifts the average is carried across the geographic and cultural spans over which tacit knowledge travels worst. At the scale of the population, the historical shift toward more, and more distant, mobility raised the centre of that distribution and did not erode its top, composition accounting for under 2% of the fall in the share of scientists reaching it and under 1% of the rise in their number, the fall tracing instead to declining rates within every mobility group and the number holding up because the workforce grew faster than those rates declined. Restricting mobility would therefore lower the centre without recovering the top. It would also slow diffusion, falling first on the peripheral

communities that mobile scientists connect to the frontier and on the reach of the work they spread, and it would leave the centres less exposed to what the most distant moves bring in; and it would narrow the two ways a move now lets a scientist recover the breadth the burden of knowledge took away — assembled in teams whose expertise a move sets further apart, or accumulated from the position between communities that a move leaves them in.

## Introduction

Scientists cross national borders in the course of their careers far more often than they did a generation ago. Removing barriers to where people may work raises aggregate output by reducing the misallocation of talent <sup>1,2</sup>, and scientists free to go where their work is best supported are more productive and more widely cited than those who stay, a mobility dividend documented across fields and eras <sup>3-7</sup>. A move does more than place a scientist where they are most useful, since it also broadens the collaborators, problems and methods within reach, carrying their search beyond the bounds it would otherwise keep <sup>8-10</sup>. New ideas are generally understood to arise from recombining existing ones <sup>11,12</sup>, so the rate at which they appear depends on the diversity of what there is to combine, and mobility supplies that variety by bringing into contact what was apart <sup>13</sup>. Over the past half century mobility has also changed in structure, since the moves careers now involve draw on a wider range of origins and destinations <sup>14,15</sup>, follow a multipolar rather than a core-periphery geography <sup>16-18</sup>, and reach across greater distances than they once did <sup>15,19</sup>. The farther a move reaches, geographically and culturally, the less the destination has in common with the origin, so mobility that has become both more common and more distant should have widened the space of combinations from which science can draw <sup>13</sup>.

Yet over the same half century science has become less likely to break with the past <sup>20,21</sup>, creative destruction has slowed across many fields <sup>22-24</sup>, and the frontier has thinned, with highly disruptive contributions failing to grow in number alongside total output so that their share has contracted <sup>20</sup>, even as a given rate of advance came to demand ever larger research inputs <sup>25</sup>. Explanations have been sought in the exhaustion of easy discoveries, in the burden of knowledge a scientist must now carry to reach the frontier, and in the slowing passage of frontier knowledge to those working behind it <sup>24,26-28</sup>, rarely in the movement of the people who do the work, which over the same period was itself being transformed.

What is known of the returns to a move speaks to this only distantly, resting on a contrast between those who cross a border and those who do not, and on productivity and citation impact rather than on departure from what came before<sup>3-7</sup>, so whether the mobility dividend reaches such work at all is unknown. An affirmative answer would not settle the matter, since the moves that have multiplied are the longer ones, and were it the shorter moves that carried a capacity to break with the past, that capacity would thin as longer ones took their place. Whether it travels with the long moves or the short ones is not obvious, since distance cuts both ways: reaching farther enlarges the pool of collaborators a scientist can draw on and the range of material there is to combine, while widening the geographic and cultural distance across which that material must travel, and what resists the written record, carried as it is in practices rather than in publications, travels badly across either<sup>13,29,30</sup>. A shorter move enters a research setting and culture more like the one left behind, easing entry into an established community at the cost of what there is to combine, so the returns to a move may not simply grow or shrink with distance but divide along it.

Here we study how far scientists move at the outset of their careers and what follows over the two decades after. We introduce a measure of early-career cultural distance, taking the six national cultural dimensions<sup>31</sup> as a proxy for the research setting and culture a move places a scientist in, and computing the mean pairwise distance across them among the countries in which a scientist held affiliations during career years 1–5. We validate it against country embeddings learned from the mobility trajectories of 4.2 million scientists<sup>32</sup>. We apply it to 584,712 scientists whose first publications appeared between 1956 and 1996, matching those who moved to those who did not on early-career performance and research environment<sup>33,34</sup>, and following productivity, impact and disruptiveness<sup>35</sup> over career years 6–25. The core finding is that the mobility dividend extends to all three, and that within it the returns divide along distance rather than growing with it, shorter moves being associated with greater impact and longer

moves with greater productivity and disruptiveness. We then trace this divergence to the way an early move reshapes the collaborations a scientist later assembles, the knowledge they draw on across fields, where in the global system it comes from, and how far they reach for it, and find the disruptiveness that comes with mobility and with distance confined to the centre of its distribution rather than extending to its extreme. Finally we return to the aggregate, holding the outcome of each group fixed and varying only the weights with which they enter the scientific population. The historical shift toward more, and more distant, mobility raises average disruptiveness rather than lowering it, while the thinning of the frontier proves to be a change occurring within mobility groups rather than a consequence of the shift between them. Mobility, on this evidence, governs where the distribution of disruptive science sits, and not whether its extreme is reached. We conclude with a discussion of what these findings imply for researchers, research systems and science policy.

## **Results**

### **Measurement of cultural distance**

To systematically characterize scientists' cross-cultural mobility, we define early-career cultural distance as the average pairwise Manhattan distance across Hofstede's six national dimensions for all international moves made within the first five years of a scientist's career (Fig. 1a). For robustness, we also constructed embedding-based cultural distance from mobility trajectories among 4.2 million scientists using three representation learning approaches (Word2vec<sup>36</sup> and two graph self-supervised learning models<sup>37,38</sup>; see Fig. S1 and Supplementary note 2), with the Word2Vec-based embedding showing the highest correspondence with the Hofstede-based measure (Spearman  $\rho = 0.69$  among mobile scientists).

The global patterns of cross-cultural mobility are visualized in Fig. 1b, with each node

representing an institution and each edge denoting an early-career international move by a mobile scientist. The color of each edge reflects the scientist's percentile of early-career cultural distance (as shown in Fig. 1c). For example, flows between the US, UK, and Australia cluster at the lower end of the distribution (red and yellow), while moves spanning East Asia or the Middle East to Western Europe or North America often fall in the higher percentiles. Overall, the empirical distribution of early-career cultural distance among mobile scientists is highly right-skewed: while most experience relatively modest cultural transitions, a substantial minority undertake much larger cross-cultural moves (Fig. 1c). Prior work shows that international mobility can amplify productivity and impact <sup>3-5</sup>, and many landmark scientists have undertaken significant cross-cultural moves early in their careers.

For example, Richard R. Ernst's move from Switzerland to the United States became decisive in his development: "Among numerous offers, I decided for Varian Associates in Palo Alto... I was extremely lucky. Weston Anderson was on his way to invent Fourier transform spectroscopy, and I could take advantage in an optimum way of my knowledge in system theory." <sup>39</sup> Similarly, Adi Shamir's arrival at MIT catalyzed one of the most important breakthroughs in cryptography: "He met Ronald Rivest and Leonard Adleman... They were inspired by a 1976 paper... Rivest, Shamir, and Adleman presented practical implementations called the 'RSA' method from the first letters of the names of the three authors..." <sup>40</sup> In economics, Amartya Sen's early career was shaped by ongoing movement and debate across India and the UK, at Cambridge he plunged into "debates... centred on macroeconomics dealing with economic aggregates for the economy as a whole", while Calcutta compelled him to confront "competing loyalties of rival political attitudes ... conflicts between substantive equity ... and universal tolerance." The tension between these arenas later crystallized into prize-winning researches that "encompassed welfare economics, economic inequality and poverty ... and rational, tolerant and democratic social choice." <sup>41</sup> These

instances underscore a broader question: how do early-career cross-cultural moves, both short and long, shape long-term academic performance at scale? We now turn to large-sample analysis to systematically address this question.

We leveraged a large-scale disambiguated dataset from the Microsoft Academic Graph, comprising the career trajectories of 584,712 scientists (See Data for details). Our analysis focuses on long-term academic performance during years 6–25, using three indicators: Publications (total publications, *productivity*), Hit 10pct (the proportion of top-10% cited papers in the same year and field, *impact*), and the D5 percentile (*disruptiveness*). Scientists with affiliations in more than one country during their first five career years are defined as mobile scientists (“Mob”), while those without cross-country moves are classified as non-mobile scientists (“non-Mob”). Within the Mob group, scientists are further divided into three equally sized subgroups—Shorter, Medium, and Longer—based on the magnitude of their early-career cultural distance. To ensure Mob and non-Mob scientists’ comparability, we employed Coarsened Exact Matching (CEM) to identify matched pairs (see Methods), controlling for early-career academic performance (i.e., early-career Publications, Hit 10pct, and D5 percentile) and environment (i.e., institutional prestige and coauthor productivity).

### **Mobility dividend and cultural divergence**

Fig. 2 shows that early-career cross-cultural mobility significantly enhances scientists’ long-term academic performance. After CEM, we observe the mobility dividend that mobile scientists (Mob) substantially outperform their non-mobile peers (non-Mob) across all measured dimensions during years 6–25. Specifically, mobile scientists publish 25.32% more papers (mean: Mob = 89.40, non-Mob = 71.34; one-sided Welch’s t-test,  $P < 0.001$ ), achieve a 7.18% higher proportion of high-impact work (Hit 10pct: Mob = 0.207 vs non-Mob = 0.193,  $P < 0.001$ ), and produce 4.06% more disruptive contributions (D5 percentile: Mob = 0.508 vs non-Mob = 0.488,  $P < 0.001$ ; Fig. 2a, d,



g). Turning to the Mob group, we find the cultural divergence that the magnitude of early-career cultural distance differentially shapes these outcomes (Fig. 2b, e, h). Using the relative ratio (each scientist's value divided by the overall sample mean) as a normalized metric, we find that scientists crossing shorter cultural distances achieve the highest long-term citation impact (mean relative ratio = 1.14), while those with longer moves exhibit superior productivity (mean relative ratio = 1.15) and disruptiveness (mean relative ratio = 1.07). Directly comparing shorter versus longer cultural distances reveals significant contrasts: scientists with longer cultural moves show a 6.11% increase in publication productivity (relative ratio: 1.15 vs 1.08; one-sided t-test,  $P < 0.001$ ) and a 9.79% increase in disruptiveness (1.07 vs 0.98;  $P < 0.001$ ). Conversely, shorter-distance movers exhibit a pronounced 20.41% higher rate of high-impact work compared to those undertaking longer moves (relative ratio: 1.14 vs 0.95;  $P < 0.001$ ).

These divergent trajectories evolve distinctly across career stages (Fig. 2c, f, i and Fig. S3). Specifically, the Mob advantage over non-Mob steadily increases for productivity and disruptiveness from early-mid to late career, while the impact advantage peaks early and subsequently decline over time. Within Mob scientists, the productivity and disruptiveness benefits of longer cultural distance and the high-impact advantage of shorter cultural distance steadily intensify. These results remain stable when varying the career observation window for performance (years 6–25; annual, 3-year, and 5-year moving averages; Supplementary Note 3.1.2; Fig. S3)

The association between early-career cultural distance and long-term academic performance proves robust to a broad array of alternative measures, estimation strategies, and sample restrictions (see Supplementary Note 3.1.1–3.1.7). The observed performance differences are not attributable to chance, as confirmed by placebo (shuffle) tests (Supplementary Note 3.1.1; Fig. S2). The results remain stable when redefining the early-career period for measuring cultural distance (years 1–3 through 1–7 instead of 1–5; Supplementary Note 3.1.2; Table S3). All patterns hold under alternative

matching algorithms and sample selection—including m: n CEM, propensity score matching <sup>34</sup> (PSM) and no-matching whole samples (Supplementary Note 3.1.3; Table S4), and are further corroborated by causal inference using both Double Machine Learning <sup>42</sup> (DML) and Doubly Robust Learner <sup>43</sup> (DRL) (Table S5).

Results are robust to multiple alternative definitions of scientific impact, including Hit at varying thresholds (top 1%, 5%, 10%) across different field level (LV0 and LV1), citation counts within fixed windows, restricting to citations within each paper's research field, and citation measures normalized by field and publication year using the MAG classification system, or by constructing reference-based peer groups via bibliographic coupling within the same publication year; and restrictions by minimum reference count (Supplementary Note 3.1.5; Tables S8–S10). For disruptiveness, the association with cultural distance persists when using multiple DI-variants (D10, D5>0, and restricting to each paper's research field's D5), applying minimum reference and citation count restrictions, and under both field-year and bibliographic coupling normalization strategies (Supplementary Note 3.1.6; Tables S11–S13). Besides, the findings are robust to early career academic distance between different countries, field difference, or temporal effects. Stepwise inclusion of country, field, and year fixed effects, as well as controls for academic distance between countries, does not change the main patterns (Supplementary Note 3.1.7; Table S14). Embedding-based measures that integrate both cultural and academic attributes likewise yield consistent results (Table S7).

Beyond long-term academic performance, we further assessed a range of downstream career outcomes to gauge the broader implications of early-career cross-cultural mobility (Supplementary Note 3.1.8; Fig. S4–S5). Compared to non-mobile peers, Mob scientists not only achieve superior research output but also advance more rapidly to team leadership roles (mean transition time: 6.19 vs. 6.43 years; +3.7%; two-sided t-test,  $P < 0.001$ ). While Mob scientists receive less U.S. funding per paper compared to

their non-mobile peers (0.2068 vs. 0.2496;  $-17.15\%$ ;  $P < 0.001$ ), they secure substantially more global funding per paper (0.0686 vs. 0.0444;  $+54.50\%$ ;  $P < 0.001$ ). They publish in higher-visibility venues (mean journal C5: 13.46 vs. 12.01;  $+12.1\%$ ;  $P < 0.001$ ), and receive significantly more media and social attention (Newsfeed mentions: 0.00667 vs. 0.00534,  $+24.9\%$ ; Tweet mentions: 0.319 vs. 0.269,  $+18.6\%$ ; both  $P < 0.001$ ). The probability of receiving major international prizes—such as the Nobel, Fields, or Turing—is 2.19 times higher among mobile scientists than non-mobile peers (0.0990% vs. 0.0452%;  $P < 0.001$ ).

Comparisons among mobile scientists reveal that crossing shorter cultural distances is associated with the fastest promotions to leadership (mean transition time: 5.78 vs. 6.53 years;  $+11.5\%$ ; two-sided t-test,  $P < 0.001$ ). Notably, those moving shorter distances secure significantly more U.S. funding per paper than those undertaking longer moves (0.2459 vs. 0.1573;  $+56.32\%$ ;  $P < 0.001$ ), whereas the opposite trend emerges for global funding (0.0537 vs. 0.0891;  $+65.92\%$ ;  $P < 0.001$ ). Scientists with shorter-distance moves also show a higher probability of being cited in a patented invention (1.4373 vs. 0.9569;  $+50.20\%$ ;  $P < 0.001$ ), and achieve greater scientific and social visibility (mean journal C5: 13.87 vs. 12.72,  $+9.0\%$ ; Newsfeed mentions: 0.00834 vs. 0.00552,  $+51.1\%$ ; tweet mentions: 0.387 vs. 0.252,  $+53.6\%$ ; all  $P < 0.001$ ). The probability of receiving a major international prize is likewise 2.11 times higher for shorter- than for longer-distance movers (0.1358% vs. 0.0645%;  $P < 0.001$ ).

Altogether, we observe striking empirical regularities that generalize across specification, sample restrictions, and alternative measures that the long-term academic advantages of early-career mobility—including the overall superiority of mobile scientists and the distinct benefits of longer-distance (enhanced productivity and disruptiveness) versus shorter-distance (greater high-impact work) moves.

## Cultural distance and collaboration

What drives these persistent performance differences? The analyses above already account for scientists' early-career academic trajectories and environments, yet the advantages of mobile scientists—and the divergence between shorter- and longer-distance movers—remain. A natural place to begin is early-career international collaboration. When scientists relocate to a new country, they become embedded in more than one national scientific community, which may make cross-border collaboration more likely. Mob scientists are accordingly more likely to engage in international collaboration during the early career: the proportion of internationally co-authored papers is more than 2.5 times higher among Mob than non-Mob scientists (Fig. S6a;  $P < 0.001$ ). We find that early-career international collaboration proportion predicts later-career team country size, and Mob scientists accordingly exhibit significantly higher team country size in years 6–25 (Fig. 3a; Fig. S6c, f;  $P < 0.001$ ). Scientists with higher early international collaboration consistently outperform those without (Fig. S6b), and the same pattern holds for later-career team country size: those in the higher group achieve greater Publications, Hit 10pct, and D5 percentile regardless of mobility status (Fig. 3b; all  $P < 0.01$ , and  $P < 0.001$  for Publications and Hit 10pct).

Greater international exposure may expand scientists' collaborative labor pool—providing a labor advantage that increases long-term productivity<sup>44</sup>. Alternatively, it may improve scientists' access to high-capital international networks, raising later team reputation and thereby supporting long-term impact<sup>45</sup>. To examine whether the evidence is consistent with these accounts, we first test whether international collaboration predicts later-career labor advantage and team reputation. Both early-career international collaboration proportion and later-career team country size are positively associated with later-career coauthor count, indicating that international collaboration is linked to an expanded labor pool (Fig. S6d, e). Mob scientists accordingly exhibit a substantially larger coauthor pool than non-Mob peers (Fig. S6f),

and this larger labor pool in turn predicts long-term Publications (Fig. S6g; Table S15). Consistent with this interpretation, once coauthor count is taken into account, the association between team country size and Publications is attenuated and no longer significant (Table S15), suggesting that the productivity advantage of broader international collaboration may operate mainly through the larger labor pool it brings. Alongside labor advantages, broader international collaboration also appears to shape later team reputation. We find that both early-career international collaboration proportion and later-career team country size are positively associated with team reputation, and Mob scientists accordingly assemble teams with significantly higher reputation in years 6–25 (Fig. 3c; Fig. S6l;  $P < 0.001$ ). Hit 10pct increases monotonically with team reputation, and this association is further amplified in broader international teams (Fig. S6m; Table S15;  $P < 0.001$ ). Altogether, these results suggest that international collaboration may contribute to the long-term advantages of Mob scientists through two channels: by expanding labor advantages that support productivity and by raising later team reputation that supports impact.

Within the mobile group, the same two pathways continue to organize the divergence between shorter- and longer-distance movers, but in different directions. Longer-distance movers show a significantly higher early-career international collaboration proportion than shorter-distance movers, by approximately 19% (Fig. S7a;  $P < 0.001$ ), and this greater early exposure carries forward into a stronger labor advantage in later career, as reflected in their larger team size (Fig. S7b, d;  $P < 0.001$ ). Larger later-career team size, in turn, predicts higher long-term Publications (Table S16; Fig. S7d). Yet despite accumulating more extensive early international collaboration, longer-distance movers do not assemble higher-reputation teams. Later-career team reputation is in fact significantly higher among shorter-distance movers (Fig. 3i;  $P < 0.001$ ), and higher team reputation again predicts higher long-term Hit 10pct within the mobile group (Table S16; Fig. S7i). These two later-career dimensions may thus map closely onto the

two main performance differences within the mobile group: the productivity advantage of longer-distance movers and the impact advantage of shorter-distance movers.

Why does greater early international exposure of longer-distance movers carry forward into labor advantages but not equally into reputational ones? In the Mob versus non-Mob comparison, international collaboration breadth predicts both outcomes equally—yet within the mobile group, the pattern diverges. The answer may lie not in how much international collaboration scientists accumulate, but in whom they become connected to through it. We found shorter-distance movers work with significantly more collaborators from Western countries and top-100 institutions during the early career (Fig. S7e, g; both  $P < 0.001$ ), and both of these features positively predict later team reputation (Fig. S7f, h). This pattern holds across both comparisons: Mob scientists likewise work with significantly more collaborators from Western countries and top-100 institutions than non-Mob peers (Fig. S6h, j), and both features are positively associated with later team reputation (Fig. S6i, k). The breadth of early international collaboration thus appears more closely tied to labor advantages, whereas access to elite collaborative networks appears more important for reputational ones—a distinction that may explain why longer- and shorter-distance movers arrive at such different long-term outcomes.

Meanwhile, entering a new country often requires rebuilding working relationships, and we find that Mob scientists are less reliant on a fixed early-career core of collaborators: their early-career dense tie ratio<sup>46</sup> is significantly lower than that of non-Mob scientists (Fig. S8a;  $P < 0.001$ ). This early tendency toward forming new ties determines later network position. Dense tie ratio is negatively associated with later-career betweenness centrality (Fig. S8b), and Mob scientists accordingly occupy significantly higher-betweenness positions<sup>47</sup> in years 6–25 (Fig. 3d;  $P < 0.001$ ). Their collaboration networks are also more globally fragmented, with greater component-size diversity and a smaller relative giant component than those of non-Mob scientists (Fig. S8d, f; both

$P < 0.001$ ), reflecting a more brokerage-oriented and structurally diverse collaborative form. Such a position is associated with better long-term performance overall (Fig. 3e; Table S17). For impact specifically, bridging positions may offer visibility across disconnected communities, while high-reputation collaborators provide within-community endorsement—both remain significant when entered together, and together help account for the impact advantage of Mob scientists (Table S17, Panel c).

Within the mobile group, one might expect longer-distance scientists, who cross the largest cultural gaps, to form the most new ties. We find the opposite: longer-distance scientists exhibit a 38.9% higher early-career dense-tie ratio than shorter-distance scientists ( $P < 0.001$ ; Fig. S9a), and the same direction holds for early collaboration strength and early team repeated-coauthor proportion (Fig. S9b, c). When the environment they enter is culturally more unfamiliar, longer-distance movers appear to lean more heavily on a tight core of trusted collaborators rather than expand outward. This early repeat-tie orientation carries forward into later-career team assembly, where longer-distance scientists exhibit a higher team repeated-coauthor proportion, stronger direct-repeat preference, and greater retention of early-career collaborators (Fig. S9g–i; all  $P < 0.001$ ).

Expansionary team assembly may open access to new collaborative opportunities, while repeat-tie organization may sustain smoother coordination at the cost of such exposure<sup>48,49</sup>. Consistent with the former, shorter-distance scientists bring in more new collaborators per paper in later career than longer-distance scientists ( $P < 0.001$ ; Fig. 3g), and team new collaborators is positively associated with team reputation (Fig. 3i). Stratifying scientists by the median split of late-career team new collaborators, those in the higher group achieve higher long-term performance regardless of mobility status (Fig. 3h; all  $P < 0.05$ , with  $P < 0.001$  for Publications and Hit 10pct). Once team reputation is controlled, however, the association between team new collaborators and Hit 10pct attenuates (Table S16, Panel c), suggesting that the impact contribution of

new collaborators may operate mainly through the team reputation they bring. Expansionary team assembly, however, may also bring greater coordination frictions, and we find team new collaborators is negatively associated with coordination efficiency (Fig. 3i). Longer-distance movers, by contrast, exhibit significantly higher coordination efficiency (Fig. 3i;  $P < 0.001$ ), a pattern also reflected in their higher mean component density (Fig. S10f;  $P < 0.001$ ). Coordination efficiency itself predicts long-term Publications; once later-career team size and coordination efficiency are jointly introduced, the association between cultural distance and Publications attenuates and is no longer significant (Table S16, Panel b), suggesting that labor advantages and coordination efficiency together may fully account for the productivity advantage of longer-distance movers. Thus, shorter-distance movers appear to trade coordination efficiency for new collaborative opportunities, gaining impact through the team reputation these opportunities bring, whereas longer-distance movers trade such exposure for smoother repeat-tie coordination, gaining productivity through the labor and coordination advantages it sustains.

Shorter-distance scientists also occupy more brokerage-oriented positions—owing to their lower early-career dense-tie ratio—and exhibit higher component-size diversity and a smaller relative giant component than longer-distance movers (Fig. S9b–f; Fig. S10a, c, e). Nonetheless, shorter-distance scientists do not translate these brokerage positions into productivity and disruptiveness advantages—so why can Mob scientists do so while shorter-distance movers cannot? We find that Mob scientists, beyond occupying brokerage positions—likewise owing to their lower early-career dense-tie ratio—also exhibit significantly higher coordination efficiency and mean component density in later career than non-Mob peers, as longer-distance movers do (Fig. S8g, h;  $P < 0.001$ ). Both remain significant predictors of productivity when jointly entered with betweenness centrality, with a positive interaction that further amplifies their effect (Table S17, Panel b). Together with labor advantages, higher coordination efficiency



and betweenness centrality account for a 69.39% reduction in the cultural distance coefficient for Publications. Bridging positions may also promote innovation by connecting heterogeneous communities. Consistent with this, we find that betweenness centrality is positively associated with later-career team expertise distance, and Mob scientists accordingly assemble teams with significantly greater expertise heterogeneity than non-Mob peers (Table S17, Panel a;  $P < 0.001$ ). When entered jointly, only expertise distance remains significant for D5 percentile (Table S17, Panel d), suggesting that brokerage contributes to disruptiveness mainly by bringing together expertise-diverse teams. Within the mobile group, shorter-distance movers occupy central network positions but tend to expand outward rather than consolidate repeat ties, and their coordination efficiency is correspondingly lower—leaving them with weaker productivity. Meanwhile, although shorter-distance movers occupy more brokerage-oriented positions, longer-distance movers nonetheless assemble teams with greater expertise distance ( $P < 0.01$ ), and betweenness again loses significance once expertise distance is controlled (Table S18, Panel d). Shorter-distance movers' brokerage position, then, is accompanied by neither efficient coordination nor heterogeneous expertise—leaving both advantages largely absent.

Thus, later-career team expertise distance may serve as one pathway through which both Mob scientists and longer-distance movers outperform their peers. But what generates this greater expertise heterogeneity? Across the Nobel biographies mentioned earlier, the act of moving was inseparable from a pivot in knowledge, and greater early-career cultural mobility may likewise expose scientists to broader pools of expertise and prompt larger field pivots. Consistent with this intuition, both Mob and longer-distance scientists exhibit significantly larger early-career coauthor expertise distance and field pivot size than their peers (Fig. S11b, g; Fig. S12a, c; all  $P < 0.001$ ), and both measures positively predict later-career team expertise distance (Fig. S11d, i; Fig. S12b, d). One might further suspect that a lower early-career dense-tie ratio, reflecting

openness to new collaborators, could also drive such pivots, and Mob scientists' lower early-career dense-tie ratios are indeed associated with larger early coauthor expertise distance and field pivots (Fig. S11a, f). Yet, within mobile scientists, shorter-distance movers, who likewise tend to form new ties, pivot less than longer-distance movers. Prior work has shown that impactful scientists tend to draw their collaborators into new research topics, suggesting that the presence of substantially higher-capital coauthors may itself exert a gravitational pull toward new fields <sup>50,51</sup>. Consistent with this, we find that early-career coauthor capital imbalance—the unevenness of scientific capital among early collaborators—positively predicts both coauthor expertise distance and field pivot size (Fig. S12f, h), and once it is accounted for, the coefficient of cultural distance on these outcomes attenuates (Table S18, Panel b–c). Longer-distance movers, who both cross larger cultural distances and enter collaboration pools of more uneven capital (Fig. S12g;  $P < 0.001$ ), thus develop broader early knowledge trajectories and go on to assemble more expertise-diverse teams, even without relying on more new ties or occupying more brokerage-oriented positions. Moreover, this same imbalance, while positively associated with later team expertise distance—which drives disruptiveness—is negatively associated with later team reputation—which sustains impact (Fig. S12i, j, k). Longer-distance movers thus outperform shorter-distance peers in disruptiveness yet fall short in impact because the same capital imbalance that pulls them toward new fields also deprives them of the reputational lift a more balanced network would provide.

### **Cultural distance and knowledge strategy**

Mobile scientists pivoted more broadly across fields early in their careers and later assembled teams with greater expertise heterogeneity. Building on the classical framework of exploration versus exploitation <sup>52</sup>, one might therefore expect them to continue venturing across fields in their own later-career work. We next examine the knowledge strategies scientists pursue following early-career mobility, and how these strategies may shape their long-term performance. We find, however, that they do not:

Mob and non-Mob scientists exhibit no significant difference in late-career reference fields pivot size (one-sided Welch's t-test,  $P = 0.30$ ). Mobile scientists do not shift their focal papers' reference fields further from their prior publications, suggesting they do not pursue exploration by reaching into more distant fields. On the contrary, they show a stronger tendency toward exploitation—drawing each paper's references from a narrower set of fields—as captured by their significantly higher reference fields imbalance than non-Mob peers ( $P < 0.001$ ; Fig. 4a). Given mobile scientists' greater early-career exploration and later-career team expertise heterogeneity, what drives their later-career exploitation? Established reputational and international capital, while supporting impact within established communities, may also encourage deeper exploitation within a narrow set of knowledge bases rather than exploration beyond them. Consistent with this view, both team reputation and team country size—greater among mobile scientists, as shown above—are positively associated with later-career reference fields imbalance (Fig. S13a, c), and the same pattern holds at the country level, where mobile scientists also show significantly higher reference countries imbalance than non-Mob peers ( $P < 0.001$ ; Fig. S13b, e, h).

Such later-career exploitation is also tied to long-term performance: when scientists are stratified by reference fields imbalance, those in the higher-imbalance group achieve higher Hit 10pct and higher D5 percentile regardless of mobility status (all  $P < 0.001$ ), and higher publications among non-Mob scientists ( $P < 0.001$ ), though this difference is not significant among Mob scientists ( $P = 0.15$ ; Fig. 4b) than their lower-imbalance peers (Fig. 4b). This concentration may be valuable for mobile scientists, whose broad early-career field pivots might otherwise be expected to depress their later impact—a “pivot penalty” whereby venturing into unfamiliar fields lowers impact<sup>53</sup> because researchers have not yet accumulated the specialized expertise on which high-impact work depends. By drawing references from a narrow set of fields, mobile scientists may deepen their specialization and expertise, offsetting the penalty their early pivots might

otherwise have carried. Moreover, because mobile scientists show this later-career concentration after broad early-career pivots, it resembles the explore-then-exploit pattern that prior work links to creative “hot streaks”<sup>54,55</sup>—periods of unusually high-impact output. Consistent with this, later-career reference fields imbalance positively predicts impact, and this association strengthens with a scientist’s early-career field pivot size (interaction term  $P < 0.001$ ; Table S19, Panel a). Meanwhile, we find that high-reputation teams may not only help draw mobile scientists toward greater exploitation but also amplify the impact returns to such exploitation. Team reputation and reference fields imbalance each significantly predict Hit 10pct, and their positive interaction indicates that concentrated knowledge use yields greater impact when supported by higher team reputation. This suggests that, for work already concentrated within a narrow set of fields, high-reputation teams may lend visibility and credibility, consistent with a “chaperone effect.”<sup>56</sup> Further, once team reputation and reference fields imbalance are accounted for, the cultural distance coefficient for Hit 10pct is reduced substantially by 92.54%, indicating that reputational capital and later-career exploitation may jointly explain a substantial part of mobile scientists’ impact advantage (Table S19, Panel b).

Reference fields imbalance does substantially capture exploitation, as reflected in its positive correlation with the relative share of references a scientist draws from their own home fields—a stricter marker of within-field exploitation—with mobile scientists scoring higher on both measures (Fig. 4c). Yet concentrating references on a few fields need not mean concentrating within one’s own; imbalance can also absorb cross-field exploration, which may in turn relate to higher disruptiveness. We therefore decompose later-career referencing into the relative shares drawn from home versus global (non-home) fields as stricter alternative measures of exploitation and exploration. For impact, these conclusions are robust to operationalizing exploitation as the reference home fields relative share rather than imbalance. Mobile scientists again show an explore-

then-exploit pattern with a higher share of references from home fields and a lower share from global fields (Fig. 4c). Home fields share positively predicts impact, with its returns again rising with team reputation, and team reputation and home fields share still reduce the cultural distance coefficient for Hit 10pct substantially by 80.6% (Table S22, Panel a). Whereas within-field exploitation fosters the specialized expertise that drives impact, an extensive literature suggests that cross-field exploration may instead foster innovation<sup>57,58</sup>. Consistent with this, reference global fields relative share positively predicts D5 percentile (Table S22, Panel b). These results indicate that mobile scientists' impact advantage is plausibly promoted by later-career exploitation, whereas this exploitation channel may be unlikely to explain their disruptiveness advantage, which is more closely linked to cross-field exploration, a strategy on which mobile scientists score lower.

Within mobile scientists, the same exploitation–exploration framework reveals that shorter- and longer-distance movers continue to follow distinct knowledge strategies in later career. Echoing their early-career differences in field pivoting, shorter-distance movers show a stronger later-career exploitation tendency, with higher reference self-citation ratio and smaller reference fields pivot size. Longer-distance movers show the reverse pattern, with lower reference self-citation ratio and larger reference fields pivot size, consistent with a stronger tendency toward exploration (Fig. 4g, h). The stricter within-field exploitation versus cross-field exploration measures show the same contrast: shorter-distance movers draw a larger share of references from home fields, whereas longer-distance movers draw more from global fields (Fig. 4k). These strategies may arise from distinct collaboration patterns between the two groups. Higher team reputation among shorter-distance movers may support focused knowledge use within established fields, while longer-distance movers' stronger repeated-collaboration tendency may lower coordination costs and sustain the trust needed for cross-field exploration (Fig. 4g–j). Moreover, longer-distance movers' greater team

expertise heterogeneity may further enable broader cross-field knowledge integration, whereas their larger team capital imbalance may pull them toward more exploratory knowledge strategies (Fig. S14). These contrasting strategies map onto the two groups' divergent long-term outcomes. When mobile scientists are stratified by the share of references drawn from global, non-home fields, those in the higher-exploration group achieve higher Publications and D5 percentile but lower Hit 10pct than their lower-exploration peers (Fig. 4l; all  $P < 0.001$ ), aligning with the impact–disruptiveness divergence that separates shorter- from longer-distance movers. Thus, shorter-distance movers' impact advantage may track their stronger reliance on home fields. Reference home fields relative share positively predicts Hit 10pct, and this association strengthens with team reputation, consistent with the same chaperone-style amplification observed among mobile scientists overall. Once team reputation and home fields share are accounted for, the cultural-distance coefficient for Hit 10pct is reduced by 41.96% (Table S20, Panel a). Longer-distance movers' disruptiveness advantage, by contrast, may track their stronger reliance on global fields. Reference global fields share positively predicts D5 percentile, but this association turns negative once team expertise distance is included, while the interaction between global fields share and team expertise distance is significantly positive (Table S20, Panel b). This suggests that cross-field exploration translates into disruptiveness mainly when embedded in expertise-heterogeneous teams, a condition longer-distance movers more often meet. Altogether, shorter-distance movers' impact advantage is linked to reputation-supported exploitation, whereas longer-distance movers' disruptiveness advantage is tied to exploration that is embedded in expertise-heterogeneous teams.

The exploitation and exploration examined so far are local, describing how scientists allocate references within individual papers. Across a whole career, however, a scientist who concentrates on narrow fields or draws heavily on home fields within individual papers (exploiting locally) may still draw on diverse fields and range widely across the

knowledge landscape (exploring globally). Turning to the career scale, Mob scientists and shorter-distance movers occupy more brokerage-oriented positions in their later careers, and by connecting otherwise separated communities, such positions may expose them to heterogeneous collaborators and fields, allowing them to integrate a broader range of fields over time. We find that the local pattern reverses at the career scale. In how they draw on knowledge across their later-career reference lists, Mob scientists spread their referencing propensity more evenly across fields than non-Mob peers and lean more heavily on fields beyond those in which they themselves have published (higher career fields propensity diversity and career non-focal fields propensity; both  $P < 0.001$ ; Fig. 4d, Fig. S15d). This broader exploration spans not only what they reference but also where they publish, with their actual research footprint covering a wider set of fields overall (higher career fields coverage;  $P < 0.001$ ; Fig. 4f). All three measures are associated with later-career betweenness centrality (Fig. S15a–c), consistent with this career exploration being linked to their brokerage positions. The same pattern holds within the mobile group, where shorter-distance movers exhibit modestly higher career exploration than longer-distance movers (Fig. S16d–f; career fields coverage and career non-focal fields propensity,  $P < 0.001$ ; career fields propensity diversity,  $P < 0.05$ ), and these measures are again associated with betweenness centrality (Fig. S16a–c). Altogether, Mob scientists and shorter-distance movers exploit locally yet explore globally, whereas longer-distance movers explore locally yet exploit globally.

As the body of accumulated knowledge grows, scientists face a rising burden of knowledge<sup>27</sup>, and we have already seen Mob and longer-distance scientists meet it at the paper level by assembling teams of greater expertise distance, pooling the specialized knowledge no single member commands alone—a solution that predicts their disruptiveness. In addition to pooling breadth within teams, a scientist may carry it personally, conducting career-wide exploration by integrating knowledge from

diverse fields and publishing across more fields over the career. As a Renaissance-like scientist, such a scientist may also meet this burden <sup>27</sup>, translating it into stronger long-term performance and a higher capacity for innovation. For Mob scientists and their non-Mob peers, stratifying by career fields propensity diversity shows that those in the higher group achieve stronger long-term performance, including disruptiveness (Fig. 4e; all  $P < 0.001$ ). Career fields propensity diversity further predicts D5 percentile even after accounting for team expertise distance, and the two channels can amplify each other (Table S21, Panel c). Within the mobile group, the same reward to Renaissance-like career exploration recurs. Career fields propensity diversity is again positively associated with D5 percentile, indicating this career exploration fosters innovation among mobile scientists as well. Once it is held constant, netting out the edge that shorter-distance movers hold on it, the disruptiveness advantage of longer-distance movers over shorter-distance movers grows larger (Table S20, Panel b). In all, cross-field exploration within papers (local exploration), which translates into disruption only when embedded in expertise-heterogeneous teams, and career-wide field breadth (global exploration), which a scientist carries personally, constitute two potential channels to long-term disruptiveness across both comparisons.

### **Cultural distance and knowledge source**

Beyond shaping how scientists integrate knowledge across fields, early-career mobility may also reposition scientists within the global scientific system and reshape where their later work draws from. A large body of work describes this system as having a core-periphery structure <sup>59,60</sup> in which positions differ in the resources, credibility, and visibility they confer. We first treat the world's leading institutions as the institutional core, marking the frontier of science, and all other institutions as its periphery. Mob scientists have more early-career exposure to this frontier, which may give them access to its resources and global visibility and leave a lasting imprint on where they later draw knowledge from. Consistent with this, Mob scientists are more likely than non-Mob



scientists to draw on knowledge from top institutions in later career, as reflected in higher career top institutions propensity (one-sided Welch's t-test,  $P < 0.001$ ; Fig. 5a), and this higher propensity is linked to their greater early-career collaboration with top institutions. The Mob advantage in career top institution propensity is more pronounced among scientists whose later-career institution lies outside the top-100, where Mob scientists keep their work anchored to the frontier even from peripheral positions and bring core knowledge into peripheral settings, and this peripheral group accounts for much of the overall higher career top institution propensity (Table S23; Supplementary Note 3.4.1).

We next use a classical, alternative core-periphery structure, treating Western countries as the geopolitical core and all other countries as its periphery. The core, the dominant national centers of global science, carries visibility advantages and reflects national citation biases<sup>59,61–63</sup>, while greater reliance on the periphery signals a broader geographic reach of knowledge sourcing. In contrast to the institutional pattern, Mob scientists are less likely than non-Mob scientists to draw on Western-country knowledge in later career, as reflected in lower career western countries propensity (one-sided Welch's t-test,  $P < 0.001$ ), and this lower propensity is associated with their lower later-career collaboration with Western-country partners, suggesting that country-level core sourcing may be more tied to the contemporaneous collaborative environment (Table S23; Fig. S18; Supplementary Note 3.4.1). Yet we observe the same bridging pattern as at the institution level, and here a fully bidirectional one. Mob scientists not only draw more on Western-country knowledge than non-Mob scientists when located in non-Western countries, against their lower overall tendency, but even draw more on non-Western knowledge when located in Western countries, and this subgroup's reversal accounts for much of their overall lower career western countries propensity (Supplementary Note 3.4.1). Combined with the institutional results, most Mob scientists are located in peripheral institutions and in core countries (Table S23),

reaching toward the core (top institution) when on the periphery (non-top institution) and toward the periphery (non-western country) when at the core (western country), indicating they occupy a context-dependent bridging position between the core and periphery of the global knowledge system. Within mobile scientists, we observe a consistent pattern across the two core–periphery measures. Shorter-distance movers have higher career top institutions propensity and higher career western countries propensity than longer-distance movers (one-sided Welch’s t-tests, both  $P < 0.001$ ), consistent with their greater exposure to top-institution and Western-country collaborators in both early and later career (Fig. 5d; Supplementary Note 3.4.1).

Drawing more heavily on top-institution knowledge may confer visibility and reflect access to elite ties formed early in scientists’ careers, potentially contributing to the long-term advantages of mobile scientists. Consistent with this, scientists with higher career top institutions propensity achieve higher performance than those with lower propensity across both the Mob and non-Mob groups (Fig. 5b; all  $P < 0.05$ , with  $P < 0.001$  for Publications and Hit 10pct). However, the advantage in D5 percentile is significantly weaker compared with Hit 10pct ( $P < 0.001$  among non-Mob scientists;  $P = 0.047$  among Mob scientists; Fig. 5b). Simultaneously, for impact, although career top institutions propensity is positively associated with Hit 10pct, it only marginally increases the attenuation of the cultural-distance effect after accounting for team reputation and paper-level local exploitation, suggesting that top-institution knowledge sourcing mainly overlaps with the visibility and credibility channels already captured by these two factors, rather than constituting a primary channel of its own (Table S24). This visibility also extends downstream, as Mob scientists’ impact is more geographically dispersed, which may be partly linked to their greater propensity to draw on top-institution knowledge, given that frontier knowledge may be more readily recognized and circulated across the global scientific system. Turning to career western countries propensity, it reflects an orientation toward Western countries as the dominant

centers of science. Drawing more on this core may raise impact not only by building on more visible and more credited knowledge but also through a national citation bias favoring these centers, while a more limited reach into peripheral knowledge may constrain disruptiveness. We find that scientists with higher career western countries propensity achieve higher Hit 10pct but lower Publications and lower D5 percentile than those with lower propensity within both the shorter and longer groups (Fig. 5e; one-sided Mann-Whitney U tests, all  $P < 0.001$ ). Career western countries propensity remains positively associated with Hit 10pct after team reputation and paper-level local exploitation are included, and negatively associated with D5 percentile after team expertise distance is accounted for (Table S24). It thus appears to channel the recognition of core-country knowledge into the impact advantage of shorter movers and a weaker reliance on it into the disruptiveness advantage of longer movers.

Beyond its position in the core-periphery hierarchy, the knowledge a scientist draws from can also be described by the breadth of this search. A broad literature holds that innovation is influenced by the reach of knowledge actors' search portfolios<sup>8,9,64,65</sup>, from familiar and usual pools that may raise impact to more distant and diverse ones that may raise disruptiveness. Embedded in different knowledge and collaboration environments after mobility, scientists with different early-career cultural distance may hold different search portfolios, related to their divergent impact and disruptiveness across both comparisons. Accordingly, we first find that longer movers draw on more geographically distant knowledge than shorter movers (Fig. 5g), associated with their weaker reliance on the Western core (Fig. 5f), their smaller number of Western later-career collaborators, and their more geographically dispersed later-career teams (Supplementary Note 3.4.2). In both groups, higher reference geographic distance goes with higher Publications and D5 percentile but lower Hit 10pct (Fig. 5h; one-sided Mann-Whitney U tests, all  $P < 0.001$ ). For impact, reference geographic distance is still negatively associated with Hit 10pct when team reputation, paper-level local

exploitation, and career western countries propensity are included, but those three already explain 48.34% of the impact difference and adding it raises this marginally to 51.71% (Table S25), so we focus on team capital, deepening field expertise, and a national citation bias as the primary channels of the impact advantage of shorter-distance movers. Turning to long-term disruptiveness, the geographic breadth that added marginal to shorter movers' impact appears to explain more of longer movers' disruptiveness advantage. After a wider team expertise distance and a more peripheral-oriented knowledge sourcing are accounted for, which together explain 10.44% of longer movers' disruptiveness advantage, reference geographic distance remains positively associated with D5 percentile and raises this share to 17.54% (Table S27). This wider geographic reach also carries through to the citation side, where longer movers show a lower career western countries citation propensity (Fig. 5i) and their work diffuses across a more geographically dispersed set of citing locations (Supplementary Note 4.1).

A wider search reaches not only across space but also across time. Reaching back to older work, captured by reference age, can signal a wider temporal search, yet earlier work suggests that a larger reference age across a career may also reflect a nostalgic reliance on familiar old work <sup>66</sup>. We find, though, that longer movers have not only higher reference age but also higher reference popularity, the prominence of the cited work, than shorter movers. This suggests that they reach far back to work that is also widely cited and thus build on classic, established foundations in the manner of standing on the shoulders of giants, rather than returning nostalgically to obscure old work or following recent prominent work. Once a wider team expertise distance, a more peripheral source orientation, and wider geographic reach are accounted for, reference age and reference popularity are both positively associated with D5 percentile, and reinforce each other through a positive interaction, indicating that standing on the shoulders of giants raises disruptiveness more when it is combined with a broader

search. This search behavior further increases the explained share of longer movers' disruptiveness advantage from 17.54% to 28.18% (Table S27), suggesting that the disruptiveness advantage of longer movers rests on wider team expertise distance, a more peripheral source orientation, and a broad search that reaches across both space and time combined with stand on the shoulders of giants. This broader search portfolio likewise helps explain the long-term disruptiveness advantage of movers. We find that Mob scientists also search broader and stand on the shoulders of giants with higher reference geographic distance, associated with the same weaker reliance on the Western core and with more geographically dispersed later-career teams, reference age, and reference popularity than their peers (Supplementary Note 3.4.2). Once a wider team expertise distance, a broader career field exploration, and a wider geographic reach are accounted for, reference age and reference popularity are again both positively associated with D5 percentile and reinforce each other through a positive interaction, raising the explained share of movers' disruptiveness advantage from 39.59% to 56.38% (Table S28). This advantage thus rests on a wider team expertise distance, a broader career field exploration, and a broad search combined with standing on the shoulders of giants.

Although standing on the shoulders of giants and searching more broadly across geography may increase average disruptiveness, the reliance on prominent, established work may also limit the most radical ideas, which sit in the tail of the disruptiveness distribution, where, straying furthest from convention, they can produce either the least or the most disruptive work of all, and a broader geographic search may further limit access to tacit knowledge, which underpins many radical breakthroughs yet travels mainly through frequent interaction, localized knowledge spillovers, and face-to-face contact within knowledge clusters<sup>29,30,67–70</sup>, and so passes less easily over distance. We indeed find that Mob and Longer scientists, despite their higher average disruptiveness, are significantly less likely to fall at the two-tailed extreme, the top and bottom of the

distribution, than their peers (29.78% and 14.65% lower; one-sided Welch's t-tests, both  $P < 0.001$ ; Fig. S19a, i), a separation between the center and the tail across both comparisons. The same center–tail separation also appears in conventionality and novelty, and in the average and dispersion of reference semantic distance (Supplementary Note 3.3). Consistent with this, reference popularity is positively associated with average disruptiveness yet negatively with the two-tailed extreme across both comparisons, and so are team geographic distance and reference geographic distance, which show the same pattern not only for disruptiveness but also for atypical novelty (Fig. S19c–e, k–m; Supplementary Note 3.3). Although tacit knowledge and the most radical ideas both resist direct measurement, a combination of prior work that has never been joined before may reflect both the radical recombination itself and, since connections already available in the codified record are seldom left unmade for long, the hard-to-codify tacit knowledge that lies behind it. We therefore use reference independence<sup>71</sup>, which contrasts the share of a paper's references that cite no other work in its reference list against the share that do, as a proxy for both. Accordingly, standing on the shoulders of giants, searching more broadly across geography, and assembling more dispersed teams are all negatively associated with reference independence, and Mob and Longer scientists draw on less of it (44.76% and 7.73% lower, respectively;  $P < 0.001$  and  $P < 0.05$ ; Fig. S19f–g, n–o). Reference independence is in turn positively associated with both mean and two-tailed extreme disruptiveness, and its effect concentrates at the tail: a one-standard-deviation increase raises average disruptiveness by 15.4%, two-tailed extreme disruptiveness by 87.2%, and upper-tail disruptiveness by 132.9% (Fig. S19h, p). Altogether, standing on the shoulders of giants and searching more broadly across geography raise the center of the disruptiveness distribution, while the reliance on prominent, established work and the frictions in tacit knowledge transmission they entail leave Mob and Longer scientists with fewer of these independent, first-time combinations, holding down its tail.

## The shifting structure of scientific mobility

The divergence documented above implies that a change in how far scientists move can bear on the aggregate record of science, since it redistributes the population across groups whose long-term outcomes differ. Both the prevalence and the distance of early-career mobility have changed over the period our sample spans. Scientists became steadily more mobile across the 1956–1996 entry cohorts (Fig. S0b), and among the 1997–2016 cohorts the share affiliated with two or more countries during career years 1–5 rose from 5.97% to 8.98%, an increase of 50.4% (Fig. 6a). Applying throughout the tercile thresholds defined above over all 1956–1996 movers, the longer-distance share rose from 19.09% to 34.93% between the 1956–1976 and 1977–1996 periods (Fig. 6b), and among the 1997–2016 cohorts it rose further from 37.13% to 54.56% while the shorter-distance share fell from 31.72% to 19.86% (Fig. 6a).

To isolate what these shifts imply on their own, we hold each group's mean outcome fixed at its observed value and vary only the weights with which the four groups enter the matched sample, so that the exercise asks what the mean long-term disruptiveness of this same population would have been had it been weighted as in another era. Three of the four resulting environments are observed directly: that of 1956–1976, in which mobility was least prevalent and shorter moves most heavily represented, lies 0.91% below the 1956–1996 baseline, and that of 1977–1996 lies 0.12% above it, an ascent of just over one percentage point traced without any calibration (Fig. 6b). Extending the same construction to 1997–2016 places that environment 1.29% above baseline, of which the change in composition among movers accounts for 0.67% and the rising mover share for 0.47%, the remainder being the interaction between two channels that move in the same direction (Fig. 6b; Supplementary Note 4.3). Holding the total sample fixed and reassigning scientists from marginal non-mobility, a 1% increase in the mover count raises the mean D5 percentile by 0.0198% when the reassigned scientists follow the observed distance composition and by 0.0471% when they are assigned entirely to

longer-distance mobility, an additional gain of 0.0272% from going farther rather than merely becoming mobile. A less mobile and less culturally distant workforce therefore corresponds to lower, not higher, mean long-term disruptiveness, and how far scientists go matters at least as much as whether they go at all, as the widening of the combinatorial space anticipates <sup>13</sup>.

We next ask whether the same shift has thinned the top, since the groups it has favoured are the ones less often found there (Fig. S19). Against an absolute threshold fixed at the 95th percentile of the raw D5 measure, computed once over the complete matched sample and applied unchanged to every cohort decade, the share of scientists clearing the bar fell from 11.89% to 4.07% while their absolute number rose from 265 to 2,459 (Fig. 6c), echoing at the level of scientists the pattern previously reported for highly disruptive papers and patents <sup>20</sup>. A counterfactual holding the four-group composition at its 1956–1966 values, and letting only within-group rates follow their observed history, tracks the actual series to within 0.14 percentage points in every decade, under 2% of the 7.82-percentage-point decline (Fig. 6c). An exact three-factor Shapley decomposition of the change in counts returns the same verdict, in that expansion of the scientific workforce contributes +4,665 scientists relative to the 1956–1966 baseline and falling within-group rates −2,454, while compositional change contributes at most 17 in any decade, or 0.76% of the net change of +2,194 (Fig. 6d). The thinning of the top is a within-group phenomenon, in that every mobility group has become less likely to clear the same absolute bar than its predecessors, irrespective of who moves or how far.

At the level of a career the cost at that same bar is substantial, with Mob scientists 22.50% less likely than matched non-mobile peers to fall in the upper 5% of the disruptiveness distribution and longer-distance movers 10.11% less likely than shorter-distance movers, a cost whose sources are documented above in the geographic breadth of search and collaboration that raises the average while impeding the transmission of tacit



knowledge and the assembly of independent recombinations (Fig. S19), and consistent with an account of cultural variance in which the diversity that supplies variants for recombination is the same diversity that lowers the fidelity with which what resists the written record is passed on<sup>72</sup>. At the level of the system, however, the contribution of any group difference to an aggregate change is the product of the difference itself and the distance the weights have actually travelled, and because the mover share and the distance composition shifted modestly against the growth of successive cohorts and the steep fall in within-group rates, a deficit of that magnitude resolves into at most 17 scientists per decade. The erosion of the share clearing the threshold traces instead to forces that lie outside mobility altogether<sup>26,27,73,74</sup>. Taken together at the scale of the population, greater and more distant mobility raises the centre of the disruptiveness distribution without materially eroding its top, so that what the structural transformation of mobility has changed is where the distribution of disruptive science sits, and not whether its extreme remains reachable.

## Discussion

For an individual scientist, the mobility dividend is real, but it divides. Moving early is associated with more publications, more highly cited work and more disruptive work than staying, yet among those who move, that dividend does not grow with cultural distance so much as split along it, and no distance delivers all of it. A shorter move enters a research setting near enough to join. Those who make one form more new ties in the early career and work with more partners from Western countries and leading institutions, and go on to assemble teams of higher standing and to deepen within familiar fields, which is where their advantage in citation impact is found; they also reach team leadership sooner than those who move farther, publish in more visible venues, see their work taken up more often in patented inventions, and are twice as likely to receive a Nobel, Fields or Turing award. A longer move recombines both the

collaborators a scientist works with and the material they work on. It returns a wider pool of collaborators, and because the setting entered is less like the one left behind, a heavier reliance on those already at hand, which makes coordination the smoother; between them these account for the advantage in output. Entering a setting less familiar, these scientists find fewer of their early collaborators in Western countries and leading institutions and those they do find differ more widely in scientific capital, so that the teams they assemble years later stand lower than a shorter move affords; that same unevenness accompanies a pivot into fields further from their own, and the pivot, together with the wider material a distant setting opens, is what carries them into the expertise-heterogeneous teams and the far-reaching search on which departure from the past rests.

The frontier receded long ago beyond what any one scientist could carry to it, and the burden of knowledge that followed was held to have killed the Renaissance man, leaving each researcher master of an ever smaller part of an ever larger whole<sup>27</sup>. What our results suggest is that a scientist may still recover breadth in either of two ways, and that both are associated with disruptive work: it may be pooled inside a team, so that one paper integrates what no member commands alone, or carried personally across a career, so that a scientist covers over the years what no single project contains. An early move supplies both. Set among unfamiliar material, mobile scientists pivot further across fields in their first years than those who stayed, and the teams they assemble two decades later draw on expertise lying correspondingly further apart; over a career they also cover more fields than their matched peers, even as the references of any one paper concentrate more narrowly. That narrowing accompanies the higher standing of the teams they assemble, and it does not come at the expense of the breadth they hold across the career, which the unfamiliar material and collaborators of the early years supply instead. Within the mobile group the two routes separate. Longer-distance movers, whose pivots are the largest, take the first, and their expertise-heterogeneous teams

reach across fields within a paper while their careers hold to fewer; shorter-distance movers, who form new ties rather than return to an early core and later stand between more communities, take the second, covering more fields over a career and deepening within each paper. Mobility does not restore the Renaissance man. It offers two substitutes for him, one assembled and one carried, and cultural distance decides which a scientist receives.

Neither mobile scientists nor longer-distance movers lift the extreme, and for the same reason in both cases. The breadth of search and collaboration that lifts average disruptiveness is carried across the geographic and cultural spans over which what resists the written record travels worst<sup>29,30</sup>, the same variance that supplies the variants for recombination lowering the fidelity with which they are passed on<sup>72</sup>; and it rests on prominent prior work, shoulders every member of a geographically dispersed team can see. Mobile scientists and longer-distance movers accordingly assemble fewer of the independent, first-time combinations on which the most extreme work depends, and reach the upper tail of the disruptiveness distribution less often than the groups they are matched against. The mobility dividend, on this evidence, raises the mean of career-level disruptiveness while leaving its extremes less often reached, and what governs the extreme is not what governs the centre. Prizes point the same way from the other side. Within the mobile group they follow the tail, in that shorter-distance movers reach it more often and are also twice as likely to receive a Nobel, Fields or Turing award; between movers and those who stayed they part from it, in that mobile scientists reach that tail less often and are honoured twice as often nonetheless, a discrepancy in which the visibility a move confers, in its venues and in the reach of the work itself, may have a part. Where a scientist goes decides where in that distribution they sit; it does not decide whether its extreme remains within reach.

At the scale of the system the same evidence answers a different question. Both the prevalence of early mobility and the distance it spans rose across the cohorts we observe,

redistributing scientists across groups whose long-term outcomes differ, and holding those outcomes fixed while varying only the weights isolates what that redistribution has done on its own. It raised the centre. A less mobile and less culturally distant workforce corresponds to lower, not higher, mean long-term disruptiveness, and when the same scientists are drawn into mobility, sending them the farthest distances adds more to the mean than making them mobile at all did, so that how far they go matters at least as much as whether they go. It did not erode the top. The share of scientists clearing a fixed absolute bar fell while their absolute number rose, and a decomposition attributes the fall to declining rates within every mobility group and the rise to the growth of the scientific workforce, leaving compositional change under a fiftieth of the fall in the share and under a hundredth of the change in the number. That the career-level and the population-level answers differ is not a contradiction, since the contribution of any group difference to an aggregate change is the product of the difference itself and the distance the weights have actually travelled, so that a deficit large enough to weigh on a career can be negligible in a population that has moved as little along that dimension as this one has. The first of two levers follows directly. Because an instrument acting on the weights between groups cannot reach a process operating within them, science policy that restricts the early movement of researchers would lower the centre without recovering the top. Cultural distance may warrant particular protection, since what a distant move returns is at least as large as what movement itself returns, and the longer moves may be the first that any contraction forecloses <sup>75,76</sup>. Universities and the agencies that fund them may count years spent abroad toward promotion and permanent appointment rather than against them, since it is in the first five years, when scientists move most, that a move sets the two decades which follow.

The costs of restriction would not fall evenly. They would fall first on the periphery <sup>77</sup>. About three-quarters of mobile scientists work mainly outside the leading institutions

in later career, and from there they keep more elite collaborators and draw more heavily on frontier knowledge than their non-mobile peers, so that restricting mobility would narrow the access of peripheral communities to the frontier itself. Science diffusion would also narrow with it, since the work of mobile scientists is taken up across a more geographically dispersed set of citing locations, and longer-distance movers carry it farthest of all, and beyond the fields in which it was published; restricting distance would cut that reach even where mobility itself survived. What slows is catch-up, not circulation alone — and it is precisely the weakening of diffusion from the frontier to those working behind it that has been identified, in science and in the wider economy alike, as a dominant force behind the slowdown documented above<sup>24,78</sup>. The core would not be spared either. Mobile scientists located inside Western countries, as about seven in ten of them are, rely less exclusively on Western sources than their non-mobile peers, while those located outside rely more, and longer-distance movers extend this furthest, holding fewer Western and elite collaborators and reaching relatively more toward peripheral sources. Restriction of either kind would cut the supply of heterogeneity on which the centres depend, at a frontier where attention is already crowded enough that citations concentrate on established work, the canon ossifies, and new contributions displace less of what precedes them<sup>28</sup>. Restriction would also cost science one of its remedies for the burden of knowledge. Mobility supplies both of the substitutes for the Renaissance man that no scientist now commands alone, and the divergence within it means the two are supplied by different moves, so that a contraction of either kind removes one of them. The binding constraint may lie less in the creation of frontier knowledge than in the capacity of the system as a whole to absorb, diffuse and recombine it<sup>24,28,62,78</sup>, and mobility, together with the distance it spans, is among the few levers that acts on that capacity directly — which is why a contraction would show up not in the extreme but in the average rate at which science advances.

The second lever is the size of the scientific workforce, which governs how much of

the extreme is realised. Over the period we observe, expansion placed nearly twice as many scientists above the fixed threshold as declining within-group rates removed, making it the one force that has kept the absolute number of frontier contributions growing even as the return to each additional scientist has fallen. Two routes lead to it, and both lie largely outside science policy as it is usually drawn. The first is to find the lost Einsteins, raising the share of a population that becomes scientists at all through early exposure to inventive work <sup>79</sup>; the second, once that share is raised, is to enlarge the population itself, and so to postpone the empty planet <sup>80</sup>. Ministries and agencies concerned with the frontier may therefore find that measures bearing on early exposure, on immigration and on the cost of raising children act on the tail of the disruptiveness distribution more directly than any instrument aimed at the scientists already inside it.

Our conclusions are not without limits. The channels we trace are associations rather than identified effects, and the reweighting holds each group's outcome fixed, describing a population reweighted rather than one that behaved differently. What survives them is a distinction worth carrying beyond mobility, namely that the conditions which raise the average of a distribution and those which extend its extreme need not be the same. The half-century this study covers is one in which the barriers to where scientists may work came down, and what came through them was not only people but the problems, methods and traditions that do not travel on the page. Free movement of that kind raises the level at which a scientific system ordinarily works, and on this evidence it has done so without costing the system its extremes. What it cannot do is manufacture them, and that limit is a ceiling rather than a price.

## Methods

### Data

Our analyses draw on four sources. Publication records, disambiguated author identifiers, institutional affiliations, fields of study and citation links come from the Microsoft Academic Graph (MAG), in the curated form provided by SciSciNet<sup>81</sup>, which contains 134,129,188 publications and 1,588,739,703 citation links and covers records through March 2022. Country-level cultural dimensions come from Hofstede Insights (2023). Citation counts for papers published after 2017, together with funder and award information, are supplemented from OpenAlex (release 27 February 2025). Funding acknowledgements to the National Science Foundation and the National Institutes of Health, patent linkages and altmetric mentions come from SciSciNet. Full details of each source and of the linkage procedures are given in Supplementary Note 1.1.

### Sample construction

We used disambiguated author data from MAG to construct longitudinal publication records for scientists whose first career year fell between 1956 and 1996. We restricted our analysis to scientists with a career length of at least 20 years and with a minimum of 10 publications in their first 20 years. To ensure consistent coverage of career stages, we required that each individual published at least one paper in each of the following intervals: years 1–5 (early career), 6–10 (early-mid), 11–15 (mid), and 16–20 (late-mid). Additionally, scientists were included only if at least one institutional affiliation in their first five career years could be assigned to a country with Hofstede cultural dimension data. The resulting sample comprises 584,712 scientists and 58,114,053 author–paper records. The vast majority of the sample (97.49%;  $n = 570,062$ ) had careers of at least 21 years, and 84.64% ( $n = 494,906$ ) had careers of 25 years or longer. Accordingly, our main analyses focus on the first 25 career years, divided into five five-year periods:

early career (years 1–5), early-mid (6–10), mid (11–15), late-mid (16–20), and late (21–25), yielding a total of 38,339,950 author–paper records for downstream analysis.

A scientist is classified as mobile (“Mob”) if affiliations in more than one country are observed during career years 1–5, and as non-mobile (“non-Mob”) otherwise, yielding 47,838 mobile and 536,874 non-mobile scientists. Within the mobile group, scientists are divided into three equally sized subgroups — Shorter, Medium and Longer — by the terciles of early-career cultural distance; all within-mobile comparisons contrast the Shorter and Longer terciles. To assess the accuracy of the author–institution–country mapping procedure, we manually validated a random sample of 200 scientists (100 mobile and 100 non-mobile) from our main dataset. For each individual, we gathered publicly available biographical information (e.g., Wikipedia, CVs) to verify their country affiliations during the first five career years, and compared these to the algorithmic assignments. This validation yielded a recall of 95%, precision of 76%, and an overall accuracy of 86% for mobility status ( $AUC = 0.875$ ). Exact country list assignments matched the ground truth for 95% of non-mobile and 70% of mobile scientists. Among the 30 mismatches observed in the mobile group, 24 were due to non-mobile scientists being incorrectly labeled as mobile. In the remaining six cases involving true mobile scientists, five had one additional country assigned compared to the ground truth (with all other countries matching), and one case involved a missing country (with the rest matching exactly). When we restrict attention to truly mobile scientists, country-list accuracy is 92% (70 of 76); among truly non-mobile scientists it is 95% (95 of 99). Because the dominant error is the misclassification of non-mobile scientists as mobile, this attenuates rather than manufactures the contrasts we report (Supplementary Note 1.2).

### **Early-career cultural distance**

We measure cultural distance using the six national culture dimensions of Hofstede and



colleagues: power distance (pdi), individualism (idv), masculinity (mas), uncertainty avoidance (uai), long-term orientation (ltowvs) and indulgence (ivr). Scores are available for 119 countries; after excluding those with missing values and imputing the indulgence score for Israel from its five closest neighbours on the remaining dimensions, 99 countries remain, together accounting for 99.07% of publication records in the sample. Each dimension is rescaled to the unit interval by its range across these countries so that no dimension dominates by virtue of its original scale. The cultural distance between two countries  $m$  and  $n$  is the sum of absolute differences across the six rescaled dimensions:

$$d(m, n) = \sum_{i \in I} |m_i - n_i|, \quad I = \{pdi, idv, mas, uai, ltowvs, ivr\} \quad (1)$$

and early-career cultural distance is the mean of this quantity over all pairs of the  $k$  countries in which a scientist held an affiliation during career years 1–5:

$$\text{Cultural Distance} = \frac{1}{k^2} \sum_{m=1}^k \sum_{n=1}^k d(m, n) \quad (2)$$

The measure equals zero for non-mobile scientists, for whom  $k = 1$ . Conclusions are unchanged under the unnormalized distance, a percentile transformation of each dimension, restriction to unordered pairs, and a Euclidean alternative (Supplementary Note 3.1.4). Because the Hofstede framework is survey-based, time-invariant and unavailable for some countries, we additionally construct behaviour-based measures by learning country embeddings from the mobility trajectories of 4.2 million scientists using three unsupervised representation learning approaches, and reproduce all substantive conclusions under each (Supplementary Note 2; Tables S7 and S29).

## Coarsened Exact Matching

To ensure comparability between mobile and non-mobile scientists, we implemented one-to-one coarsened exact matching (CEM)<sup>33</sup>. Matching was conducted based on early-career research trajectory<sup>82</sup>, including publication count in the first five years, the proportion of

highly-cited papers (top 10th percentile of citations for a given year and field), early-career D5, as well as mean coauthor publication count and the average h-index of affiliated institutions during the early career stage to reflect coauthor productivity and institutional prominence<sup>45</sup>.

All features were coarsened into quantile-based bins (see Table S1 for summary statistics and covariate balance before and after matching). Each mobile scientist was matched to a non-mobile scientist with identical coarsened covariate values using 1:1 matching without replacement. Table S1 reports pre- and post-matching means, mean differences, standardized mean differences (SMD), and two-sample t-test p-values for each covariate. After matching, all SMD values dropped below 0.02, indicating substantial covariate balance. As a robustness check, we repeated our analysis using propensity score matching (PSM) as an alternative method, and found all substantive conclusions to be robust.

## **Outcome measures**

Long-term academic performance is measured over career years 6–25 along three dimensions. Productivity is the total number of publications. Impact is the proportion of a scientist’s papers that fall in the top decile of citations among all papers published in the same year and the same MAG level-1 field (“Hit 10pct”). Disruptiveness is obtained by computing the five-year disruption index D5 for each paper, averaging the values across a scientist’s papers, and converting the resulting author-level distribution to a percentile rank across the matched sample (“D5 percentile”); Definitions and details of the D5 calculation are provided in Supplementary Note 1.5.

## **Measures of collaboration, knowledge strategy and knowledge sources**

Three families of later-career measures are used to characterize the pathways linking early-career cultural distance to long-term performance. Collaboration measures include team country size, coauthor count and per-paper team size, team reputation, betweenness centrality and other properties of the collaboration network, the number of new collaborators per paper,

coordination efficiency, and team expertise distance (Supplementary Note 3.2). Knowledge-strategy measures include the concentration and field composition of a paper's references, the size of the shift in referenced fields relative to a scientist's prior work, and career-level measures of how broadly and evenly a scientist draws on fields across years 6–25 (Supplementary Note 3.3). Knowledge-source measures include the propensity to reference work from top-100 institutions and from Western countries, the geographic distance, age and prominence of cited work, and the corresponding citation-side measures (Supplementary Note 3.4). Each is defined in full where it is first used.

### **Statistical analysis**

Group contrasts are tested with one-sided Welch's t-tests in the direction reported, and contrasts between the lower and higher halves of a measure with one-sided Mann–Whitney U tests, the latter chosen because the outcome distributions are right-skewed. Lower and higher groups are defined by a median split of the matched sample for comparisons between Mob and non-Mob scientists and of the mobile sample for comparisons between Shorter- and Longer-distance movers. Relationships between continuous measures are displayed by grouping scientists into eight equally sized bins of the horizontal-axis variable and plotting the mean of each bin, with values expressed relative to the sample mean; error bars and shaded bands are percentile bootstrap 95% confidence intervals based on 1,000 resamples.

Regressions of Publications are estimated by Poisson regression with robust standard errors; regressions of Hit 10pct and D5 percentile are estimated by ordinary least squares. To quantify the extent to which a set of later-career measures accounts for the association between early-career cultural distance and an outcome, we report the proportional reduction in the magnitude of the cultural-distance coefficient relative to its value in the corresponding baseline specification, computed from unrounded coefficients.

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[To be added after the double-blind review.]

## **Contribution**

[To be added after the double-blind review.]

## **Conflicts of interest**

The authors have no existing or potential conflicts of interest to declare.

## **Data and material availability**

The data from the analyses conducted in this study can be found within the main and supplementary files. Deidentified data necessary to reproduce all plots and statistical analyses will be made freely available. Code will be made freely available.

## References

1. Hsieh, C.-T., Hurst, E., Jones, C. I. & Klenow, P. J. The Allocation of Talent and U.S. Economic Growth. *Econometrica* **87**, 1439–1474 (2019).
2. Kerr, S. P., Kerr, W., Özden, Ç. & Parsons, C. High-Skilled Migration and Agglomeration. *Annu. Rev. Econ.* **9**, 201–234 (2017).
3. Sugimoto, C. R. *et al.* Scientists have most impact when they're free to move. *Nature* **550**, 29–31 (2017).
4. Kahn, S. & MacGarvie, M. New evidence on international postdocs in the US: Less pay, different experiences. *Res. Policy* **53**, 105077 (2024).
5. Franzoni, C., Scellato, G. & Stephan, P. The mover's advantage: The superior performance of migrant scientists. *Econ. Lett.* **122**, 89–93 (2014).
6. Holding, B. C., Acciai, C., Schneider, J. W. & Nielsen, M. W. Quantifying the mover's advantage: transatlantic migration, employment prestige, and scientific performance. *High. Educ.* **87**, 1749–1767 (2024).
7. Netz, N., Hampel, S. & Aman, V. What effects does international mobility have on scientists' careers? A systematic review. *Res. Eval.* **29**, 327–351 (2020).
8. Katila, R. & Ahuja, G. SOMETHING OLD, SOMETHING NEW: A

LONGITUDINAL STUDY OF SEARCH BEHAVIOR AND NEW PRODUCT INTRODUCTION. *Acad. Manage. J.* **45**, 1183–1194 (2002).

9. Rosenkopf, L. & Almeida, P. Overcoming Local Search Through Alliances and Mobility. *Manag. Sci.* **49**, 751–766 (2003).
10. Almeida, P. & Kogut, B. Localization of Knowledge and the Mobility of Engineers in Regional Networks. *Manag. Sci.* **45**, 905–917 (1999).
11. Weitzman, M. L. Recombinant growth. *Q. J. Econ.* **113**, 331–360 (1998).
12. Arthur, W. B. *The Nature of Technology: What It Is and How It Evolves*. (Simon and Schuster, 2009).
13. Muthukrishna, M. & Henrich, J. Innovation in the collective brain. *Philos. Trans. R. Soc. B Biol. Sci.* **371**, 20150192 (2016).
14. Zhao, X., Akbaritabar, A., Kashyap, R. & Zagheni, E. A gender perspective on the global migration of scholars. *Proc. Natl. Acad. Sci.* **120**, e2214664120 (2023).
15. Czaika, M. & Orazbayev, S. The globalisation of scientific mobility, 1970–2014. *Appl. Geogr.* **96**, 1–10 (2018).
16. Liu, T. *et al.* Spatiotemporal mobility network of global scientists, 1970–2020. *Int. J. Geogr. Inf. Sci.* **38**, 1991–2018 (2024).

17. Gui, Q., Liu, C. & Du, D. Globalization of science and international scientific collaboration: A network perspective. *Geoforum* **105**, 1–12 (2019).
18. Han, F., Zhang, R., Zhang, S. & Yuan, J. International mobility characteristics, effects of, and effects on elite scientists. *J. Informetr.* **18**, 101485 (2024).
19. Czaika, M. & De Haas, H. The Globalization of Migration: Has the World Become More Migratory? *Int. Migr. Rev.* **48**, 283–323 (2014).
20. Park, M., Leahey, E. & Funk, R. J. Papers and patents are becoming less disruptive over time. *Nature* **613**, 138–144 (2023).
21. Wu, X., Wu, L., Park, M., Leahey, E. & Funk, R. J. Is Innovation Becoming Less Disruptive? An Inventory of the Literature. Preprint at <https://doi.org/10.48550/arXiv.2602.05140> (2026).
22. Aghion, P. & Howitt, P. A model of growth through creative destruction. (1990).
23. Decker, R. A., Haltiwanger, J., Jarmin, R. S. & Miranda, J. Where has all the skewness gone? The decline in high-growth (young) firms in the US. *Eur. Econ. Rev.* **86**, 4–23 (2016).
24. Akcigit, U. & Ates, S. T. Ten facts on declining business dynamism and lessons from endogenous growth theory. *Am. Econ. J. Macroecon.* **13**, 257–298 (2021).

25. Bloom, N., Jones, C. I., Van Reenen, J. & Webb, M. Are ideas getting harder to find? *Am. Econ. Rev.* **110**, 1104–1144 (2020).
26. Gordon, R. J. *The Rise and Fall of American Growth: The U.S. Standard of Living since the Civil War.* (Princeton University Press, 2016).  
doi:10.1515/9781400873302.
27. Jones, B. F. The burden of knowledge and the “death of the renaissance man”: Is innovation getting harder? *Rev. Econ. Stud.* **76**, 283–317 (2009).
28. Chu, J. S. G. & Evans, J. A. Slowed canonical progress in large fields of science. *Proc. Natl. Acad. Sci.* **118**, e2021636118 (2021).
29. Howells, J. R. L. Tacit Knowledge, Innovation and Economic Geography. *Urban Stud.* **39**, 871–884 (2002).
30. Boschma, R. Proximity and Innovation: A Critical Assessment. *Reg. Stud.* **39**, 61–74 (2005).
31. Hofstede, G. Dimensionalizing cultures: The Hofstede model in context. *Online Read. Psychol. Cult.* **2**, 8 (2011).
32. Murray, D. *et al.* Unsupervised embedding of trajectories captures the latent structure of scientific migration. *Proc. Natl. Acad. Sci.* **120**, e2305414120 (2023).



33. Iacus, S. M., King, G. & Porro, G. Causal inference without balance checking: Coarsened exact matching. *Polit. Anal.* **20**, 1–24 (2012).
34. Rosenbaum, P. R. & Rubin, D. B. The central role of the propensity score in observational studies for causal effects. *Biometrika* **70**, 41–55 (1983).
35. Funk, R. J. & Owen-Smith, J. A Dynamic Network Measure of Technological Change. *Manag. Sci.* **63**, 791–817 (2017).
36. Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S. & Dean, J. Distributed representations of words and phrases and their compositionality. *Adv. Neural Inf. Process. Syst.* **26**, (2013).
37. Hou, Z. *et al.* GraphMAE: Self-Supervised Masked Graph Autoencoders. in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* 594–604 (ACM, Washington DC USA, 2022). doi:10.1145/3534678.3539321.
38. You, Y. *et al.* Graph contrastive learning with augmentations. *Adv. Neural Inf. Process. Syst.* **33**, 5812–5823 (2020).
39. Richard R. Ernst – Biographical. *NobelPrize.org*  
<https://www.nobelprize.org/prizes/chemistry/1991/ernst/biographical/>.
40. Adi Shamir - A.M. Turing Award Laureate.

[https://amturing.acm.org/award\\_winners/shamir\\_2327856.cfm](https://amturing.acm.org/award_winners/shamir_2327856.cfm).

41. Amartya Sen – Biographical. *NobelPrize.org*  
<https://www.nobelprize.org/prizes/economic-sciences/1998/sen/biographical/>.
42. Chernozhukov, V. *et al.* Double/debiased machine learning for treatment and structural parameters. (2018).
43. Nie, X. & Wager, S. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika* **108**, 299–319 (2021).
44. Zhang, S., Wapman, K. H., Larremore, D. B. & Clauset, A. Labor advantages drive the greater productivity of faculty at elite universities. *Sci. Adv.* **8**, (2022).
45. Li, W., Zhang, S., Zheng, Z., Cranmer, S. & Clauset, A. Untangling the network effects of productivity and prominence among scientists. *Nat. Commun.* **13**, 4907 (2022).
46. Petersen, A. M. Quantifying the impact of weak, strong, and super ties in scientific careers. *Proc. Natl. Acad. Sci.* **112**, E4671–E4680 (2015).
47. Burt, R. S. Structural Holes and Good Ideas. *Am. J. Sociol.* **110**, 349–399 (2004).
48. Guimerà, R., Uzzi, B., Spiro, J. & Amaral, L. A. N. Team Assembly Mechanisms Determine Collaboration Network Structure and Team Performance. *Science* **308**,

697–702 (2005).

49. Cummings, J. N. & Kiesler, S. Coordination costs and project outcomes in multi-university collaborations. *Res. Policy* **36**, 1620–1634 (2007).
50. Venturini, S., Sikdar, S., Rinaldi, F., Tudisco, F. & Fortunato, S. Collaboration and topic switches in science. *Sci. Rep.* **14**, 1258 (2024).
51. Zeng, A., Fan, Y., Di, Z., Wang, Y. & Havlin, S. Impactful scientists have higher tendency to involve collaborators in new topics. *Proc. Natl. Acad. Sci.* **119**, (2022).
52. March, J. G. Exploration and Exploitation in Organizational Learning. *Organ. Sci.* **2**, 71–87 (1991).
53. Hill, R. *et al.* The pivot penalty in research. *Nature* 1–8 (2025).
54. Liu, L. *et al.* Hot streaks in artistic, cultural, and scientific careers. *Nature* **559**, 396–399 (2018).
55. Liu, L., Dehmamy, N., Chown, J., Giles, C. L. & Wang, D. Understanding the onset of hot streaks across artistic, cultural, and scientific careers. *Nat. Commun.* **12**, 5392 (2021).
56. Sekara, V. *et al.* The chaperone effect in scientific publishing. *Proc. Natl. Acad. Sci.* **115**, 12603–12607 (2018).

57. Fleming, L. Recombinant Uncertainty in Technological Search. *Manag. Sci.* **47**, 117–132 (2001).
58. Uzzi, B., Mukherjee, S., Stringer, M. & Jones, B. Atypical Combinations and Scientific Impact. *Science* **342**, 468–472 (2013).
59. Gomez, C. J., Herman, A. C. & Parigi, P. Leading countries in global science increasingly receive more citations than other countries doing similar research. *Nat. Hum. Behav.* **6**, 919–929 (2022).
60. Gates, A. J., Gao, J. & Mane, I. The increasing fragmentation of global science limits the diffusion of ideas. Preprint at <https://doi.org/10.48550/arXiv.2404.05861> (2025).
61. Wang, J., Frietsch, R., Neuhäusler, P. & Hooi, R. International collaboration leading to high citations: Global impact or home country effect? *J. Informetr.* **18**, 101565 (2024).
62. Gomez, C. J. The growing concentration of national influence in global science. *Nat. Hum. Behav.* 1–17 (2026).
63. Qiu, S., Steinwender, C. & Azoulay, P. Paper tiger? Chinese science and home bias in citations. *J. Int. Econ.* **157**, 104123 (2025).
64. Laursen, K. & Salter, A. Open for innovation: the role of openness in explaining

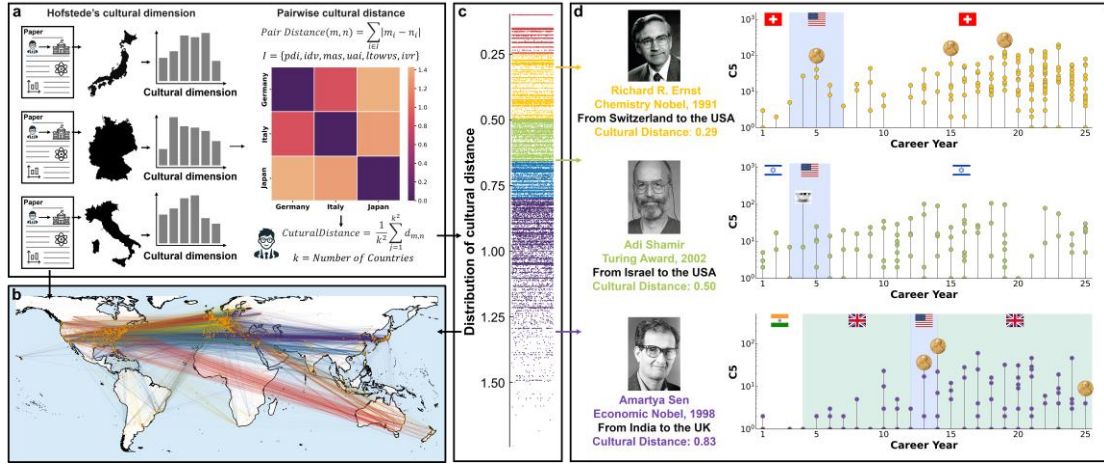
- innovation performance among U.K. manufacturing firms. *Strateg. Manag. J.* **27**, 131–150 (2006).
65. Rosenkopf, L. & Nerkar, A. Beyond local search: boundary-spanning, exploration, and impact in the optical disk industry. *Strateg. Manag. J.* **22**, 287–306 (2001).
66. Cui, H., Lin, Y., Wu, L. & Evans, J. A. Aging and the narrowing of scientific innovation. *Science* **392**, 588–591 (2026).
67. Storper, M. & Venables, A. J. Buzz: face-to-face contact and the urban economy. *J. Econ. Geogr.* **4**, 351–370 (2004).
68. Lin, Y., Frey, C. B. & Wu, L. Remote collaboration fuses fewer breakthrough ideas. *Nature* **623**, 987–991 (2023).
69. Catalini, C. Microgeography and the Direction of Inventive Activity. *Manag. Sci.* **64**, 4348–4364 (2018).
70. Boudreau, K. J. *et al.* A field experiment on search costs and the formation of scientific collaborations. *Rev. Econ. Stat.* **99**, 565–576 (2017).
71. Yu, X., Rahwan, T. & Jia, T. Knowledge Independence Breeds Disruption but Limits Recognition. *ArXiv Prepr. ArXiv250409589* <https://arxiv.org/abs/2504.09589> (2025).

72. Schimmelpfennig, R., Razek, L., Schnell, E. & Muthukrishna, M. Paradox of diversity in the collective brain. *Philos. Trans. R. Soc. B Biol. Sci.* **377**, 20200316 (2021).
73. Jones, C. I. Recipes and Economic Growth: A Combinatorial March Down an Exponential Tail. *J. Polit. Econ.* **131**, 1994–2031 (2023).
74. Akcigit, U. & Ates, S. T. What Happened to US Business Dynamism? *J. Polit. Econ.* **131**, 2059–2124 (2023).
75. Xie, Y., Lin, X., Li, J., He, Q. & Huang, J. Caught in the crossfire: Fears of Chinese–American scientists. *Proc. Natl. Acad. Sci.* **120**, e2216248120 (2023).
76. Jia, R., Roberts, M. E., Wang, Y. & Yang, E. The impact of US–China tensions on US science: Evidence from the NIH investigations. *Proc. Natl. Acad. Sci.* **121**, e2301436121 (2024).
77. Ito, R., Chavarro, D., Ciarli, T., Cowan, R. & Visentin, F. From drains to bridges: The role of internationally mobile PhD students in linking non-mobile with foreign scientists. *J. Dev. Econ.* **177**, 103577 (2025).
78. Iaria, A., Schwarz, C. & Waldinger, F. Frontier knowledge and scientific production: Evidence from the collapse of international science. *Q. J. Econ.* **133**, 927–991 (2018).

79. Bell, A., Chetty, R., Jaravel, X., Petkova, N. & Van Reenen, J. Who becomes an inventor in America? The importance of exposure to innovation. *Q. J. Econ.* **134**, 647–713 (2019).
80. Jones, C. I. The end of economic growth? Unintended consequences of a declining population. *Am. Econ. Rev.* **112**, 3489–3527 (2022).
81. Lin, Z., Yin, Y., Liu, L. & Wang, D. SciSciNet: A large-scale open data lake for the science of science research. *Sci. Data* **10**, 315 (2023).
82. Shi, D., Liu, W. & Wang, Y. Has China’s Young Thousand Talents program been successful in recruiting and nurturing top-caliber scientists? *Science* **379**, 62–65 (2023).

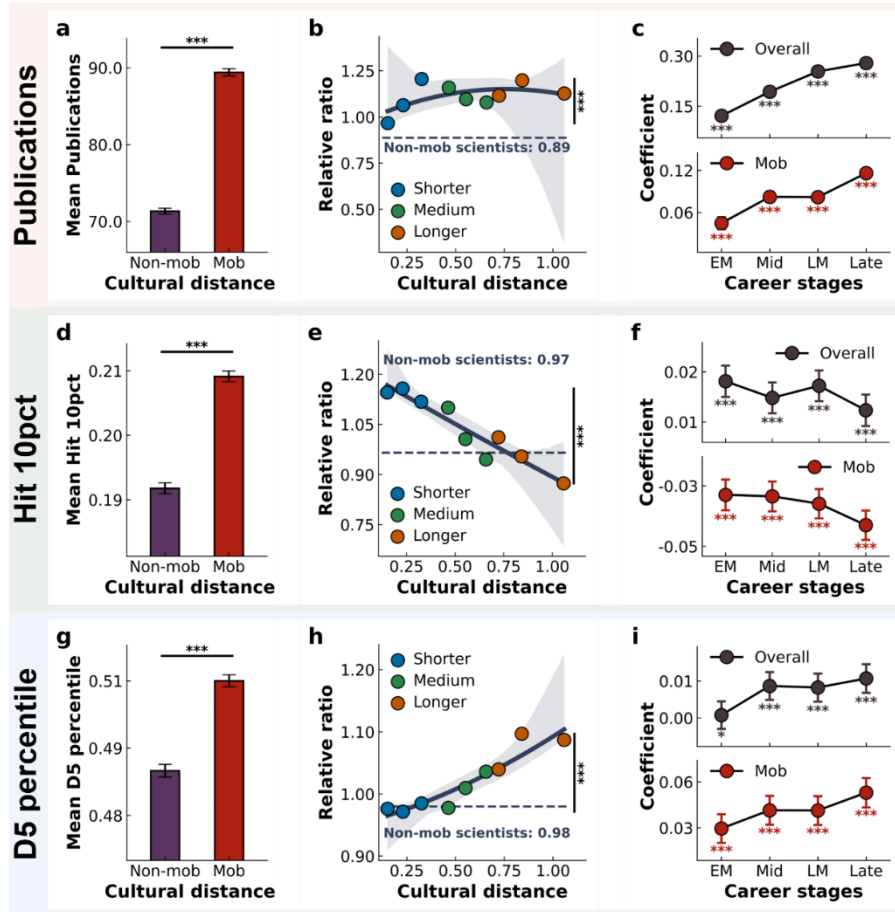
## Supplementary Information

This section includes Supplementary Notes 1–4, Figures S0–S19, Tables S1–S29, and Supplementary References.



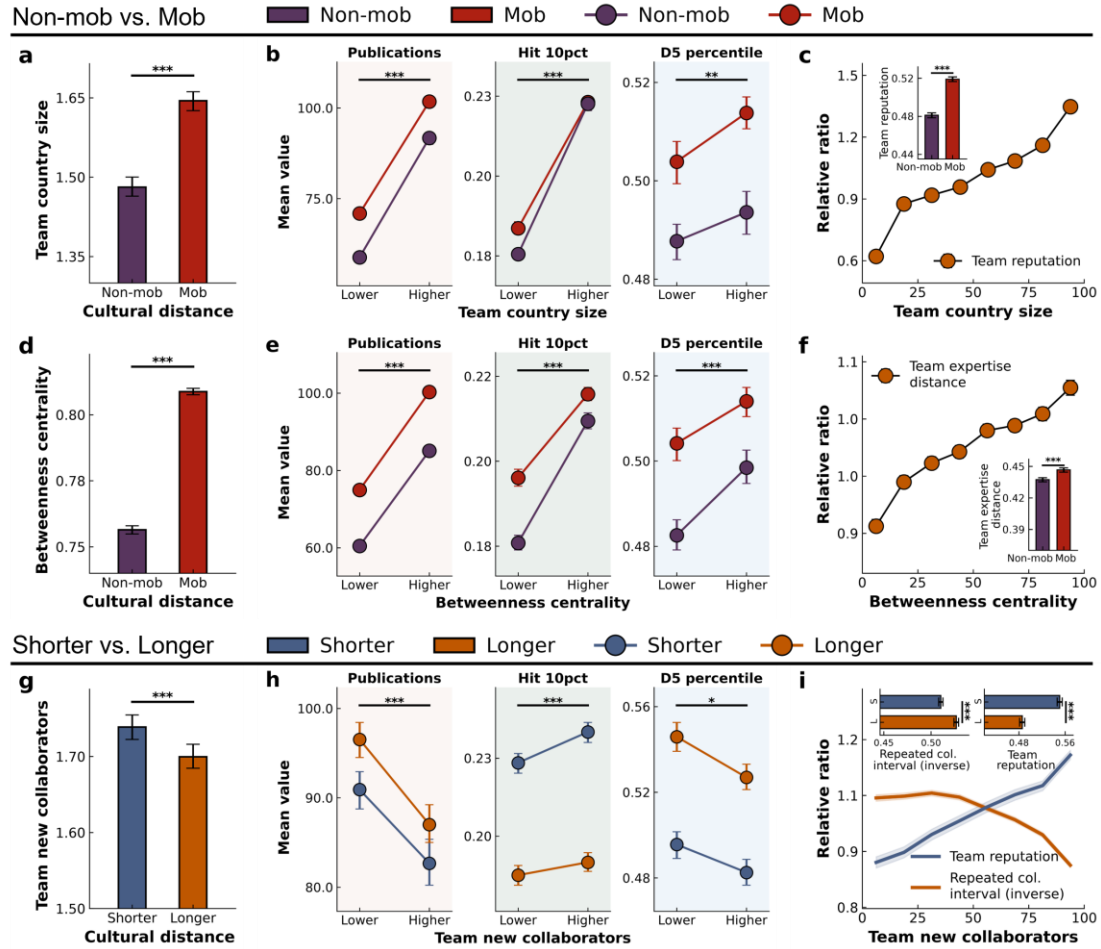
**Fig. 1 Overview of the cultural distance.** **a**, we illustrate the calculation of early cultural distance, which quantifies the pairwise country-level Hofstede cultural difference encountered by scientists during early-career (first five career years) moves. In addition to Hofstede's cultural difference, we further employ various embedding-based methods to derive a "cultural dimension" using move strengths between institutions. **b**, The geographical distribution of mobile scientists is visualized across 4,715 institutions, based on individuals whose first career year falls between 1956 and 1996. Each dot represents an institution that appears in the publication history of at least one mobile scientist. Each edge represents an individual scientist's career move between two institutions during their first five career years. For scientists with multi-institution trajectories, consecutive moves generate multiple edges. The color of each edge encodes the scientist's percentile of early-career cultural distance, as defined in panel c. **c**, The distribution of cultural distance among mobile scientists ( $n = 47,838$ ) from 1956 to 1996 is shown, with each dot representing a single scientist. For clarity, non-mobile scientists ( $n = 536,874$ ) are not displayed. The vertical axis reflects the value of cultural distance, while the horizontal axis is jittered to minimize overlapping points. Denser (darker) regions indicate more commonly observed values. To facilitate visualization, cultural distance is categorized into five percentile-based groups: 0–20th (red), 20–40th (yellow), 40–60th (green), 60–80th (blue), and 80–100th (purple) percentile. **d**, Career trajectories of three representative early mobile scientists who later received Nobel Prizes or the Turing Award. We observe that these individuals tend to publish more, receive higher citation counts, and produce prize-winning work following cross-cultural career moves.





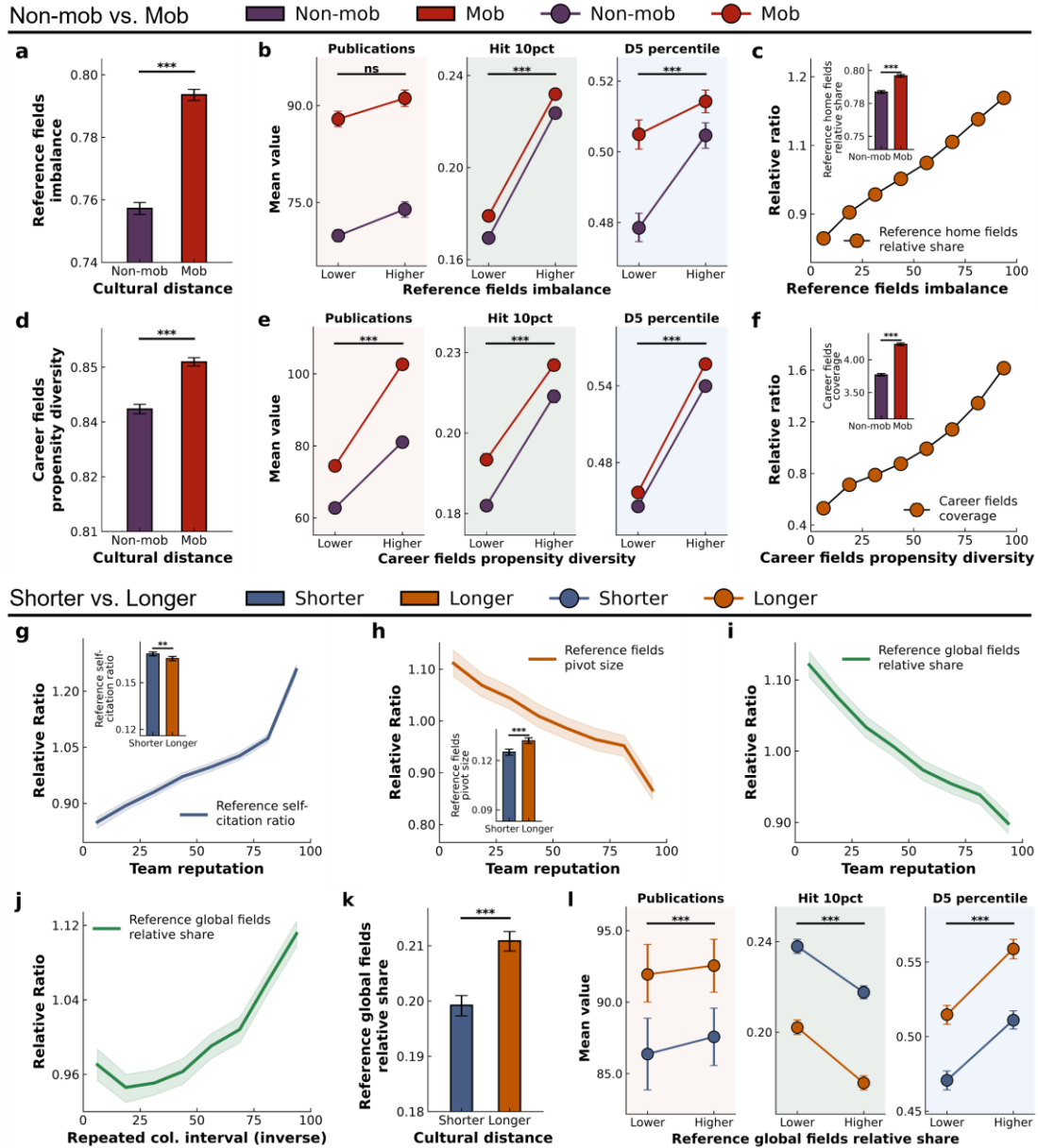
**Fig. 2 Mobility dividend and cultural divergence.** **a, d, g,** Long-term academic performance of mobile (Mob) and non-mobile (non-Mob) scientists after CEM ( $n = 92,914$ ). Mobile scientists exhibit higher mean values in productivity (Publications **a**), impact (Hit 10pct, **d**), and disruptiveness (D5 percentile, **g**) during years 6–25, compared to matched non-mobile peers. Error bars denote bootstrap 95% confidence intervals. **b, e, h,** Relative ratios for mobile scientists ( $n = 46,457$ ) stratified by early-career cultural distance (shorter, medium, longer; terciles). Points indicate group means; the lines represent quadratic fits of cultural distance versus three indicators. Shaded areas denote bootstrap 95% confidence intervals. Dashed horizontal lines show the mean value of non-Mob scientists. Shorter cultural distance has the highest long-term impact (**e**), whereas longer cultural distance is associated with greater productivity (**b**) and disruptiveness (**h**). Asterisks indicate one-sided Welch’s  $t$ -test for the mean value of shorter and longer groups. **c, f, i,** Regression coefficients for the effect of mobility status (overall) and early-career cultural distance percentile (within mobile scientists) on long-term performance across four career stages: early-mid (EM), mid (Mid), late-mid (LM), and late (Late). “Overall” panels display coefficients for a binary mobility indicator (Mob = 1 vs non-Mob = 0); “Mob” panels show regression coefficients for the effect of early-career cultural distance percentile on long-term performance among mobile scientists, estimated by Poisson regression with robust standard errors computed

via the H3 estimator for Publications and OLS regressions for Hit 10pct and D5 percentile. Error bars denote 95% confidence intervals.  $*P < 0.05$ ;  $**P < 0.01$ ;  $***P < 0.001$ .



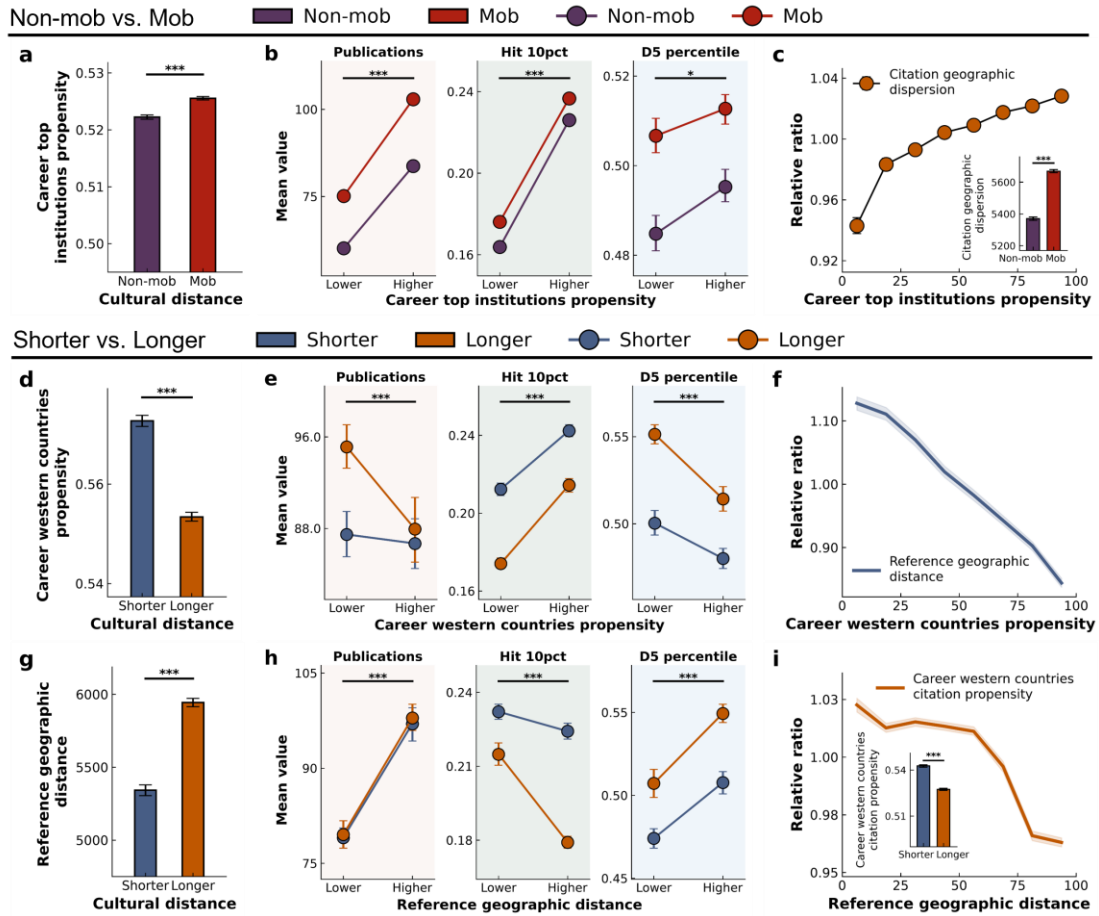
**Fig. 3 Early-career cultural distance reshapes later-career collaboration.** **a**, Mob scientists' more than twofold higher rate of early-career international co-authorship (Fig. S6a) carries forward into teams spanning more countries in years 6–25 (Fig. S6c). **b**, Scientists in the higher half of team country size achieve higher Publications, Hit 10pct and D5 percentile, regardless of mobility status. **c**, More international teams are also more reputable, and Mob scientists assemble higher-reputation teams than non-Mob peers (inset). Early international collaboration supplies two separate resources. Its sheer breadth widens the later-career collaborator pool and so the labour advantage behind Publications (Fig. S6d, e, g). Whether it reaches Western-country and top-100-institution collaborators instead determines later team reputation, which tracks Hit 10pct (Fig. S6h–m, Table S15). Within movers the two resources split along cultural distance. Longer-distance movers collaborate internationally more in early career and go on to larger later-career teams (Fig. S7a, b, d), whereas shorter-distance movers reach more Western-country and top-100-institution collaborators and go on to higher team reputation (Fig. S7e–i). **d**, Mob scientists rely less on a fixed early collaborator core (Fig. S8a) and occupy more brokerage-oriented later-career positions, as do shorter-distance movers within the mobile sample (Fig. S9a–f, S10a). **e**, Higher

betweenness centrality accompanies higher performance on all three outcomes in both groups. Beyond brokerage, Mob scientists' networks are globally more fragmented yet locally denser, and they re-engage familiar collaborators more quickly, giving them higher coordination efficiency (Fig. S8d, f–h). The labour advantage of **a**, betweenness centrality and coordination efficiency together account for 69.39% of the Mob advantage in Publications (Table S17). **f**, Brokerage also comes with expertise-heterogeneous teams, and Mob scientists assemble more expertise-distant teams (inset). It is team expertise distance rather than brokerage itself that carries through to D5 percentile (Table S17, Panel d), and it originates in the larger early-career field pivots and more expertise-diverse early collaborators that follow a move (Fig. S11a–d, g). **g**, Within movers, shorter-distance movers engage more new collaborators per paper. **h**, The higher half of team new collaborators again outperforms the lower half in both groups. **i**, Engaging new collaborators raises team reputation but lowers coordination efficiency, and the insets show shorter-distance movers higher on the former, longer-distance movers on the latter. New collaborators thus carry the reputational resource of **c** into later career for shorter-distance movers, while longer-distance movers' coordination advantage rests instead on a repeat-tie orientation that begins in early career and persists (Fig. S9a–c, g–l). Their larger teams and higher coordination efficiency fully account for their Publications advantage, and shorter-distance movers' team reputation accounts for much of their Hit 10pct advantage (Table S16, Panels b, c). Shorter-distance movers hold the more central position of **e**, yet neither productivity nor disruptiveness follows, because brokerage delivers neither without efficient coordination nor without expertise heterogeneity. Longer-distance movers' larger early field pivots and more expertise-diverse early collaborators, driven by the more uneven capital of the collaborator pools they enter (Fig. S12a–h), instead supply the greater team expertise distance behind their D5 percentile advantage (Table S18, Panel d). The local knowledge strategies and search portfolios that complete the impact and disruptiveness pathways are shown in Figs. 4 and 5. Analyses in **a–f** compare Mob ( $n = 46,457$ ) with non-Mob scientists ( $n = 46,457$ ) in the matched sample. Analyses in **g–i** compare shorter- ( $n = 15,459$ ) with longer-distance movers ( $n = 15,510$ ). Bars and points show raw sample means, and error bars and shaded bands show percentile bootstrap 95% confidence intervals (1,000 resamples). Lower and higher groups are defined by a median split of the full matched sample. In **c**, **f** and **i**, scientists are grouped into eight equally sized bins of the  $x$ -axis variable, plotted at the mean percentile of each bin. Group contrasts in **a**, **d**, **g** and the insets use one-sided Welch's  $t$ -tests, and higher-versus-lower contrasts in **b**, **e** and **h** use one-sided Mann–Whitney  $U$  tests, reporting the more conservative of the two groups. For D5 percentile the two levels are  $P = 0.006$  in non-Mob and  $P < 0.001$  in Mob in **b**, and  $P = 0.014$  in shorter and  $P < 0.001$  in longer in **h**. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



**Fig. 4 Mobile scientists and shorter-distance movers exploit locally yet explore globally, whereas longer-distance movers explore locally yet exploit globally.** **a**, Mob scientists draw each paper's references from a narrower set of fields than non-Mob peers, showing stronger local exploitation despite their larger early-career field pivots. **b**, Scientists in the higher half of reference fields imbalance achieve higher Hit 10pct and D5 percentile in both groups. **c**, Field concentration goes together with drawing more references from a paper's own home fields, and Mob scientists are higher on both (inset). This local exploitation follows from the team reputation and international capital of Fig. 3c, which favour deepening within familiar fields (Fig. S13a–c) and extend to the country level (Fig. S13d–h). Coming after broad early-career pivots, it completes an explore-then-exploit sequence, and team reputation amplifies

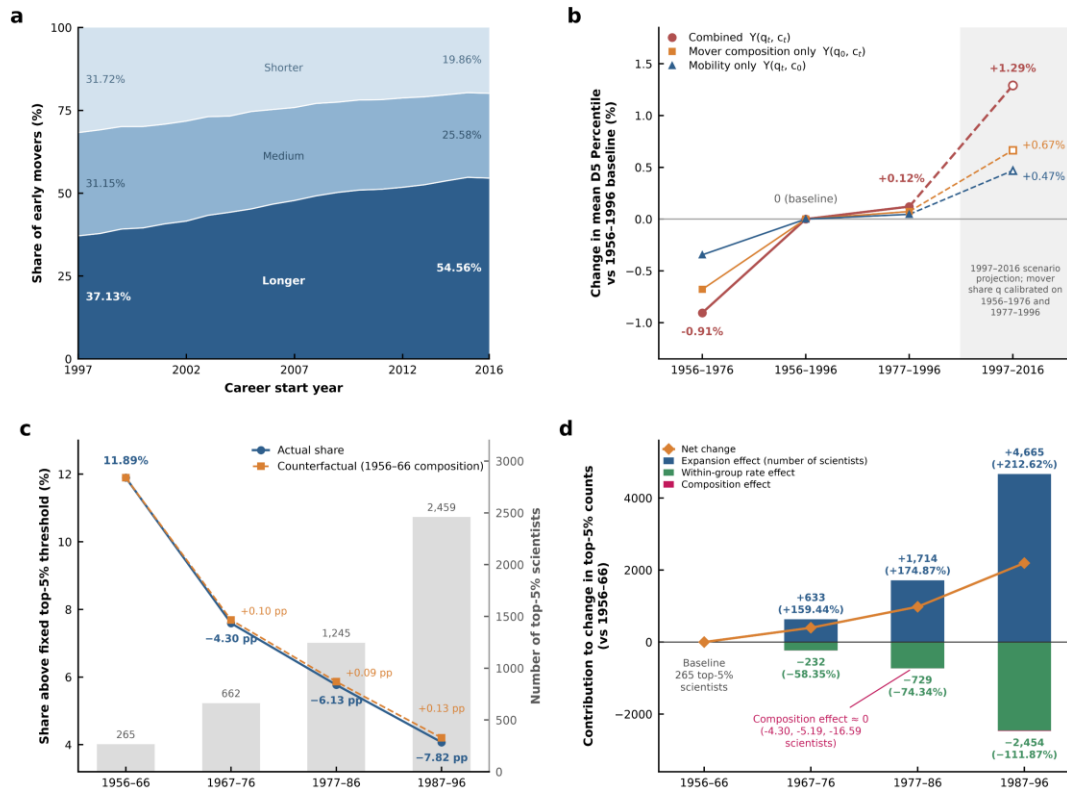
the impact returns to it: the two together account for 92.54% of the Mob advantage in Hit 10pct (Tables S19, S22). **d**, Across the whole career, however, Mob scientists spread their referencing more evenly across fields than non-Mob peers, and also draw more on fields outside those in which they publish (Fig. S15d). **e**, The higher half of career fields propensity diversity outperforms the lower half on all three outcomes in both groups. **f**, Career-level breadth in referencing tracks a wider published research footprint, and Mob scientists are higher on both (inset). All three career-level measures accompany the brokerage positions of Fig. 3d (Fig. S15a–c), so Mob scientists exploit locally yet explore globally. Career-level exploration predicts D5 percentile beyond team expertise distance, and the two amplify each other (Table S21, Panel c). **g–i**, Within movers, higher team reputation accompanies more self-citation, smaller shifts in referenced fields and less reliance on fields beyond a paper’s own, and the insets show shorter-distance movers higher on the first and longer-distance movers on the second. Their divergent local strategies thus follow the reputational split of Fig. 3c. **j**, Higher coordination efficiency instead accompanies greater reliance on global fields, so that longer-distance movers’ repeat-tie orientation lowers the frictions that cross-field integration requires (Fig. S14a, b). **k**, Longer-distance movers accordingly draw more of each paper’s references from fields beyond its own. **l**, The higher half of reference global fields relative share achieves higher Publications and D5 percentile but lower Hit 10pct in both groups, mirroring the divergence between the two groups. Local exploitation with team reputation accounts for 41.96% of shorter-distance movers’ Hit 10pct advantage, whereas local exploration carries through to D5 percentile only when embedded in expertise-heterogeneous teams (Table S20). At the career level the ordering reverses again: shorter-distance movers explore more broadly (Fig. S16d–f), consistent with their more brokerage-oriented positions (Fig. S16a–c), so that greater career-level exploration widens rather than narrows longer-distance movers’ net disruptiveness advantage. Panels **a–f** compare Mob ( $n = 46,457$ ) with non-Mob scientists ( $n = 46,457$ ); panels **g–l** compare shorter- ( $n = 15,459$ ) with longer-distance movers ( $n = 15,510$ ). Sample sizes, estimators, binning and tests are as in Fig. 3. In **b** the Publications contrast is not significant among Mob scientists ( $P = 0.153$ ) but is significant among non-Mob scientists ( $P < 0.001$ ). \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



**Fig. 5 Mobile scientists bridge core and periphery, whereas longer-distance movers source farther and diffuse wider.** **a**, Mob scientists draw more of their later-career references from top-100 institutions than non-Mob peers, tracing to their greater early-career exposure to such collaborators (Fig. S6j). The advantage is concentrated among those whose own later-career institution lies outside the top 100, where 74% of Mob scientists are located and where they nonetheless retain more top-institution coauthors, so that elite ties formed through mobility carry frontier knowledge back into peripheral settings (Table S23). At the country level the orientation reverses and becomes bidirectional: Mob scientists draw less on Western-country knowledge overall, yet more of it when located outside the West and less of it when located inside, giving them a context-dependent bridging position between core and periphery on both dimensions (Supplementary Note 3.4.1). **b**, Scientists in the higher half of career top institutions propensity achieve higher performance in both groups, most strongly for Publications and Hit 10pct and weakly for D5 percentile. Its association with Hit 10pct largely overlaps with the team reputation and local exploitation of Figs. 3 and 4, adding 0.36 percentage points beyond them, reflecting a source-side trace of earlier elite embedding rather than a separate channel (Table S24). **c**, Frontier-oriented sourcing accompanies work that is cited from a more widely dispersed set of locations, and Mob

scientists are higher on both (inset), extending their bridging position to the diffusion side (Supplementary Note 4.1). **d**, Within movers the two dimensions align rather than diverge: shorter-distance movers draw more on Western-country knowledge, consistent with their more Western and top-100-institution collaborators in both early and later career, while longer-distance movers rely relatively more on the periphery (Supplementary Note 3.4.1). **e**, Higher Western-country sourcing accompanies higher Hit 10pct but lower Publications and D5 percentile in both groups. Added to team reputation and local exploitation, it raises the share of shorter-distance movers' Hit 10pct advantage explained from 41.96% to 48.34%, so recognition of core-country knowledge constitutes a distinct source-side channel; conversely, weaker anchoring to that core adds to the team expertise distance behind longer-distance movers' D5 percentile advantage (Tables S24, S25, S27). **f**, Reliance on the Western core narrows the geographic radius of search, and **g**, longer-distance movers accordingly reach geographically farther for the knowledge they cite, in keeping with their fewer Western collaborators and more geographically dispersed teams. **h**, Wider geographic search accompanies higher Publications and D5 percentile but lower Hit 10pct in both groups. Longer-distance movers also reach farther back in time and to more prominent prior work, so that their broad search stands on the shoulders of giants. Team expertise distance, peripheral sourcing and this search portfolio together explain 28.18% of their D5 percentile advantage, and team expertise distance with career-level exploration and the same search portfolio explains 56.38% of the Mob advantage (Tables S26–S28). **i**, Farther search accompanies recognition from less Western-centred audiences, and longer-distance movers are lower on this citation-side orientation (inset). Their work also diffuses across a more geographically dispersed set of citing locations, as in **c** (Supplementary Note 4.1). Because this same reliance on prominent, established work and on geographically distant collaboration impedes the tacit knowledge and independent recombinations from which the most extreme breakthroughs arise, higher average disruptiveness does not extend to the tail (Fig. S19). Panels **a–c** compare Mob ( $n = 46,457$ ) with non-Mob scientists ( $n = 46,457$ ); panels **d–i** compare shorter- ( $n = 15,459$ ) with longer-distance movers ( $n = 15,510$ ). Sample sizes, estimators, binning and tests are as in Fig. 3. In **b** the D5 percentile contrast is  $P < 0.001$  among non-Mob and  $P = 0.047$  among Mob scientists. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .





**Fig. 6 Early cross-country mobility shifts toward longer cultural distance, and this shift raises the centre of the disruptiveness distribution without eroding its top.** **a**, Distance composition of early movers by career-start year, 1997–2016 (observed, not projected). An author is an early mover if at least two countries appear in the first five career years; early cultural distance is the cultural distance spanned by those moves. Movers are assigned to short, medium or long distance by the terciles of early cultural distance among all early movers who started between 1956 and 1996. Because the thresholds are fixed once and reused for every year, the three categories are not mechanically rebalanced to one third within each period, and the bands therefore carry information about the level of distance rather than only its rank. Bands sum to 100% within each year;  $n = 8,828$  movers in 1997,  $n = 34,259$  in 2016, and  $n = 475,602$  across the full window. The long-distance share rises from 37.13% to 54.56% and the short-distance share falls from 31.72% to 19.86%. Among all scientists with at least five observable career years and at least one affiliation country identified during career years 1–5 ( $n = 5,871,932$  pooled across the 1997–2016 cohorts), the share affiliated with two or more countries during career years 1–5 rose from 5.97% in 1997 to 8.98% in 2016, an increase of 3.01 percentage points (+50.4% relative to 1997; linear trend +0.17 percentage points per cohort year,  $r = 0.978$ ,  $P < 0.001$ ), with a pooled prevalence of 8.10% over the period. The trend is strongly upward

but not monotonic, with a small plateau after 2012 (9.06% in 2012, 8.80% in 2013, 8.81% in 2014, 8.90% in 2015, and 8.98% in 2016); the five-year observation windows of these most recent cohorts extend to 2020. This population is defined solely by five observable career years and at least one affiliation country identified, and therefore differs from the main analysis sample, which additionally requires at least 20 career years, at least 10 publications during the first 20 years, publication in each five-year career stage through year 20, and early-career affiliation coverage for countries with Hofstede cultural-dimension data; the mobility rates reported here are consequently not directly comparable with those in Fig. S0b. **b**, Scenario projection of the mean long-run D5 Percentile of the matched sample under the four mobility environments, holding the four group means fixed. The matched sample pairs early movers with marginal non-movers, i.e. authors who did not move early but whose observable early-career characteristics make them close counterfactuals to movers. Only the weights are varied through scenario function  $Y(q, c) = (1 - q)\mu_N + q(c_S\mu_S + c_M\mu_M + c_L\mu_L)$ , with  $\mu$  the fixed group means:  $q$  is the mover share of the matched sample and  $c = (c_S, c_M, c_L)$  is the distance composition among movers. Three scenarios are shown: mobility only, in which  $q$  takes its period value while  $c$  is held at the 1956–1996 baseline; mover composition only, in which  $c$  takes its period value while  $q$  is held at baseline; and combined, in which both take their period values. All values are expressed as a relative change against the 1956–1996 baseline  $Y_0$ . Filled markers and solid lines denote environments whose weights are directly observed in the matched sample, including the 1956–1996 baseline and the two historical environments, 1956–1976 and 1977–1996; open markers, dashed lines and the shaded band denote the 1997–2016 scenario, for which the composition within movers is observed in the full author population but the matched-sample mover share  $q_3 = 61.83\%$  is obtained by a two-point logit calibration of  $q$  on the population five-year mobility rate  $p$ ,  $\text{logit}(q_t) = \alpha + \beta \cdot \text{logit}(p_t)$  with  $\alpha = 1.847$  and  $\beta = 0.562$ , anchored on the two historical periods. The environment of 1956–1976 lies 0.91% below baseline and that of 1977–1996 0.12% above it. Under the 1997–2016 environment the combined scenario lies 1.29% above baseline, of which the mover-composition channel alone accounts for 0.67% and the mover-share channel alone for 0.47%. At the baseline weights, the implied elasticities are 0.0198 for the mover share and 0.0149 for the longer-distance share. A 1% increase in the number of early movers relative to its baseline level therefore raises the mean D5 Percentile by 0.0198% (percentile-bootstrap 95% confidence interval based on 10,000 replications: 0.0161–0.0235%), while a 1% increase in the number of longer-distance movers relative to its baseline level, achieved through a proportional reallocation from the other groups, raises it by 0.0149% (0.0133–0.0165%). Equivalently,

holding the total sample size fixed, reallocating scientists from marginal non-mobility so as to increase the mover count by 1% relative to baseline raises the mean percentile by 0.0198% when the reassigned scientists follow the observed mobility-distance composition. If the same reassigned scientists are instead allocated entirely to longer-distance mobility, the increase reaches 0.0471% (0.0419–0.0521%), implying an additional gain of 0.0272% (0.0236–0.0309%) from going farther rather than merely becoming mobile. Reassigned scientists are evaluated using the observed mean outcome of their destination group. **c**, Share of the matched sample above a fixed top-5% threshold, and the corresponding number of top-5% scientists, by career-start decade. The threshold  $T = 0.019756$  is the 95th percentile of the raw D5 measure computed once over the complete matched sample and then applied unchanged to all four decades, so that the bar is absolute rather than relative and the decades are directly comparable; authors with a missing raw value are treated as below the threshold, and an author is counted as top-5% when the raw value is greater than or equal to  $T$ . The blue line is the actual share and is labelled with its cumulative decline in percentage points relative to 1956–1966. The orange dashed line is a counterfactual share that holds the four-group composition at its 1956–1966 values while letting the within-group top-5% rates follow their observed history; it is labelled with the gap above the actual share, which is the contribution of compositional change to the fall in the share and never exceeds 0.14 percentage points. Grey bars, right axis, give the absolute number of top-5% scientists. The share falls from 11.89% to 4.07% while the number rises from 265 to 2,459. **d**, Exact three-factor Shapley decomposition of the change in the number of top-5% scientists, relative to the 1956–1966 baseline of 265. The number in a decade is the product of the decade sample size, the four-group composition and the within-group top-5% rates. The expansion effect is the contribution of the growing sample, the within-group rate effect that of the falling probability of clearing the fixed bar inside each group, and the composition effect that of the shifting group weights; the composition effect covers both movement between non-movers and movers and the change of distance structure within movers. For 1987–1996, the three contributions are +4,665 (+212.62%) from expansion, –2,454 (–111.87%) from the within-group rate effect, and –16.59 (–0.76%) from compositional change, summing to the net change of +2,194. The composition bars are too small to be visible at this scale and are therefore labelled directly. Percentages in parentheses are shares of the net change and may exceed 100% or be negative.

Supplementary Information for  
Mobility dividend and cultural divergence in science

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## Supplementary Note 1: Dataset and data processing

### Supplementary Note 1.1: Microsoft Academic Graph

Our analyses draw on the Microsoft Academic Graph (MAG), a large-scale bibliographic database covering publications, authors, institutional affiliations, fields of study and citation links across all areas of science. We use the 6 December 2021 edition, which contains 270,694,050 publication records, in the curated form provided by SciSciNet<sup>1</sup>. After the curation procedures implemented there, the working dataset contains 134,129,188 publications and 1,588,739,703 citation links, from which we draw publication records, author identifiers, institutional affiliations and citation relations. Although MAG ceased updating at the end of 2021, records carry publication dates extending into March 2022, and this is the effective end of the observation window throughout.

**Author name disambiguation.** MAG assigns a persistent identifier to each disambiguated author, constructed from co-authorship structure, affiliation history, publication venue and citation patterns. We use these identifiers without further modification, and reconstruct individual publication histories directly from them.

**Affiliation and country assignment.** Each author–paper record may carry one or more institutional affiliations, which we map to countries using the institution metadata provided in the database. For each scientist and each career year we collect the set of countries appearing across their author–paper records in that year, and a scientist’s early-career country set is the union of countries observed over career years 1–5. Records without an assignable institution are excluded from the country set but retained for all publication- and citation-based measures. The accuracy of this procedure is assessed by manual validation against public biographical sources (Supplementary Note 1.2).



**Fields of study.** MAG organizes topics into a hierarchical field-of-study system, of which we use the two highest levels: level 0 (L0), comprising 19 broad disciplines, and level 1 (L1), comprising 292 more granular fields. Papers may be assigned to more than one field at each level, and field-normalized measures are computed within field  $\times$  publication year cells.

**Institutional prestige.** For each institution we compute an h-index over all papers affiliated with it in MAG up to 2022 and rank institutions accordingly. The 100 highest-ranked institutions serve throughout as the operational definition of top-tier institutions, both for the early- and later-career collaborator measures (Supplementary Note 3.2) and for the knowledge-source measures (Supplementary Note 3.4).

**Supplementary linkage to OpenAlex.** Because MAG ceased updating in early 2022, the citation window is truncated for recently published papers and the coverage of funding acknowledgements is limited. We therefore link MAG records to OpenAlex (release 27 February 2025) by DOI in two places. First, we supplement citation counts for papers published after 2017, so that five-year disruption indices can be computed for a larger share of records (Supplementary Note 1.5). Second, we retrieve funder and award information for approximately 32,000 funding organizations worldwide, which extends the funding analysis beyond the two United States agencies covered by SciSciNet (Supplementary Note 3.1.8).

## **Supplementary Note 1.2: Sample construction and validation**

**Inclusion criteria.** We restrict attention to scientists whose first publication year falls between 1956 and 1996, so that at least 25 subsequent career years are observable. Among these we retain scientists who satisfy all of the following: a career length of at least 20 years; at least 10 publications during the first 20 career years; and at least one publication in each of the four five-year intervals spanning career years 1–5, 6–10, 11–15 and 16–20. The last condition ensures continuous coverage across career stages and excludes scientists whose records are too sparse for stage-specific comparison. Finally,

we require that at least one institutional affiliation during career years 1–5 can be assigned to a country with Hofstede cultural dimension data (Supplementary Note 1.3).

The resulting sample comprises 584,712 scientists and 58,114,053 author–paper records. Of these scientists, 97.49% ( $n = 570,062$ ) have careers of at least 21 years and 84.64% ( $n = 494,906$ ) of at least 25 years. Our analyses focus on the first 25 career years, divided into five five-year periods — early (years 1–5), early-mid (6–10), mid (11–15), late-mid (16–20) and late (21–25) — yielding 38,339,950 author–paper records. The choice of career years 1–5 as the early-career window is empirically motivated. As shown in Fig. S0a, the timing of scientists’ first international move is concentrated in the early career and peaks at career year 5. We therefore define early career as years 1–5 throughout. This choice is not critical to our findings: the main results are unchanged when early-career cultural distance is computed over alternative windows from years 1–3 to 1–7 (Supplementary Note 3.1.2; Table S3).

**Mobility groups.** A scientist is classified as mobile (“Mob”) if affiliations in more than one country are observed during career years 1–5, and as non-mobile (“non-Mob”) otherwise, yielding 47,838 mobile and 536,874 non-mobile scientists. Within the mobile group, scientists are further divided into three equally sized subgroups — Shorter, Medium and Longer — by the terciles of early-career cultural distance, and all within-mobile comparisons in the main text contrast the Shorter and Longer terciles.

**Manual validation of mobility classification.** Because our mobility measure depends on the completeness and accuracy of affiliation records, we manually validated a random sample of 200 scientists, comprising 100 classified as mobile and 100 as non-mobile. For each individual we assembled publicly available biographical information, including personal websites, curricula vitae and Wikipedia entries, and recorded the countries in which they held positions during their first five career years. Comparing these records with the algorithmic assignments yields a recall of 95%, a precision of 76% and an overall accuracy of 86% for mobility status, with an area under the receiver operating characteristic curve of 0.875.

Exact country lists matched the manually recorded lists for 95% of non-mobile scientists and 70% of mobile scientists. Among the 30 mismatches in the mobile group, 24 arose from scientists who were in fact non-mobile but were algorithmically labelled as mobile. Of the remaining six cases involving genuinely mobile scientists, five had one additional country assigned relative to the manual record, with all other countries matching, and one had a single country missing. Restricting attention to scientists who are genuinely mobile, country-list accuracy is 92% (70 of 76); among genuinely non-mobile scientists it is 95% (95 of 99).

The dominant error is therefore the misclassification of non-mobile scientists as mobile, which introduces genuinely non-mobile individuals into the Mob group and, within that group, assigns them cultural distances they did not in fact traverse. Both forms of contamination attenuate the contrast between groups rather than manufacture it, so the estimates we report should be read as conservative with respect to the differences documented in the main text.

### **Supplementary Note 1.3: Cultural distance**

Hofstede dimensions. We measure cultural distance using the six national culture dimensions developed by Hofstede and colleagues: power distance (pdi), individualism (idv), masculinity (mas), uncertainty avoidance (uai), long-term orientation (ltowvs) and indulgence (ivr). Each dimension places a country on a scale constructed from large-scale attitudinal surveys, and together they constitute the most widely used quantitative characterization of cross-national cultural difference.

Coverage and imputation. Country-level scores were obtained from Hofstede Insights (2023), covering 119 countries and accounting for 99.52% of all publications in MAG. After excluding countries with missing values on one or more dimensions, 98 countries remain. Because Israel accounts for a non-trivial share of the sample (0.71% of total publications) and is missing only the indulgence dimension, we impute its value as the mean of the five countries closest to it in Euclidean distance on the other five

dimensions. The final set comprises 99 countries, together accounting for 99.07% of all publication records in the sample.

Normalization and pairwise distance. Each dimension is rescaled to the unit interval by its range across the 99 countries, so that no single dimension dominates by virtue of its original scale. The cultural distance between two countries  $m$  and  $n$  is then the sum of absolute differences across the six rescaled dimensions:

$$d(m, n) = \sum_{i \in I} |m_i - n_i|, \quad I = \{pdi, idv, mas, uai, ltowvs, ivr\} \quad (1)$$

Early-career cultural distance. Let  $k$  denote the number of distinct countries in which a scientist held an affiliation during career years 1–5. Early-career cultural distance is the mean pairwise distance over all country pairs in that set:

$$\text{Cultural Distance} = \frac{1}{k^2} \sum_{m=1}^k \sum_{n=1}^k d(m, n) \quad (2)$$

The measure is zero for non-mobile scientists, for whom  $k = 1$ . Averaging over all ordered pairs rather than the upper triangle gives greater weight to scientists who traversed more countries, reflecting their broader cultural exposure; restricting to unordered pairs leaves all results unchanged, as do the unnormalized distance, a percentile transformation of each dimension, and a Euclidean alternative (Supplementary Note 3.1.4, Table S6).

**Limitations and the embedding-based alternative.** The Hofstede framework has three properties that matter here. It is survey-based and time-invariant, so it cannot capture change in national culture over the study period; it is missing for a number of countries, which restricts the sample; and it describes general societal values rather than the specific frictions scientists encounter when moving between research systems. To address these limitations we develop a behaviour-based alternative in which cultural distance is learned directly from the observed mobility trajectories of 4.2 million scientists, using three unsupervised representation learning approaches (Supplementary

Note 2). These embedding-based measures cover all countries represented in the data, and all substantive conclusions are reproduced under each of them (Tables S7 and S29).

#### **Supplementary Note 1.4: Outcome measures**

We characterize long-term academic performance over career years 6–25 along three dimensions — productivity, impact and disruptiveness — each measured at the level of the individual scientist.

**Productivity (Publications).** The total number of papers published by a scientist during career years 6–25. Because this count is right-skewed and non-negative, all regressions taking Publications as the outcome are estimated by Poisson regression with robust standard errors.

**Impact (Hit 10pct).** For each paper we determine whether its citation count falls in the top 10% among all papers published in the same year and the same MAG L1 field; papers assigned to more than one field qualify if they meet this criterion in any of their fields. Hit 10pct is then the proportion of a scientist’s papers in career years 6–25 that qualify, and therefore takes values in  $[0, 1]$ . Normalizing within field  $\times$  year removes differences in citation levels across disciplines and across the varying lengths of citation window available to papers of different vintages. We verify in Supplementary Note 3.1.5 that conclusions are unchanged under alternative thresholds (top 1% and 5%), under the coarser L0 classification, under mean-normalized citation counts, under fixed five- and ten-year citation windows, when citations are restricted to the paper’s own broad discipline, and when normalization is performed within bibliographically coupled reference groups rather than within predefined fields (Tables S8–S10).

**Disruptiveness (D5 percentile).** For each paper we compute the five-year disruption index D5 (Supplementary Note 1.5) and average these raw values across a scientist’s papers in career years 6–25. Because the resulting author-level distribution is heavily concentrated near zero with long tails in both directions, we convert it to a percentile rank across all scientists in the matched sample, and this percentile is the outcome used

throughout. We verify in Supplementary Note 3.1.6 the conclusions are robust when using multiple DI-variants (D10, D5>0, and restricting to each paper’s research field’s D5), applying minimum reference and citation count restrictions, and under both field-year and bibliographic coupling normalization strategies (Tables S11–S13). Besides, the findings are robust to early career academic distance between different countries, field difference, or temporal effects. Stepwise inclusion of country, field, and year fixed effects, as well as controls for academic distance between countries, does not change the main patterns (Supplementary Note 3.1.7; Table S14). Embedding-based measures that integrate both cultural and academic attributes likewise yield consistent results (Table S7).

### **Supplementary Note 1.5: D5 calculation**

To quantify the disruptiveness of scientific contributions, we use the five-year disruption index (D5), which is calculated following established methods for the CD (Consolidation–Disruption) index <sup>2</sup>. Specifically, for each paper, D5 is defined based on citations received during years t+1 through t+4 after publication, capturing how subsequent work builds upon the focal paper and its predecessors. We begin at t+1 rather than t because works published in the same year as the focal paper cannot be reliably ordered relative to it, so a same-year work citing only its references may in fact predate it. Formally, D5 is computed as:

$$D5 = \frac{n_i - n_j}{n_i + n_j + n_k} \quad (3)$$

where  $n_i$  is the number of future works that cite only the focal paper,  $n_j$  is the number of future works that cite both the focal paper and its references, and  $n_k$  is the number of future works that cite only the references.

Because the final available MAG snapshot ends in March 2022, papers published after 2017 do not have a complete five-year citation window. To address this limitation, we

supplemented citation data for 1,046,448 author–paper records (2.73% of the total) using OpenAlex (release date: 27 February 2025), enabling us to assign D5 values to 788,525 records (75.35% of these recent papers). As a result, only 257,923 author–paper records (0.67% of the total) remained without calculable D5 values. To assess the robustness of our results, we compared three alternative approaches for calculating author-level D5 percentiles for years 6–25: (1) supplementing post-2017 D5 values from OpenAlex, (2) assigning missing D5 as NaN, and (3) calculating D5 using available but potentially incomplete citation windows for recent papers. Author-level D5 percentiles calculated using these three methods were highly correlated (Pearson  $r = 0.98$ ), and all main findings and statistical conclusions remained unchanged regardless of the approach used.

Paper-level D5 is available only for papers with recorded references and at least one forward citation within the five-year window. Zero-reference records never enter the computation, following established practice in which the disruption index of such works is treated as undefined, and papers with no forward citations are set to missing because no subsequent citation behaviour is observed from which displacement or consolidation could be inferred. A further boundary case arises when D5 equals exactly one. Recent work has excluded all such observations, on the grounds that they cannot be reliably distinguished from incomplete coverage of citation links, which is poorest in earlier publication years<sup>3</sup>; it has also been noted in response that not every such record need reflect a data artefact<sup>4</sup>. Because our sample extends back to the 1950s, we adopt the more restrictive treatment and set these observations to missing as well. All substantive conclusions reported in the main text are unchanged when  $D5 = 1$  records are retained.

## **Supplementary Note 2: Embedding-based cultural distance**

### **Supplementary Note 2.1: Construction of mobility graph**

To compute embedding-based measures of cultural distance, we leveraged a large-scale dataset of scientist career trajectories, comprising 207,009,605 author–institution–year records. We focused on scientists who were affiliated with at least two different institutions, yielding 4,211,081 unique mobility trajectories. For each scientist, we chronologically ordered their institutional affiliations to reconstruct individual career paths. To avoid artificial ordering biases, affiliations within the same year were randomly shuffled, and consecutive duplicate affiliations were removed<sup>5</sup>. Each institution was treated as a unique token, resulting in “sentences” (trajectories) suitable for representation learning.

Based on these sentences, we utilized the Word2Vec algorithm to learn low-dimensional vector representations for each institution. This approach captures the contextual similarity of institutions based on the mobility patterns of scientists, with the underlying assumption that institutions frequently appearing together in career trajectories share latent cultural or structural features.

In parallel, we constructed a weighted, undirected institutional mobility graph to further exploit the relational structure of the data. In this graph, nodes correspond to unique institutions (23,875 in total), and an edge between two institutions indicates that at least one scientist transitioned between them during their career. The weight of each edge was determined by the log-transformed frequency of observed transitions (yielding 1,490,998 edges overall). Additionally, each node was enriched with six institution-level features, including publications, h-index, logC10 (log10 of total citations plus one), geographic coordinates (longitude and latitude), and country code (one-hot encoded). This attributed graph provided a comprehensive structure for subsequent graph-based representation learning, enabling the generation of institution embeddings that integrate both mobility connectivity and institutional characteristics.



## Supplementary Note 2.2: Unsupervised embedding models

### Supplementary Note 2.2.1: Word2vec

To capture the latent functional distances between institutions based on scientists' career trajectories, we employ the word2vec model using the skip-gram with negative sampling architecture <sup>6</sup>. This method has been adopted in the study of scientific mobility <sup>5</sup>, embeds discrete locations (institutions) into a continuous vector space by treating individual mobility trajectories as sentences and institutions as tokens. The underlying assumption is that institutions that co-occur within career trajectories share functional, cultural, or prestige-related similarities that are not fully explained by geography alone.

Let each scientist's mobility trajectory be represented as an ordered sequence of institutional affiliations,  $(a_1, a_2, \dots, a_n)$ , where  $a_i$  denotes the institution at time  $i$ . For a given target institution  $a_i$ , its "context" consists of institutions that appear within a window size  $w$  around  $i$ , i.e.,  $\{a_{i-w}, \dots, a_{i-1}, a_{i+1}, \dots, a_{i+w}\}$ . The Skip-Gram model seeks to maximize the probability of observing context institutions given the target, formalized as:

$$J = \sum_{i=1}^n \sum_{-w \leq j \leq w, j \neq 0} \log p(a_{i+j} | a_i), \quad (4)$$

where the conditional probability is parameterized as:

$$p(a_{i+j} | a_i) = \frac{\exp(\mathbf{u}_{a_{i+j}}^\top \mathbf{v}_{a_i})}{\sum_{a \in A} \exp(\mathbf{u}_a^\top \mathbf{v}_{a_i})} \quad (5)$$

Here,  $\mathbf{v}_a$  and  $\mathbf{u}_a$  are the "in" and "out" embedding vectors of institution  $a$ , respectively. To efficiently train the model, we adopt the negative sampling strategy, which replaces the softmax normalization with a simplified binary classification objective. For each observed target-context pair  $(a_i, a_{i+j})$ ,  $K$  "negative" context

institutions  $\{a_k\}_{k=1}^K$  are independently sampled from a noise distribution  $P_n$ , which is set to the institution frequency raised to the 0.75 power and normalized. The loss for each pair is then:

$$L = \log \sigma(\mathbf{u}_{a_{i+j}}^\top \mathbf{v}_{a_i}) + \sum_{k=1}^K \mathbb{E}_{a_k \sim P_n} \left[ \log \sigma(-\mathbf{u}_{a_k}^\top \mathbf{v}_{a_i}) \right] \quad (6)$$

where  $\sigma(x)$  is the sigmoid function.

Following prior research <sup>5</sup>, we use the Python gensim implementation of Word2Vec, setting the embedding dimension  $d = 300$  and context window  $w = 1$ . Prior research has shown the embedding space also reflects a mix of geographic, historic, cultural, and linguistic relationships between regions.

#### Supplementary Note 2.2.2: Graph contrastive learning.

To further fuse institutions' academic information into the embedding space, we adopt two representative graph self-supervised learning methods. Generally, self-supervised learning on graphs can be classified into two major categories: contrastive approaches and generative approaches. We first adopt graph contrastive learning (GCL <sup>7</sup>) to learn institution embeddings that capture the latent structure and embedded academic information of the scientist mobility network.

Specifically, we first implemented a simplified GCL approach, in which the attributed mobility graph is randomly augmented in each batch, and embeddings of original and augmented views are contrasted (as shown in Fig. S1c). Let the attributed institutional mobility graph be denoted as  $\mathbf{G} = (V, E, \mathbf{X}, \mathbf{A})$ , where  $V$  is the set of  $N = 23,875$  institution nodes,  $E$  is the set of observed transition edges,  $\mathbf{X} \in \mathbf{R}^{N \times 6}$  is the node feature matrix (including publication count, h-index, logC10, longitude, latitude, and country code), and  $\mathbf{A} \in \mathbf{R}^{N \times N}$  are adjacent matrix.

The core idea of GCL is to learn representations invariant to small perturbations of the

input graph. For each training mini-batch, we generate two correlated views by applying independent random augmentations: (i) node feature perturbation, where Gaussian noise is added to each node’s feature vector to encourage invariance to feature-level fluctuations. For any node  $v$ :

$$\mathbf{x}_v^{\text{aug}} = \mathbf{x}_v + \epsilon_v, \quad \epsilon_v \sim \mathcal{N}(0, \sigma^2 \mathbf{I}) \quad (7)$$

Here,  $\mathbf{x}_v$  denotes the original feature vector for node  $v$ ,  $\mathbf{x}_v^{\text{aug}}$  is the perturbed feature, and  $\epsilon_v$  is an independent Gaussian noise vector. The noise variance  $\sigma^2$  ( $\sigma = 0.1$ ) is set to match the scale of the input features; and (ii) edge perturbation:

$$E^{\text{aug}} = \{e \in E \mid u_e > 0.1\}, \quad u_e \sim \mathcal{U}(0,1) \quad (8)$$

For each edge  $e \in E$ , we sample  $u_e$  independently from a uniform distribution over  $[0,1]$ .

Only edges with  $u_e > 0.1$  are retained, i.e., each edge is retained independently with probability 0.9 (corresponding to a 10% edge dropout rate).

Let  $\mathbf{G}_{\text{orig}} = (V, E, \mathbf{X}, \mathbf{W})$  and  $\mathbf{G}_{\text{aug}} = (V, E^{\text{aug}}, \mathbf{X}^{\text{aug}}, \mathbf{A})$  be the original and augmented views (only  $\mathbf{G}_{\text{aug}}$  is perturbed). A shared graph neural network (GNN) encoder  $f_\theta$  generates node embeddings, which are subsequently passed through a multilayer perceptron (MLP) projection head  $g_\phi$ :

$$\begin{aligned} \mathbf{z}_v^{\text{orig}} &= g_\phi(f_\theta(\mathbf{G}_{\text{orig}})) \\ \mathbf{z}_v^{\text{aug}} &= g_\phi(f_\theta(\mathbf{G}_{\text{aug}})) \end{aligned} \quad (9)$$

All output vectors  $\mathbf{z}_v^{\text{orig}}$  and  $\mathbf{z}_v^{\text{aug}}$  are L2-normalized prior to similarity calculation. For each node  $v$  in the batch, the contrastive loss (normalized temperature-scaled cross-entropy, NT-Xent) is computed as:

$$\ell_v = -\log \frac{\exp(\text{sim}(\mathbf{z}_v^{\text{orig}}, \mathbf{z}_v^{\text{aug}}) / \tau)}{\sum_{u \in B} \exp(\text{sim}(\mathbf{z}_v^{\text{orig}}, \mathbf{z}_u^{\text{aug}}) / \tau)} \quad (10)$$

where  $\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^T \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$  is the cosine similarity,  $\tau$  is the temperature parameter,

and  $B$  is the batch of nodes. The total loss is the average loss of each batch.

We employed a two-layer GNN as the encoder, with both hidden and output embedding dimensions set to 128, followed by a two-layer MLP projection head. During training, subgraphs were sampled using a two-hop NeighborLoader with 10 neighbors per hop to ensure scalability for large graphs. After self-supervised learning, the institution embeddings obtained from the encoder (excluding the projection head) were used for all downstream analyses.

#### Supplementary Note 2.2.3: Graph masked autoencoders

For generative approaches, we adopt graph masked autoencoders (GMAE<sup>8</sup>), which applies masking directly to node features and learns to reconstruct these masked features from surrounding context, implicitly encoding relational information among institutions.

Same with GCL, let the attributed institutional mobility graph be denoted as  $G = (V, E, \mathbf{X}, \mathbf{A})$ . During training, a subset of nodes  $V_m \subset V$  is randomly selected with a pre-defined masking ratio (e.g., 40%), and their feature vectors are replaced with a learnable mask token vector  $[\mathbf{MASK}] \in \mathbb{R}^d$  (here,  $d = 6$ ). Formally, the masked feature matrix  $\mathbf{X}^m$  can be expressed as:

$$\mathbf{x}_i^m = \begin{cases} [\mathbf{MASK}], & \text{if } i \in V_m, \\ \mathbf{x}_i, & \text{otherwise.} \end{cases} \quad (11)$$

where  $\mathbf{x}_i$  is the original feature vector of node  $i$ .

We use a two-layer GNN as encoder  $f_E$  that consumes graph  $\mathbf{G}_m = (V, E, \mathbf{X}^m, \mathbf{A})$  with the masked features produces the node embedding matrix:

$$\mathbf{H} = f_E(\mathbf{G}_m) \in \mathbb{R}^{N \times h} \quad (12)$$

where  $N$  is the number of nodes and  $h$  is the hidden embedding dimension (set to 128).

The decoder is similarly designed as a two-layer GNN  $g_\phi$ , with the aim of reconstructing the original node features  $\mathbf{X}$  from the encoded embeddings  $\mathbf{H}$ . The reconstructed features matrix is thus:

$$\hat{\mathbf{X}} = g_\phi(V, E, \mathbf{H}, \mathbf{A}) \quad (13)$$

To encourage meaningful embeddings, we utilized the scaled cosine error (SCE) as the reconstruction loss, focusing exclusively on masked nodes. For each masked node  $i \in V_m$ , the reconstruction loss is computed as:

$$L_{\text{SCE}} = \frac{1}{|V_m|} \sum_{i \in V_m} [1 - \cos(\mathbf{x}_i, \hat{\mathbf{x}}_i)]^\gamma \quad (14)$$

where  $\mathbf{x}_i$  is the original node feature,  $\hat{\mathbf{x}}_i$  is the reconstructed node feature, and  $\gamma$  is a hyperparameter (set to 3) controlling the sensitivity towards more difficult reconstruction cases.

Same with GCL, subgraphs were sampled using a two-hop NeighborLoader with 10 neighbors per hop to ensure scalability for large graphs. Upon completion of training, embeddings for each institution were extracted from the encoder, using the trained model weights. These institution-level embeddings, derived from masked reconstruction tasks, effectively integrated the academic characteristics and weighted structural relationships inherent in the scientist mobility network.

Finally, we can get institution-level embeddings through Word2vec, GCL and GMAE,

and country-level embeddings were computed as the average embeddings of institutions within each country, and embedding-based cultural distances were calculated as mean pairwise L1 distance between country embeddings, following the same approach as Hofstede cultural distance. Each scientist's early-career embedding-based cultural distance was computed as the average of pairwise country-level distances observed during the initial five years of their career trajectory.

### **Supplementary Note 2.3: Validation of embedding-based cultural distance**

#### Supplementary Note 2.3.1: Correlation analysis

Prior work has demonstrated that Word2Vec-based embeddings derived from scientist mobility trajectories capture latent cultural, functional, and prestige-related similarities between institutions and countries<sup>5</sup>. Here, we focus on validating the embedding-based cultural distance derived from graph self-supervised learning models (GCL and GMAE), which further integrate academic attributes and relational structure of the institutional mobility network.

We assessed the validity of embedding-based cultural distance by comparing the learned embeddings with the established Hofstede cultural distance measure. Specifically, we ran GCL and GMAE using both weighted and unweighted models over five different seeds to evaluate the stability and robustness of the learned embeddings. For weighted models, we used graph convolutional networks (GCN), which natively support edge weights during message passing. For unweighted models, we adopted GraphSAGE<sup>9,10</sup>. All models were evaluated using the same setting across five runs to ensure the stability of results.

To assess the construct validity of the graph-based institution embeddings, we first compared correlation of the embedding-based and Hofstede cultural distances. As shown in Table S2, the unweighted models achieved higher Pearson and Spearman coefficients than their weighted counterparts, both in the overall sample and among mobile scientists. To further compare embedding models, we selected, for each model,

the run with the highest correlation between embedding-based and Hofstede cultural distances among mobile scientists, and included word2vec as a benchmark. The word2vec yielded the highest correspondence with Hofstede (Spearman  $\rho = 0.69$ , Pearson  $r = 0.68$ ). Both graph self-supervised models—GMAE and GCL—also demonstrated positive correlations, with unweighted variants (GMAE:  $\rho = 0.46$ ,  $r = 0.41$ ; GCL:  $\rho = 0.43$ ,  $r = 0.40$ ) outperforming their weighted counterparts (GMAE:  $\rho = 0.48$ ,  $r = 0.32$ ; GCL:  $\rho = 0.30$ ,  $r = 0.23$ ). Accordingly, we used the unweighted variants of GCL and GMAE as the primary embedding-based measures for robustness checks with respect to Hofstede cultural distance.

#### Supplementary Note 2.3.2: Interpretability of self-supervised graph embedding

To further probe what information is encoded by self-supervised graph embeddings, we focused on the unweighted variants of GCL and GMAE—both to avoid potential data leakage from edge weights and due to their higher correlations with Hofstede distances. For each model, we selected the best-performing run (see Supplementary note 3.3.1), extracted institution-level embeddings, and then trained a MLP to predict the log-transformed transition frequency (edge weight) between institutions based on the concatenated embeddings of each institutional pair.

To interpret which features these graph embeddings rely on, we applied GNNExplainer<sup>11</sup> to the trained MLPs and systematically quantified the importance of each node attribute (including publication count, h-index, logC10, geographic coordinates, and country code) for edge weight prediction. Specifically, we divided the data into five folds, trained models on each, and for each test fold applied corresponding GNNExplainer to all institutional mobility edges (approximately 1.4 million in total).

The 5-fold cross-validated MLP prediction performance for log edge weights, using the concatenated institution embeddings, was as follows: for GCL,  $R^2 = 0.28 \pm 0.03$ , RMSE =  $0.82 \pm 0.02$ , Pearson =  $0.53 \pm 0.03$ , Spearman =  $0.41 \pm 0.02$ ; for GMAE,  $R^2 = 0.23 \pm 0.00$ , RMSE =  $0.85 \pm 0.00$ , Pearson =  $0.48 \pm 0.00$ , Spearman =  $0.36 \pm 0.00$  (all correlations are

between predicted and actual values, mean  $\pm$  standard deviation).

After obtaining the 5-fold MLPs combined with GCL or GMAE encoders, we trained the GNNExplainer model for each fold using the corresponding MLP and encoder. For each target edge, we aggregated the feature importance scores across all nodes in the subgraph, summing the importance of each of the six node attributes and normalizing across features. The results indicate that “academic mass”—operationalized as institutional productivity and h-index in GCL, and as productivity and logC10 in GMAE—consistently become the most important features in both embedding frameworks. By contrast, geographic coordinates and country codes contributed substantially less (See Fig. S1e). This pattern aligns with gravity model predictions (“mass” are the primary determinants of mobility intensity) and show that the self-supervised graph embeddings encode differences in academic “mass” in addition to geographic and cultural information.

Taken together, these findings demonstrate that self-supervised graph embeddings capture a combination of geographic, cultural, and academic information. Although GCL and GMAE embeddings reflect both cultural and geographic proximity, their correlation with Hofstede-based cultural distance is lower than that of word2vec. This may be because GCL and GMAE embeddings further encode academic capital differences. While the correspondence with Hofstede decreases from word2vec to GCL/GMAE, these methods incorporate additional academic and network features, and may also reflect patterns of academic reputation and structural capital within the global scientific community that are not captured by Hofstede or word2vec alone.

Accordingly, in our main analyses, we use word2vec-based measures as a behavior-based quantification of cultural distance, supplementing traditional survey-based indicators (i.e., Hofstede). We also report results using unweighted GCL and GMAE embeddings, as well as their weighted variants, which provide a more scientific community-oriented measure of cultural distance.



## **Supplementary Note 3: Measurement**

### **Supplementary Note 3.1: Long-term academic performance**

#### Supplementary Note 3.1.1: Placebo tests

To confirm that the observed associations between early-career cultural distance and long-term academic performance are not attributable to spurious correlations, we first conducted a placebo (shuffle) test. For each regression model assessing the effect of early-career cultural distance (both in the overall sample and among mobile scientists), we generated 1,000 placebo datasets by randomly shuffling the key independent variable—early-career cultural distance—across individuals, thus breaking any genuine relationship with the outcome variable. For each shuffled dataset, we re-estimated the main regression coefficient, yielding an empirical null distribution of effect sizes under random assignment. In parallel, we generated 1,000 bootstrap samples (using the original, unshuffled data) to estimate the sampling variability and construct the confidence interval for the observed effect.

We then compared the observed (“real”) regression coefficient to the placebo (shuffle) distribution, reporting the right-tailed p-value as the proportion of shuffled coefficients exceeding the observed value. All analyses controlled for the same five early-career covariates as the main models. As shown in Fig. S2, for all outcomes—productivity, impact, and disruptiveness—the observed effect sizes are far greater than those obtained under the placebo (shuffle) distribution, with the observed coefficient consistently lying in the extreme right tail (all  $P < 0.001$ ). Moreover, the bootstrap confidence intervals for the observed effect are also clearly separated from the shuffle distributions, with minimal or no overlap between the bootstrap and shuffle distributions for all outcomes. These results provide strong evidence that the associations between early-career cultural distance and long-term academic performance are not attributable to random chance.

#### Supplementary Note 3.1.2: Observation time window

Performance observation time window: Our main analyses evaluate long-term academic performance during years 6–25 of each scientist’s career, using three primary indicators: productivity (Publications), impact (Hit 10pct), and disruptiveness (D5 percentile). To test the stability of temporal evolution of these outcomes in Fig. 2 c, f, i, we also conducted analyses using annual, 3-year, and 5-year moving-window regressions. Given that both Hit 10pct and D5 percentile require a sufficient number of publications to yield stable estimates—particularly since annual data can be sparse—we present results for these outcomes using 3-year and 5-year moving averages to enhance reliability.

Specifically, we repeated analyses using annual, 3-year, and 5-year moving-window regressions for each outcome. As shown in Supplementary Fig. S3, the main patterns—including the productivity, impact, and disruptiveness advantages of mobile scientists, and the contrasting effects of short versus long cultural distance—are robust to these alternative specifications and persist throughout career years 6–21. The distinct career-stage trajectories identified in the main text (Fig. 2c, f, i)—including the increasing advantage of mobile (Mob) over non-mobile scientists (non-Mob) in productivity and disruptiveness, the early peak and later decline in impact, as well as the persistent and differentiated effects of shorter versus longer early-career cultural distance—are preserved under all windowing strategies.

Cultural distance observation time window: To further assess the robustness of our main findings to the definition of “early-career” cultural distance, we systematically varied the time window used to calculate cross-country cultural distance during early career (from years 1–3 to 1–7 years, except for 1-5). For each window, we estimated the association between early-career cultural distance and long-term academic performance (years 6–25). Results are shown separately for all scientists and for mobile scientists only using dataset after CEM.

For each specification, we present both unadjusted models (interpreted as raw mean differences) and regressions controlling for early-career academic trajectory

(publication count, hit 10pct rate, and D5 percentile), coauthor productivity, and institutional prestige—all measured during years 1–5. Because our controls are all measured in years 1–5, we did not re-match on different windows for each alternative specification. We note that, for interpretability and comparability across specifications, we did not conduct new 1:1 matching for each alternative window; instead, we used the original CEM-matched sample based on years 1–5 covariates. This approach ensures that differences in outcomes can be attributed to the varying definition of early-career cultural distance, rather than to sample selection or changes in matching quality. Notably, as shown in Table S3, the key patterns are robust whether or not controls are included

Table S3 summarizes these results. Three patterns are robust across all windows: (1) Mob scientists outperform their non-Mob peers in productivity, impact, and disruptiveness regardless of the observation window; (2) Among Mob scientists, longer early-career cultural distance is consistently associated with higher long-term productivity and disruptiveness, while shorter cultural distance is linked to higher long-term impact; (3) The direction and magnitude of these effects are stable across both controlled and uncontrolled models.

#### Supplementary Note 3.1.3: Alternative matching methods and estimation strategies

Alternative matching methods: To assess the robustness of our main results to different matching algorithms, we systematically compared four alternative matching specifications: (1) no matching (i.e., the full sample analyzed with regression adjustment alone), (2) many-to-many coarsened exact matching (m: n CEM), (3) one-to-one coarsened exact matching (1:1 CEM, which serves as our main baseline), and (4) one-to-one propensity score matching (1:1 PSM<sup>12</sup>). For each approach, we estimated both unadjusted models (i.e., raw mean differences, labeled as “no controls”) and covariate-adjusted models (adding the same five early-career controls used in our baseline analysis).

Balance diagnostics confirm that 1:1 CEM achieves the best covariate balance, with all post-match  $|SMD| < 0.02$  for every feature (see Table S1). Balance diagnostics confirm that 1:1 CEM provides the tightest covariate balance (all post-match  $|SMD| < 0.02$ ; Table S1). Under 1:1 PSM, imbalance is modest, post-match  $|SMD|$ s span 0.04-0.07, with the smallest residual on early career publication count ( $|SMD|=0.04$ ) and the largest on early career D5 percentile ( $|SMD| = 0.07$ ). By contrast, m: n CEM has less imbalance, with post-match  $|SMD|$ s ranging from 0.08 on early career D5 percentile to 0.55 on early career publication count (other covariates fall between 0.14 and 0.32). Despite these differences in balance quality, every matching specification reproduces the same qualitative results reported in Table S4

Alternative estimation strategies: We further estimated the effect of early-career cultural distance using two doubly robust causal inference methods: Double Machine Learning (DML <sup>13</sup>) and the Doubly Robust Learner (DRL <sup>14</sup>). Double Machine Learning (DML) and the Doubly Robust Learner (DRL) are two approaches for estimating average treatment effects (ATEs) in observational data, particularly when treatment assignment and outcomes may be confounded by high-dimensional covariates. Both methods are designed to mitigate bias from model misspecification and leverage machine learning flexibility while preserving valid statistical inferences. Unlike traditional approaches such as ordinary least squares (OLS) regression—which assumes a linear relationship between treatment, covariates, and outcome—DML and DRL are explicitly designed to handle nonlinear and high-dimensional settings by leveraging machine learning models.

**DML** achieves this by orthogonalizing the estimation of the treatment effect from nuisance components (i.e., control variables) via the so-called Neyman orthogonalization. Formally, for a continuous or binary treatment  $\mathbf{D}$ , outcome  $\mathbf{Y}$ , and covariates  $\mathbf{X}$ , DML decomposes the observed data into residuals:

$$\begin{aligned}\mathbf{Y} &= \mathbf{Y} - \mathbb{E}[\mathbf{Y} | \mathbf{X}] \\ \mathbf{D} &= \mathbf{D} - \mathbb{E}[\mathbf{D} | \mathbf{X}]\end{aligned}\tag{15}$$

Here,  $\mathbb{E}[\mathbf{Y} | \mathbf{X}]$  represents the outcome model, predicting the expected outcome given covariates, while  $\mathbb{E}[\mathbf{D} | \mathbf{X}]$  is the treatment model (propensity model), predicting the probability of treatment assignment given covariates. The ATE is then estimated by regressing the residualized outcome  $\mathbf{Y}$  on the residualized treatment  $\mathbf{D}$ , yielding an estimator that is robust to misspecification in either the treatment or outcome model, so long as one is consistently estimated. In practice, DML employs sample splitting and cross-fitting—dividing data into folds (here, 5) to avoid overfitting during nuisance parameter estimation and treatment effect inference.

**DRL** further generalizes this framework by constructing so-called doubly robust score functions, which combine information from both the outcome regression model and the treatment (propensity) model.

Specifically, for a binary treatment (for multiple treatments or multi-valued treatments, the DRL framework can be extended)  $\mathbf{D}$ , let:

$$e(\mathbf{X}) = \Pr(\mathbf{D} = 1 | \mathbf{X}) \quad (16)$$

denote the propensity score model (i.e., treatment model in DML), and

$$m_d(\mathbf{X}) = \mathbb{E}[Y | \mathbf{X}, \mathbf{D} = d] \quad d \in \{0, 1\} \quad (17)$$

denote the outcome models. Collecting all nuisance models into  $\boldsymbol{\eta} = (e, m_0, m_1)$  which includes all auxiliary models, the DRL doubly robust score is:

$$\psi(\mathbf{W}_i; \theta, \boldsymbol{\eta}) = m_1(\mathbf{X}_i) - m_0(\mathbf{X}_i) + \frac{D_i[Y_i - m_1(\mathbf{X}_i)]}{e(\mathbf{X}_i)} - \frac{(1 - D_i)[Y_i - m_0(\mathbf{X}_i)]}{1 - e(\mathbf{X}_i)} - \theta, \quad (18)$$

and defines the estimator  $\hat{\theta}$  as the solution via:

$$\frac{1}{n} \sum_{i=1}^n \psi(\mathbf{W}_i; \hat{\theta}, \boldsymbol{\eta}) = 0. \quad (19)$$

where  $i$  is the index for individual observations (i.e., scientists),  $\psi(\cdot)$  is the estimating;  $\mathbf{W}_i = (Y_i, D_i, \mathbf{X}_i)$ ; and  $\hat{\theta}$  is a consistent estimator of the ATE. Once both the propensity score model  $e(\mathbf{X})$  and the outcome regression models  $m_d(\mathbf{X})$  are estimated (i.e.,  $\eta$  are fixed), the ATE  $\hat{\theta}$  can be obtained through Eq. 17. This estimator remains consistent for ATE as long as either the propensity or the outcome model is correctly specified <sup>14</sup>.

For DML estimation, we used the LinearDML implementation from the EconML library. This method orthogonalizes the estimation of treatment and outcome by removing the effect of observed confounders through sample splitting and cross-fitting. Specifically, both the treatment and outcome models in Eq. 13 were implemented using XGBoost, and 5-fold cross-validation was applied for cross-fitting. In the Overall sample, we used a binary treatment variable (Mob as 1 and non-Mob as 0) as above. In the Mob, we directly use early-career cultural distance percentile as continuous treatment. And five early-career covariates as controls.

For DRL estimation, we used LinearDRLEARNER from EconML, which further combines residuals from the outcome and treatment models into a doubly robust moment condition. This provides valid ATE estimation so long as either the outcome or treatment model is correctly specified. In the Overall sample, we used a binary treatment variable (Mob as 1 and non-Mob as 0) as above. However, for the Mob subsample, early-career cultural distance was encoded as a three-level categorical variable, reflecting short, medium, and long cultural distances among mobile scientists. Since DRL does not directly support continuous or ordinal treatments for the ATE contrast, we explicitly mapped these group labels to integer values (0, 1, 2) and estimated the ATE as the contrast between Longer (coded as 2) and Shorter (coded as 0). This approach enabled us to estimate the causal effect of the shorter and longer scientists while adhering to the requirements of the DRL implementation. All other settings, including XGBoost for both outcome and treatment models in Eq. 14 and 15,

5-fold cross-fitting, and (where relevant) sample weighting under m: n CEM, were identical to those used for DML.

As shown in Table S4 and S5, the pattern of long-term academic performance is robust across all methodological variations. Specifically, Mob scientists consistently outperform their non-Mob peers in long-term academic performance, and among mobile scientists, longer early-career cultural distance is associated with higher long-term productivity and disruptiveness, while shorter early-career cultural distance predicts greater long-term impact. These two qualitative patterns remain unchanged regardless of whether matching is applied, which matching algorithm is used, whether or not covariate controls are included (Table S4), and which estimation method is employed, including both Double Machine Learning (DML) and the Doubly Robust Learner (DRL) (Table S5).

#### Supplementary Note 3.1.4: Alternative measures of cultural distance

Alternative distance functions: In our main analysis, we used the normalized L1 distance—that is, the mean pairwise Manhattan distance across six Hofstede cultural dimensions, normalized to [0,1] by the max-min of each dimension across countries—as the primary measure of early-career cultural distance. Here, we test the robustness of our findings to alternative distance functions, normalization strategies, and computation methods. As shown in Table S6, our pattern of long-term academic performance remains consistent across alternative distance functions, normalization strategies, and computation methods.

Specifically, we compared the normalized L1 distance with both the raw (unnormalized) L1(L1 raw) and an L1 distance based on percentile transformation for each dimension (L1 pct); in both cases, the results are unchanged, indicating that our findings do not depend on max-min normalization or percentile scaling. We further tested whether restricting the calculation to upper-triangular ( $i < j$ ) pairs only (L1 tri), rather than all pairwise combinations, would affect the outcomes; again, our results were robust,

suggesting redundancy in pairwise calculations does not introduce bias. Finally, we substituted the L1 metric with a normalized L2 distance (L2), and found that all qualitative conclusions within Mob scientists held. Together, these analyses demonstrate that our findings are robust to alternative distance functions and normalization approaches. Since non-mobile scientists have a distance of zero under every metric, the binary split between mobile and non-mobile (and the overall Mob vs. non-Mob contrasts) remains exactly the same regardless of which distance function we use. Consequently, the estimates for the normalized L1 baseline—both for the full Mob vs. non-Mob comparison and for the within-Mob subgroup—are identical to those in Table S4 and are therefore not repeated here.

Alternative embedding-based cultural distance: Hofstede cultural dimension, although widely used, have several limitations: they are survey-based, time-invariant, missing for a number of countries, and reflect general societal values rather than the cultural frictions scientists encounter in the contemporary research system. To obtain a behavior-based, up-to-date proxy that is available for all countries represented in our data, we therefore re-estimated early-career cultural distance from scientists' actual mobility patterns. To address these limitations and further evaluate the robustness of our findings, we utilized the methods described in Supplementary Note 2.2, learning institution embeddings with five unsupervised models—Word2Vec, GMAE (unweighted), GCL (unweighted), GMAE (weighted) and GCL (weighted)—and computed each scientist's mean L1 pairwise embedding distance across the countries traversed in years 1–5.

As reported in Table S7, the substitution of Hofstede by any of these embedding-based measures leaves our conclusions unchanged. Mobile (Mob) scientists continue to outperform non-mobile (non-Mob) peers in long-term productivity, impact and disruptiveness, and, within the Mob group, longer early-career cultural distance is associated with higher long-term productivity and disruptiveness, while shorter cultural distance remains linked to greater long-term impact. Together, these results reinforce



that our main conclusions are not sensitive to the specific operationalization of cultural distance, whether based on Hofstede dimensions or data-driven mobility embeddings. We further show in Supplementary Note 4.4 that the evidence on the potential channels documented in Figs. 3–5 is likewise robust to these embedding-based measures (Table S29).

#### Supplementary Note 3.1.5: Alternative impact measures

Hits: Our main analyses use the fraction of a scientist’s papers that rank in the top 10 percentile of citations, normalized by publication year and research field at the MAG LV1 level (“Hit 10pct”), as our primary impact measure. To assess the robustness of our findings, we further examined whether our results depend on the specific threshold chosen for defining high-impact papers or on the granularity of field classification.

Specifically, we tested alternative top-percentile thresholds (1%, 5%, and 10%) as well as a continuous measure of normalized citation (“FY C”) <sup>15</sup>, calculated as each paper’s citation count divided by the mean citation count of all papers published in the same field and year. In addition, we explored both the LV1 and LV0 levels of field categorization in MAG to examine the potential influence of field granularity. For each scientist, the “Hit 1pct,” “Hit 5pct,” “Hit 10pct” and “FY C” metrics were computed as the proportion of their publications within a given field and year that ranked among the top 1%, 5%, or 10% of cited papers, respectively. The “FY C” measure represents the average normalized citation count, as defined above. These calculations were performed at both the field LV1 and LV0 levels, resulting in a total of seven impact indicators: Hit 1pct (LV1), Hit 5pct (LV1), Hit 10pct (LV1), FY C (LV1), Hit 1pct (LV0), Hit 5pct (LV0), and FY C (LV0). As numerous studies have pointed out, impact and disruption are often influenced by reference length, among other factors <sup>16,17</sup>. To address this issue, we report all main regression results after controlling for reference count in combination with five early-career controls in Table S8. Together, these metrics provide a comprehensive evaluation of scientific impact, accounting for different high-impact thresholds and field definitions.

To further examine the influence of reference count—particularly given that papers with very short reference lists may include non-article publications—we conducted robustness checks using different thresholds for the minimum number of cited references. Specifically, we repeated our main analyses while sequentially restricting the sample to papers citing at least 5, 10, 15, or 20 references in Table S9.

Across all these robustness checks—including varying the high-impact threshold, switching field classification levels, and applying different reference count filters—our main conclusion remains unchanged: Mob scientists exhibit a robust advantage in scientific impact compared to non-Mob peers, regardless of impact definition or sample restriction. Among Mob scientists, those with shorter early-career cultural distance consistently achieve higher long-term impact. The direction of these effects is stable across alternative specifications and sample filters, underscoring the robustness of our results to variations in impact definition, field classification, and reference list length (See Table S8-9).

Citations: To further test the robustness of our findings beyond percentile-based impact measures, we additionally employ citation counts within fixed time windows—a widely used metric in bibliometric analyses. Specifically, we focus on C5, defined as the number of citations a paper receives within five years of publication. Given our data cutoff in March 2022, we filtered the papers published after 2017. To address potential sample selection introduced by this windowing, we also compute a Full Window C5 (FW C5), which measures the total citation count up to 2022 for papers published after 2017, even though their window is shorter than 5 years. Similarly, we test the effect of a longer window using C10 (ten-year citations), filtering papers published after 2012. All results using these alternative citation windows are reported in Table S10.

To further test whether the impact advantage of Mob/Shorter scientists is primarily driven by interdisciplinary citations, we also compute main field C5 using the MAG LV0 field categorization, which restricts citations to those received only from the same broad discipline. This allows us to examine whether the observed differences in impact

persist even when focusing exclusively on within-field recognition.

Bibliographic-coupling normalized citations: Conventional impact measures, such as top-percentile hits or mean-normalized citation counts, are inherently dependent on pre-defined field categorizations (e.g., MAG FieldID), which may mask topic diversity and interdisciplinary overlap. To provide a more granular and discipline-agnostic comparison, we employ a network-based normalization strategy for C5. This approach compares the citation performance of each paper to that of its most closely related contemporaries—those that share similar bibliographic references—rather than all papers within a coarse field definition<sup>18</sup>.

Specifically, for each focal paper, we identify its “bibliographic coupling” (BC) set, defined as all papers published in the same year that share at least one reference with the focal paper. The BC-normalized C5 (BC C5) is calculated as:

$$\text{BC C5} = \frac{C5_{\text{focal}}}{\text{mean}(C5_{\text{BC set}})} \quad (20)$$

where  $\text{mean}(C5_{\text{BC set}})$  is the average five-year citation count among the BC-linked set published in the same year.

Both main field C5 and BC C5 results are summarized in Table S10. Across all these alternative measures—including citation windows, field-restricted and network-based normalizations—our primary conclusions remain unchanged: Mob scientists retain a robust advantage in scientific impact, and shorter early-career cultural distance among Mob scientists continues to predict higher long-term impact.

#### Supplementary Note 3.1.6: Alternative DI measures

DI-variants: In the main text, we report results based on the author-level five-year disruption percentile (D5 Percentile), where each scientist’s D5 score is converted to a percentile rank among all papers published in the same year and field before

aggregation. To further assess the robustness of our findings to different disruption metrics and time windows, we report several additional DI variants in Table S11. First, we present results using the original author-level D5 values (i.e., without conversion to percentiles, see Table S11, “D5 values” column). To examine the influence of the citation window, we additionally report results using the ten-year disruption index (D10) at the author level, including both the raw D10 value and its percentile rank (see Table S11, “D10 values” and “D10” columns, respectively).

Next, to test whether the level of aggregation affects our conclusions, we aggregate D5 at the author level after first converting each paper’s D5 value to a percentile within the cleaned database (i.e., restricted to papers with at least one reference and one citation, published no later than 2017; Table S11, “D5 pct” column). Furthermore, recognizing that prior work has highlighted potential biases introduced by the inclusion of  $n_k$  in the denominator, we also report results using a binary indicator for disruptive papers, defined as the proportion of a scientist’s publications for which  $n_i - n_j > 0$  (i.e.,  $D5 > 0$ ; see Table S11, “D5>0” column)<sup>19</sup>. Aside from the original D5 and D10 values as well as the already percentile-scaled D5 pct, all other disruption indicators are converted to percentiles to mitigate the skewness of their distributions.

To further test the robustness of our findings to variation in reference and citation counts, we conducted additional analyses using different thresholds for the minimum number of cited references and citing papers when calculating the disruption index. Prior studies have suggested that restricting the calculation of D5 to papers with at least a certain number of references and/or citations can improve reliability, as papers with very short reference lists or few citing works may not provide stable estimates of disruptiveness<sup>19,20</sup>. Accordingly, in Table S12, columns 1–4 sequentially restrict the sample to papers citing at least 5, 10, 15, or 20 references, while columns 5–8 further restrict the sample to papers that have at least N references and at least N citations within the observation window (i.e., both reference and citation thresholds set at 5, 10, 15, or 20, respectively).

All models include reference count and early-career controls. Across these various thresholds, the positive association between cultural distance and long-term disruptiveness—both overall and among Mob scientists—remains robust, confirming that our main conclusions do not depend on the specific criteria used for the calculation of disruption.

Across all alternative DI-variants as well as different minimum thresholds for reference and citation counts—the positive association between cultural distance and long-term disruptiveness remains consistent and robust. These results hold both in the overall sample and specifically among Mob scientists, regardless of the disruption metric used or the sample restrictions imposed (Table S11–S12). Thus, the disruption advantage associated with early-career cultural distance is robust to the choice of disruption metric, observation window, and inclusion criteria for publications analyzed.

#### Fields, Year normalized-variants:

To further assess the robustness of our main findings to different field and year normalization approaches, we respectively constructed field-year normalized and bibliographic coupling (BC) normalized DIs. For the field-year normalized D5 (FY D5), we normalized each paper's D5 value within its field and publication year, using the same cleaned database definition as in Table S11 ("D5 pct" column)—that is, including only papers with at least one reference and one citation, published no later than 2017. Specifically, within each field-year group, we first linearly rescaled all D5 values to the  $[0, 1)$  interval by subtracting the minimum and dividing by the range (max–min), and then divided each rescaled value by the group mean. This approach mitigates the influence of skewed distributions and facilitates comparability across fields and publication years.

Similarly, to ensure comparability across groups without relying on any predefined classification system, we also introduce the bibliographic coupling-based normalization for disruption index (BC D5). Specifically, we adopted the approach analogous to Eq.

18 in our methods for BC C5, where normalization is performed within clusters of bibliographically coupled papers. Same with FY D5, given that D5 values can be negative, we first linearly rescaled all D5 values within each BC cluster to the [0, 1) interval by subtracting the minimum and dividing by the range (max-min). After this linear transformation, we then divided the normalized value for each paper by the mean of the [0, 1)-scaled D5 values within its bibliographic coupling (BC) set, following the same approach as for BC C5 (see Eq. 18).

Results based on these field and year normalized DI variants are reported in Table S13. Across all specifications, the positive association between early-career cultural distance and long-term disruptiveness remains robust, both in the overall sample and among Mob scientists. In both FY DI and BC DI models, cultural distance remains a statistically significant predictor of long-term disruptiveness, regardless of normalization strategy.

#### Supplementary Note 3.1.7: Country, Reference, Field, and Year normalization

Country normalization: A major concern is that the impact of mobility may be conflated with the effect of moving to an academic “core” country (e.g., the USA). These countries may offer systematically better resources, networks, and institutional advantages. For example, both moving from China to the US and moving from China to New Zealand might involve large cultural distance, but only the US provides a strong “academic center” advantage. If we do not control for this, we cannot distinguish whether observed performance differences are due to early career cultural distance or the inherent benefits associated with migrating to a core country. Although we already partially address this concern by matching on early-career institutional prestige, here we further include country fixed effects to rigorously control for country-specific advantages. Besides, these country fixed effects also help absorb confounding factors such as economic opportunity, language, and policy environment.

To address this, we include country fixed effects (Country FE) for major science

countries (i.e., US, UK, Germany, Japan, covering ~60% scientist of the whole samples) following prior research <sup>21</sup>, coded such that a scientist receives a dummy for each country in which they had early-career experience. As a result, the estimated effect of cultural distance is net of the “academic core country” advantage and is less likely to be confounded by factors such as economic opportunity, policy, or language.

To further isolate the effect of cultural distance from differences in national scientific environments, we control for academic distance (AD) between the source and destination countries, computed as the mean pairwise country Euclidean academic distance based on normalized country-level academic metrics (country H-index, publication counts, and LogC10 per paper), where each metric is normalized to the [0,1] range using min-max normalization. This adjustment can further remove bias arising from differences in scientific environments.

In Table S7, we have already shown that our embedding-based measures of early-career cultural distance—which integrate both cultural and academic characteristics at the institutional level via GCL and GMAE—produce results that are consistent with our main findings. These graph-based embeddings combine both cultural and academic distances, as they encode not only mobility patterns but also institutional academic attributes (e.g., h-index, productivity). Both the embedding-based approach (which implicitly includes academic distance) and the regression-based approach (here which controls for early-career academic distance) yield consistent conclusions (See also Supplementary Note 2 and Tables S7-8 for details). Taken together with Table S7, we find that our core conclusions are robust regardless of how cultural distance is measured: whether using the narrow, survey-based net effect of value differences (Hofstede cultural distance), or the broader, science community-oriented cultural distance measures that incorporate academic distance and capital (as in GCL and GMAE), the results consistently indicate an independent effect of cross-cultural mobility on long-term academic performance.

Reference, fields and year normalization: A sea of prior studies have shown that

measures such as disruption scores are sensitive to the number of references in each paper. Therefore, we include the contemporaneous average reference count (measured in years 6–25 of the career) as a covariate in all regressions. This absorbs potential bias from referencing behavior and ensures our results are not artifacts of systematic differences in reference length. Although we have previously replaced field-year normalized impact and disruption metrics, we further use field fixed effects (Field FE) and year fixed effects (Year FE) models to assess the robustness of our findings.

Publication counts, impact, and disruption index vary widely across disciplines and time periods. We introduce field fixed effects (Field FE) and year fixed effects (Year FE) to account for unobserved heterogeneity due to research area and historical trends. This allows for within-field, within-cohort comparisons, minimizing the risk that results are driven by differences in field or cohort composition. Specifically, we introduce field fixed effects (Field FE) by collapsing the MAG Level-1 discipline codes into the four broad categories used by prior research <sup>22</sup>: (1) Biomedical, behavioural, and ecological sciences; (2) Engineering and computational sciences; (3) Physical and mathematical sciences; and (4) Social sciences. For each focal scientist, we first compile their early-career publication list and assign a Level-1 FieldID to each paper. These FieldIDs are then mapped to the four broad research areas above. If a single area accounts for more than half of a scientist's early-career publications, the scientist is assigned to that area. For those whose publications are evenly split between two areas, we assign both research areas (i.e., label with both categories).

As shown in [Table S14](#), we find that the effect of cultural distance on long-term academic performance remains robust after stepwise inclusion of these controls. Notably, the coefficients attenuate but largely remain significant, supporting the interpretation that cross-cultural movement has an independent association with performance, net of field, year, country, and academic environment.

Supplementary Note 3.1.8: Team leader, Fundings, Patents, Visibility, Prizes



Team leader: To gain a broader understanding of how early-career cultural distance shapes scientists' careers and look beyond long-term academic performance outcomes, we further examined scientists' transitions to team leadership positions, operationalized as their first appearance as the last author on multi-author papers.

We modeled the transition to team leadership as a birth-death (constant-hazard) process and fitted the distribution of transition times with an exponential density <sup>23</sup>:

$$p(t) = \lambda e^{-\lambda t}, \quad (t > 1) \quad (21)$$

whose memoryless property captures the roughly constant instantaneous “risk” of promotion once a scientist has entered the field. We estimated  $\lambda$  by maximum likelihood and report the corresponding average transition time  $t_{\text{avg}}$ . We restricted the analysis to transitions occurring at least two years after scientists' initial publication ( $t > 1$ ) to ensure a stable estimation of the transition rate to leadership positions (results including immediate transitions,  $t = 1$ , were identical).

We then separately estimated the exponential distribution parameters for Mob, non-Mob (CEM), and unmatched non-Mob scientists in Fig S4a, as well as shorter, medium, and longer scientists in Fig S4b.

Fig. S4 reveals early cross-cultural mobility accelerates advancement to leadership positions. Mobile scientists exhibit significantly shorter average transition times to team leader roles (Mob:  $\lambda = 0.162$ ,  $t_{\text{avg}} = 6.19$  years vs non-Mob CEM:  $\lambda = 0.155$ ,  $t_{\text{avg}} = 6.43$  years; unmatched non-Mob:  $\lambda = 0.139$ ,  $t_{\text{avg}} = 7.17$  years; Fig. S4a). Although matching substantially reduces the gap between mobile and non-mobile scientists, the mobility advantage remains significant. Among mobile scientists, those experiencing shorter cultural transitions achieved team leadership status fastest ( $\lambda = 0.173$ ,  $t_{\text{avg}} = 5.78$  years). Transition times lengthen as cultural distance increases

(Medium:  $\lambda = 0.158$ ,  $t_{\text{avg}} = 6.34$  years; Longer:  $\lambda = 0.153$ ,  $t_{\text{avg}} = 6.53$  years; Fig. S4b).

Fundings, Patents, Visibility, and Prizes: To further broaden our understanding of how early-career cultural distance shapes scientists' careers beyond long-term academic performance, we also examined a range of other career features, including funding acknowledgements (NSF and NIH), patent output, scientific (journal C5) and social visibility (newsfeed and tweet mentions), and receipt of major international prizes (Nobel, Fields, and Turing). For these analyses, we use the NSF, NIH, patent counts and Newsfeed, Tweet mentions provided by SciSciNet, an open source database <sup>1</sup>.

Fundings: We first analyzed funding acknowledgements. As shown in Fig. S5a–b, for NSF funding, Mob scientists acknowledge significantly more funding per paper than non-Mob scientists, both before and after matching (two-sided Welch's t-test,  $P < 0.001$ ). Within the Mob group, the difference between shorter and longer cultural distance is not significant. For NIH funding, however, the pattern is more complex: before matching, the Mob and non-Mob groups do not differ significantly (two-sided Welch's t-test,  $P = 0.87$ ). After matching, the (matched) non-Mob group actually shows a significantly higher mean NIH funding count per paper than the Mob group (two-sided Welch's t-test,  $P < 0.001$ ). Within the Mob group, we observe a gradient, with shorter-distance movers receiving the most NIH funding per paper, and the difference between the shorter and longer groups being significant (two-sided Welch's t-test,  $P < 0.001$ ; Fig. S5b).

After matching, the non-Mob group receives significantly more U.S. funding per paper than the Mob group (matched non-Mob: 0.2496; Mob: 0.2068; –17.15% decrease for Mob, two-sided Welch's t-test,  $P < 0.001$ ). Within the Mob group, we continue to observe a strong gradient: scientists who moved shorter distances received the most U.S. funding per paper (Shorter: 0.2459), followed by Medium (0.2174), and Longer-distance movers receiving the least (Longer: 0.1573). The difference between the

shorter and longer groups is substantial, corresponding to a 36.03% decrease in mean U.S. funding per paper for the Longer group (two-sided Welch's t-test,  $P < 0.001$ ).

Given the limited coverage of only two major U.S. funding agencies (NSF and NIH), we further linked our MAG dataset to the OpenAlex database to obtain more comprehensive funder and award information for each publication. While OpenAlex's funding coverage is limited—being more accurate and complete for articles also indexed in Scopus or WoS—it nonetheless maintains a comprehensive list of about 32,000 funder entities, allowing us to explore global funding patterns not captured by SciSciNet.

The results, shown in Fig. S5c, reveal a consistent pattern: whether or not matching is performed, Mob scientists receive significantly more global funding per paper than non-Mob scientists (Mob: 0.0686 vs. Non-mob: 0.0444; +54.50% increase; two-sided Welch's t-test,  $P < 0.001$ ). Within the Mob group, we observe a clear positive gradient with respect to cultural distance: Longer-distance movers receive the highest mean global funding per paper (Longer: 0.0891), followed by Medium (0.0629) and Shorter (0.0537) distance groups. The difference between longer and shorter groups is substantial (+65.92% increase; two-sided Welch's t-test,  $P < 0.001$ ). Notably, this trend is the inverse of that observed for U.S. funding (NSF+NIH), where shorter-distance movers tended to receive more support. This suggests scientists with longer early cultural distance being associated with greater access to diverse global funding sources, as captured by the broader set of funders in OpenAlex.

Overall, Mob scientists and longer-distance movers acknowledge more funding per paper, based on the roughly 32,000 funders covered after linking MAG to OpenAlex. Their lower NSF and NIH funding is consistent with their lower likelihood of being located in Western countries and top-100 institutions in later career (Table S23).

Patents: We next analyzed patent output (Fig. S5d). Before matching, Mob scientists have a significantly higher mean patent count per paper than non-Mob scientists (two-

sided Welch's t-test,  $P < 0.001$ ). However, after matching, this pattern reverses: the matched non-Mob group demonstrates a higher patent count per paper than the Mob group (two-sided Welch's t-test,  $P = 0.14$ ). Within the Mob group, shorter scientists produce the most patents per paper, and this is significantly higher than those with longer cultural distance (two-sided Welch's t-test,  $P < 0.001$ ). Specifically, the mean patent count per paper for the shorter group is 1.4373 compared to 0.9569 for the longer group, representing an increase of 50.2% for shorter movers.

Visibility: We further examined scientific and social visibility (Fig. S5e–g). For these outcomes, results were consistent regardless of matching; therefore, we focus on the comparisons between Mob and matched non-Mob scientists.

Scientific visibility was measured by the mean C5 of journals (or conferences) in which papers were published. Here, the C5 metric is defined as the average number of C5 citations received by articles published in the same venue and year—specifically, for each journal in each year, the journal C5 is calculated using the average C5 citations for all papers published in that journal in that year (for years after 2017, all journal C5 values are set to NaN due to incomplete citation windows). Each paper's journal C5 is assigned based on the value corresponding to its venue and publication year, providing a proxy for the scientific prestige and visibility of the publication venue.

As shown in Fig. S5d, Mob scientists published in journals with significantly higher journal C5 compared to matched non-Mob scientists (mean: Mob 13.46, matched non-Mob 12.01; two-sided Welch's t-test,  $P < 0.001$ ). This represents an increase of 12.07% in scientific visibility for the Mob group. Within the Mob group, shorter-distance movers again published in the most prestigious venues (Shorter: 13.87; Longer: 12.72), an increase of 9.04% for shorter vs. longer, with the difference also being statistically significant (two-sided Welch's t-test,  $P < 0.001$ ).

Social visibility was measured by the number of Newsfeed and Tweet mentions per paper (Fig. S5e–f). Mob scientists received significantly more Newsfeed mentions per

paper than matched non-Mob scientists (Mob: 0.00667; matched non-Mob: 0.00534; two-sided Welch's t-test,  $P < 0.001$ ), an increase of 24.91%. Similarly, for Tweet mentions, Mob scientists also outperformed their matched non-Mob peers (Mob: 0.319; matched non-Mob: 0.269; two-sided Welch's t-test,  $P < 0.001$ ), an increase of 18.59%. Within the Mob group, those with shorter cultural distance transitions had the highest social visibility (Newsfeed: Shorter 0.00834, Longer 0.00552; an increase of 51.09%. Tweet: Shorter 0.387, Longer 0.252; an increase of 53.57%; both comparisons, two-sided Welch's t-test,  $P < 0.001$ ).

Prize: Finally, we examined the fraction of scientists in each group who received major international prizes—specifically, the Nobel Prize, Fields Medal, and Turing Award (Fig. S5h).

To identify prize recipients, we compiled lists of winners for the Nobel Prize, Fields Medal, and Turing Award from official prize websites<sup>24</sup> and Wikipedia. We then performed simple text-based matching to determine whether each scientist's name appeared in these lists, flagging them as potential recipients. For further accuracy, we manually verified each match using institution and web records, as well as H-index in MAG up to 2022. The final, manually checked statistics are reported in Fig. S5g. Because results were consistent regardless of matching, we focus on the matched (CEM) data.

For the Nobel Prize, 0.0753% of Mob scientists received the award, compared to 0.0366% of non-Mob scientists, representing a 2.06-fold difference. Within the Mob group, shorter-distance movers had the highest share of laureates at 0.1164%, which is 3.01 times that of longer-distance movers (0.0387%). For the Fields Medal and Turing Award considered together, the proportion is 0.0258% for Mob and 0.0087% for non-Mob scientists (2.97-fold difference), and among Mob scientists, the rate is 0.0323% for shorter-distance versus 0.0258% for longer-distance movers, a 1.25-fold difference. When considering all three major international prizes combined, 0.0990% of Mob scientists received at least one award, compared to 0.0452% among non-Mob

scientists—a 2.19-fold increase; within Mob, the fraction for shorter-distance movers (0.1358%) is 2.11 times that of longer-distance movers (0.0645%). These results indicate that early-career mobility—especially for shorter scientists—is associated with a substantially higher probability of receiving the world’s top scientific honors.

In summary, beyond long-term academic performance, we assessed a constellation of additional career outcomes including transitions to team leadership, funds, patent, scientific and social visibility, and the probability of receiving major international scientific prizes (Fig. S4–S5). Compared to matched non-mobile scientists, those with early-career cross-cultural moves exhibited significantly shorter average transition times to team leader roles (6.19 vs. 6.43 years;  $P < 0.001$ ; +3.73%), published in more visible journals (mean C5: 13.46 vs. 12.01;  $P < 0.001$ ; +12.07%), received more social media attention (Newsfeed mentions per paper: 0.00667 vs. 0.00534;  $P < 0.001$ ; +24.91%; Tweet mentions: 0.319 vs. 0.269;  $P < 0.001$ ; +18.59%), and had a 2.19-fold higher likelihood of to receive a major international prize (0.0990% vs. 0.0452%).

Finer-grained comparisons among mobile scientists revealed that those with shorter early-career cultural distance transitions achieved leadership positions more rapidly (mean transition time: 5.78 vs. 6.53 years;  $P < 0.001$ ; +11.49%), produced more patents per paper (mean: 1.4373 vs. 0.9569;  $P < 0.001$ ; +50.2%), and published in higher-visibility venues (mean journal C5: 13.87 vs. 12.72;  $P < 0.001$ ; +9.04%). They also received higher social visibility (Newsfeed mentions per paper: 0.00834 vs. 0.00552;  $P < 0.001$ ; +51.09%; Tweet mentions: 0.387 vs. 0.252;  $P < 0.001$ ; +53.57%), and were 2.11 times as likely to be awarded a major international prize (0.1358% vs. 0.0645%).

Collectively, these findings suggest that early-career mobility, and especially mobility across shorter cultural distances, is linked to a broad array of advantages on later career.

## **Supplementary Note 3.2: Collaboration**

### **Supplementary Note 3.2.1: Labor advantages and Team reputation**

After matching and controlling for a series of early-career features, the most salient

remaining difference between mob and non-mob scientists is their involvement in international collaborations during early career—a direct manifestation of cross-border movement. By design, mob scientists are characterized by higher rates of cross-national/cultural moves and thus have more opportunities to engage in early international collaboration.

Labor advantages (non-Mob vs Mob): To examine this fundamental distinction, we compared early-career international collaboration proportion between the two groups. Fig. S6a shows Mob scientists exhibit a significantly higher proportion of internationally coauthored papers (0.1732) than non-mob scientists (0.0679, two-sided t-test  $p < 0.001$ ), with an increase of over 2.5-fold. Stratifying scientists by whether they engaged in any early-career international collaboration (i.e., international collaboration proportion  $> 0$ ), regardless of mobility status, those with such experience consistently produce more long-term publications, achieve higher Hit 10pct, and have higher D5 percentiles (Fig. S6b). We use the presence or absence of early-career international collaboration rather than the median split, as fewer than half of the sample have a nonzero early-career international collaboration proportion.

Previous research pointing that larger teams drive greater productivity through increased labor input and resources<sup>25,26</sup>, here we hypothesize that the higher early-career international collaboration proportion among Mob scientists enables them to access a broader pool (and more international) of collaborators in the later career stage, thereby conferring labor advantages over non-mob scientists. We first observe that, across all scientists, team country size (the average number of countries per paper, years 6–25) is positively correlated with coauthor count in the same period (Fig. S6d), consistent with the notion that more international teams bring in a larger pool of collaborators. Furthermore, mob scientists exhibit significantly higher team country size and coauthor count in years 6–25 than non-mob scientists ( $p < 0.001$ ; see Fig. 3a, S6f), and coauthor count in late career stage is itself positively correlated with long term Publications (Fig. S6g).

To probe the origins of these differences, we test that whether the higher early-career international collaboration proportion among mob scientists enables them to access a broader (and more international) pool of collaborators, thus driving up both team country size and coauthor count in later career stages. Our data support this mechanism: early-career international collaboration proportion is positively correlated with both team country size and coauthor count ( $p < 0.001$ ; see Fig. S6c, e).

Fig. 3b and Table S15, Panels a and b, further confirm that both team country size and coauthor count in the later stage are significant predictors of long-term Publications. This finding holds both in regression analyses—after controlling for cultural distance and early-career controls—and in stratified analyses based on the median split of team country size and coauthor count. Notably, when we introduce coauthor count in column Panel b (3), the coefficient on team-country size turns negative and loses significance once the interaction term is added in column Panel (4). This sign reversal implies that the team country size contributes to productivity primarily through the larger pool of collaborators. In other words, Mob scientists' superior output is driven higher early-career international collaboration proportion allow them to accumulate a substantially bigger labor advantage.

Labor advantages (Shorter vs Longer): Consistent with the patterns observed in the Mob vs. non-Mob comparison, we next examine how labor advantages unfold among scientists crossing shorter versus longer early-career cultural distances.

Longer-distance movers (longer scientists) consistently display a higher mean proportion of early-career international collaboration research compared to their shorter-distance counterparts (0.1861 vs. 0.1559; a relative increase of 19.38%,  $p < 0.001$ ; Fig. S7a). Furthermore, in the subsequent stages of their careers, these scientists go on to form significantly larger labor advantages (as measured by per paper team size in the later career stage and super team papers, i.e., papers with over 29 authors, are excluded), with mean later-career team sizes of 5.15 for longer-distance movers versus 4.78 for shorter-distance movers—an increase of 7.77% (Fig. S7d). This



trend mirrors the mechanism observed above: early international collaboration enables them to access a broader pool (and more international) of collaborators, thereby accumulating larger labor advantages later in the career. The results remain unchanged regardless of when super team papers are included.

In line with this, we again find a positive association between early-career international collaboration and later-career labor advantages (Fig. S7b). Across both shorter and longer groups, scientists working in larger teams consistently achieve higher long-term Publications (Table S16, Panel a–b). This relationship remains robust in regression models adjusting for early-career controls and cultural distance (Table S16, Panel b). Here, we use per-paper team size as the primary proxy for labor advantages in later career stages. Longer scientists likewise exhibit a higher coauthor count during later career (mean: 384.3 for longer vs. 349.0 for shorter). Because this difference is statistically weaker (one-sided Welch’s t-test,  $P < 0.05$ ) than the corresponding difference in per-paper team size, we retain per-paper team size as the primary measure. Moreover, results above remain unchanged if we use coauthor count instead of per-paper team size as the measure for later labor advantage.

Thus, the labor advantage mechanism—whereby higher early-career international collaboration proportion predicts larger labor advantages in later career—holds consistently both for Mob vs. non-Mob and within the mobile scientist for longer-versus shorter-distance scientists.

Team reputation (non-Mob vs Mob): We next sought to disentangle the mechanisms through which the impact advantage of Mob scientists arises. Previous work has established that international collaboration is associated with higher impact<sup>21,27</sup> and we also observed this pattern: scientists with international collaboration experience—whether indicated by engaging in any early-career international collaboration (i.e., collaboration proportion  $> 0$ ) or by belonging to teams with larger country size in the later career stage—consistently achieve higher long-term research impact (Hit 10pct) than those without such experience ( $p < 0.001$  in Fig3 b and S6b; see also FigS6 m).

Building on prior evidence that team reputation enhances research impact <sup>28</sup>, here we further ask whether this effect may operate through the reputation of the teams that Mob scientists are able to assemble in the later stage of their careers and where are this later career stage reputation comes from.

We first confirm that, in years 6–25, team country size is positively correlated with team reputation (measured as the mean cumulative number of top-1% cited papers among all coauthors, for each focal paper, up to the year prior; ranked because of right skew distribution; Fig. 3c). This suggests that scientists who participate in more internationally diverse teams tend to access collaborators with greater scientific capital. Turning to scientific impact, both stratified and regression analyses ( $p < 0.001$ ; Fig. S6m; Table S15, Panel a and c) demonstrate that team reputation is a predictor of long-term Hit 10pct. The effect is robust across approaches: regardless of mobility status, scientists in the higher team reputation group consistently achieve greater Hit 10pct, and the positive relationship holds in regressions after accounting for cultural distance and other early-career control. Furthermore, the interaction between team country size and team reputation is significantly positive ( $p < 0.001$ ), indicating that the benefit of collaborating with reputable teams is amplified in more international contexts.

Having established the central role of team reputation, we next probe its origins. Our analyses reveal that mob scientists possess a higher average number of per paper early-career coauthors from Western countries (US, Canada, UK, EU27) and from top-100 global institutions (using affiliations' h-index rankings up to 2022 calculated in MAG) (Fig. S6h, S6j, all two-sided  $p < 0.001$ ), both of which are associated with later team reputation (Fig. S6g and S6i). Notably, early-career international collaboration proportion also predicts later team reputation (Fig. S6l). These results point to early international exposure—not only in terms of the number of international collaboration research but also the prestige of early-career collaborators—as foundational for assembling high-reputation teams in the future.

Team reputation (Shorter vs Longer): In terms of internal differences among mobile

scientists, we next examine how team reputation brings the impact advantage of shorter- versus longer-distance movers. We first find that team reputation in the later career stage is significantly higher for shorter-distance scientists compared to their longer-distance counterparts (Fig. S7i). Interestingly, while longer-distance movers have a higher proportion of early-career international collaborations—which does translate into greater labor advantages through larger team sizes later in their careers—this increase in collaboration does not lead to higher team reputation in the long term, unlike the association observed between early-career international collaborations and reputation in the mob vs non-mob comparison.

To probe this discrepancy, we further investigate the composition of early-career collaborative networks. We find that shorter-distance movers, assemble teams with a higher average number of per paper coauthors from Western countries and from top-100 global institutions (Fig. S7e, S7g, all  $p < 0.001$ ). Both Western and elite early-career collaborators are, echoed with non-Mob vs Mob, positive associated with later team reputation (Fig. S7f, S7h). These findings suggest that access to elite networks—manifested by shorter-distance moves—providing higher team reputation among these scientists. Consistent with the patterns observed in the mob vs non-mob comparison, we also find that team reputation remains a robust predictor of long-term research impact within both shorter- and longer-distance groups. Both stratified mean comparisons and regression models (Fig. S7i, Table S16, Panel a and c) demonstrate a positive association between team reputation and Hit 10pct, regardless of cultural distance group.

Taken together, these findings indicate that the long-term impact advantage of Mob scientists—particularly those with shorter early-career cultural distances—is not simply a direct result of early-career international collaboration, but is also driven by early access to high-reputation, elite collaborative networks, commonly concentrated in Western countries and top-tier institutions. In sum, early elite collaborative networks and team reputation emerges as a central mechanism through which early-career cross-

border/cultural mobility translates into enduring long-term scientific impact.

### **Supplementary Note 3.2.2: Team assembly and Network structure**

Team assembly (non-Mob vs Mob): Beyond the early-career features and international collaboration proportion that have been matched and controlled, a crucial distinction between non-Mob and Mob, as well as between Shorter- and Longer-distance scientists, lies in the formation of new collaborative ties. By necessity, Mob scientists must establish new research partnerships after relocating internationally; similarly, Longer- and Shorter-distance movers/scientists encounter different degrees of cross-cultural/country differences, which may, in turn, lead to variation in their tendencies to forge new collaborations in new environments versus maintain existing ties. Such structural differences could potentially influence their team assembly and collaboration networks structure in later career.

To capture individual scientists' tendency to form new (as opposed to repeated/old) collaborations in early career, we measure the early-career dense tie ratio (Fig. S8a)—the fraction of early-career coauthors who appear on every focal scientist's paper, relative to all coauthors in early career<sup>29</sup>. A lower dense tie ratio suggests that a scientist is less reliant on a fixed set of “core” collaborators and is more likely to build new collaborative relationships during their early career. The results indicate a marked difference: Mob scientists exhibit a significantly lower early dense tie ratio (mean = 0.0167) compared to non-Mob scientists (mean = 0.0448), a decrease of 62.72% (two-sided t-test,  $p < 0.001$ ). This suggests that Mob scientists are less likely to rely on “super ties” and more likely to build fresh collaborative links when starting a new research trajectory abroad.

Such team assembly tendency in early-career networks has lasting consequences. We further find that early-career dense tie ratio is negatively associated with later-career betweenness centrality (years 6–25, betweenness centrality was obtained from unweighted collaboration network of focal scientists, excluding super-team papers; see

Fig. S8b), suggesting that those who forged more new ties early on are more likely to occupy the structural bridges in their later career's networks.

Extensive researches have repeatedly demonstrated that occupying a “bridging” (high-betweenness) position in the collaboration network allows scientists to connect heterogeneous communities, thus enhancing their scientific performance<sup>30,31</sup>. To test whether Mob scientists' superior long-term output can be attributed to this bridging position, we examined the betweenness centrality of Mob and non-Mob scientists' networks in the late-career stage, Fig. 3d). Mob scientists consistently display significantly higher betweenness centrality than their non-Mob counterparts ( $p < 0.001$ ). We also observe, regardless of mobility status, higher betweenness centrality, and in turn, is associated with consistently higher long-term Publications, Hit 10pct, and D5 percentiles (Fig. 3e and Table S17).

Thus, Mob scientists' early less dense tie ratio—building new collaborative connections in unfamiliar settings—appears to pave the way for their later emergence as knowledge brokers, ultimately contributing to superior long-term academic performance.

To further test whether the effects of team reputation and network position (betweenness centrality) operate as distinct channels in explaining long-term scientific impact among Mob scientists, we included both variables simultaneously in regression models (see Table S17, Panel c, column (3)). Both team reputation and betweenness centrality remain strongly significant predictors of Hit 10pct when entered together, suggesting that their effects are distinct. The interaction term in column (4) is significantly negative, indicating a degree of substitutability between these two mechanisms, yet the main effects remain robust and dominant. In addition, to quantify the extent to which these mechanisms account for the effect of early-career cultural distance, we compared the coefficient for cultural distance on Hit 10pct before and after adjusting for both team reputation and betweenness centrality (see Table S4 Panel b, column (6) and Table S17 Panel c, column (3)). The coefficient decreases from 0.0134 to 0.0019, representing an 85.82% reduction and the significance level drops from

$p < 0.001$  to  $p < 0.05$ . This substantial attenuation indicates that these two mechanisms together explain the majority of the impact of early-career cultural distance on long-term scientific impact.

#### Team assembly (Shorter vs Longer):

We next examined differences between shorter and longer scientists in their early-career strategies of expanding new collaborative ties versus consolidating existing ones. Would Longer-distance scientists, like Mob scientists compared to non-Mob, expand more new collaborative ties due to larger geographic/cultural moves, or instead rely more heavily on established ties due to unfamiliar environments? Conversely, might Shorter-distance scientists more readily form new collaborations because of the relative familiarity of their environments?

To address these questions, we first measured the early-career dense tie ratio—the fraction of early-career coauthors appearing on every focal scientist's paper ("super ties"). Interestingly, Longer-distance scientists exhibited a significantly higher early-career dense tie ratio (mean = 0.0200) than Shorter-distance scientists (mean = 0.0144, two-sided t-test,  $p < 0.001$ ; Fig. S9a), suggesting that Longer-distance scientists relied more on stable, core collaborators in early career, whereas Shorter-distance scientists showed greater openness to forming new collaborations.

To further characterize these patterns, we introduced two additional measures. First, we calculated early-career collaboration strength, defined as the mean number of collaborations between each scientist and their early-career coauthors (up to career year 5) to capture the tie strength and stability in early career. Longer-distance scientists showed significantly greater collaboration strength (mean = 1.83) than Shorter-distance scientists (mean = 1.72; two-sided t-test,  $p < 0.001$ ; Fig. S9b).

Second, following prior literature<sup>32</sup>, we classified collaborators of each focal scientist into three distinct groups for every early-career paper after year one: (1)  $n_p$  (repeat

collaborators who previously coauthored with the focal scientist), (2)  $n_q$  (indirect collaborators who previously collaborated only with focal scientist's prior coauthors), and (3)  $n_f$  (new collaborators, who neither directly nor indirectly collaborated with the focal scientist previously) to further measure the tendency to repeat collaboration and fresh collaboration. We then computed early team repeated coauthor proportion, defined as  $\frac{n_p}{n_p + n_q + n_f}$  for each paper. A higher value indicates a greater reliance on repeat collaborations. Longer-distance scientists demonstrated significantly higher repeated coauthor proportion (mean = 0.415) than Shorter-distance scientists (mean = 0.374, two-sided t-test,  $p < 0.001$ ; Fig. S9c).

Taken together, these results indicate that Longer-distance scientists exhibited a stronger preference in early career toward repeated and stable collaborative relationships, as evidenced by their significantly greater dense tie ratio, collaboration strength, and repeated coauthor proportion. Additionally, these three indicators negatively correlated with later-career betweenness centrality (years 6–25; Fig. S9d–f), suggesting scientists with stronger early-career reliance on repeated collaborators were less likely to occupy bridging positions in collaboration networks later in their careers. Indeed, Shorter-distance scientists exhibited significantly higher later-career betweenness centrality (mean = 0.826) compared to Longer-distance scientists (mean = 0.803; two-sided t-test,  $p < 0.001$ ; Fig. S10a). This pattern is consistent with our findings in the non-Mob versus Mob comparison as well as previous literature, where higher betweenness centrality is associated with greater long-term Hit 10pct (Fig. S10h).

Moreover, Longer-distance scientists continued to exhibit stronger tendencies toward repeat collaborations in later career. Early team repeated coauthor proportion positively correlated with both later-career team repeated coauthor proportion and later-career team direct repeat collaboration preference. We compute the team's propensity to prefer direct over indirect, i.e., the latter metric is defined as:

$$\text{Team direct repeat collaboration preference} = \frac{n_p}{n_p + n_q + n_f} - \frac{n_q}{n_p + n_q + n_f} \quad (22)$$

A larger value (bounded in  $[-1,1]$ ) indicates a stronger tendency to reuse established collaborators rather than to pull in “friends-of-friends”; values near zero imply indifference, while negative values suggest an inclination towards indirect or entirely new partnerships. The repeated-coauthor proportion  $\frac{n_p}{n_p + n_q + n_f}$  captures how often a scientist reuses any prior tie, but it does not distinguish between (a) reproducing the exact same relationships and (b) cautiously expanding the ego-network via trusted intermediaries. Subtracting the indirect-repeat share isolates this distinction.

Empirically, Longer-distance scientists show both a higher team repeated coauthor proportion and a higher direct repeat collaboration preference (both  $p < 0.001$ ; Fig. S9g–j), whereas their friend-of-friend expansion is smaller—consistent with their reliance on a tight core of familiar collaborators late in their careers. Therefore, longer-distance movers/scientists not only showed greater reliance on repeat collaborations in their early careers, but this tendency also persisted into later stages. Longer-distance scientists exhibited significantly smaller network expansions, both at first-order (direct new ties) and second-order (indirect ties or friends-of-friends), compared to Shorter-distance scientists, who expanded more widely at both levels.

Moreover, Longer-distance scientists maintained deeper collaborative relationships with their early-career coauthors. Specifically, 31.12% of their later-career collaborations involved early-career coauthors, whereas for Shorter-distance scientists, this proportion was significantly lower at 28.64%—a decrease of approximately 8% ( $p < 0.001$ ; Fig. S9k). Additionally, Longer-distance scientists published more papers on average with each retained early-career collaborator (6.91 papers) compared to Shorter-distance scientists (5.87 papers;  $p < 0.001$ ; Fig. S9l).

To ensure these differences were not driven merely by publication volume, we



calculated the percentile rank of each collaborator's publication count among all later-career collaborators. Even controlling for this factor, Longer-distance scientists continued to demonstrate significantly stronger collaborations with their early-career partners (mean percentile: 0.684 vs. 0.678,  $p < 0.01$ ; Fig. S9l).

We further categorized scientists based on median splits of early-career collaboration strength, early team repeated coauthor proportion, and later-career team direct repeat collaboration preference. Those with higher scores consistently achieved significantly higher long-term Publications but lower long-term Hit 10pct (Fig. S9m–o). This finding aligns with the observed patterns for longer scientists, suggesting that differing team assembly strategies between longer- and shorter- movers/scientists may contribute to divergent career outcomes in productivity and impact.

Indeed, Shorter-distance scientists appeals significantly more new collaborators per paper in later career (mean = 1.74 vs. 1.70;  $p < 0.001$ ; Fig. 3g), consistent with their broader network expansions. Although direction remains uncertain—whether more new collaborations generate higher team reputation or high team reputation attracts more new collaborators—a positive correlation exists between later-career team new collaborators and team reputation. However, such an expansion-oriented team assembly—i.e., increasing the number of unfamiliar new collaborators—may come at the cost of reduced coordination efficiency<sup>32,33</sup>. While direct measurement of transaction costs is infeasible, we operationalize coordination efficiency as the per-paper inverse of the average interval between repeated collaborations with all coauthors in the late career. This indicator captures how quickly and frequently scientists are able to re-engage familiar collaborators, reflecting the smoothness of team assembly and ongoing coordination. Indeed, we observe a negative correlation between team new collaborators and coordination efficiency (Fig. 3i), and coordination efficiency is significantly higher for longer-distance scientists compared to shorter-distance scientists ( $p < 0.001$ ; Fig. S10g).

We therefore indicate that shorter scientists, owing to smaller geographic/cultural

moves, more easily adapted to their new environments with lower adaptation and integration costs, facilitating new collaborations with high-capital (high-reputation) partners. Indeed, shorter scientists averaged significantly more top-100 affiliated coauthors per paper (mean = 1.00) compared to longer scientists (mean = 0.79;  $p < 0.001$ ). However, their expansive collaborative strategies entailed greater coordination costs (Fig. S10g). Conversely, longer scientists, facing higher adaptation and integration costs in unfamiliar environments, relied more heavily on stable collaborations, thereby minimizing coordination costs in the later career.

To explore whether these differences in team assembly along with labor advantages and team reputation, might explain disparities between shorter and longer scientists in Publications and Hit 10pct, we conducted regression analyses (Table S16). For Publications, including both team size and coordination efficiency reversed the effect of cultural distance from positive to negative (columns 3–4 in Panel b). Together, these two channels may fully account for the positive association between early-career cultural distance and long-term Publication, as the coefficient for cultural distance becomes negative and statistically insignificant after their inclusion, which means that scientists with longer early-career cultural moves achieve higher long-term productivity primarily through greater labor advantages and lower coordination costs with repeated collaborators.

Regarding Hit 10pct, while the number of new collaborators positively predicted impact in the baseline model (Panel c, column 1), this relationship became negative after controlling for team reputation (column 3), suggesting that shorter scientists may have enhanced their impact by leveraging new collaborations to build higher-reputation teams. However, caution is warranted as direction remains unclear.

Supporting this interpretation, Mob scientists also had significantly fewer new collaborators per paper (mean = 1.72) but higher coordination efficiency (Fig. S8h) compared to non-Mob scientists (mean = 1.81). This pattern mirrors the behaviors of longer-distance scientists, providing another potential channel for the long-term

productivity advantage of the Mob group. However, Mob scientists do not suffer from the reputation penalty associated with long-distance moves; instead, they maintain significantly higher team reputations than their non-mobile peers. This is likely attributable to their early-career accumulation of international collaboration experiences and elite access (Note 3.2.1), which, when combined with the efficient team assembly patterns observed here, allows Mob scientists to simultaneously outperform in both long-term productivity and impact.

Collectively, these results suggest that contrasting team assembly—balancing network expansion, capital access, and coordination costs—represent potential channel to the differences in long-term productivity and impact between shorter and longer scientists.

Network structure (non-Mob vs Mob): Prior research suggests that a dominant giant component and lower entropy—reflecting a more integrated, less fragmented structure—can increase innovations in the context of corporate patenting and greater organizational integration is generally seen as beneficial for cumulative innovation within firms <sup>34</sup>. So here, we ask what is the structure of collaboration networks among Mob scientists who become network bridges and whether Mob scientists’ superior in productivity can be ascribed to it?

We next probe the internal organization of late-career collaboration networks among scientists in different groups. To quantify the degree of network fragmentation and diversity, we calculate the component size diversity using normalized entropy. Specifically, component size diversity was defined as:

$$H = -\frac{1}{\ln K} \sum_{i=1}^K p_i \ln p_i \quad (23)$$

where  $K$  is the number of valid components (i.e., connected components remaining after removing the focal scientists and after filtering as described below).  $p_i = \frac{n_i}{N}$  is the proportion of nodes in component  $i$ , where  $n_i$  is the number of nodes in

component  $i$ , and  $N$  is the total number of nodes in all valid components. Only components whose size exceeds 5% of the total number of nodes are included in the calculation. Small components (below this threshold) are excluded to minimize noise from isolated or minor subgroups, ensuring the entropy reflects substantive structural diversity. A higher value of normalized entropy indicates that the collaboration network is more fragmented and balanced—that is, the sizes of the major connected subgroups are more evenly distributed, reflecting greater structural diversity. Conversely, a lower value indicates a network dominated by one or a few large components, with most nodes concentrated in a single group, thus reflecting less diversity and higher integration.

And we also introduce relative giant component as:

$$\text{Relative giant component} = \frac{n_{\text{giant}}}{N} \quad (24)$$

where  $n_{\text{giant}}$  is the number of nodes in the largest connected component (after removing the focal author), and  $N$  is the total number of nodes in all components. All components are included in this calculation (no 5% filtering).

We first find late-career betweenness centrality is positively correlated with component size diversity (Fig. S8c) and negatively correlated with the relative giant component (Fig. S8e). Meanwhile, Mob scientists exhibit significantly greater component size diversity (mean = 0.5164) than non-Mob scientists (mean = 0.4869), representing a relative increase of 6.1% ( $p < 0.001$ ) (Fig. S8d). In parallel, the relative giant component (the share of collaborators in the largest connected group) is smaller for Mob scientists (mean = 0.7283) than non-Mob (mean = 0.7514), a decrease of 3.1% ( $p < 0.001$ ) (Fig. S8f). Equation (22) defines the relative giant component as the proportion of nodes in the largest connected component. A higher value of the relative giant component indicates that a larger share of the network is consolidated into a single, dominant group, signifying greater organizational integration and connectivity. A lower value suggests

that the network is more fragmented, with nodes spread across multiple smaller groups rather than being concentrated in one giant component.

Here, we observe the opposite pattern: Mob scientists possess more balanced and fragmented networks, but still achieve higher long-term publication output. One may ask: why do Mob scientists, whose networks are more balanced (higher entropy) and less centralized (smaller giant component), nevertheless achieve higher long-term Publications? While we have shown that greater labor advantages among Mob scientists account for much of this effect, alternative explanations may exist.

A closer look at the structure of Mob scientists' networks in the later career stage reveals additional clues. We use mean component density computed as:

$$\text{mean component density} = \frac{1}{K} \sum_{j=1}^K \left( \frac{2e_j}{n_j(n_j-1)} \right) \quad (25)$$

where  $n_j$  is the number of nodes in component  $j$ , and  $e_j$  is the number of edges within component  $j$ . This metric calculates the average density across all connected components in the network (after removing the focal author). All components are included in this calculation (no 5% filtering). A higher mean component density indicates that, on average, members within each connected component are more closely linked to one another, reflecting greater internal cohesion within subgroups. Interestingly, we find that, despite globally increased network fragmentation, Mob scientists exhibit locally higher mean component density (mean = 0.5390) compared to non-Mob (mean = 0.5253,  $p < 0.001$ ; see Fig. S8g).

Besides examining component density, we also considered whether Mob scientists might benefit from lower collaboration transaction costs and higher coordination efficiency in repeated collaborations. We observe Mob scientists have greater coordination efficiency in later career stage (measured by the inverse of the average interval between repeated collaborations, Fig. S8h: mean = 0.5170 for Mob vs. 0.4966

for non-Mob,  $p < 0.001$ ). These findings indicate that Mob scientists, while operating in more fragmented and diverse networks, are able to efficiently coordinate with familiar coauthors and maintain frequent interactions within each subgroup where tight-knit clusters are connected by bridging individuals, i.e., focal scientists themselves.

Table S17, Panels a and b, further support that temporal coordination efficiency are positive associated with long-term publications. This finding holds robustly across both regression analyses—after controlling for cultural distance and early-career controls—and in stratified analyses based on the median split of it. Besides, when we introduce coordination efficiency in Table S17 Panel b column (3), the coefficient on coordination efficiency remains positive, and both variables retain statistical significance when the interaction term is added in column (4). This pattern suggests that coordination efficiency contribute to productivity, and its effects are further amplified for scientists occupying network-bridge positions (i.e., with higher betweenness centrality). An analogous pattern holds for locally mean component density with respect to Publications: when we substitute mean component density in place of coordination efficiency, we observe the same positive effect on long-term publication output and a similarly positive interaction with betweenness centrality across all model specifications.

To quantify the extent to which the publication advantage of Mob scientists is jointly associated with labor advantages, network brokerage, and coordination efficiency, we next include these three mechanisms in the same regression framework. When later-career unique coauthor count is introduced alone with early-career cultural distance and controls, the coefficient on cultural distance decreases from 0.2310 (baseline in Table S4) to 0.1816, corresponding to a 21.39% reduction. When later-career betweenness centrality is introduced alone, the coefficient decreases to 0.1312, corresponding to a 43.22% reduction. When unique coauthor count and betweenness centrality are jointly included, the coefficient further decreases to 0.1136, corresponding to a 50.81% reduction. Finally, when coordination efficiency is additionally introduced, all three

coefficients remain positive and statistically significant ( $P < 0.001$ ) and the coefficient on cultural distance decreases to 0.0707, corresponding to a total reduction of 69.39% relative to the baseline. These results suggest that Mob scientists' superior long-term publication output is not driven by labor advantages alone, but by the joint combination of a larger collaborator pool, stronger brokerage positions, and higher coordination efficiency. In other words, Mob scientists not only bridge more communities, but also coordinate more efficiently within them, allowing their larger labor pool to be translated into substantially higher long-term productivity. This interpretation is also consistent with the broader structural patterns documented above: Mob scientists exhibit greater global network diversity and brokerage, yet also higher local cohesion, as reflected in greater component size diversity, mean component density and coordination efficiency.

Network structure (Shorter vs Longer): We next further take closer look at differences between shorter and longer scientists in their late-career's coauthor network structure. Based on prior theory, we still first hypothesize that a larger relative giant component and lower entropy with higher internal cohesion may facilitate greater coordination efficiency within scientific teams<sup>34</sup>. We observe that, akin to the patterns between Mob and non-Mob, within mobile scientists, betweenness centrality is positively correlated with component size diversity and negatively correlated with the relative giant component (Fig. S10b–e). Shorter scientists exhibit significantly greater component size diversity, while longer-distance scientists have a significantly larger relative giant component ( $p < 0.001$ ) and higher mean component density ( $p < 0.05$ ; Fig. S10g). Unlike non-Mob vs Mob scientists' comparison, as hypothesized, coordination efficiency is positively correlated with the size of the relative giant component (Fig. S10f). These network structural features align with observed behavioral differences in team assembly: shorter scientists' more open collaborations and bridge position brings higher fragment component, thereby associated with higher impact while longer scientists' preference for stable, repeated ties fosters and forms internally cohesive giant components that promote coordination efficiency and, ultimately, linked to long-term

productivity.

Taken together, the network-structure results reveal a clear divergence across the two comparisons. Relative to non-Mob scientists, Mob scientists occupy more brokerage-oriented positions in later-career collaboration networks, with higher betweenness centrality, greater component size diversity, and a smaller relative giant component, while simultaneously exhibiting greater local cohesion and coordination efficiency, as reflected in higher mean component density and repeat-collaboration efficiency. Within the mobile sample, shorter-distance movers likewise occupy more brokerage-oriented and structurally fragmented network positions, whereas longer-distance movers exhibit more cohesive and repeat-tie-oriented network structures, with a larger relative giant component and higher coordination efficiency. In this sense, betweenness centrality primarily captures scientists' ability to connect otherwise separated communities, whereas coordination efficiency and mean component density capture the extent to which collaboration is tightly organized and efficiently sustained within communities. This distinction also helps clarify why shorter-distance movers, despite higher betweenness centrality, do not achieve higher long-term publication output: connecting more communities does not by itself generate greater productivity without the larger labor advantages and stronger within-community coordination that characterize longer-distance movers. These differences help organize the productivity and impact patterns documented above, but they do not by themselves explain why scientists with greater early-career cultural mobility are more likely to achieve higher long-term disruptiveness.

### **Supplementary Note 3.2.3: Team expertise**

Currently, we have unveiled a bundle of channels—labor advantages, team reputation, team assembly, and network structure—that jointly illuminate how early-career cross-border/cultural mobility enhances long-term productivity (Publications) and impact (Hit 10pct). Yet, one crucial dimension remains elusive: what explains the persistent advantage in long-term disruptiveness (D5 percentile)? Why is it that those who have



greater early-career cultural distances (i.e., Mob scientists and longer scientists) are more likely to break with the past and set new trajectories in their fields compared with scientists with shorter early-career cultural distances?

To resolve this puzzle, we revisit examples in Fig. 1d: Richard Ernst's move from Switzerland to Palo Alto was not merely a change of location, but a decisive entry into the emerging paradigm of Fourier transform spectroscopy, fundamentally disrupting the field of NMR. Similarly, Adi Shamir's relocation from Israel to MIT brought him into direct contact with a new circle of mathematical thinkers, enabling the synthesis that birthed the RSA cryptographic method. Amartya Sen's career, shaped by three major moves—from India to the UK, then to the US, and back to the UK—embodied continual engagement with diverse economic, philosophical, and political traditions, culminating in a Nobel-winning synthesis of welfare economics and social choice. Their major scientific breakthroughs and “hot streaks”<sup>35</sup> often follow periods of geographic/cultural movement. Therefore, might mobility precede creative peaks and breakthrough achievements in science?

If so, what underlies this link between movement and breakthroughs as well as hot streaks? Recent research also confirms across domains—including scientists, artists, and film directors—breakthrough periods (“hot streaks”) are tightly coupled to shifts in creative and topic strategy<sup>36</sup>. Thus, it may not be simply the act of moving itself, but also the exposure to new ideas, techniques, and influences that mobility affords, especially in scientific centers (UK and USA in prior examples), which sets the stage for disruptive innovation?

Previous research also reveals that global artist migration hubs consistently overlap with centers of artistic innovation, style change, and collective creativity<sup>37</sup>. Artists who moved to these hubs may more likely to be at the forefront of stylistic revolutions and creative bursts. This phenomenon is also exemplified in artistic history: Vincent van Gogh's relocation from the Netherlands to Paris in 1886 introduced him abruptly to Impressionist color palettes, triggering the profound stylistic reinvention that

culminated in masterpieces like *The Starry Night*. Similarly, Picasso's arrival in Paris transitioned his Blue Period into Cubism.

Might this process also operate in science? When scientists move to new country, especially in global migration/science hub, they encounter unfamiliar ideas and methods compared with their prior location. This exposure may force scientists to integrate formerly distant ideas with new ones, and pivoting into fresh fields. Therefore, longer cultural moves (i.e., Mob scientists and longer scientists) may trigger the field shifts necessary for initiating scientific hot streaks and, subsequently, producing highly disruptive breakthroughs.

We therefore posit a potential channel in science: early-career cross border/cultural mobility—by relocating scientists to new expertise environments—push them to adapt to new collaborators and knowledge and pivot into new research area—thus increasing their probability of producing highly disruptive breakthroughs. The question then becomes: can we demonstrate that mobility-induced shifts in scientists' knowledge space and team expertise provide the missing link that connects early-career mobility to enhanced long-term disruptiveness (D5 percentile)? In what follows, we examine this hypothesis.

#### Team expertise (non-Mob vs Mob):

Our earlier analysis reveals that Mob scientists exhibit a significantly lower early-career dense tie ratio compared to their non-Mob counterparts, which may show their exposure to fresh collaborators after moving. We next test whether it can translate into pivoting into new areas and diverse team expertise capital.

We observe this dense tie ratio is negatively associated with early-career field pivot size (measured as one minus the mean cosine similarity between the MAG FieldID level 0 distributions of a focal paper's references and those of the scientist's cumulative prior publications<sup>38</sup>, with first-year papers excluded since no prior publication record exists), suggesting that scientists who form more new ties are also those more likely to switch

research domains early in their careers (Fig. S11a).

Mob scientists indeed exhibit a larger early-career field pivot size (mean: 0.1346 vs. 0.1303,  $p < 0.001$ ; a relative increase of 3.31%), reflecting a greater willingness or necessity to change research directions after relocation (Fig. S11b). Crucially, a higher early-career field pivot size is positively correlated with both increased team expertise distance in later career (Fig. 3f, S11d) and with greater betweenness centrality in the late-career collaboration network (Fig. S11f). Here, team expertise distance is computed as the mean pairwise Euclidean distance (L2 normalized) between the knowledge vectors of all coauthors (each vector based on the distribution of MAG FieldLV0 fields up to the publication year of the focal paper), excluding super team paper. Notably, our findings remain consistent when we employ an alternative metric of team expertise diversity: specifically, by first L1-normalizing each coauthor's knowledge vector, aggregating these to form a composite knowledge vector for the team, and then calculating the entropy of this mixed vector <sup>39</sup>. Consistently, mob scientists assemble teams with significantly higher expertise distance in their later careers compared to non-mob peers (mean: 0.4466 vs. 0.4373,  $p < 0.001$ ; a relative increase of 2.13%), and similarly exhibit greater team expertise diversity (mean: 0.2431 vs. 0.2381,  $p < 0.001$ ; a relative increase of 2.08%). Team expertise diversity in later career is also positively correlated with early-career field pivot size (Pearson  $r = 0.410$ ,  $p < 0.001$ ; Spearman  $\rho = 0.536$ ,  $p < 0.001$ ), and is linked with long-term disruptiveness (D5 percentile; Pearson  $r = 0.180$ ,  $p < 0.001$ ; Spearman  $\rho = 0.175$ ,  $p < 0.001$ ). When we exclude the focal scientist themselves and focus purely on coauthor only expertise distance, the results remain robust: mob scientists still exhibit significantly higher coauthor expertise distance (mean: 0.4367 vs. 0.4292,  $p < 0.001$ ; a relative increase of 1.75%) and higher coauthor only expertise diversity (mean: 0.2184 vs. 0.2160,  $p < 0.001$ ; a relative increase of 1.12%) compared to non-mob peers.

This pattern suggests that early-career mobility drives both knowledge pivot and exposure to more diverse expertise networks, positioning mob scientists as bridges

between disconnected heterogeneous knowledge communities. We also observe that the early-career dense tie ratio is negatively associated with the early-career coauthor expertise distance (pairwise L2-normalized Euclidean distance of all coauthor knowledge vectors up to year 5), and mob scientists consistently show higher early-career coauthor expertise distance than non-mob (mean: 0.4785 vs. 0.4479,  $p < 0.001$ ; a relative increase of 6.83%; Fig. S11g), which means, after moving, Mob scientists face has the more diverse expertise coauthors in the early career.

These early-career features are not only distinctive of mob scientists but also predictive of future disruptiveness. Scientists with higher early-career field pivot size and coauthor expertise distance consistently achieve significantly higher long-term D5 percentile, regardless of mobility status (Fig S11c, h). This finding supports the hypothesis that early exposure to diverse knowledge pools and field pivots provides a channel for long-term disruptiveness. Further, in late career, team expertise distance is itself significantly positively associated with long-term D5 percentile (Fig. S11j). Stratified analysis in Table S17a and S17d confirms that, within both non-mob and mob groups, teams with higher late-career expertise distance achieve greater disruptiveness ( $p < 0.001$ ).

Regression analyses further confirm this conclusion: after controlling for cultural distance and other early-career controls, team expertise distance remains a significant predictor of long-term D5 percentile (D5 percentile, Table S17, Panel d). Notably, when both betweenness centrality and team expertise distance are included in the same model (column 3), only team expertise distance remains statistically significant, whereas the effect of betweenness centrality loses statistical significance. This pattern suggests that the disruptive advantage conferred by mobility is largely channeled through the assembly of heterogeneous, high-expertise teams, rather than through network bridging alone. When both terms and their interaction are entered simultaneously (column 4), the interaction term turns strongly positive and highly significant. This result indicates a synergistic relationship: brokerage and expertise heterogeneity, when combined, substantially amplify disruptiveness.

Taken together, these results demonstrate that mob scientists—through exposure to diverse expertise and early-career field pivoting—are more likely to position themselves at the intersections of heterogeneous knowledge communities, assembling teams with broader expertise and achieving higher long-term disruptiveness.

#### Team expertise (Shorter vs Longer):

Shifting our lens to the comparison between shorter- and longer-distance scientist, we find that scientists undertaking longer early-career moves consistently demonstrate greater early field pivot size (mean: 0.1419 vs. 0.1349, representing an increase of 5.18%,  $p < 0.001$ ; Fig. S12a) and higher early coauthor expertise distance (mean: 0.4785 vs. 0.4480, an increase of 6.81%,  $p < 0.001$ ; Fig. S12c), compared to their shorter counterparts. Both early field pivot size and early coauthor expertise distance are, in turn, positively correlated with later-career team expertise distance, which is itself significantly higher for longer movers (mean: 0.4562 vs. 0.4489,  $p < 0.01$ ; a relative increase of 1.61%). Using team expertise diversity as an alternative metric, the result remains robust: longer movers also show significantly higher team expertise diversity in their later careers (mean: 0.2484 vs. 0.2439,  $p < 0.001$ ; a relative increase of 1.84%).

Team expertise diversity is significantly correlated with early-career field pivot size (Pearson  $r = 0.414$ ,  $p < 0.001$ ; Spearman  $\rho = 0.545$ ,  $p < 0.001$ ), and is also positively associated with long-term disruptiveness (D5 percentile; Pearson  $r = 0.174$ ,  $p < 0.001$ ; Spearman  $\rho = 0.170$ ,  $p < 0.001$ ). Same with non-Mob and Mob comparison, when we exclude the focal scientist and focus purely on coauthor expertise distance, the results remain robust: longer scientists still exhibit significantly higher coauthor expertise distance (mean: 0.4452 vs. 0.4362,  $p < 0.001$ ; a relative increase of 2.06%) and higher coauthor expertise diversity (mean: 0.2225 vs. 0.2172,  $p < 0.001$ ; a relative increase of 2.44%) compared to shorter scientists.

However, this raises a key puzzle. As shown earlier, longer scientists also exhibit a higher early-career dense tie ratio and a greater propensity for repeated collaborations,

both indicators associated with lower field pivot and coauthor expertise distance in the non-Mob vs. Mob comparison. Why, then, do longer movers display the opposite pattern, achieving greater field and expertise pivots despite their preference for stable collaborations?

We first propose that the answer lies in the difference of their mobility: by crossing greater early-career cultural and geographic distances, longer scientists are exposed to new knowledge environments and collaborator pools that are more distant from their previous networks compared with shorter counterpart. Thus, even when relying on “super ties,” the new collaborators encountered by longer movers tend to be further from their original knowledge space, enabling larger knowledge pivots.

However, it may not tell the whole story. Recent studies have demonstrated that impactful scientists are more likely to involve their collaborators in new research topics, effectively acting as catalysts for topic switches within their networks<sup>40,41</sup>. Building on this evidence, we hypothesize a potential mechanism underlying the greater field pivot observed among longer-distance scientists: following moves that span larger cultural distances, scientists are more likely to enter collaboration pools where the scientific capital/impact of their coauthors is, on average, substantially higher than in their previous networks, compared to those who move shorter distances. As a result, these newly formed ties with more prominent collaborators may further facilitate knowledge pivoting and transitions into new research fields. In contrast, scientists who move shorter distances are likely to experience a smaller increase in collaborator capital, resulting in more modest field and knowledge pivots.

To further probe this channel, we first examine whether longer scientists experience larger jump in coauthors scientific capital, using early-career coauthor capital imbalance—measured as the coefficient of variation in cumulative coauthor citation impact (Log C5) up to career year 5 of focal scientists (results are robust to Gini and other inequality metrics). We find that longer scientists display substantially greater capital imbalance among early-career collaborators than shorter movers (mean: 0.5525

vs. 0.4821, an increase of 14.62%,  $p < 0.001$ ; Fig. S12g). Then might this larger jump can bring larger field pivot in early career after moving and thus surrounded around larger heterogeneous network brought by larger knowledge difference between coauthors before and after moving? We then observe greater early-career coauthor capital imbalance is positively associated with both early field pivot size and early coauthor expertise distance (Fig. S12g, h). Stratified analyses further reveal that scientists with higher coauthor capital imbalance consistently achieve significantly higher long-term D5 percentile but lower Hit 10pct, regardless of mobility status (Fig. S12j).

These findings suggest that early-career coauthor capital imbalance may act as a potential channel through which longer scientists achieve greater early knowledge pivots and, then, higher disruptiveness in their careers. We also find coauthor capital imbalance is itself positively correlated with later team expertise distance and negatively with later team reputation (Fig. S12i, k), which may explain its association with higher long-term D5 percentile but lower Hit 10pct. In other words, although greater early-career capital imbalance facilitates broader field switching and the accumulation of diverse expertise—thereby increasing the potential for long-term disruptiveness—this same imbalance, reflecting a relative disadvantage in coauthor capital, may ultimately limit the probability of accumulating impact and lose the “chaperone effect” later in the career.

Regression analyses confirm these conclusions (Table S18). Panel a and d confirm that higher late-career team expertise distance is significantly associated with increased long-term disruptiveness (D5 percentile), for both shorter and longer scientists. When both team expertise distance and betweenness centrality are included, only the former remains statistically significant (Panel d, column 3), while their interaction term is again positive and significant indicating that the synergy of brokerage and expertise diversity amplifies disruptiveness—paralleling patterns observed in the non-Mob vs. Mob comparison. This result also explains why, despite shorter scientists and Mob scientists

both achieving higher betweenness centrality (i.e., occupying more bridging network positions) and opener collaboration structure, their long-term D5 percentile remains lower than that of longer scientists. The reason may be that these bridging positions do not necessarily connect sufficiently heterogeneous knowledge communities unless accompanied by higher team expertise distance. Thus, expertise diversity—rather than network position alone—plays the predominant role in enabling highly disruptive science in our analysis.

Panels b and c (Table S18) further reveal that after jointly controlling for early-career cultural distance and coauthor capital imbalance, the effect of cultural distance on both early field pivot size and early coauthor expertise distance diminishes (columns 3) and even reverses sign when interaction terms are included (columns 4), while the interaction remains significantly positive. These results underscore coauthor capital imbalance as a key driver for longer movers' greater knowledge pivot and expertise diversity, independent of cultural distance itself.

It is also worth noting that, across all specifications, early-career dense tie ratio and coauthor capital imbalance act as two largely distinct channels: the former is always negatively associated and the latter positively associated with early field pivot size and coauthor expertise distance, both in non-Mob vs. Mob and shorter vs. longer comparisons. Interestingly, while Mob scientists exhibit lower coauthor capital imbalance compared to non-Mob peers, they nevertheless achieve greater early knowledge pivots through the dense tie ratio channel. Thus, multiple routes exist by which early-career cultural mobility can generate the conditions necessary for long-term disruptiveness through knowledge recombination and the formation of fresh, heterogeneous, and high-expertise teams.

### **Supplementary Note 3.3: Knowledge strategy**

Innovation—whether in science or technology—has long been conceptualized as a process of “creative destruction”<sup>42</sup>, wherein new ideas and approaches supplant the old.



Building on this foundation, subsequent research has highlighted the central role of knowledge recombination in enabling such creative renewal <sup>43,44</sup>. Within this dynamic, both organizations and individual scientists confront a persistent tension between exploration—venturing beyond familiar domains—and exploitation—deepening established trajectories <sup>45</sup>. A growing body of empirical work demonstrates that balance of this exploration–exploitation trade-off shape creative outcomes, determining not only the impact but also the disruptive potential of scientific and technological contributions <sup>38,46</sup>.

In the preceding section (Supplementary Note 3.2), we have shown how early-career cultural mobility shapes the labor advantages, team reputation, team assembly and network structure as well as team expertise, thereby affecting long-term academic performance. However, what remains less understood is how these structural differences in collaborations impact specific knowledge strategies—namely, whether and how scientists with different early-career cultural distances utilized distinct approaches to knowledge exploration and exploitation. Moreover, do these behavioral strategies affected by the collaboration patterns in their careers, contributing to divergent trajectories in long-term academic performance?

Therefore, in this section (Supplementary Note 3.3), we connect the dots between collaboration and knowledge strategy of focal scientists. And we further first dissect the distinct propensity to exploitation and exploration among focal scientists.

### **Supplementary Note 3.3.1: Exploitation vs. exploration**

#### Exploitation vs. exploration (non-Mob vs Mob):

In the previous section, we found that Mob scientists—those with early-career international moves—tend to have more high-reputation collaborators in their early careers (e.g., more coauthors from western countries and top-100 institutions) and assemble teams with greater country size and reputation in later career stages. This pattern suggests that Mob scientists may be embedded in more mature and knowledge-

rich networks, potentially favoring an exploitation strategy in fields level—leveraging deep knowledge accumulation within established fields. At the same time, we also observed that Mob scientists demonstrate larger early-career field pivot size and assemble teams with greater expertise distance and diversity later on, pointing to a possible inclination toward exploration and venturing into new domains. Which of these knowledge strategies in fields level—exploitation or exploration—do Mob scientists predominantly adopt in the long run?

To address this, we first measure fields exploitation using the coefficient of variation (CV) of reference field distributions (reference fields imbalance), normalized for field size by dividing the frequency of each field by its cumulative publication count up to the year of the focal paper. Domain exploration, on the other hand, is captured by the late-career reference fields pivot size.

Our analyses reveal that there is no significant difference in the degree of domain exploration (reference fields pivot size, years 6–25) between Mob and non-Mob scientists (Mob = 0.1253, non-Mob = 0.1261;  $p = 0.30$ ), indicating that both groups pursue field exploration at similar levels. However, when it comes to exploitation, a striking difference emerges: Mob scientists exhibit significantly higher reference fields imbalance compared to their non-Mob counterparts (means: Mob = 0.794, non-Mob = 0.757;  $p < 0.001$ ; see Fig. 4a) during years 6–25. This pattern suggests that Mob scientists with higher team reputation, despite their greater team expertise diversity, are more likely to draw heavily from a focused set of fields in their referencing behavior, consistent with a stronger exploitation strategy.

Further, stratifying scientists by the median split of late-career reference fields imbalance, we find that those in the higher imbalance group achieve higher Hit 10pct and D5 percentile regardless of mobility status (all  $P < 0.001$ ), and higher publications among non-Mob scientists ( $P < 0.001$ ), although this difference is not significant among Mob scientists ( $P = 0.15$ ; Fig. 4b). Moreover, reference fields imbalance is positively correlated with the normalized relative share of home fields in references (reference

home fields relative share, measured as the sum of field-normalized frequencies for the focal papers' home fields divided by the sum for home plus non-home fields, thus capturing the focal scientist's tendency to draw upon focal papers' disciplinary knowledge; correspondingly, reference global fields relative share is measured as the sum of field-normalized frequencies for non-home fields divided by the sum for home plus non-home fields) and Mob scientists also exhibit a significantly higher normalized relative share of home fields in their references compared to non-Mob peers (means: Mob = 0.796, non-Mob = 0.784;  $p < 0.001$ ; Fig. 4c), reinforcing the interpretation that mob scientists with greater reliance on self-disciplinary knowledge.

To probe whether this local exploitation is linked to structural features of Mob scientists' teams, we examined the association between team reputation, team country size, and fields exploitation. We find that both team reputation and team country size are positively correlated with later-career reference fields imbalance and the relative share of home fields in references (Fig. S13a–c). Moreover, this pattern generalizes to the country level: both reference countries imbalance (calculated analogously to fields imbalance, using the normalized frequency of reference countries) and reference home countries relative share are also significantly higher for Mob scientists (all  $p < 0.001$ ; Fig. S13e, h). These two measures are likewise positively correlated (Fig. S13d). Stratifying scientists by the median split of reference countries imbalance, we also observe that those with higher imbalance—irrespective of mobility group—consistently achieve greater long-term academic performance ( $p < 0.001$ ; Fig. S13f). These patterns echo our earlier findings at the field level, indicating that Mob scientists with greater team reputation and access to well-established knowledge networks, draw more intensively from their own field and country's knowledge base and exhibit higher tendency to local exploitation.

But how exactly does such local exploitation contribute to long-term academic performance? Previous research has highlighted the importance of a sequential exploration–exploitation dynamic, whereby an initial period of broad exploration

followed by later deep exploitation is associated with the emergence of “hot streaks” in creative careers <sup>36</sup>. In line with this, our findings reveal that Mob scientists not only display greater early-career exploration, as indicated by higher early-career field pivot size (Fig. S12a), but also subsequently exhibit stronger reference fields exploitation in their later careers. As shown in Table S19 (Panel a), both early-career field pivot size and later-career reference fields imbalance are positively associated with long-term impact (Columns 1–3). The interaction between these two variables is significantly positive (Column 4), the results suggest that a pattern of initial exploration followed by focused exploitation may represent a potential channel to higher impact and “hot streak” after moves we observed in the examples shown in Fig1. d among Mob scientists.

We further examined whether the enhanced visibility afforded by high-reputation collaborators acts as a “chaperone effect <sup>47</sup>” not only boosting the visibility and prestige of focal scientists but also facilitating deeper field-specific knowledge accumulation for focal scientists. Considering the positive relationship between team reputation and knowledge accumulation, does team reputation further enhance how effectively accumulated knowledge translates into scientific impact? Indeed, regression models including both reference fields imbalance and team reputation—as well as their interaction—reveal that each factor significantly and positively predicts long-term impact, with a notably positive interaction effect (Table S19, Panel b, Column 1-4). To quantify the extent to which these mechanisms account for the mobility advantage, we compared the coefficient for cultural distance before and after adjustment. The coefficient decreases from 0.0134 (baseline) to 0.0010 (Column 3), representing a 92.54% reduction, and loses statistical significance. This substantial drop indicates that the interplay of team reputation and focused knowledge exploitation statistically accounts for the impact advantage of mobile scientists. Thus, deep exploitation of home field knowledge and the visibility conferred by prestigious collaborative networks appear to function as complementary pathways, jointly enhancing long-term scientific impact.

Combined with collaboration channels discussed above, early-career cross

border/cultural moves expose Mob scientists to novel expertise environments, catalyzing field exploration and the accumulation of expertise diversity. Crucially, this period of mobility also establishes connections with high-impact collaborative networks, which provide both reputational capital and structural resources for subsequent knowledge exploitation. This sequential dynamic—initial exploration followed by later focused deepening—may ultimately facilitates both sustained impact and the emergence of “hot streaks” following international moves.

#### Exploitation vs. exploration (Shorter vs Longer):

Parallel to our comparison between non-Mob and Mob scientists, we next investigate whether the magnitude of cultural distance—shorter versus longer—induces divergent knowledge strategies that underpin their distinct long-term performance trajectories. Specifically, we ask whether shorter-distance scientists—benefiting from early exposure to elite collaborators from western countries and top-100 institutions, and subsequently assembling teams with higher reputation—are more inclined toward a strategy of local exploitation at the paper level. Conversely, longer-distance scientists, despite having higher rates of early international collaboration proportions, generally lack access to such elite networks and thus form teams with lower reputation. Instead, driven by larger early-career field pivot size and the assembly of teams with greater expertise distance in the later career, do they exhibit a stronger tendency toward exploration? Concurrently, shorter-distance scientists involve a higher team new collaborators, associated with lower coordination efficiency in the later career. In contrast, longer-distance scientists, despite fewer new collaborators and lower team reputation, maintain higher coordination efficiency. How do these distinct patterns of team assembly and coordination further shape their divergent local knowledge strategies?

Consistent with our previous findings, scientists with higher team reputation tend to exhibit exploitation strategies at the paper level. As shown in Fig. 4g, h, k, shorter-distance scientists exhibit a significantly higher reference self-citation ratio—defined

as the ratio of references whose author lists overlap with the focal paper's author list—compared to longer-distance movers (mean: Shorter = 0.1683 vs. Longer = 0.1653; one-sided Welch's t-test,  $P < 0.01$ ; Fig. 4g), representing a 1.80% relative increase in exploiting their own past knowledge. By contrast, longer-distance scientists display a higher propensity for paper-level exploration in their later career stages. They exhibit a larger reference fields pivot size (mean: Shorter = 0.1251 vs. Longer = 0.1321; one-sided Welch's t-test,  $p < 0.001$ ; Fig. 4h), reflecting a 5.60% relative increase, and a greater reference global fields relative share (mean: Shorter = 0.1992 vs. Longer = 0.2109; one-sided Welch's t-test,  $p < 0.001$ ; Fig. 4k), representing a 5.87% relative increase. Together, these patterns indicate a divergence in paper-level (local) knowledge strategy: shorter-distance movers emphasize local exploitation by drawing more heavily on familiar fields and established knowledge bases, whereas longer-distance movers engage more in unfamiliar-field knowledge exploration and exhibit larger shifts in the referenced field space.

We next probe the origins of these divergent local knowledge strategies. Fig. 4g-i indicates that team reputation is more conducive to deepening within familiar knowledge bases, rather than promoting paper-level exploration. Consistent with this view, among Mob scientists, later-career team reputation is positively associated with the reference self-citation ratio (Pearson  $r = 0.294$ ,  $p < 0.001$ ; Spearman  $r = 0.331$ ,  $p < 0.001$ ), but negatively associated with both the reference fields pivot size (Pearson  $r = -0.096$ ,  $p < 0.001$ ; Spearman  $r = -0.063$ ,  $p < 0.001$ ) and the reference global fields relative share (Pearson  $r = -0.117$ ,  $p < 0.001$ ; Spearman  $r = -0.097$ ,  $p < 0.001$ ). These patterns suggest that shorter-distance scientists' exploitation-oriented local knowledge strategy may be rooted in their early access to elite collaboration networks, which later enables the formation of higher-reputation teams.

A complementary question is why longer-distance scientists exhibit stronger paper-level exploration. One plausible channel, beyond team reputation, is transaction costs and relational trust in collaboration. Although shorter-distance scientists assemble

higher-reputation teams and bring in more new collaborators, such an expansion-oriented team assembly may increase transaction costs and weaken shared routines and mutual trust, potentially discouraging the integration of knowledge across fields and reinforcing reliance on familiar field knowledge<sup>33,48</sup>. Using coordination efficiency, as defined above, as an inverse proxy for transaction costs (higher coordination efficiency indicates more frequent and timely re-engagement with prior coauthors, and thus lower transaction costs and greater relational trust), we find that coordination efficiency is slightly positively associated with the reference global fields relative share (Pearson  $r = 0.083$ ,  $p < 0.001$ ; Spearman  $r = 0.101$ ,  $p < 0.001$ ). This pattern is consistent with the idea that repeated interaction may build shared routines and relational trust and may lower transaction costs, potentially increasing cross-disciplinary knowledge integration, whereas lower coordination efficiency may reinforce reliance on familiar field knowledge at the paper level. Conversely, longer-distance scientists—despite lower team reputation—may mitigate these barriers by collaborating more with familiar, repeated coauthors, leveraging relational trust to facilitate cross-field integration at the paper level.

A second, and potentially more direct, mechanism for longer-distance scientists' paper-level exploration is the greater team expertise distance, which may be shaped by earlier knowledge pivots. As we discussed above, longer-distance movers exhibit larger early-career field pivot size and later assemble teams with greater expertise distance. In line with this, Fig. S14a–b shows that later-career team expertise distance is strongly positively associated with both the reference fields pivot size (Pearson  $r = 0.707$ ,  $p < 0.001$ ; Spearman  $r = 0.771$ ,  $p < 0.001$ ) and the reference global fields relative share (Pearson  $r = 0.560$ ,  $p < 0.001$ ; Spearman  $r = 0.565$ ,  $p < 0.001$ ). Together, these results suggest that longer-distance scientists' paper-level exploration reflects a joint process: early large field pivots place them in more heterogeneous expertise environments, and sustained collaboration with familiar coauthors helps keep collaboration frictions low, which may jointly support larger shifts in referenced field space and greater tendency

to cross-disciplinary knowledge exploration at paper level.

Additionally, motivated by the earlier evidence that longer-distance movers exhibit larger early-career field pivots and greater coauthor capital imbalance, we further hypothesize that longer-distance movers, consistent with their lower exposure to elite networks compared to their shorter-distance counterparts, may continue to exhibit greater team capital imbalance in later career stages. We measure this imbalance using the per-paper coefficient of variation of coauthors cumulative citation impact (Log C5) up to the year prior to the focal paper's publication (results are robust when using the Gini coefficient as an alternative metric). As shown in Fig. S14c, longer-distance scientists exhibit a significantly higher team capital imbalance than shorter-distance scientists (mean: Longer = 0.5535 vs. Shorter = 0.5431; one-sided Welch's t-test,  $p < 0.001$ ), corresponding to an approximately 1.92% relative increase.

Although we cannot disentangle whether this higher imbalance arises because longer-distance scientists collaborate more with particular pools of coauthors in the later career (e.g., potentially including collaborators from their origin networks) or because they themselves occupy a relatively lower position in scientific capital, we find that greater later-career team capital imbalance is consistently associated with a more exploration-oriented local knowledge strategy. Specifically, later career team capital imbalance is negatively associated with the reference self-citation ratio (Pearson  $r = -0.258$ ,  $p < 0.001$ ; Spearman  $r = -0.290$ ,  $p < 0.001$ ), but positively associated with both the reference fields pivot size (Pearson  $r = 0.181$ ,  $p < 0.001$ ; Spearman  $r = 0.173$ ,  $p < 0.001$ ) and the reference global fields relative share (Pearson  $r = 0.179$ ,  $p < 0.001$ ; Spearman  $r = 0.173$ ,  $p < 0.001$ ) (Fig. S14d–f). These correlations suggest that teams with more uneven distributions of collaborator capital tend to drawing less on the authors' own prior knowledge base while exhibiting larger shifts in referenced field space and greater reliance on global fields knowledge at the paper level.

Ultimately, the divergent local knowledge strategies between shorter- and longer-distance movers appear to be shaped by differences in their collaborative ecologies.



While shorter-distance scientists' emphasis on local exploitation is linked to their integration into elite networks and higher-reputation teams, the stronger exploration-oriented strategy of longer-distance scientists is associated with a multifaceted set of factors: larger expertise distances may provide the cognitive breadth, sustained relational trust (as captured by higher coordination efficiency) is associated with lower barriers to cross-disciplinary knowledge integration, and higher team capital imbalance is additionally associated with a departure from familiar knowledge bases. Taken together, these patterns illustrate how specific collaboration characteristics may provide a plausible basis for the observed divergence in local knowledge strategy between shorter- and longer-distance scientists. Notably, although Mob scientists also exhibit larger team expertise distances and higher coordination efficiency, their substantially higher team reputation (vs. non-Mob) may offset these exploration-favoring conditions, aligning with a more exploitation-oriented local knowledge strategy. We next examine how these divergent knowledge strategies—characterized by varying degrees of global field exploration at the paper level—predict long-term scientific performance.

As shown in Fig. 4l, within both the Shorter- and Longer-distance groups, scientists in the higher reference global fields relative share group (median split) achieve significantly higher long-term productivity and disruptiveness as well as lower impact than those in the low group (one-sided Mann–Whitney U tests; all  $P < 0.001$ ). Building on the patterns discussed above, we consistently observe that team reputation in the later career stage is positively associated with a more exploitation-oriented paper-level knowledge strategy in both the Mob vs. non-Mob and the Shorter vs. Longer comparisons in the later career. In the non-Mob vs. Mob analysis, we further showed that the joint inclusion of local exploitation and team reputation accounts for most of the mobility advantage in long-term impact, corresponding to a 92.54% reduction in the cultural-distance coefficient (Table S19).

We therefore test whether the impact advantage of Shorter-distance scientists may similarly be associated with the joint role of team reputation and local exploitation.

Table S20 supports this interpretation. Consistent with Fig. 4l, we operationalize exploitation using reference home fields relative share (defined as  $1 - \text{reference global fields relative share}$ ), because Longer-distance scientists exhibit stronger global-field referencing yet lower long-term impact; using the reference home fields relative share provides directionally consistent coefficient interpretation for exploitation without altering the underlying information. In Panel a, reference home fields relative share is positively associated with long-term impact (Hit 10pct), and team reputation remains a strong positive predictor (all  $p < 0.001$ ). The interaction term is positive and significant (Panel a; columns (1), (2), and (4), all  $p < 0.001$ ), indicating that these two factors are jointly associated with higher long-term impact. Moreover, when both team reputation and reference home fields relative share are included, the coefficient on cultural distance attenuates from  $-0.0317$  (baseline; Table S4) to  $-0.0184$ , corresponding to a 41.96% reduction in magnitude. Taken together, these results suggest that the higher long-term impact observed among Shorter-distance scientists is to a considerable extent (41.96%) statistically associated with their access to higher-reputation teams—shaped by early integration into elite networks—and with a more exploitation-oriented local knowledge strategy. While team reputation and reference home fields relative share are positively correlated, indicating partial overlap between team reputation and local knowledge exploitation, both variables remain significant when entered jointly, and their positive interaction further suggests that the impact returns to exploitation are amplified in higher-reputation team contexts.

We further examine the relationship between local knowledge strategy and long-term disruptiveness in the later career stage. Consistent with the patterns observed in Fig. 4l, Table S20 Panel b demonstrates that among mobile scientists, a local exploration strategy—operationalized as the reference global fields relative share—is significantly and positively associated with long-term disruptiveness (Column 1,  $p < 0.001$ ). However, upon introducing team expertise distance into the model, the coefficient for reference global fields relative share flips to negative (Column 2). Given the strong

positive correlation we previously observed between team expertise distance and reference global fields relative share (Fig. S14b), this sign reversal suggests that the two measures may capture substantially overlapping information relevant to long-term disruptiveness. Once team expertise distance is accounted for, the independent marginal contribution of local exploration strategy attenuates substantially. Notably, when the interaction term between team expertise distance and reference global fields relative share is included, it is significantly positive (Column 4,  $p < 0.001$ ), indicating that local exploration strategy is more associated with long-term disruptiveness when it is embedded in teams with more heterogeneous expertise. These results indicate that longer-distance movers' disruptiveness advantage may be closely linked to assembling higher expertise distance teams, and the combination of high team expertise distance and exploration-oriented local knowledge strategy is synergistically associated with higher long-term disruptiveness.

In summary, the shorter–longer comparison points to a clear divergence in paper-level knowledge strategy that mirrors their distinct long-term performance outcomes. Shorter-distance movers—who are more strongly embedded in elite, high-reputation collaborative networks—systematically exhibit a more local exploitation-oriented knowledge strategy (e.g., higher reference self-citation ratio and higher reference home fields relative share), which is closely tied to higher team reputation and, in turn, helps account for their long-term impact advantage (Table S20 Panel a). By contrast, longer-distance movers exhibit a more local exploration-oriented knowledge strategy (larger reference fields pivot size and higher reference global fields relative share), which aligns with their lower team reputation, larger expertise heterogeneity and lower transaction costs implied by more repeated collaborations. Moreover, this exploration-oriented local knowledge strategy is significantly positively associated with long-term disruptiveness, but its positive association attenuates after accounting for team expertise distance, indicating that expertise heterogeneity may constitute the more central pathway; further, their combination is synergistically associated with higher

long-term disruptiveness (Table S20 Panel b). Overall, these findings suggest that shorter-distance scientists are more strongly associated with an impact-oriented pathway that couples elite reputation with local exploitation, whereas longer-distance scientists are more strongly associated with a disruptiveness-oriented pathway that couples heterogeneous expertise with exploration-oriented local knowledge strategy.

### **Supplementary Note 3.3.2: Career knowledge structure**

Having established that early-career mobility enables Mob scientists to both explore new knowledge domains—via increased early-career field pivot size—and access more prestigious collaborative networks, which in turn facilitates later-career local field exploitation and elevated team reputation, we have shown how a sequential pattern of initial exploration followed by focused exploitation can promote “hot streaks” and sustained long-term impact.

However, it remains unclear how these dynamics manifest at the scale of an entire scientific career. Specifically, when viewed across their full trajectory, do Mob scientists continue to exhibit a local exploitation strategy—focusing their research within a narrow set of fields and minimizing domain switches, as suggested by their referencing behavior at the paper level? Or, conversely, does their position as network brokers—characterized by higher betweenness centrality in later career and the assembly of teams with greater expertise distance and diversity—allow Mob scientists to bridge heterogeneous knowledge communities and draw diverse knowledge from a broader spectrum of scientific fields? Moreover, do these long-term career behaviors represent a potential channel underlying Mob scientists' superiority in long-term academic performance?

In the following section, we address this question by systematically comparing the field exploration versus exploitation tendencies of Mob and non-Mob scientists across the entire time window of their careers.

#### Career knowledge structure (non-Mob vs Mob):

Although numerous indicators exist to measure a scientist's propensity for exploration versus exploitation at the career level, we begin with the most direct metric: the number of unique research fields (career fields coverage) that each scientist engages with during the later stages of their career. A broader coverage can be inferred to a greater likelihood of field switching and exploration, whereas a narrower focus suggests local exploitation within established fields. We find Mob scientists exhibit significantly greater career fields coverage (Mob = 4.23, non-Mob = 3.77;  $p < 0.001$ ; Fig. 4f), representing a relative increase of 12.33% compared with non-Mob peers. Moreover, career fields coverage is positively correlated with betweenness centrality in scientists' later-career collaboration networks (Fig. S15b), suggesting that occupying bridging positions may facilitate broader engagement across diverse research fields.

However, simply participating in more fields does not guarantee the breadth and exploration in the knowledge actually integrated into knowledge/conceptual space in the later career stage. To rigorously capture how evenly a scientist draws upon knowledge from across the diverse fields encountered throughout their career, namely, career knowledge exploration, we introduce a measure inspired by prior research. Here, we treat each scientist as a focal entity (and source entity) and, for every field appearing among the references in their publications during years 6–25, we quantify the scientist's propensity to draw from that field across their entire career. For each focal scientist, we calculate the area under the receiver operating characteristic curve (AUC) for every referenced field, which reflects the probability that a reference to that field is as likely to be drawn as from any other field the scientist has engaged with (AUC = 0.5 denotes random preference, AUC > 0.5 indicates over-preference).

Specifically, for each scientist  $s$ , let the set of fields referenced across all late-career papers be  $\mathcal{F} = \{f_1, f_2, \dots, f_n\}$ , where  $n \in [1, 19]$ . For each paper  $j$  authored by scientist  $s$ , let  $N_{j,f}$  denote the number of references made to field  $f$  in that paper. Aggregating across all papers, for each field  $f$ , we can obtain a vector of referencing

frequencies:

$$\{N_{j_1,f}, N_{j_2,f}, \dots, N_{j_{l_f},f}\} \quad (26)$$

where  $l_f$  is the number of papers by scientist  $s$  that contain references to field  $f$ .

Then, the AUC-based career fields propensity  $P_{s,f}$  for scientist  $s$  toward field  $f$  defined as:

$$P_{s,f} = \frac{1}{l_f \cdot L_{-f}} \sum_{a=1}^{l_f} \sum_{b=1}^{L_{-f}} \mathbb{I}(N_{a,f} > N_{b,-f}) \quad (27)$$

where:  $N_{a,f}$  is the  $a$ -th referencing frequency to field  $f$  (from the set above),  $N_{b,-f}$  is the  $b$ -th referencing frequency to any field in the set above other than  $f$ ,  $l_f$  is the number of papers with references to field  $f$ ,  $L_{-f}$  is the total number of referencing frequencies to all fields except  $f$ ,  $\mathbb{I}(\cdot)$  is the indicator function, which is 1 if the condition inside holds, and 0 otherwise.

This formulation makes it clear that for each field, all referencing frequencies (one per paper) are compared to all frequencies from other fields, and the AUC thus measures the tendency of a scientist to reference a particular field relative to all others occurs in the reference lists of their publication records in the later career stage.

After obtaining the AUC-based career fields propensity  $P_{s,f}$  for each referenced field  $f$  of scientist  $s$ , we define the diversity of field propensity by computing the normalized Shannon entropy of the propensity distribution. Specifically, for each scientist  $s$ , we first normalize the propensity scores across all referenced fields  $\mathcal{F}$  as  $f$  follows:

$$\tilde{P}_{s,f} = \frac{P_{s,f}}{\sum_{f' \in \mathcal{F}} P_{s,f'}} \quad (28)$$

where  $\tilde{P}_{s,f}$  denotes the normalized propensity of scientist  $s$  toward field  $f$ , and  $\mathcal{F}$  is the set of all unique fields referenced by scientist  $s$  during their late career.

The career fields propensity diversity for scientist  $s$  is then computed as:

$$H_{sp} = -\frac{1}{\ln |\mathcal{F}|} \sum_{f \in \mathcal{F}} \tilde{P}_{s,f} \ln \tilde{P}_{s,f} \quad (29)$$

where  $|\mathcal{F}|$  is the number of unique fields referenced by scientist  $s$  during their late career. A higher value of  $H_{sp}$  indicates that the scientist draws upon a more diverse set of fields with relatively even propensity, reflecting greater career-level knowledge integration and exploration. In contrast, a lower  $H_{sp}$  implies strong preference or exploitation within a limited set of fields.

Our results reveal that Mob scientists exhibit significantly greater career fields propensity diversity compared to their non-Mob counterparts (mean: Mob = 0.856, non-Mob = 0.844,  $p < 0.001$ , see Fig. 4d), reflecting a 1.5% relative increase. This finding indicates that Mob scientists, over the course of their late-career, not only engage with a wider range of scientific fields but also distribute their referencing propensity more evenly across these fields, suggesting broader and more balanced knowledge integration at the career level.

To further quantify the extent to which scientists draw upon knowledge outside their main research domains, we calculated the career non-focal fields propensity, defined as the proportion of total referencing propensity that is attributed to fields not directly represented among the focal scientist's own late-career publication fields:

$$\text{career non-focal fields propensity} = \frac{\sum_{f \in \mathcal{F}_{\text{non-focal}}} P_{s,f}}{\sum_{f \in \mathcal{F}_{\text{focal}}} P_{s,f} + \sum_{f \in \mathcal{F}_{\text{non-focal}}} P_{s,f}} \quad (30)$$

where  $\mathcal{F}$  denotes the set of all unique fields referenced by scientist  $s$  during their late career (e.g., years 6-25),  $\mathcal{F}_{\text{focal}} \subset \mathcal{F}$  is the set of fields in which scientist  $s$  has authored publications during this period,  $\mathcal{F}_{\text{non-focal}} = \mathcal{F} \setminus \mathcal{F}_{\text{focal}}$  denotes the set of referenced fields in which the scientist has not authored any papers, and  $P_{s,f}$  is the career fields propensity of scientist  $s$  toward field  $f$ .

Again, Mob scientists display a significantly higher mean career non-focal fields propensity than non-Mob scientists (mean: Mob = 0.2508 vs. non-Mob = 0.2403; one-sided Welch's t-test,  $P < 0.001$ , Fig. S15d), representing a 4.39% increase relative to non-Mob scientists. Notably, this measure is positively correlated with overall career fields propensity diversity (Fig. 4f), reinforcing the interpretation that Mob scientists not only cross more fields and draw balanced knowledge but also draw more substantially from out-of-domain knowledge sources.

All three indicators—career fields coverage, career fields propensity diversity, and career non-focal fields propensity—not only are significantly higher among Mob scientists but are also each positively correlated with later-career betweenness centrality (Fig. S15a-c). Stratified analysis by the median of each indicator demonstrates that, regardless of mobility status, scientists in the higher group achieve superior long-term academic performance (all  $p < 0.001$ ; Fig. 4e, S15e, f). These results collectively suggest that, while Mob scientists may display a tendency toward local exploitation within individual fields at the paper level, at the global (career-wide) level they exhibit a distinct pattern of exploration. Specifically, their positions as network brokers—connecting disparate, heterogeneous knowledge communities—are associated with both broader field coverage and more even, integrative use of knowledge across diverse



scientific domains, including greater utilization of out-of-focal-field knowledge. Furthermore, based on prior works<sup>36,49</sup>, we also calculated a network-based indicator: topic diversity. We constructed a unweighted co-citation network of late-career references and applied community detection to identify research topics and then calculated the normalized entropy of the topic frequency distribution. Consistent with our career exploration results, Mob scientists exhibit significantly higher career-wide topic diversity compared to non-mobile peers (Fig. S15g).

This duality—local exploitation within fields, yet global exploration across the full scope of career knowledge integration—may echo our earlier findings on network structure. Mob scientists’ higher mean component density in coauthor collaboration networks (i.e., more tightly-knit local clusters). Overall, these patterns highlight that Mob scientists, through their brokerage positions and extensive domain coverage, combine deep field-specific expertise with the ability to bridge and integrate knowledge from multiple areas, thereby establishing a potential channel for superior long-term academic performance.

But here we also raise another question: While the combination of early-career local exploration and later-career local exploitation (the sequential exploration–exploitation pattern) represents one potential channel underlying long-term impact, it remains unclear whether global exploration measured at the career level could explain remaining differences in long-term productivity and disruptiveness observed between Mob and non-Mob scientists.

Previous research suggests increased field switching later in a career (career years  $\geq 12$  in prior research) positively correlates with publication productivity. To test this in our context, and determine whether global/career knowledge exploration—captured by career fields propensity diversity—predicts long-term productivity, we conducted regression analyses in Table S21, Panel a. Our results confirm career fields propensity diversity robustly predicts long-term Publications, complementary to labor advantages. When including later career coauthor count, career fields propensity diversity each

variable significantly predicts productivity ( $p < 0.001$ , Table S21, Panel a, columns 3). However, the interaction between career fields propensity diversity and labor advantage is significantly negative ( $p < 0.001$ , Table S21, Panel a, columns 4), suggesting that larger teams may face higher costs for field switching. Similar results are observed when replacing career fields propensity diversity with career fields coverage: both indicators remain significant predictors of productivity after including labor advantages. These findings indicate global knowledge exploration may act as a potential channel associated with increased productivity.

Our prior analyses demonstrated that career fields propensity diversity is positively correlated with network brokerage positions (betweenness centrality, Fig. S15 a), while both coordination efficiency and betweenness centrality independently and synergistically promote long-term productivity (see Table S17). We further investigated whether global/career knowledge exploration and coordination efficiency also can serve as independent and potentially complementary channels. The results (Table S21, Panel b, columns 3–4) show that both variables retain significantly positive effects ( $p < 0.001$ ), and their interaction is positive while marginally significant ( $p = 0.06$ ), indicating that scientists with global knowledge exploration combined with high coordination efficiency are associated with higher publications, which may act as potential channels to mobile scientists' long-term productivity advantages.

In terms of D5 percentile, we also observe team expertise distance positively correlates with both career fields propensity diversity (Pearson  $r = 0.359$ ,  $p < 0.001$ ; Spearman  $\rho = 0.358$ ,  $p < 0.001$ ) and career non-focal fields propensity (Pearson  $r = 0.214$ ,  $p < 0.001$ ; Spearman  $\rho = 0.190$ ,  $p < 0.001$ ). However, causal directions remain uncertain: it is unclear whether assembling teams with greater expertise distance facilitates broader knowledge integration, thereby promoting disruptiveness, or whether early-career field exploration positions scientists centrally within collaboration networks, subsequently supporting broader field coverage and diverse career knowledge space, thereby leading to diverse team assembly and eventually promoting disruptiveness. Alternatively, these

may represent distinct yet synergistic channels enhancing disruptiveness.

To further clarify these relationships, we conducted regressions including later career team expertise distance, career fields propensity diversity, and their interactions (Table S21, Panel c). Regarding disruptiveness, both team expertise distance and career fields propensity diversity significantly predict higher D5 percentile independently, and their interaction is significantly positive ( $p < 0.001$ , Table S21, Panel c, columns 1–4), indicating synergy. This suggests the combination of maintaining broad, balanced knowledge exploration and assembling diverse expertise teams may act as two distinct potential channels to long-term disruptiveness. These findings collectively demonstrate global exploration measured at the career level is robustly associated with enhanced long-term productivity and disruptiveness, complementary to labor advantages and team expertise diversity. While causal interpretations remain beyond our analysis, these persistent associations underscore mobility-induced career exploration as a potential factor shaping long-term academic performance.

#### Career knowledge structure (Shorter vs Longer):

Building on the career-level evidence in comparison between non-Mob versus Mob scientists above, we next examine whether the magnitude of early-career cultural distance also shapes scientists' career/global-level knowledge integration in the late career stage (years 6–25). Supplementary Note 3.2 documents systematic differences in collaboration ecology: shorter-distance movers are more strongly embedded in elite, high-reputation collaboration networks and occupy more brokerage-oriented positions, whereas longer-distance movers tend to assemble teams with greater expertise distance and exhibit higher coordination efficiency and a greater reliance on repeated ties. Consistent with these ecological differences, our paper-level analyses further reveal a divergence in local knowledge strategy, with shorter-distance movers leaning toward exploitation and longer-distance movers leaning toward exploration. The remaining question is whether these paper-level differences persist, reverse, or reconfigure when knowledge strategy is measured at the career scale.

On the one hand, shorter-distance scientists exhibit a lower early dense-tie ratio and subsequently adopt a more open team-assembly pattern in the later career stage, positioning themselves in more brokerage-oriented positions within collaboration networks. Although these scientists also gain greater access to elite, high-reputation collaborators—which is positively associated with a more exploitation-oriented local knowledge strategy at the paper level—such brokerage-oriented and expansionary collaboration positions may simultaneously facilitate access to diverse knowledge communities and thereby enable broader career-wide field exploration, paralleling the pattern observed among Mob scientists in the Mob vs. non-Mob comparison. On the other hand, although longer-distance scientists display larger early-career field pivot sizes and later assemble teams with greater expertise distance—features associated with stronger paper-level local exploration—multiple early team-assembly indicators in Supplementary Note 3.2.2 (Fig. S9) suggest a more cohesive, repeat-tie collaboration ecology that persists into later career stages. Importantly, these early repeat-tie indicators are consistently negatively associated with later-career brokerage positions. Given the link between brokerage and career-wide exploration observed in the non-Mob versus Mob comparison, longer-distance movers' lower brokerage positions may constrain career-wide/global field exploration, even as their cohesive and repeated collaborations can support cross-field integration at the paper level.

To address this question, we apply the same three career-level indicators introduced above—career fields coverage, career fields propensity diversity, and career non-focal fields propensity—to the shorter–longer comparison. Together, these measures capture not only how many fields scientists engage with over years 6–25, but also how evenly they integrate knowledge across these fields and the extent to which they draw upon knowledge outside their focal publication domains at the career level. Notably, unlike paper-level referencing measures that reflect *local* knowledge choice within individual publications, these *career-level/global* indicators reflect a scientist's long-run pattern of knowledge integration across the entire late-career trajectory.

As shown in Fig. S16d–f, shorter-distance scientists exhibit systematically higher career-level exploration across all three indicators. First, shorter-distance movers engage with a broader set of research domains in late career, as measured by career fields coverage (mean: Shorter = 4.370, Longer = 4.259; one-sided Welch's t-test,  $t = 4.26$ ,  $p < 0.001$ ), corresponding to a relative increase of 2.62% compared with longer-distance movers. Second, shorter-distance movers also display higher career fields propensity diversity (mean: Shorter = 0.8616, Longer = 0.8584; one-sided Welch's t-test,  $t = 2.27$ ,  $p < 0.05$ ), reflecting a 0.37% relative increase. Third, they draw more heavily on out-of-domain knowledge at the career scale, as indicated by career non-focal fields propensity (mean: Shorter = 0.2562 vs. Longer = 0.2520; one-sided Welch's t-test,  $p < 0.001$ ), corresponding to a relative increase of 1.64%. Consistent with these indicators, the network-based measure of career-wide topic diversity is also higher among shorter-distance movers (mean: Shorter = 0.8943, Longer = 0.8896; one-sided Welch's t-test,  $t = 2.11$ ,  $p < 0.05$ ; Fig. S16g), corresponding to a relative increase of 0.53%.

We next relate these career knowledge-structure differences to collaboration ecology. As shown in Fig. S16a–c, among mobile scientists, later-career betweenness centrality is positively correlated with career fields coverage (Pearson  $r = 0.257$ ,  $P < 0.001$ ; Spearman  $\rho = 0.264$ ,  $P < 0.001$ ), career fields propensity diversity (Pearson  $r = 0.157$ ,  $P < 0.001$ ; Spearman  $\rho = 0.217$ ,  $P < 0.001$ ), and career non-focal fields propensity (Pearson  $r = 0.174$ ,  $P < 0.001$ ; Spearman  $\rho = 0.195$ ,  $P < 0.001$ ). This pattern parallels the non-Mob versus Mob comparison and supports the interpretation that brokerage-oriented network positions are associated with broader career-wide knowledge exploration and integration. Taken together, these results indicate that, although shorter-distance movers are more exploitation-oriented at the paper level, their more brokerage-oriented collaboration networks are associated with broader career-wide knowledge exploration, whereas longer-distance movers' more repeat-tie collaboration ecology is associated with lower brokerage and correspondingly weaker career-level exploration.

We further examine whether career-level exploration is associated with long-term academic outcomes, and whether it operates as a potential channel—alongside paper-level knowledge strategy—in shaping scientists’ impact and disruptiveness.

As shown in Table S20, Panel a, column (3), career-level exploration—measured by career fields propensity diversity—remains positively associated with Hit 10pct after controlling for team reputation and paper-level exploitation (reference home fields relative share) ( $p < 0.001$ ). Its inclusion yields a further slight attenuation of the cultural-distance coefficient from  $-0.0184$  (column (2)) to  $-0.0181$  (column (3)) ( $p < 0.001$ ), an additional 1.63% reduction in magnitude (42.90% in total relative to the baseline specification in Table S4). Together with the positive association between brokerage positions and career-level exploration (Fig. S16a–c), this pattern is consistent with the interpretation that shorter-distance movers’ greater career-wide exploration may operate as an additional channel—alongside team reputation and paper-level exploitation—linking early-career cultural distance to long-term impact.

We next test whether career-level exploration remains positively associated with long-term disruptiveness within mobile scientists. As shown in Table S20, Panel b, column (3), after simultaneously controlling for team expertise distance and paper-level exploration (reference global fields relative share), career-level exploration (career fields propensity diversity) remains positively associated with D5 Percentile ( $p < 0.001$ ). Moreover, once career-level exploration is introduced, the estimated association between early-career cultural distance and D5 Percentile becomes more positive (Table S20, Panel b: the cultural-distance coefficient shifts upward from column (2) to column (3); both positive and significant), while the model fit improves substantially ( $R^2$  increases from 0.080 in column (2) to 0.096 in column (3)). Together, these patterns suggest that career-level exploration may operate as an additional channel—beyond team expertise distance and paper-level exploration, which is closely aligned with expertise heterogeneity—associated with differences in long-term disruptiveness between shorter- and longer-distance movers. In particular, because shorter-distance

movers exhibit higher career-level exploration in the mean comparisons (Fig. S16d), accounting for this advantage reduces the extent to which career-level exploration compresses the observed disruptiveness gap, thereby revealing a larger net disruptiveness advantage for longer-distance movers conditional on comparable career-level exploration.

Taken together, our analyses to this point indicate two parsimonious, structurally grounded channels most centrally associated with long-term disruptiveness among mobile scientists. First, an expertise–local exploration channel, where longer-distance movers assemble teams with greater expertise distance and, correspondingly, exhibit stronger paper-level exploration; these two components are synergistically associated with higher D5 Percentile (Table S20, Panel b, column (4): positive interaction;  $p < 0.001$ ), while the standalone paper-level exploration association attenuates once team expertise is controlled, consistent with substantial overlap between the two. Second, a brokerage–career exploration channel, where career-level exploration—positively linked to brokerage positions (Fig. S16a–c)—may be independently and positively associated with D5 Percentile after controlling for team expertise distance and paper-level exploration. Although longer-distance movers exhibit lower career-level exploration on average than shorter-distance movers, the expertise–local exploration channel appears to have a larger effective contribution for long-term disruptiveness, consistent with longer-distance movers’ higher D5 percentiles overall.

A similar two-pathway interpretation also applies to the Mob versus non-Mob comparison. Relative to non-Mob scientists, Mob scientists assemble teams with higher expertise heterogeneity in the later career stage and exhibit stronger brokerage-oriented collaboration networks, as reflected by higher betweenness centrality and higher career-level knowledge exploration. At the same time, Mob scientists display a more exploitation-oriented local referencing pattern (higher reference fields imbalance and lower reference global fields relative share (mean: Mob = 0.2041 vs. non-Mob = 0.2164; one-sided Welch’s  $t$ -test,  $p < 0.001$ ; a 5.66% relative decrease)). This combination

implies that, although Mob scientists may realize weaker complementarity between team expertise heterogeneity and paper-level exploration than longer-distance movers, their advantage in team expertise distance together with their stronger brokerage–career exploration profile provides two structurally grounded channels that are jointly consistent with Mob scientists’ higher long-term disruptiveness. For long-term impact, an analogous unifying potential channel also emerges across both comparisons. In the Mob versus non-Mob comparison, the joint inclusion of team reputation and a more exploitation-oriented local knowledge strategy is associated with a substantial attenuation of the mobility advantage in Hit 10pct (92.54% reduction). In the Shorter versus Longer comparison, the same two factors likewise account for a sizable share of the impact gap (41.96% reduction). Taken together, these results suggest a consistent organizing framework across comparisons: long-term impact differences are most closely aligned with team reputation coupled with local exploitation, whereas long-term disruptiveness differences are most closely aligned with team expertise heterogeneity coupled with paper-level exploration, complemented by brokerage-linked career-level exploration.

Additionally, the collaboration-ecology evidence in Supplementary Note 3.2 echoes the multi-scale knowledge-strategy patterns documented in Supplementary Note 3.3. At the career level, brokerage-oriented network positions provide a consistent structural counterpart of career-wide exploration: later-career betweenness centrality is positively associated with all three career-level exploration indicators (Fig. S15a–c, S16a–c), and correspondingly, Mob scientists exhibit higher career-level exploration than non-Mob peers, while within mobile scientists, shorter-distance movers—who occupy higher-betweenness positions—also exhibit systematically higher career-level exploration. At the paper level, by contrast, exploration versus exploitation patterns are more closely aligned with team-level channels—especially expertise heterogeneity, team reputation, and coordination efficiency—as documented in Supplementary Note 3.3.1. Taken together, these results suggest a coherent local–global configuration across groups:



Mob scientists and shorter-distance movers tend to combine a more exploitation-oriented local (paper-level) strategy with stronger global (career-level) exploration, whereas longer-distance movers tend to exhibit a more exploration-oriented local strategy but weaker career-level exploration on average.

Finally, one apparent nuance is that reference fields imbalance, our paper-level exploitation indicator based on the coefficient of variation (CV) of the field-size-normalized reference-field distribution, can exhibit positive associations with both impact and disruptiveness in some specifications; for example, it is positively associated with Hit 10pct and D5 Percentile in the Mob vs non-Mob comparison (Fig. 4b). This may appear to blur the usual exploitation–exploration contrast at the paper level, motivating a clarification of what the CV-based imbalance measure does and does not capture. We note that this indicator does not uniquely map onto a pure paper-level exploitation-versus-exploration contrast: for instance, a high imbalance can arise from heavy concentration on a single dominant field, but it can also arise from uneven mass split across multiple fields, which may partly coincide with cross-field referencing rather than strict within-field deepening. For instance, consider toy reference profiles where the first element corresponds to the focal paper’s home field and the remaining elements correspond to non-home fields. Both  $[10, 1, 1, 1]$  and  $[5, 5, 0, 0]$  can yield high imbalance, yet the latter places substantial mass on a single non-home field and thus would imply a higher degree of cross-field integration than the former, even though the CV is similarly large. In this sense, reference fields imbalance can mix exploitation-like concentration with certain exploration-like configurations in the reference space, potentially generating sign patterns that appear less clean when compared with measures that explicitly separate “home” versus “global” field reliance.

Consistent with this interpretation, when we use reference global fields relative share (and its complement, reference home fields relative share) to operationalize paper-level exploration and exploitation, the empirical patterns are directionally coherent across

both comparisons: paper-level exploitation aligns with higher impact and lower disruptiveness, whereas paper-level exploration aligns with higher disruptiveness and lower impact. Specifically, Table S22 (Mob vs non-Mob) corroborates the same directionally interpretable patterns as Table S20 (Shorter vs Longer) when paper-level exploitation and exploration are operationalized using reference home/global fields relative share. In Panel a, using reference home fields relative share as the paper-level exploitation measure, it is positively associated with long-term impact (Hit 10pct;  $p < 0.001$ ). This positive association remains significant after additionally controlling for team reputation and career fields propensity diversity ( $p < 0.001$ ). Team reputation and reference home fields relative share together attenuate the cultural-distance coefficient from the baseline specification in Table S4 (0.0134) to 0.0026 (column (2)), an 80.6% reduction in magnitude; further including career fields propensity diversity attenuates it to 0.0020 (column (3)), corresponding to an 85.07% reduction. Moreover, the interaction between reference home fields relative share and team reputation is significantly positive (column (4),  $p < 0.001$ ). Taken together, these results mirror Table S20 Panel a and support the same interpretation: paper-level exploitation is aligned with higher long-term impact, whereas paper-level exploration (the complement) is aligned with lower impact.

Panel b exhibits a parallel structure to Table S20 Panel b. Using reference global fields relative share as the paper-level exploration measure, it is positively associated with long-term disruptiveness (D5 Percentile) in the baseline specification (column (1),  $p < 0.001$ ). Consistent with substantial overlap between paper-level exploration and team expertise heterogeneity, once team expertise distance is introduced, the standalone association of reference global fields relative share becomes negative (columns (2)–(3),  $p < 0.001$ ). In addition, the interaction between reference global fields relative share and team expertise distance is significantly positive (column (4),  $p < 0.001$ ), indicating that paper-level exploration is more strongly associated with higher disruptiveness when embedded in teams with greater expertise distance. Overall, the positive

coefficients on reference fields imbalance in some specifications are consistent with a measurement-mixing nuance rather than a distinct local exploitation to disruptiveness mechanism. Using reference home/global relative-share measures resolves this ambiguity and yields a consistent organizing pattern across comparisons.

Consistent with this measurement nuance, the association between reference fields imbalance (CV) and D5 Percentile is statistically significant but small in magnitude. In the Mob vs non-Mob comparison, reference fields imbalance is positively correlated with D5 Percentile (Pearson  $r = 0.034$ ,  $p < 0.001$ ; Spearman  $r = 0.040$ ,  $p < 0.001$ ). The same pattern holds within mobile scientists in the shorter vs longer comparison: reference fields imbalance remains positively correlated with D5 Percentile, yet the association is even smaller (Shorter vs Longer among mobile scientists: Pearson  $r = 0.015$ ,  $p < 0.001$ ; Spearman  $r = 0.018$ ,  $p < 0.001$ ). In both comparisons, these correlations remain below 0.05 in absolute value, and within mobile scientists (i.e., Shorter vs Longer) they are below 0.02. By contrast, the two primary channels show substantially larger correlations with D5 Percentile in both comparisons. Team expertise distance exhibits a consistently stronger positive association with D5 (Mob vs non-Mob: Pearson  $r = 0.159$ ,  $p < 0.001$ ; Spearman  $r = 0.155$ ,  $p < 0.001$ ; Shorter vs Longer among mobile scientists: Pearson  $r = 0.151$ ,  $p < 0.001$ ; Spearman  $r = 0.148$ ,  $p < 0.001$ ). Similarly, career-level exploration, measured by career fields propensity diversity, is also more strongly and robustly positively correlated with D5 (Mob vs non-Mob: Pearson  $r = 0.169$ ,  $p < 0.001$ ; Spearman  $r = 0.207$ ,  $p < 0.001$ ; Shorter vs Longer among mobile scientists: Pearson  $r = 0.185$ ,  $p < 0.001$ ; Spearman  $r = 0.219$ ,  $p < 0.001$ ).

Conventionality, Novelty, and Disruptiveness: Beyond where scientists search (local exploration versus exploitation) and how they allocate attention (global exploration versus exploitation), recombination can also be characterized by how unusual the resulting reference combinations are<sup>44</sup>. Following prior work, we operationalize this dimension using paper-level conventionality and novelty, which are based on the statistical atypicality of pairwise combinations of cited journals relative to their

expected co-occurrence in the global citation network. Each paper thus yields a distribution of z-scores over cited-journal pairs, where lower-tail values capture rare (novel) combinations and central values capture typical (conventional) combinations. Specifically, a paper is classified as exhibiting “high tail novelty” if its 10th percentile z-score falls below 0, and it is classified as exhibiting “high median conventionality” if its 50th percentile z-score is in the upper half of all papers. Given that novelty follows a heavy-tailed distribution, we distinguish between shifts in the center of the distribution (median conventionality) and changes in the extreme tail (tail novelty). We then examine how these two dimensions co-vary with scientists’ collaboration ecologies. For consistency with existing studies, we rely on the pre-computed conventionality and novelty measures provided by SciSciNet. Finally, we aggregate the per-paper median conventionality and tail novelty scores to each scientist’s late-career trajectory.

As shown in Fig. S17, across both mobility comparisons we observe a consistent divergence between conventional grounding and tail novelty in recombination outcomes. Using the journal-pair-based conventionality and novelty measures from SciSciNet, mobile scientists (Mob) and longer-distance movers exhibit higher median conventionality yet lower tail novelty, whereas non-mobile scientists and shorter-distance movers exhibit the opposite pattern. Concretely, in the Mob versus non-Mob comparison, Mob scientists display a higher probability of high median conventionality than non-Mob scientists (mean: Mob = 0.5630 vs. non-Mob = 0.5206; one-sided Welch’s t-test,  $P < 0.001$ ), corresponding to an 8.14% increase. In contrast, Mob scientists exhibit a lower probability of high tail novelty (mean: Mob = 0.4731 vs. non-Mob = 0.4966; one-sided Welch’s t-test,  $P < 0.001$ ), corresponding to a 4.74% decrease. A parallel pattern holds within mobile scientists in the shorter–longer comparison: longer-distance movers show a higher probability of high-median conventionality than shorter-distance movers (mean: Longer = 0.5871 vs. Shorter = 0.5347; one-sided Welch’s t-test,  $P < 0.001$ ), corresponding to a 9.81% increase, while exhibiting a lower

probability of high-tail novelty (mean: Longer = 0.4566 vs. Shorter = 0.4799; one-sided Welch’s t-test,  $P < 0.001$ ), corresponding to a 4.86% decrease.

To further validate that this center–tail (median conventionality–tail novelty) divergence is not an artifact of defining novelty/conventionality via journal IDs, we also compute an embedding-based semantic robustness measure. Following prior work<sup>50</sup> that uses semantic representations of scientific text to characterize knowledge relatedness and recombination in the reference space, we conduct an embedding-based robustness check that does not rely on journal IDs or co-citation baselines. For each focal paper, we retrieve the titles of all referenced papers and map each title into a vector representation using a SciBERT<sup>51</sup> model. We then compute cosine similarity for all unordered pairs of referenced titles within the same reference list, yielding a within-paper distribution of pairwise semantic similarities. Intuitively, this distribution summarizes whether a paper’s references concentrate within a semantically cohesive region or instead mix semantically close and far elements.

We summarize this within-paper distribution using two complementary measures. First, we define Reference Semantic Similarity as the mean of all pairwise cosine similarities among referenced titles within a paper. Higher values indicate that the referenced items are, on average, semantically more cohesive (i.e., more strongly grounded within a narrower semantic region). Second, we define Reference Semantic Diversity as the dispersion of pairwise similarities within a paper, operationalized as the coefficient of variation of pairwise similarities. Higher values indicate more uneven mixing of semantically near and far references, corresponding to a more dispersed semantic recombination profile. For each scientist, we aggregate these paper-level measures by averaging across all of their papers in the late career stage. Again, these semantic proxies reproduce the same qualitative ordering as the median conventionality–tail novelty pattern. Within mobile scientists, longer-distance movers exhibit a slightly higher Reference Semantic Similarity than shorter-distance movers (mean: Longer = 0.7954 vs. Shorter = 0.7905; one-sided Welch’s t-test,  $P < 0.001$ ), corresponding to a

0.62% increase, and simultaneously exhibit a lower Reference Semantic Diversity (mean: Longer = 0.0826 vs. Shorter = 0.0855; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 3.32% decrease. In the Mob versus non-Mob comparison, Mob scientists likewise show lower semantic dispersion, with a lower Reference Semantic Diversity than non-Mob scientists (mean: Mob = 0.0844 vs. non-Mob = 0.0854; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 1.13% decrease. For Reference Semantic Similarity, the Mob–non-Mob difference is directionally consistent (mean: Mob = 0.7926 vs. non-Mob = 0.7925) but not statistically significant (one-sided Welch's t-test,  $P = 0.31$ ). Because the semantic measures serve only as an alternative operationalization, in what follows we focus on the conventionality-novelty framework.

Overall, the qualitative conclusion is consistent across the journal-pair and semantic frameworks: Mob scientists and longer-distance movers tend to exhibit stronger conventional grounding but weaker tail novelty (and correspondingly more centered and less dispersed semantic recombination), whereas non-Mob scientists and shorter-distance movers exhibit the opposite center–tail profile. These results indicate that early-career cultural mobility does not uniformly increase or decrease novelty; instead, it shifts the typical and extreme components of recombination in opposite directions across groups, motivating an account of why conventional grounding and tail novelty diverge under different mobility groups.

We next explore the origins of the divergent knowledge-recombination patterns observed across the two mobility comparisons. A natural starting point is a canonical account from prior work<sup>38</sup>, which emphasizes that pivoting is associated with more atypical recombinations: larger pivot size tends to predict a higher likelihood of tail novelty and a lower likelihood of conventional grounding. Consistent with this view, in both comparisons we observe that late-career reference fields pivot size is negatively associated with median conventionality and positively associated with tail novelty. In the Mob versus non-Mob comparison, reference fields pivot size is negatively correlated with median conventionality (Pearson  $r = -0.267$ ,  $P < 0.001$ ; Spearman  $\rho =$

$-0.321$ ,  $P < 0.001$ ) and positively correlated with tail novelty (Pearson  $r = 0.155$ ,  $P < 0.001$ ; Spearman  $\rho = 0.242$ ,  $P < 0.001$ ). The same pattern holds within mobile scientists in the shorter–longer comparison: reference fields pivot size is negatively correlated with median conventionality (Pearson  $r = -0.291$ ,  $P < 0.001$ ; Spearman  $\rho = -0.348$ ,  $P < 0.001$ ) and positively correlated with tail novelty (Pearson  $r = 0.181$ ,  $P < 0.001$ ; Spearman  $\rho = 0.266$ ,  $P < 0.001$ ). However, this association does not reconcile the group-level center–tail ordering documented above. For instance, within mobile scientists, shorter-distance movers exhibit a smaller reference fields pivot size on average, yet they show a higher probability of high tail novelty and a lower probability of high median conventionality than longer-distance movers. In the Mob versus non-Mob comparison, Mob scientists exhibit a directionally smaller reference fields pivot size than non-Mob scientists, which is directionally consistent with Mob scientists' higher median conventionality and lower tail novelty; however, the Mob–non-Mob difference in reference fields pivot size is not statistically significant (one-sided Welch's  $t$ -test,  $P = 0.30$ ), limiting pivot size as a group-level explanation. In addition, later career team expertise distance exhibits the same canonical association pattern as pivoting—positively associated with tail novelty and negatively associated with median conventionality (all  $P < 0.001$ ) across both comparisons. However, Mob scientists and longer-distance movers assemble teams with greater expertise distance, which would predict higher tail novelty and lower median conventionality, a pattern that does not align with the observed center–tail patterns. These patterns motivate us to search for alternative collaboration-ecology channels that can jointly account for the observed divergence between the center and the tail of recombination outcomes.

We therefore return to differences in collaboration ecology. Prior work has highlighted the role of scientific status and reputational capital in shaping conventional grounding<sup>52</sup>. In our data, team reputation in the later career stage is positively associated with a higher probability of high median conventionality in both comparisons (Mob vs non-Mob: Pearson  $r = 0.100$ ,  $P < 0.001$ ; Spearman  $\rho = 0.095$ ,  $P < 0.001$ ; shorter vs longer:

Pearson  $r = 0.108$ ,  $P < 0.001$ ; Spearman  $\rho = 0.103$ ,  $P < 0.001$ ). This association is directionally consistent with the Mob versus non-Mob contrast, where Mob scientists assemble higher-reputation teams and also exhibit higher median conventionality. However, the same reputation channel does not provide a unified account across both mobility groups: within mobile scientists, shorter-distance movers also assemble higher-reputation teams, yet they exhibit lower median conventionality than longer-distance movers. Moreover, team reputation is also positively associated with the probability of high tail novelty in both comparisons (Mob vs non-Mob: Pearson  $r = 0.166$ ,  $P < 0.001$ ; Spearman  $\rho = 0.163$ ,  $P < 0.001$ ; shorter vs longer: Pearson  $r = 0.171$ ,  $P < 0.001$ ; Spearman  $\rho = 0.167$ ,  $P < 0.001$ ). These patterns suggest that team reputation is more consistent with a broad uplift/capacity association with both conventional grounding and tail novelty, rather than a directionally discriminative channel that can reconcile why mobility regimes differ in opposite directions at the center versus the tail.

Accordingly, neither pivoting/expertise heterogeneity nor team reputation alone can explain the systematic separation between median conventionality and tail novelty observed across both mobility comparisons. We therefore turn to collaboration-ecology mechanisms that map more directly onto center-versus-tail behavior. We first examine whether open versus closed team assembly provides a more coherent account of the center–tail separation in recombination outcomes. As documented in Supplementary Note 3.2.2, mobility groups differ systematically in their tendency to incorporate unfamiliar collaborators in the late career stage, captured by team new collaborators (the mean number of new collaborators per paper in years 6–25). Conceptually, an influx of new collaborators may increase tail novelty through a variance mechanism: to the extent that new ties expose teams to less familiar inputs, recombinant search is associated with greater uncertainty and dispersion in outcomes<sup>43</sup>. New collaborators may expand access to fresh information channels, enlarging the opportunity set of atypical pairings, while simultaneously weakening shared routines and increasing coordination frictions. This erosion of shared routines may also reduce conventional



grounding in typical recombinations. When collaborators lack a common repertoire and mutual knowledge, teams may be less able to coordinate around—or less inclined to adopt—well-established, mainstream integration templates, lowering the likelihood that typical recombinations remain anchored in highly conventional mixtures. The combination of novel inputs and higher coordination uncertainty may widen the distribution of recombination outcomes, thereby increasing the frequency of extreme, low-probability recombinations. At the same time, it may shift typical recombinations away from highly conventional mixtures.

Consistent with this account, we find that team new collaborators is positively associated with the probability of high tail novelty and negatively associated with the probability of high median conventionality in the late-career stage. In the Mob versus non-Mob comparison, team new collaborators is positively correlated with tail novelty (Pearson  $r = 0.204$ ,  $P < 0.001$ ; Spearman  $\rho = 0.314$ ,  $P < 0.001$ ). Its association with median conventionality is weak in Pearson correlation but negative and significant in rank correlation (Pearson  $r = -0.005$ ,  $P = 0.11$ ; Spearman  $\rho = -0.068$ ,  $P < 0.001$ ). A parallel pattern holds within mobile scientists in the shorter–longer comparison: team new collaborators is positively correlated with tail novelty (Pearson  $r = 0.245$ ,  $P < 0.001$ ; Spearman  $\rho = 0.355$ ,  $P < 0.001$ ) and negatively correlated with median conventionality (Pearson  $r = -0.024$ ,  $P < 0.001$ ; Spearman  $\rho = -0.085$ ,  $P < 0.001$ ). Together, these results suggest that expansionary, open collaboration ecologies—characterized by greater incorporation of unfamiliar collaborators—are associated with more frequent tail novelty events and weaker conventional grounding at the center of the distribution.

We next examine a second potential collaboration-ecology channel that maps directly onto center-versus-tail behavior: team geographic distance and tacit-knowledge frictions. A long tradition in the innovation and economic-geography literature emphasizes that tacit knowledge is costly to transfer across distance and that geographically dispersed collaboration can raise communication frictions in interactive learning and knowledge integration<sup>27,53,54</sup>. Motivated by these accounts, we measure

team geographic distance as the average pairwise geographic distance among coauthors' affiliations (in kilometers) for each scientist's late-career papers (years 6–25), and examine its association with median conventionality and tail novelty. We observed Mob scientists exhibit substantially greater team geographic distance than non-Mob scientists (mean: Mob = 1772.46 km vs. non-Mob = 1220.42 km; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 45.23% increase. A parallel pattern holds within mobile scientists: longer-distance movers exhibit greater team geographic distance than shorter-distance movers (mean: Longer = 1862.67 km vs. Shorter = 1740.18 km; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 7.04% increase.

Conceptually, greater team geographic distance may selectively constrain extreme, low-probability recombinations that require high-bandwidth, tacit coordination to integrate and validate, thereby attenuating tail novelty. At the same time, geographically dispersed teams may rely more heavily on codified and broadly legible integration templates, shifting typical recombinations toward stronger conventional grounding at the center of the distribution. We next test these predictions by examining whether team geographic distance is negatively associated with tail novelty and positively associated with median conventionality across both mobility comparisons. This channel is distinct from repeat-tie-based coordination efficiency discussed above, as it operates through spatial dispersion and tacit-knowledge transfer frictions rather than through relational trust and shared routines. Consistent with these predictions, team geographic distance is positively associated with median conventionality and negatively associated with tail novelty in both comparisons. In the Mob versus non-Mob comparison, team geographic distance is positively correlated with median conventionality (Pearson  $r = 0.187$ ,  $P < 0.001$ ; Spearman  $\rho = 0.185$ ,  $P < 0.001$ ) and negatively correlated with tail novelty (Pearson  $r = -0.083$ ,  $P < 0.001$ ; Spearman  $\rho = -0.037$ ,  $P < 0.001$ ). A parallel pattern holds within mobile scientists in the shorter–longer comparison: team geographic distance is positively correlated with median conventionality (Pearson  $r = 0.159$ ,  $P < 0.001$ ; Spearman  $\rho = 0.163$ ,  $P < 0.001$ ) and negatively correlated with tail novelty

(Pearson  $r = -0.088$ ,  $P < 0.001$ ; Spearman  $\rho = -0.056$ ,  $P < 0.001$ ). Together, these results suggest that greater geographic dispersion is associated with stronger conventional grounding at the center of the distribution and fewer tail novelty events, consistent with a tacit-knowledge friction channel.

Taken together, the evidence above indicates that the center–tail separation in recombination outcomes is not readily reconciled by canonical pivoting and team expertise heterogeneity accounts or team reputation alone. By contrast, two collaboration-ecology channels map more directly and coherently onto the center-versus-tail decomposition: expansionary team assembly, captured by team new collaborators, is associated with more frequent tail novelty and weaker conventional grounding, whereas geographic dispersion, captured by team geographic distance, is associated with stronger conventional grounding and fewer tail novelty events, consistent with tacit-knowledge frictions in distant collaboration. Across both mobility comparisons, these two channels also help organize the group-level center–tail configuration. Mobile scientists (Mob) in the Mob–non-Mob comparison and longer-distance movers in the shorter–longer comparison tend to operate in more repeat-tie-oriented and more geographically dispersed collaboration ecologies. Reliance on repeated collaborators can sustain shared routines and relational trust, and—consistent with our evidence that these groups also assemble more expertise-diverse teams—may support paper-level cross-field knowledge integration. Yet, when coupled with greater geographic distance, the higher tacit-knowledge and communication frictions associated with spatial dispersion—together with a comparatively weaker influx of unfamiliar collaborators—are consistent with stronger conventional grounding in typical recombinations and fewer extreme tail-novelty events. By contrast, shorter-distance movers are embedded in comparatively less geographically dispersed teams and exhibit more expansionary incorporation of unfamiliar collaborators. This more open team assembly can inject fresh perspectives and widen the opportunity set of atypical pairings, while closer geographic proximity may lower tacit-knowledge

transfer frictions, together aligning with weaker conventional grounding at the center and more frequent tail novelty. Having established that collaboration ecology is systematically associated with the center–tail profile of recombination, we next examine how this distributional separation relates to long-term disruptiveness in scientists’ late-career trajectories.

### **Supplementary Note 3.4: Knowledge source**

#### **Supplementary Note 3.4.1: Core and periphery**

In Supplementary Note 3.3, we characterized how scientists search and recombine knowledge at multiple scales. At the paper level, we examined two contrasts—Mob vs. non-Mob scientists and, within the mobile sample, shorter- vs. longer-distance movers—and documented systematic differences in local exploration versus exploitation. At the career level, we quantified how broadly and evenly scientists integrate knowledge across fields (e.g., career fields coverage, career fields propensity diversity, and career non-focal fields propensity). Together, these analyses show that early-career cultural mobility is associated with multi-scale differences in knowledge strategy, spanning both local (paper-level) search behavior and global (career-level) knowledge integration.

Yet an important dimension remains unexplored: where the knowledge that scientists draw upon comes from in the global scientific system. Field-based indicators capture topical breadth, but they do not reveal whether scientists disproportionately source ideas from the scientific “core” versus the “periphery,” nor whether they systematically orient toward particular global knowledge lineages (e.g., elite institutions or dominant geopolitical centers). This distinction matters because knowledge sources shape not only the opportunity set of ideas but also visibility, recognition, and diffusion: peripheral sources may be less recognized or diffuse more slowly, whereas core sources may confer greater credibility and faster uptake<sup>22</sup>. Naturally, cross-border mobility can facilitate exchanges across core and peripheral knowledge infrastructures, potentially

leaving persistent imprints on scientists’ knowledge sourcing.

Motivated by our collaboration-ecology results in Supplementary Note 3.2, we hypothesize that early-career cultural mobility can reshape scientists’ access to and reliance on different global knowledge infrastructures. In the Mob–non-Mob comparison, Mob scientists accumulate more early-career international exposure and more early-career collaborators from Western countries and Top-100 institutions (Supplementary Note 3.2.1), both of which are strongly associated with later team reputation and impact. Importantly, similar heterogeneity also exists within mobile scientists: shorter-distance movers are more strongly embedded in elite collaborative networks (e.g., more Western and Top-100 early-career collaborators and higher later team reputation), whereas longer-distance movers are less likely to access such elite channels on average. These early network differences suggest that mobility—and the magnitude of cultural distance—may leave persistent imprints on the institutional and geopolitical origins of the work scientists cite, beyond the topical composition of their research. Below, we therefore examine how early-career cultural distance predicts scientists’ late-career knowledge sources and how these sourcing differences relate to long-term academic performance.

Core and periphery (non-Mob vs Mob): Here, we compute career-level knowledge-source propensities using the same AUC-based procedure as in Eqs. (24)–(28), but replacing referenced fields with referenced knowledge sources (institutions and countries). For each scientist  $s$  and each source  $u$  (institution or country) that appears in the reference lists of their late-career papers (years 6–25), we construct the per-paper referencing frequency sequence  $\{N_{1,u}, N_{2,u}, \dots\}$  and compute the source propensity  $P_{s,u}$  following the AUC definition in Eq. (25). We then aggregate  $P_{s,u}$  across sources to obtain two career-level core-sourcing propensities. First, we define career top institutions propensity by classifying institutions as top institutions if they are in the global top list ranked by institution h-index (computed in MAG up to 2022;

top 100), averaging  $P_{s,u}$  over referenced top institutions and over referenced non-top institutions, and define career top institutions propensity as mean propensity over top institutions divided by the sum of the mean propensities over top and non-top institutions. Second, we define career western countries propensity analogously by grouping referenced countries into western (US, Canada, UK, EU27) versus non-western sets, computing the mean source propensity  $P_{s,u}$  within each set, and define career western countries propensity as the relative share of propensity assigned to western countries.

We first compare these two measures between Mob and non-Mob scientists. Interestingly, the two dimensions display diverging patterns. Mob scientists exhibit a significantly higher career top institutions propensity than non-Mob scientists (mean: Mob = 0.5244 vs. non-Mob = 0.5218; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 0.51% increase. By contrast, Mob scientists exhibit a significantly lower career western countries propensity than non-Mob scientists (mean: Mob = 0.5586 vs. non-Mob = 0.5639; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 0.94% decrease. These results indicate that, in their later careers, Mob scientists exhibit a stronger propensity toward top-institution knowledge sources, yet a weaker propensity toward western-country knowledge sources. This asymmetric pattern suggests that institution-level and country-level core knowledge sourcing may arise through partially distinct channels.

We next probe the collaborative origins of these two career-level sourcing propensities. Building on Supplementary Note 3.2.1, we revisit whether the composition of later-career collaborators can account for scientists' subsequent knowledge-source propensities. We observe that the mean number of per paper later-career coauthors from top-100 institutions (super team papers, i.e., papers with over 29 authors, are excluded) is positively associated with career top institutions propensity (Pearson  $r = 0.254$ ,  $P < 0.001$ ; Spearman  $\rho = 0.422$ ,  $P < 0.001$ ). Similarly, the mean number of per paper later-

career coauthors (super team papers, i.e., papers with over 29 authors, are excluded) from western countries is positively associated with career western countries propensity (Pearson  $r = 0.139$ ,  $P < 0.001$ ; Spearman  $\rho = 0.307$ ,  $P < 0.001$ ). These patterns suggest that later-career collaborative exposure remains an important proximal channel through which scientists' knowledge-sourcing orientations are shaped.

This later-career collaborator composition readily explains the country-level result. Compared with non-Mob scientists, Mob scientists have a lower mean number of per paper later-career coauthors from western countries (mean: Mob = 2.4015 vs. non-Mob = 2.5672; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 6.46% decrease, and correspondingly also exhibit a lower career western countries propensity. In other words, scientists' propensity toward western-country knowledge sources may depend on whether they continue to collaborate with western-country partners in the later stage of their careers. However, the institution-level result cannot be explained as directly. Although the mean number of later-career coauthors from top-100 institutions per paper is also lower among Mob scientists than among non-Mob scientists (mean: Mob = 0.9113, non-Mob = 0.9808; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 7.09% decrease, Mob scientists nevertheless exhibit a higher career top institutions propensity. This mismatch suggests that later-career collaborator composition alone is insufficient to explain why Mob scientists continue to show a stronger propensity toward top-institution knowledge sources.

We therefore further examine whether this institution-level pattern may depend on scientists' own later-career institutional environments. Specifically, we stratify scientists by whether the modal institution in their later career belongs to the top-100 institutions. The larger difference arises among scientists whose later-career modal institution does not belong to the top-100 group: within this subgroup, Mob scientists exhibit a significantly higher career top institutions propensity than non-Mob scientists (mean: Mob = 0.5223 vs. non-Mob = 0.5189; mean gap = 0.0033; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 0.64% increase. Among scientists whose later-career

modal institution is itself a top-100 institution, the difference is smaller (mean: Mob = 0.5308 vs. non-Mob = 0.5287; mean gap = 0.0021; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 0.40% increase. While this is consistent with the general expectation that scientists located at top institutions show higher top-institution knowledge propensity overall, the Mob–non-Mob gap is markedly larger outside the top-100 institutional stratum. In fact, the mean gap in the non-top-100 group is 56.0% larger than that in the top-100 group. This pattern suggests that Mob scientists' higher top-institution knowledge propensity is not simply a reflection of current elite institutional placement; rather, it is especially visible among those located in non-top institutions, consistent with a stronger orientation toward top-institution knowledge even outside elite institutional environments. One possible reason why Mob scientists located outside top-100 institutions retain a stronger top-institution knowledge propensity is that mobility may allow them to carry elite collaborative ties into later peripheral positions. Although Mob scientists have fewer top-institution coauthors per paper overall in later career, this aggregate pattern masks an important conditional contrast: among scientists whose later-career modal institution is outside the top-100, Mob scientists have more top-institution coauthors per paper than non-Mob scientists (Table S23). This suggests that elite ties formed or activated through early mobility may remain available even after scientists move into non-top institutional environments, helping them sustain a stronger orientation toward top-institution knowledge sources from peripheral institutional positions.

To further assess which later-career institutional stratum accounts for more of the overall Mob advantage in career top-institution propensity, we conducted a symmetric Kitagawa decomposition of the aggregate mean difference using the stratified sample sizes and means reported in Supplementary Table S23. This decomposition separates the overall Mob–non-Mob gap into two parts. The first is the within-location contribution, calculated by multiplying the Mob–non-Mob mean gap within each institutional stratum by the average share of Mob and non-Mob scientists in that stratum.



This component captures how much each stratum-specific Mob advantage contributes to the aggregate gap. The second is the compositional contribution, calculated from differences in the distribution of Mob and non-Mob scientists across institutional strata, weighted by the average outcome level in each stratum. This component captures how much of the aggregate gap is attributable to Mob and non-Mob scientists being distributed differently across top-100 and non-top institutions. The decomposition shows that the overall Mob advantage in career top-institution propensity is mainly attributable to the positive within-location contribution. The total within-location contribution is approximately +0.0030, whereas the compositional contribution is small and negative, approximately -0.0003. This negative compositional contribution reflects the fact that Mob scientists are less concentrated in top-100 institutions, where the baseline level of career top-institution propensity is slightly higher. Within the positive within-location component, the non-top institution stratum contributes approximately +0.00247, accounting for about 81% of the total positive within-location contribution, whereas the top-100 institution stratum contributes approximately +0.00058, accounting for about 19%. Thus, the overall Mob advantage in career top-institution propensity is mainly driven by scientists whose later-career institutions lie outside the top 100, rather than by current placement in elite institutions.

We next trace this pattern back to early-career collaboration exposure. Consistent with our earlier findings, Mob scientists accumulate more early-career collaborators from top-100 institutions, and this early-career exposure is positively associated with career top institutions propensity in the later career stage. Specifically, the mean number of per paper early-career coauthors from top-100 institutions is positively associated with career top institutions propensity (Pearson  $r = 0.177$ ,  $P < 0.001$ ; Spearman  $\rho = 0.275$ ,  $P < 0.001$ ). Likewise, the mean number of per paper early-career coauthors from western countries is positively associated with career western countries propensity (Pearson  $r = 0.102$ ,  $P < 0.001$ ; Spearman  $\rho = 0.228$ ,  $P < 0.001$ ). These results suggest that early-career elite exposure may leave a persistent imprint on later knowledge-sourcing

orientations.

Taken together, these findings point to an asymmetry between career institution-level and country-level knowledge sourcing. At the country level, later-career western countries propensity appears to be more tightly linked to contemporaneous collaboration patterns: whether scientists continue to collaborate with western-country coauthors predicts whether they continue to show a stronger propensity toward western-country knowledge sources. At the institution level, however, early-career exposure to top-institution collaborators appears to generate a more persistent imprint. Even when Mob scientists are no longer located at top institutions in the later career stage, they still exhibit a stronger propensity toward top-institution knowledge sources. These results raise the possibility that early exposure to top-institution networks forms a persistent elite linkage, and potentially a degree of institutional lock-in, that continues to shape knowledge-sourcing orientations long after scientists have moved beyond those immediate environments.

We next examine whether this country-level pattern also varies with scientists' later-career geographic environments. Specifically, we stratify scientists by whether their modal affiliation in the later career stage is located in western countries. Among scientists whose later-career modal affiliation is located in non-western countries, Mob scientists exhibit a significantly higher career western countries propensity than non-Mob scientists (mean: Mob = 0.5384 vs. non-Mob = 0.5307; mean gap = 0.0078; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 1.46% increase. By contrast, among scientists whose later-career modal affiliation is located in western countries, the pattern reverses: Mob scientists exhibit a significantly lower career western countries propensity than non-Mob scientists (mean: Mob = 0.5668 vs. non-Mob = 0.5731; mean gap = -0.0063; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 1.09% decrease. The non-western stratum again points to a possible collaborator-based explanation for how Mob scientists remain connected to core knowledge sources from peripheral locations. Although Mob scientists have fewer western-country coauthors per paper

overall in later career, this aggregate pattern masks an important conditional contrast: among scientists whose later-career modal affiliation is located in non-western countries, Mob scientists have more western-country coauthors per paper than non-Mob scientists (Table S23). This suggests that western collaborative ties formed or activated through early mobility may remain available after scientists move into non-western environments, helping them sustain a stronger orientation toward western-country knowledge sources from peripheral country positions. The western stratum reveals the reverse side of this bridging pattern. Among scientists located in western countries, Mob scientists show lower career western countries propensity than non-Mob scientists, indicating that they are less exclusively oriented toward western knowledge sources. Further, Mob scientists located in western countries also have fewer western country coauthors per paper than non-Mob scientists (Table S23). Thus, rather than being more tightly confined to western country collaborators and knowledge sources after entering the core, Mob scientists exhibit a less western-exclusive collaboration and knowledge sourcing pattern, suggesting that early mobility may help scientists located in the core remain less dependent on western-country collaborators and knowledge sources, enabling a reverse flow from periphery to core (Table S23). These results suggest that when located in non-western contexts, Mob scientists exhibit a stronger propensity toward western-country knowledge sources than non-Mob scientists, whereas within western environments they appear less exclusively oriented toward western-country knowledge sources.

To assess which later-career country stratum accounts for Mob scientists' overall lower career western countries propensity—or equivalently, their higher propensity toward non-western country knowledge sources—we applied the same symmetric Kitagawa decomposition using the stratified sample sizes and means reported in Supplementary Table S23. The stratified means already suggest that the western-country stratum is especially important. Among scientists located in non-western countries, Mob scientists have higher western-country propensity than non-Mob scientists, which would work

against a higher non-western-country propensity. By contrast, among scientists located in western countries, Mob scientists have lower western-country propensity than non-Mob scientists, directly contributing to their higher non-western-country propensity. The decomposition confirms this interpretation while also showing that the aggregate pattern has both within-location and compositional sources. The total within-location contribution to Mob scientists' higher non-western-country propensity is positive, approximately +0.0028. This positive within-location contribution is mainly driven by scientists located in western countries: within this stratum, Mob scientists' lower western-country propensity contributes approximately +0.0047 to non-western-country propensity, partly offset by a negative contribution of approximately  $-0.0020$  among scientists located in non-western countries. The compositional contribution is also positive, approximately +0.0026, reflecting the fact that Mob scientists are more concentrated in non-western later-career locations, where baseline non-western-country propensity is higher. Thus, Mob scientists' overall higher non-western-country propensity reflects two forces: their greater presence in non-western environments and, more importantly for the within-location pattern, the less western-exclusive knowledge sourcing of Mob scientists located within western countries.

In this sense, Mob scientists may occupy a distinctive position in the global knowledge system, though the institutional and geopolitical dimensions differ in form. At the institution level the pattern is one-directional. Mob scientists remain more oriented toward top-institution knowledge across settings, and most clearly from peripheral positions, where about 74% of Mob scientists are located in later career (Table S23), without a corresponding move away from top-institution knowledge among those at top institutions. At the country level the pattern is two-directional. When located in non-western environments, Mob scientists draw more on western-country knowledge than non-Mob scientists, whereas when located in western environments, where about 71% of Mob scientists are located in later career (Table S23), they are less exclusively oriented toward western-country knowledge and draw more on non-western sources.

Taken together, Mob scientists tend to occupy peripheral positions institutionally yet core positions geopolitically, and from each they reach toward the opposite pole, drawing core institutional knowledge into the periphery and non-western knowledge into the core. This is consistent with their occupying a context-dependent bridging position between core and periphery in later-career knowledge sourcing.

Additionally, we examine paper-level normalized relative shares of knowledge sources. Analogous to the field-based measures above, reference top institutions relative share is defined as the sum of institution-normalized frequencies for top-100 institutions divided by the sum for top-100 plus non-top-100 institutions; reference western countries relative share is defined analogously at the country level. At the overall mean level, these relative-share measures provide a complementary view of Mob scientists' later-career knowledge sourcing. More importantly, the stratified results further sharpen the interpretation above. At the institution level, among scientists whose later-career modal institution does not belong to the top-100 group, Mob scientists exhibit a substantially higher reference top institutions relative share than non-Mob scientists (mean: Mob = 0.5727 vs. non-Mob = 0.5579; mean gap = 0.0148; one-sided Welch's t-test,  $P < 0.001$ ), whereas among scientists located in top-100 institutions, the difference becomes negligible and slightly reverses in sign (mean: Mob = 0.6897 vs. non-Mob = 0.6903; one-sided Welch's t-test,  $P = 0.58$ ). This suggests that Mob scientists' stronger linkage to top-institution knowledge is disproportionately driven by those located outside top institutions themselves, rather than merely reflecting current placement within elite institutional environments. At the country level, a parallel pattern also emerges: among scientists located in non-western environments, Mob scientists exhibit a higher reference western countries relative share than non-Mob scientists (mean: Mob = 0.4854 vs. non-Mob = 0.4363; mean gap = 0.0491; one-sided Welch's t-test,  $P < 0.001$ ), whereas among those located in western environments, Mob scientists exhibit a lower reference western countries relative share (mean: Mob = 0.6966 vs. non-Mob = 0.7213; mean gap = -0.0247; one-sided Welch's t-test,  $P < 0.001$ ). Thus, both

propensity- and share-based measures support the same conclusion. At the institution level, Mob scientists' stronger linkage to top-institution knowledge is driven by those located outside top institutions, with no offsetting shift away from top-institution knowledge among those within them. At the country level, Mob scientists draw more on western knowledge when located in non-western settings, yet are less exclusively oriented toward it when located in western settings. Both measures therefore support the same context-dependent bridging pattern between core and periphery.

We next ask whether these career-level core–periphery sourcing orientations are also associated with long-term academic performance, particularly long-term impact and disruptiveness. As discussed above, career top institutions propensity may reflect scientists' persistent linkage to elite institutional knowledge sources, even when they are no longer located in top institutions themselves. In this sense, it may represent a source-side manifestation of earlier embedding in elite collaborative environments. Because prior work has shown that early embedding in high-status or influential networks can predict later scientific success, a stronger career top institutions propensity may likewise be associated with higher long-term performance. Meanwhile, career western countries propensity may be more closely related to visibility, recognition, and diffusion within dominant scientific audiences. We therefore begin by testing whether these two career-level source orientations are associated with long-term academic performance. Stratifying scientists within both the Mob and non-Mob groups by the median of career top institutions propensity, we find that scientists in the higher-propensity group consistently achieve higher Publications and Hit 10pct than those in the lower-propensity group (Fig. 5b; one-sided Mann–Whitney U tests, all  $P < 0.001$ ). For D5 percentile, scientists in the higher-propensity group also tend to exhibit greater values, but the statistical evidence is weaker compared with Publications and Hit 10pct. This contrast is significant in both groups, but the significance is statistically weaker within the Mob group (mean: higher = 0.5127 vs. lower = 0.5067; one-sided Mann–Whitney U test,  $P = 0.047$ ) than within the non-Mob group (mean: higher = 0.4952 vs.

lower = 0.4848;  $P < 0.001$ ). Combined with the results in Supplementary Note 3.2.1, this pattern is consistent with the interpretation that career top institutions propensity may partly reflect a knowledge source-side manifestation of earlier success signals, i.e., earlier embedding in elite collaborative environments and the subsequent accumulation of scientific advantages associated with such networks<sup>55</sup>. In other words, its positive associations with multiple long-term performance outcomes may arise because scientists who entered elite collaborative environments earlier in their careers were already on trajectories of broader scientific success.

To further assess this interpretation, we include career top institutions propensity in regressions predicting long-term impact (Hit 10pct), and then compare its contribution with the two main channels identified above, namely later-career team reputation and paper-level local exploitation (reference fields imbalance). Career top institutions propensity is positively associated with Hit 10pct when entered together with early career cultural distance and controls (coef = 0.8876,  $P < 0.001$ ). However, after team reputation and reference fields imbalance are progressively introduced, its coefficient attenuates substantially, from 0.8876 to 0.1998 and then to 0.1042 (all  $P < 0.001$ ), suggesting potentially overlap with the two pathways identified above. Consistent with this, relative to the substantial attenuation already documented in Table S19—where team reputation and reference fields imbalance jointly reduce the coefficient on cultural distance from 0.0134 (Table S4) to 0.0010, corresponding to a 92.54% reduction—career top institutions propensity contributes only a very small additional amount of explanatory reduction. When all three variables are included together, the coefficient on cultural distance declines only marginally further, to 0.00095, corresponding to a 92.90% total reduction relative to the baseline. Thus, adding career top institutions propensity increases the explained share by only 0.36 percentage points beyond the two main pathways. We therefore do not treat career top institutions propensity as an additional primary channel. Rather, we interpret it more cautiously as a potential knowledge source-side manifestation of broader success trajectories, while continuing

to regard later-career team reputation and paper-level local exploitation as the principal pathways underlying the long-term impact advantage of Mob scientists.

Moreover, as discussed above, career western countries propensity captures scientists' tendency to draw upon knowledge originating from western countries, and may therefore be more closely related to visibility, recognition, and diffusion within dominant scientific audiences. We next include career western countries propensity in regressions predicting long-term impact. When entered together with early-career cultural distance and controls, career western countries propensity is positively associated with Hit 10pct (coef = 0.1704,  $P < 0.001$ ). Because Mob scientists exhibit a lower career western countries propensity on average, while this measure itself is positively associated with impact, introducing career western countries propensity slightly increases the coefficient on cultural distance relative to the baseline specification in Table S4 (coef = 0.0137 vs. 0.0134,  $P < 0.001$ ), consistent with the descriptive patterns above. After later-career team reputation is additionally introduced, the coefficient on career western countries propensity remains positive though somewhat attenuated (coef = 0.1359,  $P < 0.001$ ), and it remains positive after paper-level local exploitation (reference fields imbalance) is further added (coef = 0.1218,  $P < 0.001$ ). Overall, from the specification including career western countries propensity alone to the full specification including all three variables, its coefficient declines by 28.53%. By comparison, the coefficient on cultural distance rises only modestly when career western countries propensity is introduced, from 0.0134 to 0.0137, an increase of 2.61%, whereas the two principal channels identified above—later-career team reputation and paper-level local exploitation—reduce the coefficient on cultural distance from 0.0134 to 0.0010, corresponding to a 92.54% reduction (Table S19). Thus, although western-oriented knowledge sourcing propensity represents a complementary source-side dimension related to scientific visibility and recognition, its contribution is small relative to the two principal pathways already identified, and does not alter our interpretation that later-career team reputation and paper-level local exploitation remain



the dominant channels underlying the long-term impact advantage of Mob scientists. Consistent patterns also emerge when examining paper-level knowledge-source shares. Specifically, reference top institutions relative share is positively associated with long-term impact (Pearson  $r = 0.216$ ,  $P < 0.001$ ; Spearman  $\rho = 0.232$ ,  $P < 0.001$ ), and reference western countries relative share displays a similar positive association (Pearson  $r = 0.185$ ,  $P < 0.001$ ; Spearman  $\rho = 0.200$ ,  $P < 0.001$ ). These results provide additional paper-level knowledge source evidence that scientists drawing more heavily on core institutional and western knowledge sources tend to achieve higher impact. Together with the career-level propensity measures above, these findings suggest that core-oriented knowledge sourcing is positively aligned with long-term impact outcomes.

We next examine whether these career-level source orientations are also associated with disruptiveness. Compared with the patterns observed for long-term impact, however, the evidence for D5 percentile is substantially weaker. For example, median-split comparisons based on career top institutions propensity yield weaker differences in disruptiveness, including weaker significant contrasts in some groups compared with impact (e.g., within the Mob group; one-sided Mann–Whitney U test,  $P = 0.047$ ). In addition, median-split comparisons based on career top institutions propensity within the short- and long-distance mobility groups also show weaker or inconsistent differences: the contrast remains significant within the shorter-distance group (one-sided Mann–Whitney U test,  $P < 0.01$ ), but is not significant within the longer-distance group (one-sided Mann–Whitney U test,  $P = 0.70$ ). In parallel, evidence based on paper-level knowledge source relative share is also weak. The relationship between reference top institutions relative share and D5 percentile is close to zero and directionally inconsistent across correlation measures (Pearson  $r = 0.004$ ,  $P = 0.29$ ; Spearman  $\rho = -0.005$ ,  $P = 0.11$ ). These results suggest that the relationship between knowledge sources and disruptiveness is weaker and less consistent than that observed for long-term impact. Overall, these results indicate that, across both career-level propensities and paper-level

source shares, Mob scientists display a context-dependent bridging pattern in later-career knowledge sourcing. Mob scientists occupy different positions across the two dimensions, peripheral at the institution level yet core at the country level, and in each they reach toward the opposite pole, drawing core institutional knowledge into the periphery from their peripheral institutional positions and non-western knowledge into the core from their core country positions. Whichever dimension is considered, Mob scientists hold a bridging position between core and periphery in the global knowledge system. Meanwhile, the two career-level knowledge-source propensities appear to capture different dimensions of later-career scientific advantage. Career top institutions propensity is more consistent with a source-side manifestation of broader success trajectories, likely reflecting the long-run imprint of earlier embedding in elite collaborative environments. By contrast, career western countries propensity is more closely related to visibility and recognition within mainstream scientific communication.

Core and periphery (Shorter vs Longer): Building on the collaboration-ecology patterns documented above, we next ask whether the magnitude of early-career cultural distance may also leave a persistent imprint on scientists' later-career orientation toward core versus peripheral knowledge sources within the mobile sample. Given that shorter-distance scientists are more strongly embedded in elite collaborative networks—having more early-career coauthors from western countries and top-100 institutions, as well as higher later-career team reputation—we hypothesize that they may also exhibit a stronger propensity to draw upon knowledge from core scientific sources in the later career stage. We test this hypothesis using the same career-level knowledge source propensity measures introduced above. Unlike the Mob vs non-Mob comparison, where institution-level and country-level source propensities display a more context-dependent pattern, the shorter–longer comparison reveals a directionally aligned core–periphery gradient across both dimensions. Shorter-distance scientists exhibit a significantly higher career top institutions propensity than longer-distance scientists

(mean: Shorter = 0.5259 vs. Longer = 0.5228; one-sided Welch's t-test,  $p < 0.001$ ), corresponding to a 0.59% increase. A parallel pattern holds at the country level: shorter-distance scientists also exhibit a significantly higher career western countries propensity than longer-distance scientists (mean: Shorter = 0.5645 vs. Longer = 0.5500; one-sided Welch's t-test,  $p < 0.001$ ), corresponding to a 2.63% increase. Thus, within mobile scientists, shorter-distance movers are more strongly oriented toward core knowledge sources, whereas longer-distance movers rely relatively more on peripheral knowledge sources. This pattern is consistent with the collaboration ecology documented above. As shown in Supplementary Note 3.2.1, shorter-distance scientists have a higher mean number of per paper early-career coauthors from western countries and from top-100 institutions, and the same pattern persists into the later career stage, where they continue to collaborate more with coauthors from these core environments. In particular, shorter-distance scientists exhibit a significantly higher mean number of per paper later-career coauthors from top-100 institutions than Longer-distance scientists (mean: Shorter = 1.0790, Longer = 0.7895; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 36.67% increase. They also exhibit a significantly higher mean number of per paper later-career coauthors from western countries (mean: Shorter = 2.5495 vs. Longer = 1.9674; one-sided Welch's t-test,  $p < 0.001$ ), corresponding to a 29.59% increase. Consistent with this, later-career coauthors from top-100 institutions are positively associated with career top institutions propensity (Pearson  $r = 0.285$ ,  $p < 0.001$ ; Spearman  $r = 0.435$ ,  $p < 0.001$ ), and later-career coauthors from western countries are positively associated with career western countries propensity (Pearson  $r = 0.206$ ,  $p < 0.001$ ; Spearman  $r = 0.351$ ,  $p < 0.001$ ). Early-career exposure shows the same direction: the mean number of per paper early-career coauthors from top-100 institutions is positively associated with career top institutions propensity (Pearson  $r = 0.183$ ,  $p < 0.001$ ; Spearman  $r = 0.262$ ,  $p < 0.001$ ), and the mean number of per paper early-career coauthors from western countries is positively associated with career western countries propensity (Pearson  $r = 0.130$ ,  $p < 0.001$ ; Spearman  $r = 0.235$ ,  $p < 0.001$ ). Together, these results suggest that shorter-distance scientists' stronger

orientation toward core knowledge sources is closely aligned with their greater and more persistent exposure to western-country and top-institution collaboration networks. We next ask whether these core-oriented knowledge-source propensities are associated with long-term academic performance. Here, the institution- and country-level source measures appear to capture related but slightly different dimensions. Career top institutions propensity is more closely aligned with persistent linkage to elite knowledge infrastructures and can be viewed, as in the Mob vs non-Mob comparison, as a source-side manifestation of earlier embedding in elite collaborative environments. By contrast, career western countries propensity is more closely related to later reliance on knowledge from dominant scientific communication centers and thus may be more directly aligned with visibility and recognition in mainstream scientific audiences. Consistent with this interpretation, stratified analyses based on the median split show that scientists in the higher-propensity groups consistently achieve higher Hit 10pct than those in the lower-propensity groups (Fig. 5e; one-sided Mann–Whitney U tests, all  $p < 0.001$ ). Regression analyses further support this interpretation. For career top institutions propensity, the variable is positively associated with Hit 10pct when entered together with cultural distance and controls (coef = 1.1658,  $p < 0.001$ ). After later-career team reputation is introduced, the coefficient for career top institutions propensity attenuates to 0.2648 ( $p < 0.001$ ), and remains positive after paper-level local exploitation, measured by reference home fields relative share, is further included (coef = 0.2232,  $p < 0.001$ ). At the same time, the coefficient on cultural distance attenuates from the baseline specification in Table S4, where it is -0.0317, to -0.0189 and then to -0.0183, corresponding to a 42.12% reduction in magnitude in the fuller specification. This attenuation is only slightly larger than the 41.96% reduction already explained by later-career team reputation and local exploitation in Table S20. Thus, adding career top institutions propensity provides only a very small additional reduction beyond the two principal pathways identified earlier. In this sense, as in the Mob vs non-Mob comparison, career top institutions propensity is best interpreted not as an additional primary channel, but rather as a source-side manifestation of elite knowledge

environments that is substantially overlapped with team reputation and local exploitation in Shorter vs Longer comparison. For career western countries propensity, the pattern is somewhat different. It is likewise positively associated with Hit 10pct when entered together with cultural distance and controls (coef = 0.2336,  $p < 0.001$ ), and remains positively associated after later-career team reputation is introduced (coef = 0.1317,  $p < 0.001$ ) and after paper-level local exploitation is further added (coef = 0.1250,  $p < 0.001$ ). Moreover, when career western countries propensity, later-career team reputation, and local exploitation are jointly included, the coefficient on cultural distance attenuates from the baseline value of -0.0317 to -0.0164, corresponding to a 48.34% reduction in magnitude. This attenuation is larger than both the 41.96% reduction explained by team reputation and local exploitation alone and the 42.12% reduction obtained when replacing career western countries propensity with career top institutions propensity. These results suggest that, within the shorter–longer comparison, orientation toward western country knowledge sources constitutes a more distinct source-side pathway linked to long-term impact. This pattern is consistent with the idea that shorter-distance scientists, who remain more strongly oriented toward western country knowledge sources, may enjoy greater visibility, recognition, and access to mainstream scientific audiences. Taken together, these results indicate that stronger reliance on core institutional and especially western country knowledge sources is aligned with the long-term impact advantage observed among shorter-distance scientists.

We next turn to long-term disruptiveness. Within the shorter–longer comparison, stratified analyses based on the median split of career western countries propensity show that scientists in the higher-propensity group consistently achieve lower D5 percentile than those in the lower-propensity group (Fig. 5e; one-sided Mann–Whitney U tests,  $p < 0.001$ ). Consistent with this pattern, career western countries propensity is negatively associated with D5 percentile within mobile scientists (Pearson  $r = -0.032$ ,  $p < 0.001$ ; Spearman  $r = -0.056$ ,  $p < 0.001$ ). A similar negative pattern is also observed

for the paper-level reference western countries relative share (Pearson  $r = -0.023$ ,  $p < 0.001$ ; Spearman  $r = -0.034$ ,  $p < 0.001$ ). In regressions predicting D5 percentile, team expertise distance is positively associated with disruptiveness, and when introduced alone, the coefficient on cultural distance decreases from the baseline value of 0.0479 (Table S4) to 0.0458, corresponding to a 4.38% reduction in magnitude. Career western countries propensity, when introduced alone, is negatively associated with disruptiveness (coef = -0.1383,  $p < 0.001$ ), and the coefficient on cultural distance decreases to 0.0450, corresponding to a 6.12% reduction in magnitude. When both team expertise distance and career western countries propensity are included simultaneously, both variables remain significant (team expertise distance: coef = 0.1747,  $p < 0.001$ ; career western countries propensity: coef = -0.1513,  $p < 0.001$ ), and the coefficient on cultural distance further decreases to 0.0429, corresponding to a 10.40% reduction in magnitude. Thus, within the shorter–longer comparison, both team expertise distance and lower reliance on western-country knowledge sources contribute to explaining the higher long-term disruptiveness of longer-distance scientists, with the latter constituting an additional source-side pathway beyond the principal expertise-distance channel.

By contrast, in the Mob vs non-Mob comparison, career western countries propensity plays a much more limited role for disruptiveness. When entered alone with early-career cultural distance and controls, it is negatively but not significantly associated with D5 percentile (coef = -0.0018,  $p = 0.92$ ). Earlier analyses showed that the two principal pathways associated with long-term disruptiveness in the Mob vs non-Mob comparison are team expertise distance and career-level exploration. After these two principal pathways are included, the coefficient on cultural distance attenuates from 0.0197 to 0.0160, corresponding to an 18.80% reduction in magnitude. When career western countries propensity is further introduced on top of these two pathways, its coefficient remains negative but non-significant (coef = -0.0010,  $p = 0.95$ ), and the coefficient on cultural distance changes only trivially, from 0.0160 to 0.016045. Taken together, these results suggest that the relationship between career western countries

propensity and long-term disruptiveness in the Mob vs non-Mob comparison is too weak and unstable to be considered a primary pathway. We therefore do not treat it as an explanatory channel for disruptiveness in this comparison.

In summary, core–periphery knowledge sourcing provides a further lens through which to understand how early-career cultural mobility shapes long-term scientific trajectories. Our results show that mobility is associated not only with different collaboration ecologies and knowledge strategies, but also with systematic differences in orientation toward core versus peripheral knowledge sources in the global scientific system. Importantly, the two mobility comparisons display different source-side configurations. In the Mob vs non-Mob comparison, Mob scientists show a higher career top-institution propensity and a lower career western-country propensity, equivalently a higher non-western-country propensity, than non-Mob scientists. Because most Mob scientists end up in non-top institutions yet in western countries in later career, these orientations together cast them as context-dependent bridges between core and periphery. From their typically peripheral institutional positions they channel core institutional knowledge into the periphery, and from their typically core country positions they channel non-western knowledge into the core, consistent with their broader bridging role documented above. In the shorter–longer comparison, by contrast, the pattern is more directionally aligned: shorter-distance movers are more strongly oriented toward core knowledge sources, whereas longer-distance movers rely relatively more on peripheral knowledge sources. These source-side differences mirror the broader divergence already documented in long-term academic outcomes, with core-oriented sourcing more closely aligned with long-term impact and peripheral-oriented sourcing more closely aligned with long-term disruptiveness.

This naturally raises a further question. Knowing whether scientists draw more from core or peripheral sources tells us where they search, but not yet how far they search across the global knowledge space. In particular, the negative association between western country knowledge orientation and long-term disruptiveness among longer-

distance movers suggests that peripheral sourcing may reflect a broader geographic search strategy in the knowledge they cite. We therefore next move beyond core–periphery source orientation itself and examine the geographic span of referenced knowledge more directly.

#### **Supplementary Note 3.4.2: Search portfolios**

A longstanding theme in the management and innovation-search literature is that innovative outcomes depend not only on whether actors or organizations search, but also on how their search portfolios are allocated across different knowledge pools<sup>56</sup>. Classic work has emphasized that search varies in familiarity and distance: organizations may exploit proximate, familiar knowledge or explore more distant pools, and such differences in search location can systematically shape innovation outcomes<sup>57</sup>. Prior studies further suggest that actors can overcome locally bounded search through bridges such as alliances and mobility, while broader and deeper external search may improve performance up to a point<sup>56,58</sup>. In the previous section, we showed that, especially within the shorter–longer comparison, orientation toward western versus non-western knowledge sources is systematically associated with long-term disruptiveness. We therefore next examine the most direct geographic implication of this pattern by measuring the geographic distance of referenced knowledge itself.

Search portfolios (non-Mob vs Mob): We next move beyond core–periphery source orientation and examine how far scientists search across geographic space in their referenced knowledge. Here, we first focus on the geographic radius of search, i.e., reference geographic distance, measured as the average geographic distance between the focal paper and its cited references. Specifically, for each focal paper, we construct all pairwise geographic distances between the focal paper’s affiliations and the affiliations observed in its reference list, and then take the average. A larger value indicates that the focal paper draws upon knowledge from geographically more distant sources. We find that Mob scientists exhibit a significantly larger reference geographic distance than non-Mob scientists (mean: Mob = 5522.47 km vs. non-Mob = 4850.40



km; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 13.86% increase. Thus, relative to non-mobile scientists, Mob scientists search across geographically broader knowledge environments in the later career stage. This larger geographic search radius is consistent with the source-orientation patterns documented above, but it also points to a broader search process. In the previous section, we showed that Mob scientists exhibit both a lower career western countries propensity and a lower mean number of per paper later-career coauthors from western countries. Correspondingly, career western countries propensity is negatively associated with reference geographic distance (Pearson  $r = -0.243$ ,  $p < 0.001$ ; Spearman  $r = -0.256$ ,  $p < 0.001$ ), and the mean number of per paper later-career coauthors from western countries is also negatively associated with reference geographic distance (Pearson  $r = -0.366$ ,  $p < 0.001$ ; Spearman  $r = -0.373$ ,  $p < 0.001$ ). These patterns indicate that scientists who remain less oriented toward western-country knowledge sources and western-country collaborators tend to search across geographically more distant knowledge environments. This pattern is consistent with the idea that moving beyond dominant knowledge pools may enlarge the geographic scope of search. At the same time, mobility may also expand scientists' exposure to geographically broader collaborative environments, which in turn can widen the spatial range of the knowledge they subsequently draw upon. Consistent with this interpretation, later-career team geographic distance is positively associated with reference geographic distance (Pearson  $r = 0.363$ ,  $p < 0.001$ ; Spearman  $r = 0.367$ ,  $p < 0.001$ ), suggesting that scientists embedded in more geographically dispersed collaboration teams also tend to search more broadly across geographic knowledge space. These results suggest that Mob scientists' broader geographic search radius may arise from both weaker reliance on western-core knowledge sources and greater exposure to geographically wider collaborative environments.

We next examine how this geographic search radius is related to long-term academic performance. Stratified analyses based on the median split of reference geographic distance show that scientists in the higher-distance group consistently achieve lower

Hit 10pct but higher Publications and D5 percentile than those in the lower-distance group (one-sided Mann–Whitney U tests, all  $P < 0.001$ ). Given that the long-term impact advantage of Mob scientists has already been shown above to be primarily associated with team reputation and local exploitation, here we focus primarily on long-term disruptiveness. We then test whether reference geographic distance constitutes an additional search-side pathway associated with the higher long-term disruptiveness of Mob scientists. Earlier analyses showed that the two principal disruptiveness-related pathways in the Mob vs non-Mob comparison—team expertise distance and career-level exploration—reduce the coefficient on cultural distance from 0.0197 to 0.0160, corresponding to an 18.80% reduction in magnitude relative to the baseline. When reference geographic distance is introduced alone, it is positively associated with D5 percentile (coef = 0.000007,  $p < 0.001$ ), and the coefficient on cultural distance decreases from 0.0197 to 0.0152, corresponding to a 22.82% reduction. When it is introduced together with team expertise distance, the coefficient further decreases to 0.0141, corresponding to a 28.55% reduction. Finally, when reference geographic distance, team expertise distance, and career-level exploration are jointly included, the coefficient on cultural distance decreases to 0.0119, corresponding to a total reduction of 39.58% relative to the baseline. These results suggest that geographic search radius provides a substantial additional search-side pathway associated with long-term disruptiveness. Conceptually, this pattern is consistent with the innovation-search view that drawing on more geographically distant knowledge pools may enlarge the opportunity set of unfamiliar combinations and expose scientists to ideas, techniques, and contexts that are less available in more proximate search environments. Taken together, this pattern suggests that, alongside team expertise distance and career-level exploration, geographic search radius constitutes a third major pathway associated with the disruptiveness advantage of Mob scientists.

However, geographic distance captures only one dimension of search portfolios. Classic work in the management and innovation-search literature suggests that innovative

outcomes depend not only on how far actors search across space, but also on how their search is distributed across other dimensions of the opportunity space<sup>56,57</sup>. Once one considers the spatial span of search, a natural next question is whether scientists also differ in the temporal span of the knowledge they draw upon—namely, whether they search older versus newer knowledge. In parallel, prior work in science studies has emphasized that search is also differentiated by the popularity or attention profile of the knowledge being recombined, for example whether scientists draw more heavily on already prominent versus less prominent prior work<sup>59</sup>. Thus, these dimensions suggest that scientific search is better understood not through geographic distance alone, but through a broader search-portfolio profile. We therefore next examine whether Mob and non-Mob scientists differ systematically not only in the geographic distance of their referenced knowledge, but also in the age and popularity composition of the knowledge they search.

Mob scientists exhibit a significantly higher reference age than non-Mob scientists (mean: Mob = 9.2793 vs. non-Mob = 9.0880; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 2.11% increase. Likewise, Mob scientists exhibit a significantly higher reference popularity than non-Mob scientists (mean: Mob = 96.7719 vs. non-Mob = 87.0511; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to an 11.17% increase. Combined with their larger reference geographic distance documented above, these results suggest that Mob scientists differ from non-Mob scientists in the broader profile of the knowledge they search: they draw upon knowledge that is geographically farther, older, and more popular.

We next examine whether these search-portfolio dimensions are associated with long-term disruptiveness. When introduced separately with early-career cultural distance and controls, both reference age (coef = 0.004762,  $P < 0.001$ ) and reference popularity (coef = 0.000340,  $P < 0.001$ ) are positively associated with D5 percentile. When the two measures are jointly introduced, both remain significantly positive (reference age: coef = 0.007387,  $P < 0.001$ ; reference popularity: coef = 0.000387,  $P < 0.001$ ). We then

examine how these search-portfolio dimensions contribute to explaining the disruptiveness advantage associated with cultural distance. When reference geographic distance is introduced alongside team expertise distance and career-level exploration, the coefficient on cultural distance further decreases to 0.0119. When reference age is additionally introduced, the coefficient decreases slightly further to 0.0115. Finally, when reference popularity is also included, the coefficient decreases to 0.0086, corresponding to a total reduction of 56.38% relative to the baseline. These results suggest that mobility-related differences in search portfolios extend beyond geographic search radius alone. Rather, Mob scientists differ systematically in the broader profile of the knowledge they search—how far, how old, and how prominent it is—and these dimensions jointly constitute a substantial search-side pathway associated with their long-term disruptiveness advantage.

Taken together, these findings suggest that Mob scientists differ from non-Mob scientists not only in where they search, but also in the broader profile of the knowledge they search. They draw upon knowledge that is geographically farther, older, and more prominent, consistent with a broader search portfolio spanning wider and more established knowledge environments. Although the relationship of reference age and reference popularity to disruptiveness is likely complex, these dimensions jointly constitute an important part of the search-side profile associated with the long-term disruptiveness advantage of Mob scientists.

Search portfolios (Shorter vs Longer): We next examine whether the source-orientation differences documented above within the mobile sample are also reflected in how far scientists search across geographic space in their referenced knowledge. In the previous section, we showed that longer-distance movers exhibit a lower career western countries propensity on average. A natural next question is therefore whether these scientists also search across geographically more distant knowledge environments in the later career stage.

We first focus on the geographic radius of search, i.e., reference geographic distance.

We find that longer-distance movers exhibit a significantly larger reference geographic distance than shorter-distance movers (mean: Longer = 5944.1742 km vs. Shorter = 5343.2471 km; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to an 11.25% increase. Thus, relative to shorter-distance movers, longer-distance movers search across geographically broader knowledge environments in their referenced knowledge.

This larger geographic search radius is consistent with the source-orientation patterns documented above. In particular, because longer-distance movers exhibit a lower career western countries propensity on average, one natural expectation is that they may also be less anchored to western-core knowledge environments and therefore search across geographically more distant sources. Consistent with this expectation, career western countries propensity is negatively associated with reference geographic distance (Pearson  $r = -0.2310$ ,  $p < 0.001$ ; Spearman  $r = -0.2680$ ,  $p < 0.001$ ), and the mean number of per paper later-career coauthors from western countries is also negatively associated with reference geographic distance (Pearson  $r = -0.4080$ ,  $p < 0.001$ ; Spearman  $r = -0.4280$ ,  $p < 0.001$ ). At the same time, later-career team geographic distance is positively associated with reference geographic distance (Pearson  $r = 0.3630$ ,  $p < 0.001$ ; Spearman  $r = 0.3670$ ,  $p < 0.001$ ), suggesting that scientists embedded in more geographically dispersed collaboration teams also tend to search more broadly across geographic knowledge space. Taken together, these results suggest that longer-distance movers' broader geographic search radius may arise from both weaker reliance on western-core knowledge sources and greater exposure to geographically broader collaborative environments.

We next examine how this geographic search radius is related to long-term academic performance. Stratified analyses based on the median split of reference geographic distance show that scientists in the higher-distance group consistently achieve lower Hit 10pct but higher Publications and D5 percentile than those in the lower-distance group (Fig. 5h; one-sided Mann–Whitney U tests, all  $P < 0.001$ ). This pattern suggests that broader geographic search is associated with a lower impact but more

disruptiveness-oriented performance profile. Given that the long-term impact advantage of shorter-distance movers has already been partially explained above by team reputation, local exploitation, and career western countries propensity, we next ask whether reference geographic distance provides an additional clue to the divergence between impact and disruptiveness in this comparison. Consistent with the stratified results, when introduced alone with early-career cultural distance and controls, reference geographic distance is negatively associated with Hit 10pct (coef = -0.000004,  $P < 0.001$ ), and the coefficient on cultural distance attenuates from -0.0317 to -0.0289, corresponding to an 8.80% reduction in magnitude. Earlier analyses showed that career western countries propensity, later-career team reputation, and local exploitation jointly reduce the coefficient on cultural distance to -0.0164, corresponding to a 48.34% reduction relative to the baseline. When reference geographic distance is further introduced, the coefficient decreases to -0.0153, corresponding to a total reduction of 51.71% relative to the baseline. This additional attenuation suggests that geographic search radius constitutes an additional search-side pathway associated with the lower long-term impact of longer-distance movers.

We then test whether reference geographic distance constitutes an additional search-side pathway associated with the higher long-term disruptiveness of longer-distance movers. When introduced alone with early-career cultural distance and controls, reference geographic distance is positively associated with D5 percentile (coef = 0.000008,  $P < 0.001$ ), consistent with the descriptive patterns above, and the coefficient on cultural distance decreases from 0.0479 to 0.0427, corresponding to a 10.76% reduction. We next examine whether this geographic search-radius measure adds explanatory power beyond the two principal pathways identified above, namely team expertise distance and career western countries propensity. When team expertise distance, career western countries propensity, and reference geographic distance are jointly included, the coefficient on cultural distance decreases to 0.0395, corresponding to a 17.54% reduction relative to the baseline. Compared with the 10.44% reduction

explained by team expertise distance and career western countries propensity alone, this additional attenuation suggests that reference geographic distance provides an additional search-side pathway associated with the higher long-term disruptiveness of longer-distance movers.

Longer-distance movers may also differ from shorter-distance movers in other dimensions of their search portfolios beyond geographic distance. We therefore next examine whether they differ systematically not only in the geographic distance of their referenced knowledge, but also in the age and popularity composition of the knowledge they search. Longer-distance movers exhibit a significantly higher reference age than shorter-distance movers (mean: Longer = 9.5667 vs. Shorter = 9.0855; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 5.30% increase. Likewise, longer-distance movers exhibit a significantly higher reference popularity than shorter-distance movers (mean: Longer = 97.3431 vs. Shorter = 93.8725; one-sided Welch's t-test,  $P < 0.01$ ), corresponding to a 3.70% increase. Combined with their larger reference geographic distance documented above, these results suggest that longer-distance movers differ from shorter-distance movers in the broader profile of the knowledge they search: they draw upon knowledge that is geographically farther, older, and more popular.

We next examine whether these search-portfolio dimensions are associated with long-term disruptiveness. When introduced separately with early-career cultural distance and controls, both reference age (coef = 0.002832,  $P < 0.001$ ) and reference popularity (coef = 0.000332,  $P < 0.001$ ) are positively associated with D5 percentile. When the two measures are jointly introduced, both remain significantly positive (reference age: coef = 0.005689,  $P < 0.001$ ; reference popularity: coef = 0.000373,  $P < 0.001$ ). We then examine how these search-portfolio dimensions jointly contribute to explaining the disruptiveness advantage associated with cultural distance. When team expertise distance, career western countries propensity, reference geographic distance, reference age, and reference popularity are jointly included, the coefficient on cultural distance decreases to 0.0344, corresponding to a total reduction of 28.18% relative to the

baseline. Compared with the 17.54% reduction explained by team expertise distance, career western countries propensity, and reference geographic distance together, this further attenuation suggests that mobility-related differences in search portfolios extend beyond geographic search radius alone. Rather, longer-distance movers differ systematically in the broader profile of the knowledge they search—how far, how old, and how prominent it is—and these dimensions jointly constitute a substantial search-side pathway associated with their long-term disruptiveness advantage.

In summary, these findings suggest that longer-distance movers differ from shorter-distance movers not only in where they search, but also in the broader profile of the knowledge they search. They draw upon knowledge that is geographically farther, older, and more prominent, consistent with a broader search portfolio spanning wider and more established knowledge environments. Although the relationship of reference age and reference popularity to disruptiveness is likely complex, these dimensions jointly constitute an important part of the search-side profile associated with the long-term disruptiveness advantage of longer-distance movers.

However, search portfolios characterize only one side of scientists' later-career knowledge dynamics: where they draw knowledge from. An equally important question is where their work is subsequently recognized and cited. We therefore next turn from the reference side to the citation side and examine whether early-career cultural mobility is also associated with systematic differences in the institutional, geopolitical, and geographic profiles of the audiences that cite scientists' later-career work.

At the career level, we construct citation-side source propensities using the same AUC-based procedure as above, but replacing referenced sources with citing-paper sources. Specifically, we define career top institutions citation propensity and career western countries citation propensity to capture the relative propensity of scientists' later-career papers to be cited by papers affiliated with top-100 institutions and western countries, respectively. In the Mob vs non-Mob comparison, the citation-side pattern is again asymmetric across institutional and country dimensions. Mob scientists exhibit a



significantly higher career top institutions citation propensity than non-Mob scientists (mean: Mob = 0.5144 vs. non-Mob = 0.5113; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 0.59% increase. By contrast, Mob scientists exhibit a significantly lower career western countries citation propensity than non-Mob scientists (mean: Mob = 0.5363 vs. non-Mob = 0.5397; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 0.62% decrease. Thus, consistent with the source-side patterns documented above, Mob scientists remain more strongly connected to top-institution citation audiences while being less exclusively concentrated within western-country citation audiences. In other words, Mob scientists are not only more likely to remain connected to top-institution citation audiences, but are also less exclusively concentrated within western-country citation audiences in the later career stage.

Within the mobile sample, the shorter–longer comparison displays a more directionally aligned pattern. Shorter-distance movers exhibit a significantly higher career top institutions citation propensity than longer-distance movers (mean: Shorter = 0.5175 vs. Longer = 0.5104; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 1.40% increase. Likewise, shorter-distance movers also exhibit a significantly higher career western countries citation propensity than longer-distance movers (mean: Shorter = 0.5426 vs. Longer = 0.5273; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 2.89% increase. These results suggest that, within mobile scientists, shorter-distance movers are more strongly recognized by core institutional and western-country citation audiences, whereas longer-distance movers are cited relatively more by non-core audiences. This pattern is also consistent with the broader collaboration and source-side results above, where shorter-distance movers are more strongly embedded in elite and western-centered scientific environments.

We next further examine the geographic radius of citation itself. Here, citation geographic distance is measured as the average geographic distance between the focal paper and the papers that cite it. A larger value indicates that a focal paper is recognized by geographically more distant citing papers and therefore reaches a broader global

citation audience. In the Mob vs non-Mob comparison, Mob scientists exhibit a significantly larger citation geographic distance than non-Mob scientists (mean: Mob = 5160.76 km vs. non-Mob = 4691.89 km; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 9.99% increase. This pattern is directionally consistent with the reference-side results above, where Mob scientists also exhibit significantly larger reference geographic distance. Taken together, these results suggest that Mob scientists not only search across geographically broader knowledge environments, but are also recognized across a wider geographic citation space in later career. By contrast, within the shorter–longer comparison, citation geographic distance differs only weakly between the two groups. Shorter-distance movers exhibit a slightly larger, but not significantly different, citation geographic distance than longer-distance movers (mean: Shorter = 5280.10 km vs. Longer = 5270.54 km; one-sided Welch's t-test,  $P = 0.63$ ). At first glance, this pattern may appear inconsistent with the reference-side results above, where longer-distance movers exhibit significantly larger reference geographic distance than shorter-distance movers. In other words, longer-distance movers search farther in geographic knowledge space, yet this does not translate into a significantly larger geographic span of citation audiences.

To reconcile this pattern, we note that citation geographic distance may reflect at least two distinct components simultaneously. First, it may carry a search-side imprint: scientists who draw upon geographically more distant knowledge may also diffuse to geographically more distant audiences. Consistent with this interpretation, reference geographic distance is positively associated with citation geographic distance in both comparisons (Mob vs non-Mob: Pearson  $r = 0.754$ ,  $p < 0.001$ ; Spearman  $r = 0.664$ ,  $p < 0.001$ ; shorter vs longer: Pearson  $r = 0.775$ ,  $p < 0.001$ ; Spearman  $r = 0.682$ ,  $p < 0.001$ ). Second, citation geographic distance may also reflect an audience-side visibility effect: scientists whose work is more strongly connected to high-visibility scientific environments may be cited from a geographically broader set of locations. Consistent with this interpretation, career top institutions propensity is positively associated with

citation geographic distance in both comparisons (Mob vs non-Mob: Pearson  $r = 0.110$ ,  $p < 0.001$ ; Spearman  $r = 0.172$ ,  $p < 0.001$ ; shorter vs longer: Pearson  $r = 0.105$ ,  $p < 0.001$ ; Spearman  $r = 0.161$ ,  $p < 0.001$ ). As discussed in Supplementary Note 3.4.1, career top institutions propensity can be interpreted as a source-side manifestation of earlier embedding in elite and globally visible knowledge environments. Scientists with stronger later-career top-institution knowledge orientation may therefore occupy a more globally visible position, making their work more likely to be recognized and cited from geographically distant locations.

This helps explain why the citation-side geographic pattern does not fully mirror the reference-side geographic pattern within the shorter–longer comparison. Although longer-distance movers exhibit a larger reference geographic distance, shorter-distance movers simultaneously exhibit a stronger career top institutions propensity, a stronger top-institution citation orientation, and higher team reputation, all of which are consistent with greater global visibility. Thus, the absence of a significant difference in citation geographic distance may reflect two offsetting forces: longer-distance movers search farther in geographic knowledge space, whereas shorter-distance movers are more strongly embedded in globally visible elite scientific environments and may therefore attract citations from geographically broader audiences. In this sense, citation geographic distance should not be interpreted simply as a mirror image of reference geographic distance. Rather, it reflects both where scientists search and how broadly their work is recognized across the global scientific audience.

Overall, the source-side and citation-side results together suggest that early-career cultural mobility is associated not simply with stronger attachment to the scientific core, but with a broader bridging position across core and non-core knowledge environments. In the Mob vs non-Mob comparison, Mob scientists retain a stronger orientation toward top-institution knowledge and citation audiences, consistent with the long-run imprint of earlier embedding in elite networks, yet they are simultaneously less exclusively concentrated within western-country sources and citing audiences and are connected to

geographically broader search and citation spaces. This pattern suggests that Mob scientists remain linked to core scientific infrastructures while also extending beyond them. Within the mobile sample, the shorter–longer comparison reveals a different configuration. Shorter-distance movers remain more strongly oriented toward core knowledge and citation audiences, including both top institutions and western countries, consistent with their higher team reputation and stronger embeddedness in globally visible elite networks. By contrast, longer-distance movers are relatively less tied to these core audiences. Although they search farther in the reference space, this does not translate into a significantly larger citation geographic distance, likely because shorter-distance movers’ stronger embeddedness in influential and globally visible networks broadens the geographic reach of their citation audiences and offsets the search-side advantage of longer-distance movers.

Finally, the evidence across Supplementary Notes 3.2–3.4 reveals that early-career cross-border/cultural mobility does not generate a single uniform advantage, but instead reorganizes scientists’ later-career collaboration, knowledge strategy, and knowledge sources into distinct and internally coherent configurations. In the Mob versus non-Mob comparison, these channels largely accumulate in the same direction: mobile scientists begin with greater early-career international exposure, more early international collaboration, and greater access to western-country and top-institution collaborators, and these early advantages are progressively translated into broader later-career collaborator pools, stronger brokerage positions, higher coordination efficiency, greater team expertise heterogeneity, broader career-level knowledge exploration, wider search portfolios, and a more bridging-like position between core and non-core knowledge environments. By contrast, within the mobile sample, the shorter-versus-longer comparison reveals a clearer divergence rather than simple accumulation. Shorter-distance movers are more strongly embedded in elite and western-centered collaborative environments, which later translate into higher team reputation, stronger local exploitation, stronger orientation toward core knowledge and citation audiences,

and ultimately higher long-term impact. Longer-distance movers, by contrast, begin with greater early international collaboration and larger cultural jumps, which later translate into larger labor advantages, more stable and efficiently coordinated collaboration structures, greater team expertise heterogeneity, broader and more distant search portfolios, and weaker anchoring to western-core knowledge sources, thereby aligning more strongly with long-term productivity and disruptiveness. Thus, the key result is not that one type of mobility is uniformly “better,” but that different magnitudes of cultural mobility route scientists into different bundles of scientific resources and different long-term performance profiles. For long-term productivity, the dominant channel is the joint accumulation of labor advantages, brokerage, and coordination efficiency, but the internal structure of this pathway differ across the two comparisons. In the Mob versus non-Mob comparison, the channel begins with higher early-career international collaboration, which is associated with larger later-career collaborator pools and greater labor advantages. These broader labor pools are then combined with stronger brokerage positions, rooted in lower early dense-tie ratios and greater willingness to form new ties, and with higher coordination efficiency that allows mobile scientists to organize collaboration more effectively within the communities they bridge. Quantitatively, these three channels jointly account for a large share of the publication advantage: later-career collaborator pool, betweenness centrality, and coordination efficiency together reduce the cultural-distance coefficient by 69.39%. Importantly, the network structural evidence clarifies why this combination works. Mob scientists do not merely occupy more fragmented and brokerage-oriented collaboration structures, with higher component size diversity and a smaller relative giant component; they also exhibit stronger within-component cohesion, as reflected in higher mean component density and coordination efficiency. In other words, Mob scientists connect more communities while also coordinating more efficiently inside them. Within the shorter-versus-longer comparison, Shorter-distance movers occupy more brokerage-oriented and fragmented structures, yet they do not achieve higher long-term publication output. The reason is that brokerage alone is insufficient. Once labor advantages and

coordination efficiency are taken into account, the publication difference is fully explained by the longer-distance movers' stronger later-career team size and more efficient repeated collaboration. That is, in this comparison the productivity pathway is not driven by broader brokerage positions, but by the combination of larger labor input and stronger within-team coordination. This may be why the shorter-longer contrast does not mirror the Mob-non-Mob contrast: shorter-distance movers are better connected across communities, but longer-distance movers are better able to convert collaboration into sustained output. For long-term impact, a different channel emerges, centered on elite embedding, team reputation, local exploitation, and, especially in the shorter-versus-longer comparison, stronger attachment to core knowledge and wider search portfolios. In the Mob versus non-Mob comparison, the pathway begins early: mobile scientists not only collaborate more internationally, but also gain more exposure to western-country and top-institution collaborators. These early elite linkages persist into later career through stronger team reputation, stronger orientation toward top-institution knowledge sources, and a more durable connection to core scientific infrastructures. At the paper level, these reputational advantages are associated with a more exploitation-oriented local knowledge strategy, including stronger reliance on home fields and familiar knowledge bases. The impact advantage therefore does not come from mobility alone, but from a layered sequence in which early international and elite exposure later becomes high-reputation teams and more focused local exploitation under high-visibility conditions. Quantitatively, this pathway is especially strong in the Mob versus non-Mob comparison, where team reputation and local exploitation together account for 92.54% of the Hit 10pct advantage. Within the shorter-versus-longer comparison, the same pathway holds, but here the divergence is more explicit. Early on, shorter-distance movers do not dominate in international collaboration volume; indeed, longer-distance movers show higher early international collaboration and later stronger labor advantages. Yet shorter-distance movers accumulate more western-country and top-institution collaborators, which then feed into higher team reputation. These results indicate more international collaboration does not

automatically generate more impact, because what matters for impact is not internationality alone but the kind of networks into which scientists are embedded. Later in the career, these differences become visible not only in team reputation and local exploitation, but also in source-side knowledge sourcing. Shorter-distance movers are more strongly oriented toward western country knowledge sources and toward top-institution knowledge sources, consistent with greater visibility in core scientific communication. Accordingly, team reputation and local exploitation explain 41.96% of the impact difference, and when the stronger orientation toward western country knowledge sources and wider geographic radius of search is further incorporated, the explained share rises to 51.71%. Thus, across both comparisons, impact is most closely aligned with a pathway in which early elite exposure is translated into later team reputation, more exploitation-oriented local knowledge use, and stronger recognition from core scientific audiences. For long-term disruptiveness, however, the main pathway shifts away from reputation and toward expertise heterogeneity, local and global exploration-oriented knowledge strategy, broader search portfolios, and, in part, weaker confinement to core western knowledge environments. In the Mob versus non-Mob comparison, early mobility lowers dense-tie dependence and increases early field pivots and early exposure to heterogeneous coauthor expertise. These early shifts then propagate into later-career teams with greater expertise heterogeneity, which in turn support more disruptive science. At the same time, mobile scientists also occupy stronger brokerage positions and display broader career-level knowledge exploration, so that disruptiveness here is tied not only to who is on the team, but also to how widely scientists range across knowledge domains over the career. Search-side evidence in 3.4 adds a further layer: mobile scientists search across more geographically distant, older, and more prominent knowledge, and these broader search portfolios explain additional variance in disruptiveness. Quantitatively, team expertise heterogeneity, career-level exploration, reference geographic distance, reference age, and reference popularity together explain 56.38% of the disruptiveness advantage in the Mob versus non-Mob comparison. Within the shorter-versus-longer comparison, the disruptiveness pathway

again reveals a different internal logic. Longer-distance movers begin with larger early field pivots and stronger early coauthor capital imbalance, and these early conditions later translate into greater team expertise heterogeneity and more exploration-oriented local knowledge strategies. They also search farther across geographic space and draw on older and more popular knowledge, while remaining less anchored to western-country knowledge sources. These channels together align them more strongly with disruptiveness. Here another important tension is resolved. At the paper level, longer-distance movers are more exploration-oriented, yet at the career level they show weaker exploration than shorter-distance movers. This is not a contradiction, because paper-level exploration and career-level exploration are generated by different structural conditions. Paper-level exploration is more tightly tied to expertise heterogeneity and the capacity to integrate cognitively distant knowledge within teams, whereas career-level exploration is more closely associated with brokerage-oriented positions. Thus, shorter-distance movers can be more exploratory at the career level because they occupy stronger brokerage positions, while longer-distance movers can still be more exploratory at the paper level because they assemble more heterogeneous teams. In the shorter-versus-longer comparison, longer-distance movers are less oriented toward western country knowledge sources, which is negatively associated with impact but positively aligned with disruptiveness. This suggests that their disruptiveness is linked not only to heterogeneous teams and broader search, but also to a weaker dependence on dominant western-core knowledge environments. Quantitatively, in this comparison team expertise heterogeneity, weaker western-country orientation, and broader search portfolios together explain 28.18% of the disruptiveness advantage. Overall, the mechanisms documented in Supplementary Notes 3.2–3.4 therefore converge on a single broader conclusion. Early-career cultural mobility is consequential not merely because it relocates scientists geographically, but because it reorganizes the architecture of their subsequent careers across multiple, connected layers. It reshapes early collaboration exposure, which then reshapes later labor advantages, elite embedding, team assembly styles, brokerage positions, coordination efficiency, expertise



composition, local and global knowledge strategies, knowledge-source orientation, and search portfolios. In the Mob versus non-Mob comparison, these layers tend to accumulate into a broad bridging configuration that supports simultaneous advantages in productivity, impact, and disruptiveness. In the shorter-versus-longer comparison, the same layers separate into two different bundles: a shorter-distance pathway centered on elite embedding, team reputation, local exploitation, and stronger orientation toward core knowledge sources, and thus higher impact; and a longer-distance pathway centered on labor advantages, stable coordination, expertise heterogeneity, broader and more distant search, weaker western-core anchoring, and thus higher productivity and disruptiveness. Put differently, mobility does not simply make scientists more successful. It changes the kind of scientific success to which they are most strongly connected.

## **Supplementary Note 4: Additional analysis**

### **Supplementary Note 4.1: Knowledge diffusion**

In Supplementary Note 3.4.2, we examined citation geographic distance, defined as the average geographic distance between a focal paper and the papers that cite it, and showed that early-career cultural mobility is associated with differences in the global visibility of scientists' later-career work. However, citation geographic distance captures only one aspect of citation-side geography: how far the citing papers are located from the focal paper itself. It does not reveal whether the citing papers themselves are geographically concentrated or instead dispersed across a broader global space. We therefore next examine knowledge diffusion more directly by focusing on citation geographic dispersion, defined as the mean pairwise geographic distance among the papers that cite a focal paper. In this sense, the two measures capture different dimensions of citation-side geography. Citation geographic distance reflects how far the work reaches from the focal paper, whereas citation geographic dispersion reflects how widely the citing papers are geographically distributed relative to one another. For example, a focal paper may be cited by papers that are, on average, far from the focal paper itself, yet those citing papers may still cluster within a relatively limited set of places. Conversely, the average citing distance from the focal paper may not be especially large, while the citing papers themselves are spread across a broader set of locations. Compared with citation geographic distance, which is closer to global visibility, citation geographic dispersion more directly captures the geographic breadth with which knowledge diffuses across downstream audiences.

We first compare citation geographic dispersion across the two mobility contrasts. In the Mob versus non-Mob comparison, Mob scientists exhibit a significantly higher citation geographic dispersion than non-Mob scientists (mean: Mob = 5669.1062 vs. non-Mob = 5370.5021; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 5.56% increase (Fig. 5c). Within the mobile sample, longer-distance movers likewise exhibit

a significantly higher citation geographic dispersion than shorter-distance movers (mean: Longer = 5744.5845 vs. Shorter = 5715.9052; one-sided Welch's t-test,  $P < 0.05$ ), corresponding to a 0.50% increase. Combined with the citation geographic distance results in Supplementary Note 3.4.2, these patterns suggest that mobility is associated not only with broader citation-side visibility, but also with broader geographic diffusion. More specifically, in the Mob versus non-Mob comparison, Mob scientists exhibit both greater citation geographic distance and greater citation geographic dispersion, indicating that their later-career work is cited not only from more distant locations but also across a more spatially dispersed audience. Within the shorter–longer comparison, by contrast, the two citation-side geographic dimensions diverge: although citation geographic distance does not differ significantly between shorter- and longer-distance movers, citation geographic dispersion is significantly greater for longer-distance movers. Shorter-distance movers appear to retain greater visibility in core and high-status scientific environments, consistent with their stronger embeddedness in elite collaborative networks, stronger orientation toward top-institution and western-country citation audiences, and greater citation geographic distance, as documented in Supplementary Note 3.4.2. Longer-distance movers, by contrast, appear less anchored to these highly visible core audiences and search farther, yet their later-career work diffuses across a more geographically dispersed set of citing locations.

We next continue to examine whether this citation-side diffusion pattern is also reflected in the field composition of citing papers. In other words, beyond geographic diffusion, does scientists' later-career work also diffuse differently across fields? To address this question, and to parallel the multi-scale knowledge-strategy patterns documented on the reference side in Supplementary Note 3.3, we examine citation-side field diffusion at both the paper and career levels. At the paper level, the two mobility comparisons again reveal distinct but fully corresponding patterns relative to Supplementary Note 3.3. In the Mob versus non-Mob comparison, where the reference-side difference was primarily reflected in a stronger local concentration pattern

measured by reference fields imbalance, we observe the same direction on the citation side. Specifically, Mob scientists exhibit a significantly higher citation fields imbalance than non-Mob scientists (mean: Mob = 0.6503 vs. non-Mob = 0.6323; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 2.84% increase. This result suggests that the later-career work of Mob scientists is cited in a more field-concentrated manner at the paper level, consistent with the stronger local exploitation pattern documented on the reference side. Within the mobile sample, by contrast, the paper-level difference follows the same home-versus-global field logic as in Supplementary Note 3.3. Longer-distance movers exhibit a significantly higher citation global fields relative share than shorter-distance movers (mean: Longer = 0.2669 vs. Shorter = 0.2605; one-sided Welch's t-test,  $P < 0.01$ ), corresponding to a 2.47% increase. This pattern indicates that the later-career work of longer-distance movers diffuses more strongly beyond the focal paper's home fields, consistent with their more exploration-oriented paper-level knowledge strategy on the reference side. Thus, at the paper level, citation-side field diffusion closely echoes the reference-side knowledge-strategy patterns: Mob scientists exhibit stronger field concentration than non-Mob scientists, whereas longer-distance movers exhibit broader cross-field diffusion than shorter-distance movers.

At the career level, we then examine whether the same logic extends from individual papers to scientists' later-career trajectories as a whole. To do so, we compute career citation fields propensity diversity to capture how broadly and evenly a scientist's later-career work is cited across different fields over the career. Here again, the results largely parallel the career-level knowledge-strategy patterns documented in Supplementary Note 3.3. In the Mob versus non-Mob comparison, Mob scientists exhibit a significantly higher career citation fields propensity diversity than non-Mob scientists (mean: Mob = 0.8824 vs. non-Mob = 0.8696; one-sided Welch's t-test,  $P < 0.001$ ), corresponding to a 1.48% increase. Within the mobile sample, shorter-distance movers exhibit a directionally higher career citation fields propensity diversity than longer-distance movers, although the difference is not statistically significant (mean: Shorter = 0.8852

vs. Longer = 0.8849; one-sided Welch's t-test,  $P = 0.84$ ), corresponding to a 0.03% increase. This direction nevertheless remains consistent with the career-level pattern documented on the reference side. These results indicate that, although Mob scientists are more concentrated at the paper level and shorter-distance movers are more exploitation-oriented at the paper level, the later-career work of these two groups is nevertheless cited more broadly and evenly across fields over the career as a whole. In this sense, the citation side reproduces the same local–global configuration documented on the reference side: Mob scientists and shorter-distance movers are more local or concentrated at the paper level, yet more diverse and exploratory at the career level.

Overall, the citation-side diffusion results are fully consistent with the multi-scale knowledge-strategy patterns documented above. In the Mob versus non-Mob comparison, mobility is associated with both broader geographic diffusion and a local–global field configuration in which citation patterns are more concentrated at the paper level but more diverse at the career level. Within the mobile sample, the shorter–longer comparison displays the same layered structure: longer-distance movers exhibit broader geographic diffusion and greater paper-level cross-field diffusion, whereas shorter-distance movers exhibit directionally broader career-level field diffusion. These findings suggest that early-career cultural mobility shapes not only where scientists draw knowledge from and who cites them, but also how their later-career work diffuses across geographic and field space. More broadly, they reinforce the interpretation that different magnitudes of cultural mobility are associated with distinct downstream diffusion profiles: some are more strongly aligned with broader geographic spread, whereas others are more strongly aligned with broader long-run diffusion across fields.

#### **Supplementary Note 4.2: Tail innovation**

In Supplementary Note 3.3, we showed that novelty exhibits a clear center–tail separation across mobility groups. Specifically, although Mob scientists and longer-distance movers display stronger conventional grounding at the center of the recombination distribution, non-Mob scientists and shorter-distance movers exhibit a

higher incidence of tail novelty. Here, we ask whether an analogous center–tail separation also appears in disruptiveness. Our main analyses above have already shown that Mob scientists and longer-distance movers achieve higher mean D5 percentile. The key question in this section is therefore whether this higher average disruptiveness also extends to the tail of disruptive outcomes. To address this, we move beyond mean D5 percentile and focus primarily on two-tailed 10% extreme disruptiveness, defined as whether a scientist’s author-level D5 value falls in either the top 5% or the bottom 5% of the full distribution. We find that the answer is no. In the Mob versus non-Mob comparison, Mob scientists exhibit a significantly lower probability of two-tailed 10% extreme disruptiveness than non-Mob scientists (mean: Mob = 0.0825 vs. non-Mob = 0.1175; one-sided Welch’s t-test,  $P < 0.001$ ), corresponding to a 29.78% decrease. Within the mobile sample, longer-distance movers likewise exhibit a significantly lower probability of two-tailed 10% extreme disruptiveness than shorter-distance movers (mean: Longer = 0.0783 vs. Shorter = 0.0917; one-sided Welch’s t-test,  $P < 0.001$ ), corresponding to a 14.65% decrease. The same pattern remains at stricter thresholds, and the deficit widens monotonically as the threshold tightens (two-tailed 5%: Mob = 0.0379 vs. non-Mob = 0.0622, a 39.16% decrease; Longer = 0.0348 vs. Shorter = 0.0424, a 17.87% decrease; two-tailed 2%: Mob = 0.0132 vs. non-Mob = 0.0269, a 50.98% decrease; Longer = 0.0113 vs. Shorter = 0.0158, a 28.53% decrease; all one-sided Welch’s t-tests,  $P < 0.001$ ). Thus, as in the novelty results above, disruptiveness also exhibits a clear center–tail separation across mobility groups: scientists with greater early-career cultural mobility achieve higher average disruptiveness, but not a higher incidence of two-tailed extreme disruptive outcomes. The same pattern holds at the upper tail alone. In the Mob versus non-Mob comparison, Mob scientists exhibit a lower upper-tail 5% probability than non-Mob scientists (mean: Mob = 0.0437 vs. non-Mob = 0.0564; one-sided Welch’s t-test,  $P < 0.001$ ), a 22.50% decrease, and within the mobile sample longer-distance movers likewise exhibit a lower probability than shorter-distance movers (mean: Longer = 0.0434 vs. Shorter = 0.0483; one-sided Welch’s t-test,  $P < 0.05$ ), a 10.11% decrease. The deficit again widens at

stricter cutoffs (upper 2.5%: Mob = 0.0200 vs. non-Mob = 0.0301, a 33.68% decrease; Longer = 0.0180 vs. Shorter = 0.0237, a 24.10% decrease; upper 1%: Mob = 0.0073 vs. non-Mob = 0.0127, a 42.23% decrease; Longer = 0.0062 vs. Shorter = 0.0094, a 34.02% decrease; all  $P < 0.001$ ).

These differences are also estimated in linear probability models with heteroskedasticity-robust standard errors under four control sets: the five early-career matching covariates alone; those covariates with later-career productivity added, which we include because averaging over more papers compresses the dispersion of career-level disruptiveness; those covariates with the controls applied to the main outcomes in Supplementary Note 3.1.7 added (later-career reference count, field and year fixed effects); and all of them entered together. In the Mob versus non-Mob comparison the coefficient is negative and significant under every control set for two-tailed 10% extreme disruptiveness ( $-0.0338$ ,  $-0.0296$ ,  $-0.0228$  and  $-0.0203$ ), for the upper 5% ( $-0.0121$ ,  $-0.0103$ ,  $-0.0081$  and  $-0.0069$ ) and for the upper 1% ( $-0.0049$ ,  $-0.0040$ ,  $-0.0032$  and  $-0.0026$ ; all  $P < 0.001$ ). Within the mobile sample the same holds for two-tailed 10% extreme disruptiveness ( $-0.0123$ ,  $-0.0110$ ,  $-0.0112$  and  $-0.0114$ ;  $P = 0.007$ ,  $0.016$ ,  $0.012$  and  $0.011$ ) and for the upper 1% ( $-0.0037$ ,  $-0.0035$ ,  $-0.0033$  and  $-0.0033$ ;  $P = 0.007$ ,  $0.011$ ,  $0.018$  and  $0.017$ ). For the upper 5% the contrast is not significant under the early-career covariates alone ( $-0.0030$ ,  $P = 0.38$ ) or with later-career productivity added ( $-0.0024$ ,  $P = 0.49$ ), and becomes significant once field, year and reference count are included, whether alone ( $-0.0080$ ,  $P = 0.017$ ) or together with later-career productivity ( $-0.0081$ ,  $P = 0.016$ ). The upper 5% is the least restrictive of the upper-tail thresholds, and since the deficit widens monotonically as the threshold tightens, it is where the contrast between shorter- and longer-distance movers is weakest; at the upper 1% it is significant under every specification.

We first consider the most direct paper-level explanation, namely whether local exploration and exploitation in the reference space can account for the tail-disruptiveness pattern. As discussed in Supplementary Note 3.3.1, paper-level

exploration is captured by reference global fields relative share, which reflects the extent to which a focal paper draws on knowledge from non-home fields. However, the evidence indicates that reference global fields relative share does not provide a clear or coherent explanation for two-tailed tail disruptiveness across the two mobility comparisons. In both comparisons, its association with two-tailed 10% extreme disruptiveness is weak and directionally inconsistent across correlation measures (Mob vs non-Mob: Pearson  $r = 0.013$ ; Spearman  $r = -0.008$ ; shorter vs longer: Pearson  $r = 0.008$ ; Spearman  $r = -0.007$ ). We also note that alternative paper-level local exploitation indicators do not yield a more coherent picture. For example, reference fields imbalance is negatively associated with two-tailed tail disruptiveness, but it cannot generate a consistent interpretation across the two mobility comparisons, because the direction of local exploitation differs between them: in the Mob versus non-Mob comparison, mobile scientists are more locally exploitative, whereas in the shorter-versus-longer comparison, shorter-distance movers are more locally exploitative. Thus, although paper-level field strategies may still relate to average disruptiveness in some specifications, they do not provide a stable and unified account of the observed center–tail separation.

We next turn to the team level and examine whether team expertise distance, as our main field-based indicator of team-level knowledge heterogeneity, can account for the same pattern. Here the evidence is more consistent. Specifically, later-career team expertise distance is positively associated with mean D5 percentile (Mob vs non-Mob: Pearson  $r = 0.159$ ,  $p < 0.001$ ; Spearman  $r = 0.155$ ,  $p < 0.001$ ; shorter vs longer: Pearson  $r = 0.151$ ,  $p < 0.001$ ; Spearman  $r = 0.148$ ,  $p < 0.001$ ), but negatively associated with two-tailed 10% extreme disruptiveness (Mob vs non-Mob: Pearson  $r = -0.071$ ; Spearman  $r = -0.071$ ; shorter vs longer: Pearson  $r = -0.056$ ; Spearman  $r = -0.054$ ; all  $P < 0.001$ ). Thus, greater team-level knowledge heterogeneity is associated with higher average disruptiveness but a lower incidence of two-tailed extreme disruptive outcomes. Because Mob scientists and longer-distance movers also exhibit greater team expertise



distance, this pattern is consistent with the observed center–tail separation across mobility groups.

Therefore, the field-based account is informative but not fully coherent as a principal explanation. At the team level, greater expertise distance is consistently associated with higher mean disruptiveness and lower tail disruptiveness, in line with the group-level center–tail separation. However, the most direct paper-level indicators of local exploration and exploitation do not provide equally coherent evidence across the two mobility comparisons. We therefore do not treat the field-based account as the principal explanation for the tail-disruptiveness pattern. This raises a further question. If local field exploration cannot coherently explain the center–tail separation in disruptiveness, could the explanation instead lie in the geographic changes in collaboration and search induced by mobility? In Supplementary Note 3.3, we found that the center–tail separation in novelty could be more coherently understood through geography-related mechanisms. We therefore next examine whether the same may also hold for disruptiveness: namely, whether mobility-induced differences in geographic collaboration span and geographic search range can help explain why mobility groups achieve higher average disruptiveness without a parallel increase in tail disruptiveness.

We first examine reference geographic distance, which captures the geographic radius of knowledge search in the reference space. Larger reference geographic distance indicates that focal papers draw upon knowledge originating from geographically more distant sources. As shown above in Supplementary Note 3.4.2, Mob scientists and longer-distance movers exhibit broader geographic search in their later-career reference portfolios. Here, we further show that reference geographic distance is positively associated with mean D5 percentile (Mob vs non-Mob: Pearson  $r = 0.065$ ,  $p < 0.001$ ; Spearman  $r = 0.079$ ,  $p < 0.001$ ; shorter vs longer: Pearson  $r = 0.078$ ,  $p < 0.001$ ; Spearman  $r = 0.093$ ,  $p < 0.001$ ), but negatively associated with two-tailed 10% extreme disruptiveness (Mob vs non-Mob: Pearson  $r = -0.087$ ; Spearman  $r = -0.098$ ; shorter vs longer: Pearson  $r = -0.060$ ; Spearman  $r = -0.070$ ; all  $P < 0.001$ ). These results suggest

that broader geographic search is associated with higher average disruptiveness but a lower incidence of two-tailed extreme disruptive outcomes. Because Mob scientists and longer-distance movers also exhibit greater reference geographic distance, this pattern provides a coherent source-side explanation for the observed center–tail separation. Conceptually, broader search can improve innovative performance and boundary-spanning exploration can generate more influential outcomes, yet the broader search literature also shows diminishing returns and curvilinear patterns as search expands, rather than a monotonic rise in extreme outcomes<sup>56</sup>. In parallel, economic-geography work emphasizes that geographically distant knowledge exchange is harder to integrate because tacit knowledge travels less easily and coordination over distance becomes more demanding<sup>53,60</sup>.

We next examine team geographic distance, which captures the geographic span of later-career collaboration teams. Here a complementary possibility arises. As in our discussion in Supplementary Note 3.3, broader geographic search may enlarge the opportunity set of knowledge combinations, but wider team geographic span may also increase scientific communication costs and reduce the efficiency of tacit knowledge transmission within the team. If so, then mobility-induced geographic expansion may simultaneously promote average disruptiveness while constraining the emergence of extreme disruptive outcomes. Consistent with this interpretation, team geographic distance is positively associated with mean D5 percentile in both comparisons (Mob vs non-Mob: Pearson  $r = 0.048$ ,  $p < 0.001$ ; Spearman  $r = 0.043$ ,  $p < 0.001$ ; shorter vs longer: Pearson  $r = 0.035$ ,  $p < 0.001$ ; Spearman  $r = 0.034$ ,  $p < 0.001$ ), but negatively associated with two-tailed 10% extreme disruptiveness (Mob vs non-Mob: Pearson  $r = -0.056$ ; Spearman  $r = -0.095$ ; shorter vs longer: Pearson  $r = -0.035$ ; Spearman  $r = -0.058$ ; all  $P < 0.001$ ). Thus, wider team geographic span is likewise associated with higher average disruptiveness but lower tail disruptiveness. Taken together, reference geographic distance and team geographic distance provide a set of geography-related variables that consistently align with the center–tail separation in both disruptiveness

and atypical novelty across the two comparisons.

Having identified the geographic expansion associated with mobility, we next probe the mechanisms through which it may reorganize the disruptiveness distribution. Geographic distance increases the frictions of communication and coordination, and thereby weakens the transmission and integration of tacit knowledge, which underpins many radical breakthroughs yet travels mainly through frequent interaction and face-to-face contact. Wider geographic search and collaboration may therefore enlarge the accessible knowledge space while making it harder to access and absorb tacit knowledge from which the most extreme disruptive outcomes arise. Tacit knowledge, however, is not directly observable. What we observe at the reference level is not tacit knowledge itself, but an outcome that may depend on it: the ability to combine prior work that has not previously shared a common knowledge base. We therefore next examine reference independence<sup>61</sup> as a paper-level proxy for the extent to which cited knowledge is combined in a structurally independent manner. For each focal paper, a reference is classified as ind-type if it cites no other work in the same reference list and as dep-type otherwise, and reference independence is measured as the difference between the numbers of ind-type and dep-type references divided by their sum, ranging from  $-1$  to  $+1$ . A high value indicates that a paper draws together streams of prior work that had not previously entered into dialogue. A combination of prior work that has never been joined before may reflect, since connections already available in the codified record are seldom left unmade for long, the hard-to-codify tacit knowledge that lies behind it.

In the Mob versus non-Mob comparison, non-Mob scientists exhibit higher reference independence than Mob scientists (mean: non-Mob = 0.1084 vs. Mob = 0.0599; one-sided Welch's t-test,  $P < 0.001$ ), whereas within the mobile sample, shorter-distance movers exhibit higher reference independence than longer-distance movers (mean: Shorter = 0.0689 vs. Longer = 0.0636, a 7.73% difference; one-sided Welch's t-test,  $P < 0.05$ ). If wider geographic search and collaboration increases communication

frictions and weakens tacit knowledge transfer, then broader search and geographically more dispersed teams may become less able to assemble sufficiently independent combinations of prior work. Consistent with this possibility, both geographic measures are negatively associated with reference independence: team geographic distance (Mob vs non-Mob: Pearson  $r = -0.164$ ; Spearman  $r = -0.222$ ; shorter vs longer: Pearson  $r = -0.126$ ; Spearman  $r = -0.164$ ) and reference geographic distance (Pearson  $r = -0.101$ ; Spearman  $r = -0.137$ ; Pearson  $r = -0.035$ ; Spearman  $r = -0.068$ ; all  $P < 0.001$ ), indicating that geographically more dispersed teams and searches tend to assemble less independent reference combinations (Fig. S19f, n).

At the same time, reference independence is positively associated with mean D5 percentile (Mob vs non-Mob: Pearson  $r = 0.266$ ; Spearman  $r = 0.249$ ; shorter vs longer: Pearson  $r = 0.292$ ; Spearman  $r = 0.282$ ), with two-tailed 10% extreme disruptiveness (Pearson  $r = 0.290$ ; Spearman  $r = 0.261$ ; Pearson  $r = 0.273$ ; Spearman  $r = 0.240$ ) and with upper-tail 5% disruptiveness (Pearson  $r = 0.305$ ; Spearman  $r = 0.261$ ; Pearson  $r = 0.293$ ; Spearman  $r = 0.248$ ; all  $P < 0.001$ ). The mean-level association nevertheless cannot explain the group-level difference in average disruptiveness, because the group ordering runs in the opposite direction: non-Mob scientists and shorter-distance movers exhibit higher reference independence, even though Mob scientists and longer-distance movers exhibit higher average disruptiveness. Further, a one-standard-deviation increase in reference independence raises mean D5 percentile by 15.4% of its baseline, but the probability of two-tailed 10% extreme disruptiveness by 87.2% and of upper-tail 5% disruptiveness by 132.9%; within the mobile sample the corresponding figures are 16.1%, 91.0% and 137.2% (Fig. S19h, p). On this measure the tail response is roughly an order of magnitude larger than the mean response, in both comparisons. Thus, the explanatory value of reference independence lies mainly in the tail, especially the upper tail, rather than in the mean. Although broader geographic search may expand the accessible opportunity set of knowledge, the most outstanding disruptive breakthroughs appear to depend more strongly on whether scientists are able to

assemble sufficiently independent reference combinations.

Another possible pathway operates through the higher reference popularity which the Mob and Longer scientists carry. Although standing on the shoulders of giants may increase average disruptiveness, it may also limit the most radical ideas, which sit in the tail of the disruptiveness distribution, where, straying furthest from convention, they can produce either the least or the most disruptive work of all. Consistent with the expected pattern, reference popularity is positively associated with mean D5 percentile (Mob vs non-Mob: Pearson  $r = 0.118$ ; Spearman  $r = 0.167$ ; shorter vs longer: Pearson  $r = 0.120$ ; Spearman  $r = 0.170$ ) but negatively associated with two-tailed 10% extreme disruptiveness (Pearson  $r = -0.145$ ; Spearman  $r = -0.266$ ; Pearson  $r = -0.123$ ; Spearman  $r = -0.223$ ) and with upper-tail 5% disruptiveness (Pearson  $r = -0.076$ ; Spearman  $r = -0.133$ ; Pearson  $r = -0.067$ ; Spearman  $r = -0.108$ ; all  $P < 0.001$ ; Fig. S19c, k). This reliance may itself be partly a consequence of geographic expansion: highly visible, codified and widely legible work is especially useful as a common foundation for dispersed teams, and both team geographic distance (Mob vs non-Mob: Pearson  $r = 0.098$ ; Spearman  $r = 0.216$ ; shorter vs longer: Pearson  $r = 0.068$ ; Spearman  $r = 0.147$ ) and reference geographic distance (Pearson  $r = 0.083$ ; Spearman  $r = 0.174$ ; Pearson  $r = 0.053$ ; Spearman  $r = 0.115$ ; all  $P < 0.001$ ) are positively associated with reference popularity. Although the most radical ideas also resist direct measurement, a combination of prior work that has never been joined before may function as a proxy for the radical ideas. We further find that reference popularity leaves the same trace on reference independence as the geographic measures do, and a stronger one (Mob vs non-Mob: Pearson  $r = -0.314$ ; Spearman  $r = -0.518$ ; shorter vs longer: Pearson  $r = -0.308$ ; Spearman  $r = -0.482$ ; all  $P < 0.001$ ; Fig. S19f, n). To quantify how much of the group-level tail gap the two pathways account for, we decompose it in linear probability models, entering each separately and then jointly. In the Mob comparison, reference popularity together with the two geographic measures accounts for 37.2% of the two-tailed 10% gap and reference independence alone for 47.8%, while entering both

accounts for 61.0%; for the upper-tail 5% gap the corresponding shares are 35.2%, 101.5% and 91.6%, with the group coefficient reduced to insignificance ( $P = 0.22$ ). Within the mobile sample the two sets account for 44.5% and 11.4% of the two-tailed gap and 61.9% and 25.2% of the upper-tail gap, and entering both adds nothing beyond the first (43.9% and 60.4%), indicating that in this subsample the two pathways are largely collinear rather than that reference independence is uninformative there. All models are linear probability models with heteroskedasticity-robust standard errors.

Overall, these results suggest that the center–tail disruptiveness pattern may be coherently explained by a geographic expansion that carries two constraints. Broader geographic search and wider collaborative span enlarge the accessible knowledge space and are associated with higher average disruptiveness, matching the group-level center–tail separation observed for Mob scientists and longer-distance movers, and this pattern parallels the center–tail separation in novelty discussed above. The same expansion, however, increases the frictions that impede the transmission and integration of tacit knowledge which underpins tail disruptiveness. It also shifts dispersed teams toward more visible and established prior work, which sustains the center while bounding how far from convention a work can travel. Both constraints leave the same downstream trace, a reduced capacity to combine prior work that had not shared previously common knowledge base, and reference independence registers it: it declines with both geographic measures and, more strongly, with reliance on popular work; its association with disruptiveness is roughly an order of magnitude stronger at the tail than at the mean, so that it accounts for the tail rather than the center. In this sense, early-career cultural mobility appears not simply to raise the entire disruptiveness distribution, but rather to reorganize its center and tail through later-career differences in geographic search, collaboration and knowledge integration.

In both mobility comparisons, scientists with greater early-career cultural mobility appear to benefit from broader geographic search and wider collaborative span, which can lift average disruptiveness by expanding the reachable knowledge space, but these

same conditions may also increase communication frictions and reduce the capacity to integrate highly independent knowledge combinations, thereby limiting the emergence of the most extreme disruptive outcomes. This interpretation may also help contextualize why shorter-distance movers, despite lower average disruptiveness than longer-distance movers, can still be more strongly connected to the most exceptional scientific recognition, namely Nobel Prizes, documented above. One possible reason is that the upper tail of disruptiveness appears more closely linked to the capacity to assemble sufficiently independent and radical combinations, whereas prize outcomes likely also depend on additional mechanisms such as visibility, audience reception, and cumulative recognition. Given the rarity of such prizes, we do not pursue that connection further here.

#### **Supplementary Note 4.3: The changing structure of early-career mobility and the decline and thinning of disruptive science**

Removing barriers to where people may work raises aggregate output by reducing the misallocation of talent <sup>62,63</sup>. The claim originates as one about individuals, that anyone may enter any occupation for which they are suited, and international mobility extends it to systems. If scientists can go where their work is best supported and most productively combined, science as a whole should gain <sup>64-68</sup>. The past half century has been an extended test of that proposition. Scientists have become steadily more mobile across the earlier cohorts (1956-1996) in our main sample (Fig. S0b). The share affiliated with two or more countries during career years 1–5 also rose from 5.97% among the 1997 entry cohort to 8.98% among the 2016 cohort, a 50.4% increase (Fig. 6). Science appears to have gained accordingly. Mobile scientists are more productive and more highly cited than those who stay <sup>68</sup>, immigrant researchers contribute disproportionately to invention <sup>69,70</sup>, and the émigrés who fled Nazi Germany raised US patenting in their fields by roughly a third <sup>71</sup>. Our own results agree. Matched against comparable non-mobile peers, mobile scientists publish 25.32% more and produce 7.18% more high-impact work.

Over the same decades, mobility has not only grown but has also changed shape. Scholarly migration has globalized in reach as well as expanded in volume, with origin and destination countries becoming steadily more diversified over the past two decades, with international flows shifting from a core–periphery toward a multipolar geography, and with world migration broadening in span even where its volume has not risen <sup>72–77</sup>. Our data show the same within scientific careers. Using cultural-distance terciles fixed to the 1956–1996 mover distribution, the shift toward longer-distance mobility was already visible between 1956–1976 and 1977–1996 (short: 47.54% to 31.74%; long: 19.09% to 34.93%; Fig. 6b), and continued among the 1997–2016 cohorts, where the share crossing the largest cultural distances rose from 37.13% to 54.56%, while the share crossing the smallest fell from 31.72% to 19.86% (Fig. 6a). What science asks of those who move is therefore no longer what it asked a generation ago, and an account that records only whether a border was crossed cannot register the change. Recording how far scientists move reveals that the dividend does not fall evenly but diverges. Longer-distance movers gain in productivity, entering unfamiliar environments where they lean on larger collaborator pools and on the efficient coordination that repeated ties afford. Shorter-distance movers gain in impact, adapting more readily to their new settings, drawing in more new collaborators, entering elite networks and assembling teams of higher reputation. Where scientists go shapes not only how much they gain but which kind of gain they receive.

Gains of this kind follow from talent finding its use, and the existing literature has focused primarily on them, leaving it less clear whether and why mobility also promotes innovation itself. A second body of theory holds that movement matters for science not only because it places people where their talent is least misallocated, but because it moves ideas. Knowledge and innovation advance through recombination, in that new ideas are formed by joining existing ones in configurations not previously tried <sup>78,79</sup>, and much of what scientists know is tacit or difficult to codify, and is therefore transmitted more readily through the movement and interaction of people than through



the written record <sup>80</sup>. Movement is therefore among the principal means by which the raw material of recombination reaches places it would not otherwise reach, and by which scientists themselves search beyond the environments in which their own search would otherwise remain confined <sup>57,58</sup>. The historical record bears this out. Economic historians attribute the intellectual transformation of early modern Europe to a transnational community across which scholars and their writings circulated freely, and in which heterodox thinkers could carry their thinking across borders <sup>81</sup>; the Huguenots expelled from France in 1685 raised the long-run productivity of the Prussian manufactories that received them <sup>82</sup>; and the record of intellectual migration over two millennia shows notable individuals disproportionately converging on a small number of cultural centres <sup>37</sup>, where intellectual and cultural production was itself highly concentrated. Individual careers show the same pattern, in science as in art, from Ernst and Shamir to van Gogh and Picasso, where the moves that preceded a decisive turn were inseparable from a pivot in the material being combined (Fig. 1d).

But not every move brings the same material into contact. Two scientists may both cross a border and yet enter research cultures that differ from their own to very different degrees, and cultural-evolutionary theory holds that this difference is consequential. On that account a population's rate of innovation depends not only on its size and how interconnected it is, but on the cultural variance it contains <sup>83</sup>. Mobility supplies interconnection; the variance it adds depends on what it connects, since linking two similar populations or research cultures enlarges the pool of variants far less than linking two dissimilar ones. The prediction that follows is a gradient rather than a threshold, in which the innovative returns to moving plausibly rise with cultural distance and a workforce whose mobility has grown more diverse may become more innovative for that reason alone. Existing evidence is consistent with it. Rising cultural diversity raised innovation in US counties between 1850 and 1940, operating through the range of ideas, skills and perspectives available for recombination <sup>84</sup>. Similar patterns appear across other domains of creative work, where travelling farther

broadens the range of culture that listeners explore <sup>85</sup> and the cultural distance of a designer's foreign experience shapes the originality of what they go on to produce <sup>86</sup>. Here too our results agree. Mobile scientists exhibit 4.06% higher disruptiveness than matched non-mobile peers, and among mobile scientists disruptiveness rises with cultural distance, with longer-distance movers 9.79% above shorter-distance movers.

Over the same decades, however, papers and patents have become less disruptive, and an inventory of 105 studies finds broadly consistent evidence across scientific articles, patents, products, legal precedent and the arts, recovered by citation-based, text-based, firm-displacement and embedding-based measures alike <sup>50,87</sup>. A parallel weakening appears in the wider economy, where creative destruction has also slowed, with new firms and technologies displacing incumbents at lower rates than before <sup>42,88,89</sup>. Moreover, frontier advances have grown scarcer and harder to produce. The number of highly disruptive contributions has not grown with total output, so that their share among all work has contracted <sup>50</sup>, and sustaining a constant rate of advance now requires far more researchers than it once did <sup>90</sup>. Ideas, in this sense, have become harder to find, and what it costs to produce a breakthrough has risen.

The two records are difficult to reconcile, since science has grown more mobile and more culturally distant in its mobility, and both changes should, on the evidence above, have pushed toward more innovative work. Yet disruptiveness has declined and the frontier has thinned over the same period. This raises the possibility that the changing composition of mobility has altered aggregate patterns of innovation, and our own evidence supplies a candidate mechanism, particularly for the thinning. Although mobile scientists and longer-distance movers achieve higher average disruptiveness, they reach the extreme of the disruptiveness distribution less often than their peers, by 29.78% and 14.65% respectively (Supplementary Note 4.2). If the population has shifted toward exactly those groups, the aggregate consequence for the top could be adverse even as the average rises. We therefore ask what the structural shift has done to the average level of disruptive science, and what it has done to its extreme.

To isolate what these shifts in scientific mobility alone imply, we hold each group's mean outcome fixed at its observed value and vary only the weights with which the four groups enter the matched sample. The matched sample pairs early movers with marginal non-movers, that is, scientists who did not move early but whose observable early-career characteristics make them close counterfactuals to those who did. Within it two weights are free to vary, the mover share of the sample and the distance composition among movers, and everything else is held at its observed value. The exercise therefore asks what the mean long-term disruptiveness of this same population would have been had it been weighted as in another era, and not what would have happened had different scientists behaved differently.

Three of the four resulting environments are observed directly. The environment of 1956–1976, in which mobility was least prevalent and shorter-distance moves most heavily represented, lies 0.91% below the 1956–1996 baseline; that of 1977–1996 lies 0.12% above it (Fig. 6b). Moving from the first of these to the second, with the forty-year average as the reference point, corresponds to an increase of 1.03%. The observed record therefore already traces a monotone ascent of just over one percentage point without any calibration, and a less mobile and less distant scientific workforce corresponded to a lower mean long-term disruptiveness.

Extending the same construction to 1997–2016 places that environment 1.29% above baseline. In separate one-channel counterfactuals the change in composition among movers produces a 0.67% increase and the rising mover share a 0.47% increase. These do not sum to the combined figure, and the residual of 0.15% is the interaction between them. When only the mover share rises, the scientists newly counted as mobile are assumed to distribute across distances as movers did at baseline, and so take the baseline average mover outcome. When both weights change together, those same scientists distribute according to the new composition, which contains a substantially larger share of longer-distance moves and a correspondingly higher mean. The interaction is the product of the two changes and is positive because both move in the same direction.

Only this last environment involves calibration, since its matched-sample mover share is obtained by two-point logit anchoring of the matched-sample mover share on the population five-year mobility rate, fitted on the two historical periods (details in Fig. 6b). Its direction continues an ascent the observed points have already established, and the observed portion of the series spans 1.03 of the 2.20 percentage points across which the four environments range.

At baseline weights a 1% increase in the number of early movers raises the mean D5 percentile by 0.0198% (percentile-bootstrap 95% confidence interval from 10,000 replications, 0.0161–0.0235%), while a 1% increase in the number of longer-distance movers, achieved through proportional reallocation from the other three groups, raises it by 0.0149% (0.0133–0.0165%). Because the two elasticities are defined relative to different baseline populations, they are not directly comparable. Placing them on a common footing requires holding the total sample fixed and reassigning the same scientists from marginal non-mobility. Raising the mover count by 1% in this way yields 0.0198% when the reassigned scientists take the observed distance composition of movers, and 0.0471% (0.0419–0.0521%) when they are assigned entirely to longer-distance mobility. The difference of 0.0272% (0.0236–0.0309%) is the return to distance as distinct from the return to movement, and on this comparison it exceeds the return to movement itself. Reassigned scientists are evaluated at the observed mean outcome of their destination group throughout. These magnitudes are proportional changes in the sample mean, computed within our matched sample on a disruption measure constructed under the most restrictive exclusion criteria available, and are therefore conservative and internally referenced.

The answer to the first question is unambiguous, and it holds whatever the aggregate trend is. The compositional transformation of scientific mobility has pushed the mean upward rather than downward, so it cannot be among the causes of a decline in average disruptiveness. Mobility is a load-bearing positive contributor to average disruptiveness rather than a drag on it, and it would have had to act substantially in the opposite

direction to have been implicated. Within this positive effect the two channels are of comparable weight, and once placed on a common reallocation scale the distance channel is somewhat the larger, so that how far scientists go matters at least as much as whether they go at all. The channels through which this operates are the ones documented above. Longer-distance movers assemble teams of greater expertise heterogeneity, search farther across geographic and temporal space, draw more on peripheral sources, and produce work that diffuses across a more geographically dispersed set of citing locations (Supplementary Notes 3.2–3.4, 4.1). A lift at the centre of the distribution is what broader circulation and recombination of knowledge would be expected to produce, arising from the growth of mobility and its lengthening together, and it is distinct from, and does not require, an extension of the extreme.

The second question is whether the same shift has thinned the tail. Against an absolute threshold fixed at the 95th percentile of the raw D5 measure computed once over the complete matched sample and applied unchanged to every cohort decade, the share of scientists clearing the bar fell from 11.89% to 4.07%, while their absolute number rose from 265 to 2,459 (Fig. 6c). This echoes, at the level of scientists rather than works and against an absolute rather than a relative criterion, the pattern previously reported for highly disruptive papers and patents, in which a thinning frontier advance coexists with an absolute count that does not fall. Because the threshold is fixed once on the raw measure and applied unchanged across decades, the comparison does not depend on the shape of the contemporaneous distribution and provides an independent line of evidence for that pattern. A counterfactual holding the four-group composition at its 1956–1966 values, and letting only within-group rates follow their observed history, tracks the actual series to within 0.14 percentage points in every decade, under 2% of the 7.82-percentage-point decline. An exact three-factor Shapley decomposition of the change in counts returns the same verdict. Relative to the 1956–1966 baseline, expansion of the scientific workforce contributes +4,665 scientists and falling within-group rates –2,454, while compositional change contributes at most 17 in any decade,

or 0.76% of the net change of +2,194 (Fig. 6d). The thinning of the top is a within-group phenomenon. Every mobility group has become less likely to clear the same absolute bar than its predecessors, and this holds irrespective of who moves or how far.

At the level of a career the tail cost is real, with mobile scientists 29.78% less likely than matched non-mobile peers to fall at the two-tailed extreme and 22.50% less likely at the upper tail alone, and longer-distance movers 14.65% and 10.11% less likely than shorter-distance movers on the same two measures. Its sources through a potential geographic channel are documented above: the geographic breadth of search and collaboration that raises the average also impedes the transmission of tacit knowledge and the assembly of the independent recombinations on which the most extreme work depends (Supplementary Note 4.2). The same duality is anticipated by the theory invoked earlier, in whose later development cultural trait diversity is described as fuel for recombination and simultaneously as a burden on communication, and thus on the transmission of what cannot easily be written down <sup>91</sup>. Our findings agree with that account. Mobile scientists, and especially longer-distance movers, crossed greater cultural distances early in their careers than their counterparts, while also reaching farther geographically in the collaborators they retained, the teams they assembled and the knowledge they drew on. Both literatures converge on what follows: greater geographic and cultural spans widen the pool of variants available for recombination, which lifts the average, but they also lower the fidelity with which knowledge is transmitted, most of all the part that resists codification, leaving both groups with fewer of the independent, first-time combinations that the extreme requires <sup>91</sup>.

At the level of the system, however, this cost is not what governs the tail. The contribution of any group difference to an aggregate change is the product of the difference itself and the distance the weights have actually travelled, and because the mover share and the distance composition shifted modestly against the growth of successive cohorts and the steep fall in within-group rates, a deficit of that magnitude resolves into at most seventeen scientists per decade. At the system level the growth in

the number clearing the threshold is driven by expansion, and the erosion of the share clearing it by the fall in within-group rates, which traces to forces that lie outside mobility altogether. The burden of knowledge required to reach the frontier has risen <sup>92</sup>, the most accessible discoveries in any field are made first, so that later work must reach further for less <sup>93</sup>, and the number of additional attempts required to surpass an existing maximum grows steeply with the number already made <sup>94</sup>, and the diffusion of frontier knowledge to those working behind it has slowed <sup>95</sup>. The asymmetry between the two scales is not a qualification of either result; it is the reason a system-level question about mobility cannot be answered by aggregating individual-level returns.

Taken together, the two answers describe an unusually favourable configuration. Greater and more distant mobility raises the centre of the disruptiveness distribution and widens the geographic and disciplinary range over which knowledge is sourced and diffused, without materially eroding the top. The gain is real, and the cost that was feared does not materialise at the scale of the system. Mobility offers a broad innovation dividend. What it does not offer is a breakthrough premium, and that limit is a ceiling rather than a price. What the structural shift has changed is where in the distribution science accumulates, not whether its extreme remains reachable.

Two levers follow for policy, and neither substitutes for the other. The first is to sustain mobility, and in particular not to penalise distance. Restricting mobility could not restore the top, because compositional change is not what depleted it, and a policy acting on group weights cannot reach a process operating inside groups. Restricting it in either direction, whether shorter or longer moves, would lower the average and knowledge diffusion without recovering the tail. It would, however, forgo a gain at the centre that is visible in the observed historical record, in which the least mobile and least distant environment we can construct sits nearly a percentage point below baseline, and it would forgo the wider search portfolios and broader diffusion that longer-distance movers disproportionately supply. Similar evidence on the outcomes of restriction points the same way, with liberalised short-term entry raising innovative activities <sup>96</sup>,

whereas tightened access to skilled visas disperses research activity in ways that raise coordination costs for innovation <sup>97</sup>. Restriction would be ineffective on one count and costly on the other. That asymmetry is what our evidence supports, and it requires no claim about where the aggregate trend is heading. The second follows from the decomposition itself. Because the growth in the absolute number of top-tier contributions is driven overwhelmingly by expansion, continued investment in the size and breadth of the scientific workforce, rather than the redistribution of scientists across mobility regimes, is the operative instrument on the tail. Mobility governs where the distribution sits; expansion governs how much of its extreme is realised. Expansion cannot prevent the share above the threshold from falling, since that follows from the decline in within-group rates, but it remains the one force that keeps the absolute number of frontier contributions growing even as the return to each additional scientist declines. Enlarging that pool further is therefore the operative response, for example by widening early exposure to invention, which prior work identifies as a large and unrealised source of future inventors <sup>98</sup>.

The costs of restriction would not fall evenly, because mobile scientists occupy a bridging position in the global knowledge system that non-mobile scientists do not. Most of them work outside the top-100 institutions in later career, yet from those positions they retain more elite collaborators and draw more heavily on frontier institutional knowledge than their non-mobile peers, and their work is subsequently taken up across a more geographically dispersed set of citing locations (Supplementary Notes 3.4.1, 4.1). They carry frontier knowledge outward, and longer-distance movers carry it farthest, diffusing more widely still and beyond the fields in which it was published. A contraction of mobility would therefore act first on follower systems and peripheral communities <sup>99</sup>. Restricting mobility as a whole would narrow their access to frontier knowledge, and restricting long-distance mobility in particular would narrow the geographic reach over which that knowledge travels <sup>100,101</sup>; both would reduce the recombination opportunities such access affords, slowing catch-up rather than merely



slowing circulation. The same diagnosis has been reached in the wider economy, where the weakening of knowledge diffusion from the frontier to those working behind it has been identified as a dominant force behind the decline in business dynamism documented above <sup>95</sup>. The bridge runs in the other direction as well. Located inside Western countries, mobile scientists rely less exclusively on Western sources than their non-mobile peers, and located outside them they rely more; longer-distance movers extend this further, holding fewer Western and elite collaborators, relying relatively more on peripheral sources, and reaching farther across geographic and temporal space for the work they cite (Supplementary Notes 3.4.1, 3.4.2). What a contraction would remove here is heterogeneity rather than access. The centres would continue to host leading scientists while the channels through which unfamiliar problems, methods and contexts reach them grew narrower, and attention at the frontier is already crowded enough that citations concentrate on established work, the canon ossifies, and new contributions become less able to displace what precedes them <sup>102</sup>. This connects our findings to a broader concern in discussions of declining research productivity, namely that the binding constraint may lie less in the creation of frontier knowledge than in the capacity of the wider system to absorb, diffuse and recombine it at scale <sup>95,102–104</sup>.

Our evidence identifies what has not caused the thinning of the top rather than what has, and that boundary should be read strictly. The within-group decline in the probability of clearing a fixed absolute bar is large, general and, on our decomposition, essentially independent of mobility; explaining it will require mechanisms operating inside groups rather than between them, and the candidates named above are not mutually exclusive. Three limitations qualify the analysis. The fixed threshold is defined on the raw D5 measure and inherits its known sensitivities, though defining it once on the raw measure and applying it unchanged across decades avoids era-specific re-ranking and ensures that changes in the share above the threshold are not mechanically induced by contemporaneous percentile normalisation. The direction and sign of the scenario results are observed, but the magnitude reported for 1997–2016 depends on a calibrated

mover share and should be read accordingly. And the scenario holds group means fixed by construction, so it describes the consequence of reweighting an observed population, not a counterfactual world in which the same scientists would have behaved differently had they moved.

#### **Supplementary Note 4.4: Robustness to potential channels using alternative embedding-based measures of early career cultural distance**

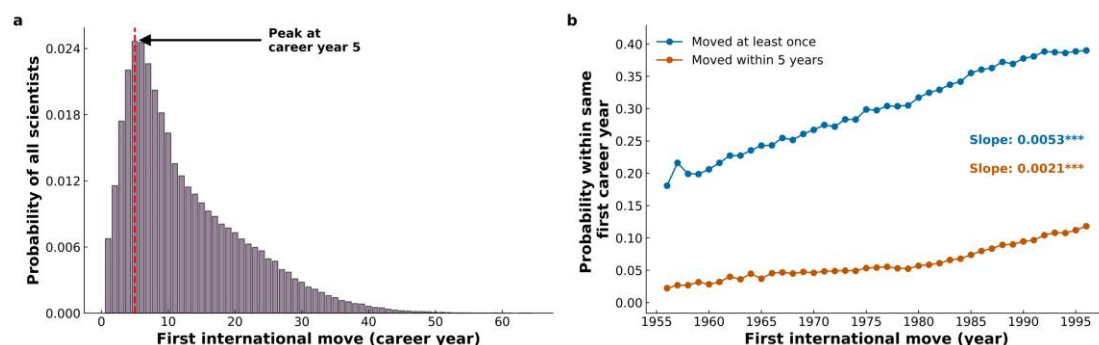
To assess whether the evidence on potential channels documented in Figs. 3–5 depends on the Hofstede-based operationalization of early-career cultural distance, we repeat the corresponding group comparisons using the five country-embedding measures introduced in Supplementary Note 2.2, validated in Supplementary Note 2.3, and applied to long-term performance outcomes in Supplementary Note 3.1.4. Because these measures cover all countries represented in our data, they additionally identify 66 scientists (0.07% of the matched sample) whose early-career moves are assigned zero distance under the Hofstede measure but non-zero distance under all five embedding-based measures. The five measures agree completely on these cases, indicating that the discrepancy primarily reflects differences in country coverage rather than instability across embedding methods.

We then reproduce the two group comparisons used in the main text and report the corresponding differences in group means for all 19 variables displayed in Figs. 3–5 (Table S29). For the Mob versus non-Mob comparison, each embedding-based distance is converted into a binary indicator, with positive distance denoting Mob and zero distance denoting non-Mob. Because all five embedding measures classify exactly the same scientists as mobile, they generate an identical Mob versus non-Mob split and therefore identical mean differences across the five specifications (Table S29, Panel a); this comparison accordingly tests whether the results are sensitive to the broader country coverage of the embedding measures rather than to the choice of embedding model. For the within-Mob comparison, following the main-text analysis, scientists are ranked by early-career cultural distance under each measure, with the bottom 33%

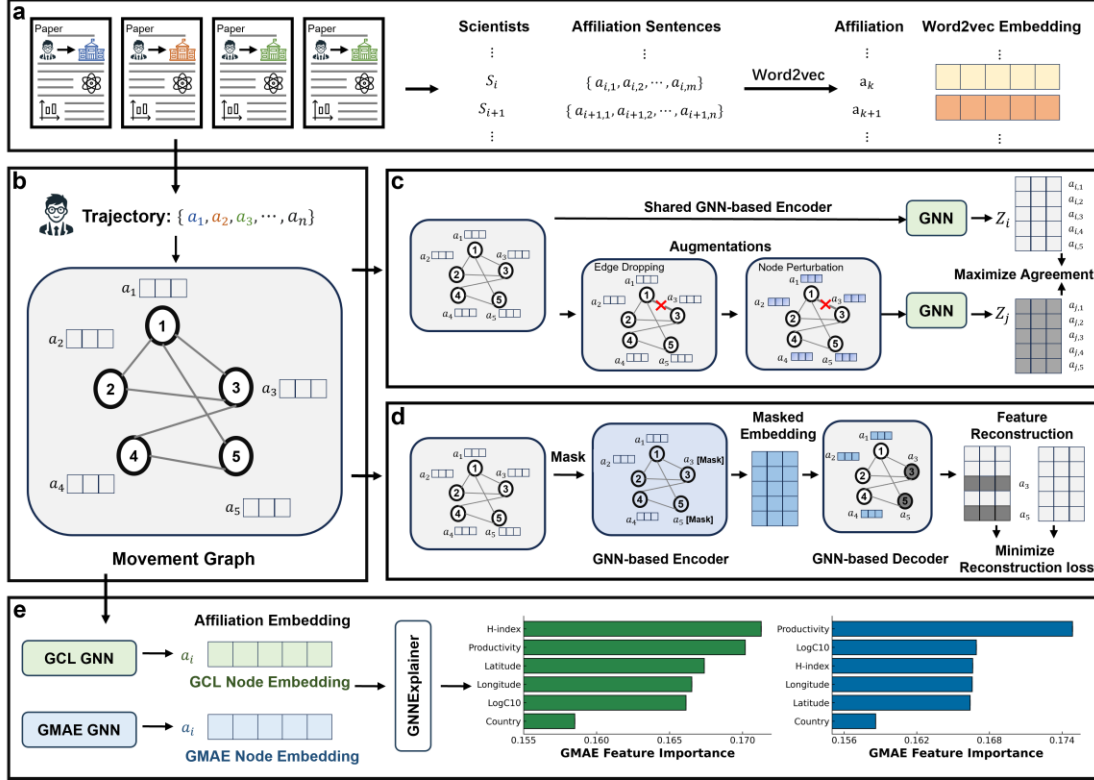
classified as shorter-distance movers and the top 33% as longer-distance movers (Table S29, Panel b). For each comparison, Table S29 reports the relative difference in group means and evaluates whether each alternative measure reproduces the direction reported in the main text at  $P < 0.05$  using one-sided Welch's t-tests.

Across the 19 variables and five alternative measures, 94 of 95 tests (99%) meet this criterion. All 10 variables in the Mob versus non-Mob comparison are reproduced under all five measures, indicating that including the 66 additional movers identified only by the embedding measures leaves these patterns unchanged. Within the shorter- versus longer-distance comparison, eight of the nine variables are reproduced by all five measures, whereas the reference self-citation ratio is reproduced by four of the five. Together, these results show that the evidence on the potential channels documented in Figs. 3–5 is not sensitive to the Hofstede-based operationalization of cultural distance and remains nearly unchanged when cultural distance is derived from country embeddings with broader geographic coverage.

## Supplementary Figures

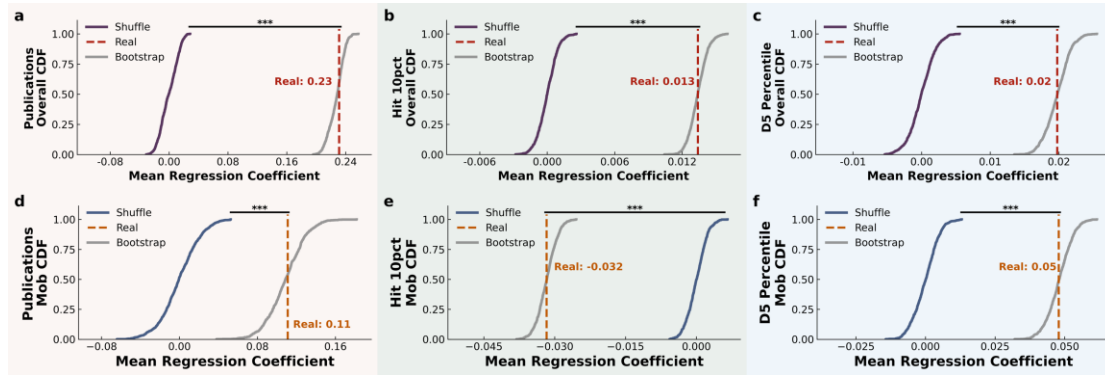


**Fig. S0 Patterns and trends in early-career international/cross-cultural mobility among scientists.** **a**, Probability density function of the timing of scientists' first international/cross-cultural move during their career, based on 203,522 scientists. The distribution peaks at career year 5 (red dashed line), indicating that the majority of first international moves occur within the first five years after career start. Based on this, we define the “early-career” window as the first 5 years; results using alternative windows are consistent (see Table S3). **b**, Trends in international mobility over time. For scientists whose first career year falls between 1956 and 1996, each data point represents the probability of having moved internationally at least once up to 2022 (blue), or within the first 5 years (orange), calculated among all scientists ( $n = 584,712$ ) with the same first career year. Both probabilities show a significant upward trend over time (slopes: 0.0053 and 0.0021 per year, respectively;  $P < 0.001$ , estimated by linear regression), reflecting an increasing prevalence of international mobility in the scientific workforce. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

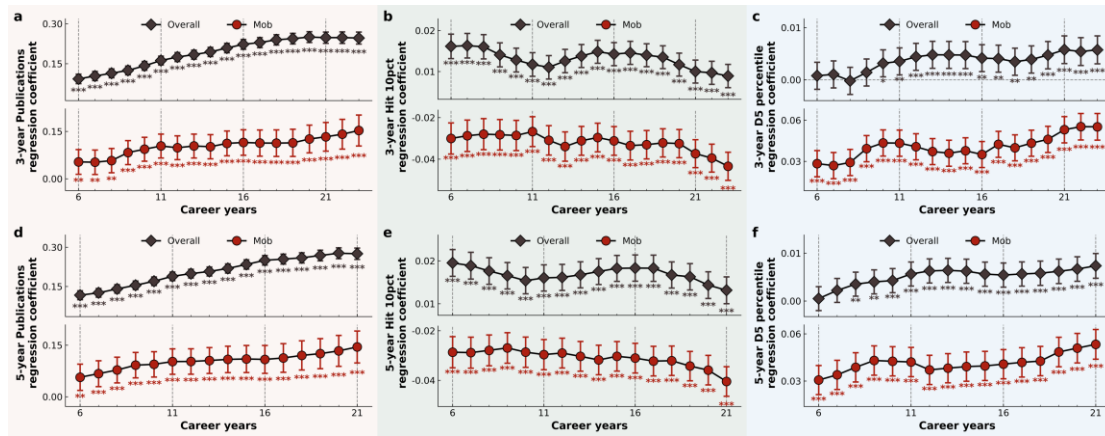


**Fig. S1 Overview of embedding-based cultural distance computation.** **a**, We analyzed a large-scale dataset comprising 207,009,605 author–institution–year records, and constructed 4,211,081 mobility trajectories from scientists who had records at least two different institutions. Each trajectory was built by chronologically ordering the institutional affiliations of an individual scientist. To avoid artificial ordering, affiliations within the same year were randomized, and consecutive duplicate affiliations were removed. Then, each trajectory can be expressed as an ordered sequence of institutions  $S_i = (a_{i,1}, a_{i,2}, \dots, a_{i,n})$  and each institution  $a_k$  was treated as a unique token in the sequence, and initial affiliation embeddings were learned using the Word2Vec algorithm based on these trajectories<sup>5</sup>. **b**, We constructed a weighted, undirected graph where nodes represent institutions and edges correspond to observed transitions in scientists’ career paths, with edge weights (transformed into log) reflecting transition frequencies (23,875 nodes and 1,490,998 edges). Each node was further enriched with features, including institution publications, h-index, logC10, longitude, latitude, and country (one-hot encoded). This attributed movement graph served as the input for graph-based representation learning. To obtain robust institution embeddings, we adopted two mainstream graph learning frameworks. **c**, Simplified graph contrastive learning (GCL). The movement graph is one-sided augmented through random edge dropping or node perturbation by adding Gaussian noise. Contrastive loss was calculated within each mini-batch, using the

normalized representations of corresponding nodes as positive pairs and all other nodes within the batch as negatives. A shared GNN-based encoder is used to maximize the agreement between augmented graph views, encouraging the model to learn meaningful, context-aware embeddings. **d**, Simplified graph masked autoencoder (GMAE). We also employed a GMAE framework, where parts of the node features are randomly masked by replacing them with learnable mask token and reconstructed via a GNN-based encoder–decoder architecture. The model is trained to minimize the reconstruction loss of the masked features, leading to embeddings that capture both the local and global structure of the affiliation graph. **e**, We used GNNExplainer to assess the interpretability of the learned institution embeddings and to quantify the importance of node features. Panel shows explanations for the edge weight prediction task (using GCL, GMAE unweighted self-supervised embedding combined with MLP). We split dataset into 5-fold and each target fold as test set and the rest folds as training set so that we can get the whole explanation for each edge. The results indicates that academic “mass”—reflected by h-index and productivity in GCL, and by productivity and LogC10 in GMAE—consistently dominates both embedding frameworks, while geographic coordinates and country codes contribute substantially less. This pattern, robust across models, aligns with gravity-model predictions, underscoring that institutional scale and output are the primary determinants of mobility intensity between institutions. Finally, to obtain country-level cultural distances, institution embeddings within each country were averaged to yield a country vector, serving as an embedding-based “cultural dimension.” The cultural distance between countries was then quantified as L1 distance between their embedding vectors. Each scientist’s early-career embedding-based cultural distance was calculated as the mean pairwise country-level distance across all observed moves during their first five career years. In this metric, a higher value indicates less frequently traveled, more “cultural distance” pathways, while a lower value reflects more commonly traversed, culturally proximate moves.

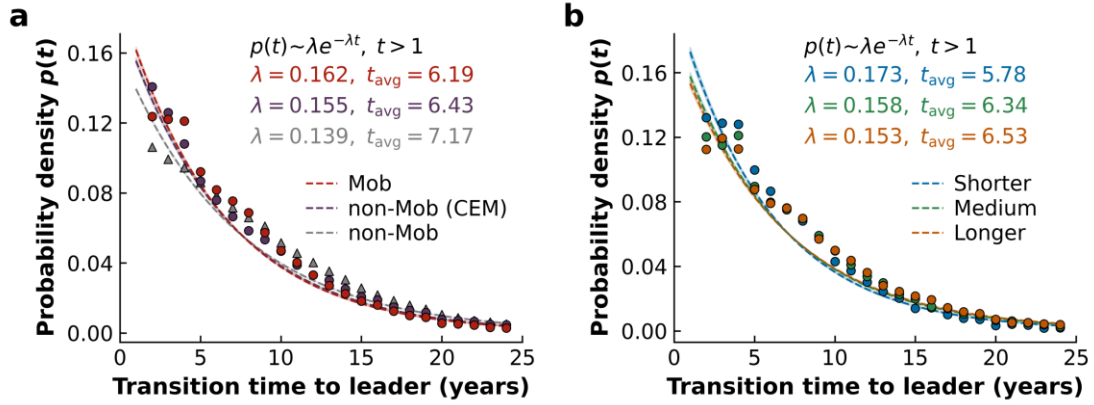


**Fig. S2 Placebo test of the association between early-career cultural distance and long-term scientific performance.** For each outcome—(a–c) overall sample; (d–f) mobile scientists only—plots show the cumulative distribution function (CDF) of mean regression coefficients from 1,000 placebo (shuffle), bootstrap, and observed (real) regressions estimating the effect of early-career cultural distance on long-term productivity (Publications, a, d), impact (Hit 10pct, b, e), and disruptiveness (D5 percentile, c, f), controlling the five CEM-matching covariates. The dashed vertical line indicates the real regression coefficient (with value annotated). The solid black line and asterisks indicate the right-tailed p-value based on the shuffled distribution. In all cases, the real effect size lies far outside the null distributions generated by shuffling.  $*P < 0.05$ ;  $**P < 0.01$ ;  $***P < 0.001$ .

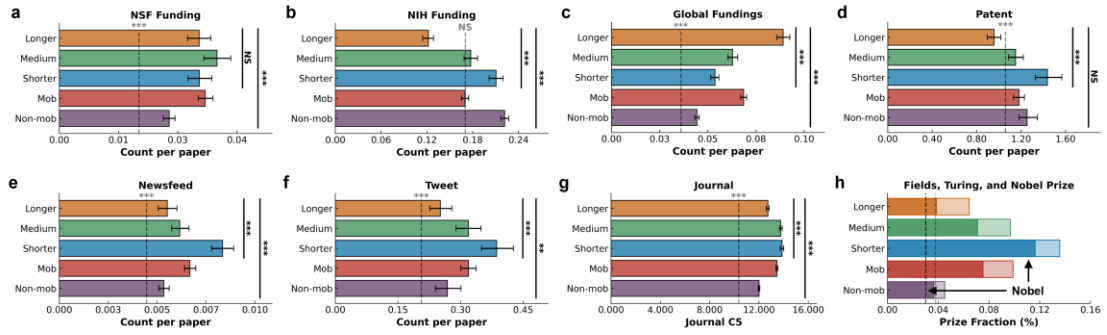


**Fig. S3 Temporal trajectories of long-term academic performance across alternative time windows.** Panels **a–c** report 3-year moving-window regression coefficients (with controls, i.e., the five CEM-matching covariates.), and panels **d–f** report 5-year moving-window coefficients for the three performance outcomes: Publications (**a, d**), Hit 10 pct (**b, e**), and D5 percentile (**c, f**). For each window, points show the estimated mobility effect (black: overall mobile versus non-mobile; red: early-career cultural distance among mobiles) at each career year, with 95% confidence intervals. Across all window lengths and career stages, the observed trajectories consistently align with the patterns described in the main text (Fig. 2). \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

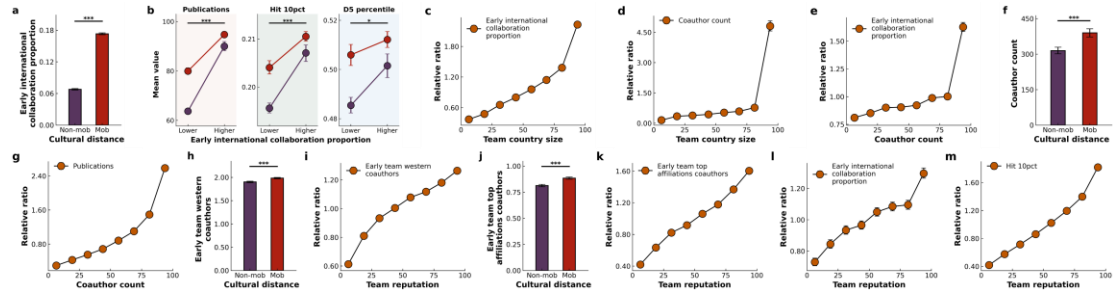




**Fig. S4 Early-career cultural distance and transition time to team leader.** **a**, Probability density of transition times to team leader for Mob scientists (red circles), non-Mob (CEM) scientists (purple circles), and unmatched non-Mob scientists (grey triangles). Exponential fits illustrate that Mob scientists transition to team leader significantly faster ( $\lambda = 0.162$ ,  $t_{\text{avg}} = 6.19$  years). **b**, Probability density of transition times for mobile scientists grouped by early-career cultural distance (Shorter, Medium, Longer). Scientists with shorter cultural distance transitions reach leadership positions most rapidly ( $\lambda = 0.173$ ,  $t_{\text{avg}} = 5.78$  years). Curves represent fitted exponential probability densities; symbols indicate observed empirical distributions. Shaded areas denote bootstrap 95% confidence intervals.



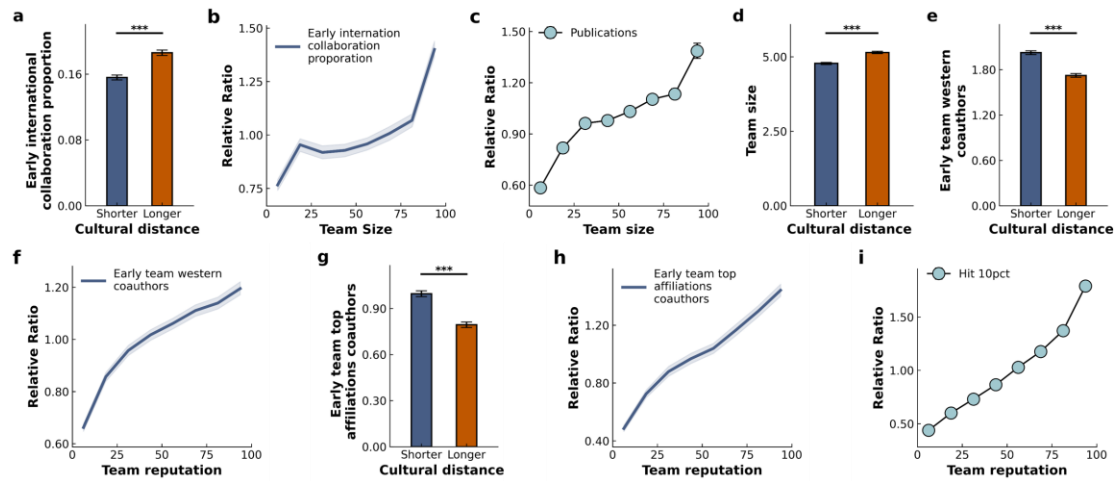
**Fig. S5 Fundings, Patents, Visibility, and Prizes.** Panels show outcomes for different career groups: non-Mob, Mob, and mobile scientists further divided into shorter, medium, and longer early-career cultural distance. (a) Mean NSF funding acknowledgements per paper. (b) Mean NIH funding acknowledgements per paper. (c) Mean all global fundings acknowledgements per paper. (d) Mean number of patents linked per paper. (e) Mean Journal C5 per paper. (f) Mean number of Newsfeed mentions per paper. (g) Mean number of Tweet mentions per paper. (h) Fraction of scientists receiving major international prizes: the darker bars denote Nobel Prize only, lighter bars indicate the combined Nobel, Fields, and Turing prizes. For the non-Mob (unmatched) group, the corresponding fractions are shown as vertical (dashed) lines—where the darker line indicates the Nobel fraction and the lighter line represents the combined prize fraction. In panels a–g, the vertical dashed lines indicate the mean value for the unmatched non-mob group. Gray significance markers indicate two-sided t-tests between the unmatched non-Mob and Mob groups. Black lines above the bars represent statistical significance of group comparisons: long lines indicate two-sided t-tests between Mob and non-Mob groups, short lines indicate two-sided t-tests between shorter and longer groups. Asterisks denote statistical significance (\* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ ; NS: not significant). Error bars indicate 95% confidence intervals.



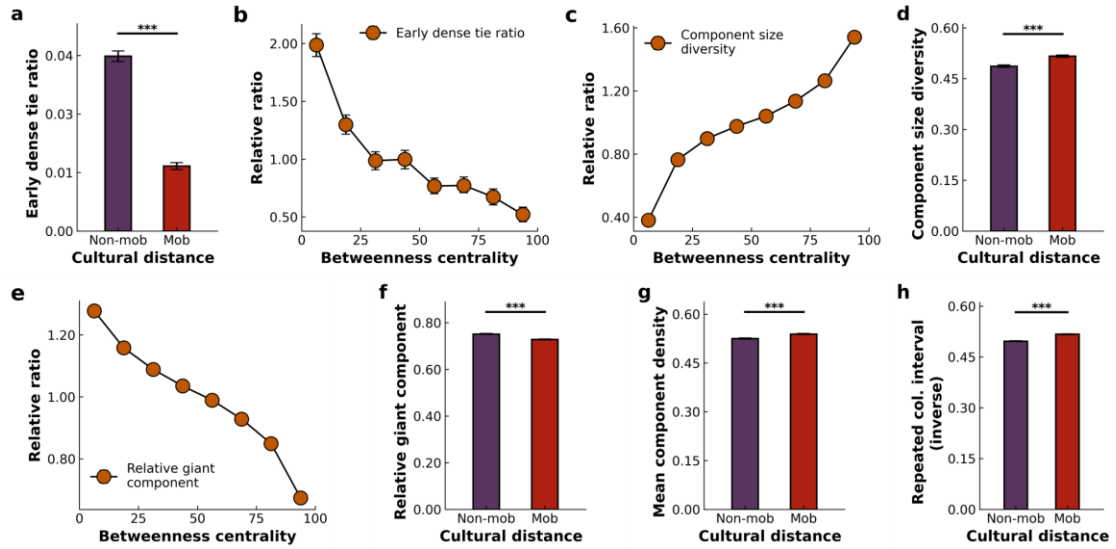
**Fig. S6 Labour advantages and team reputation for non-Mob and Mob scientists.**

**a**, Early-career international collaboration proportion. **b**, Later-career performance for scientists with and without any early-career international collaboration, shown separately for Non-mob and Mob scientists. **c**, Later-career team country size as a function of early-career international collaboration proportion. **d**, Later-career coauthor count as a function of team country size. **e**, Later-career coauthor count as a function of early-career international collaboration proportion. **f**, Later-career coauthor count. **g**, Publications as a function of coauthor count. **h**, Mean number of early-career coauthors from Western countries per paper. **i**, Later-career team reputation as a function of early Western coauthors. **j**, Mean number of early-career coauthors from top-100 institutions per paper. **k**, Later-career team reputation as a function of early top-institution coauthors. **l**, Later-career team reputation as a function of early-career international collaboration proportion. **m**, Hit 10pct as a function of team reputation. Together these panels show that early international collaboration widens the later collaborator pool that supports productivity, whereas the elite composition of those early collaborators supports later team reputation and impact. In **b**, groups are defined by whether the proportion exceeds zero rather than by a median split, because fewer than half of scientists collaborate internationally in early career. Relative ratios are values divided by the sample mean, with scientists grouped into eight equally sized bins of the  $x$ -axis variable. Bars show group means and error bars and shaded bands show bootstrap 95% confidence intervals.

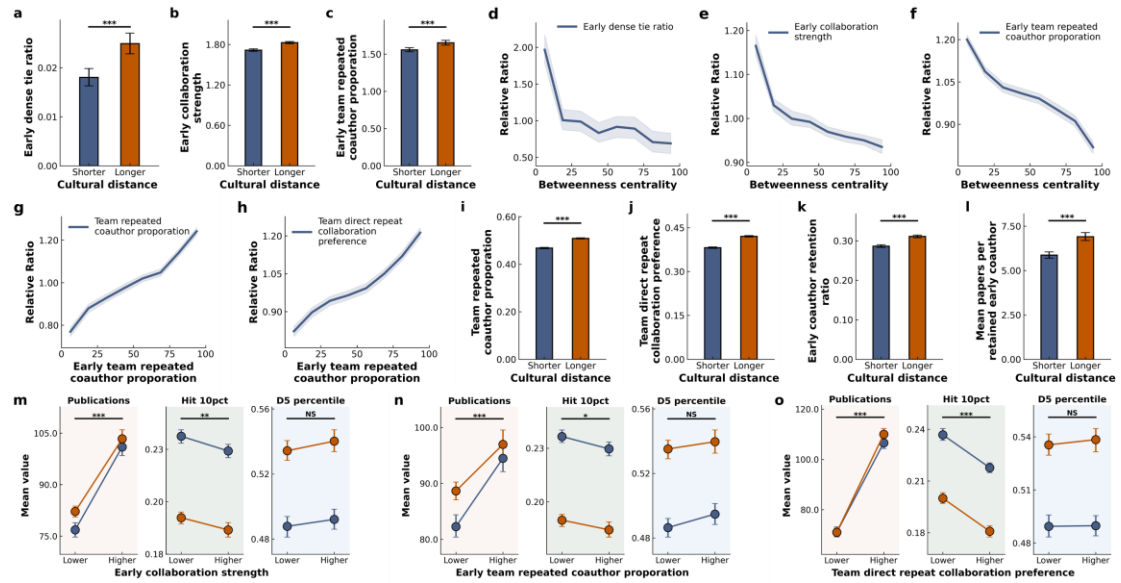
\*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



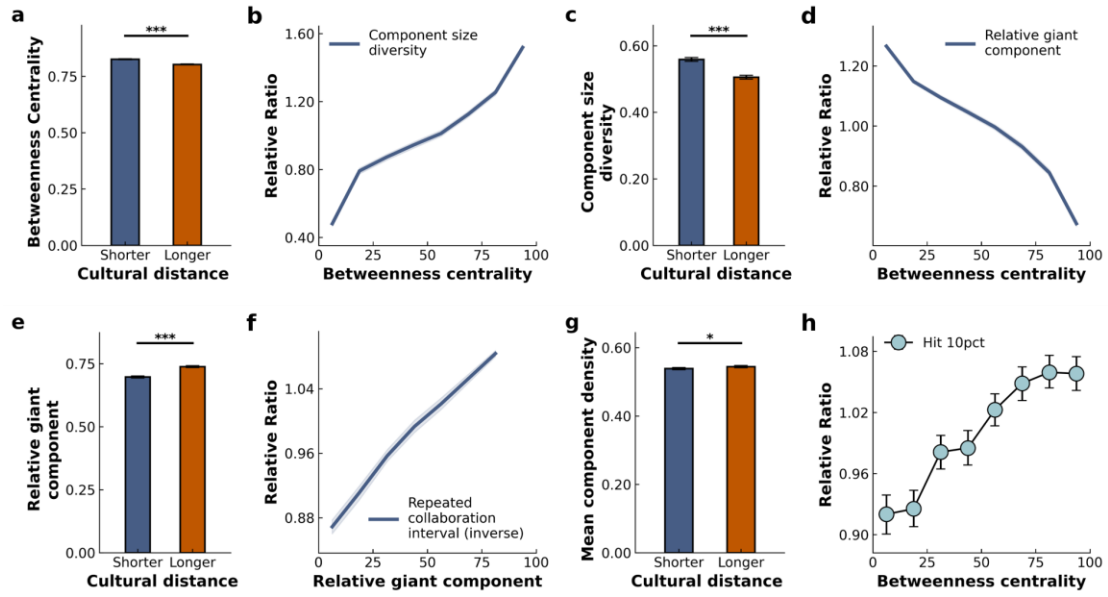
**Fig. S7 Labour advantages and team reputation for shorter- and longer-distance movers.** **a**, Early-career international collaboration proportion. **b**, Later-career team size as a function of early-career international collaboration proportion. **c**, Publications as a function of team size. **d**, Later-career team size. **e**, Mean number of early-career coauthors from Western countries per paper. **f**, Later-career team reputation as a function of early Western coauthors. **g**, Mean number of early-career coauthors from top-100 institutions per paper. **h**, Later-career team reputation as a function of early top-institution coauthors. **i**, Hit 10pct as a function of team reputation. The panels show that the two resources identified in Fig. S6 divide along cultural distance: longer-distance movers collaborate internationally more and assemble larger later-career teams, whereas shorter-distance movers reach more Western-country and top-100-institution collaborators and assemble higher-reputation teams. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



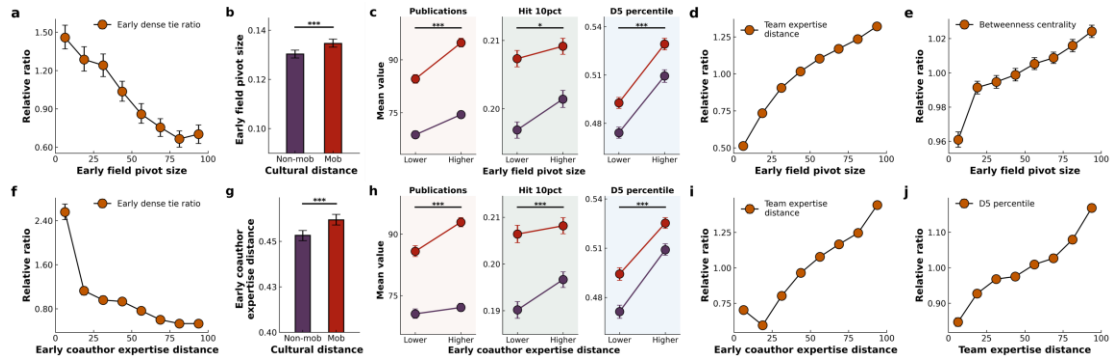
**Fig. S8 Team assembly and network structure for non-Mob and Mob scientists.** **a**, Early-career dense tie ratio. **b**, Later-career betweenness centrality as a function of early dense tie ratio. **c**, Component size diversity as a function of betweenness centrality. **d**, Component size diversity. **e**, Relative giant component as a function of betweenness centrality. **f**, Relative giant component. **g**, Mean component density. **h**, Coordination efficiency, measured as the inverse of the mean interval between repeated collaborations. Mob scientists rely less on a fixed early collaborator core and later occupy more brokerage-oriented positions in networks that are globally more fragmented yet locally denser and more efficiently coordinated. All network measures are computed on the later-career collaboration network excluding super-team papers. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



**Fig. S9 Team assembly for shorter- and longer-distance movers.** **a**, Early-career dense tie ratio. **b**, Early-career collaboration strength. **c**, Early team repeated coauthor proportion. **d–f**, Later-career betweenness centrality as a function of each of the three early measures in **a–c**. **g**, **h**, Later-career team repeated coauthor proportion (**g**) and team direct repeat collaboration preference (**h**) as functions of early team repeated coauthor proportion. **i**, Later-career team repeated coauthor proportion. **j**, Team direct repeat collaboration preference. **k**, Early coauthor retention ratio, the share of later-career collaborations involving early-career coauthors. **l**, Mean papers per retained early coauthor. **m–o**, Later-career performance for the lower and higher halves of early collaboration strength (**m**), early team repeated coauthor proportion (**n**) and later team direct repeat collaboration preference (**o**). Longer-distance movers rely more on stable, repeated ties from early career onward, which persists into later career and is consistently associated with lower brokerage but higher productivity and lower impact. Conventions are as in Fig. S6. NS, not significant; \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

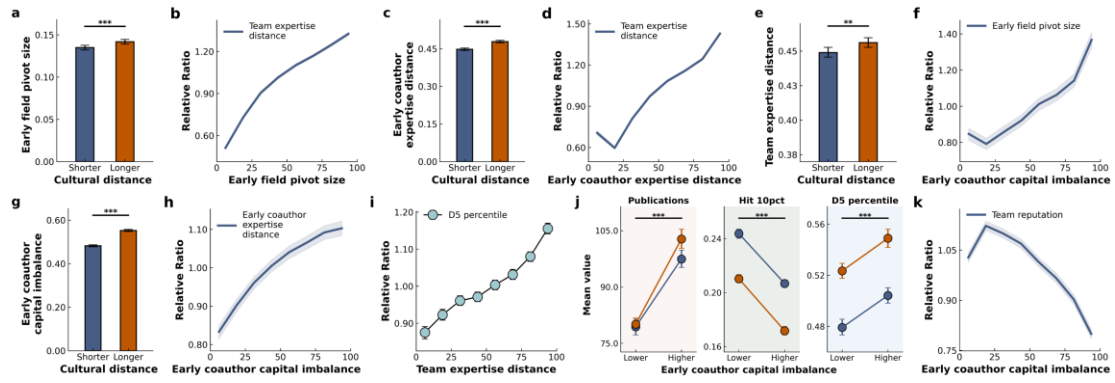


**Fig. S10 Network structure for shorter- and longer-distance movers.** **a**, Later-career betweenness centrality. **b**, Component size diversity as a function of betweenness centrality. **c**, Component size diversity. **d**, Relative giant component as a function of betweenness centrality. **e**, Relative giant component. **f**, Coordination efficiency as a function of the relative giant component. **g**, Mean component density. **h**, Hit 10pct as a function of betweenness centrality. Shorter-distance movers occupy more brokerage-oriented and more fragmented network positions, whereas longer-distance movers form larger, more cohesive giant components in which repeated collaboration is more efficiently sustained. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

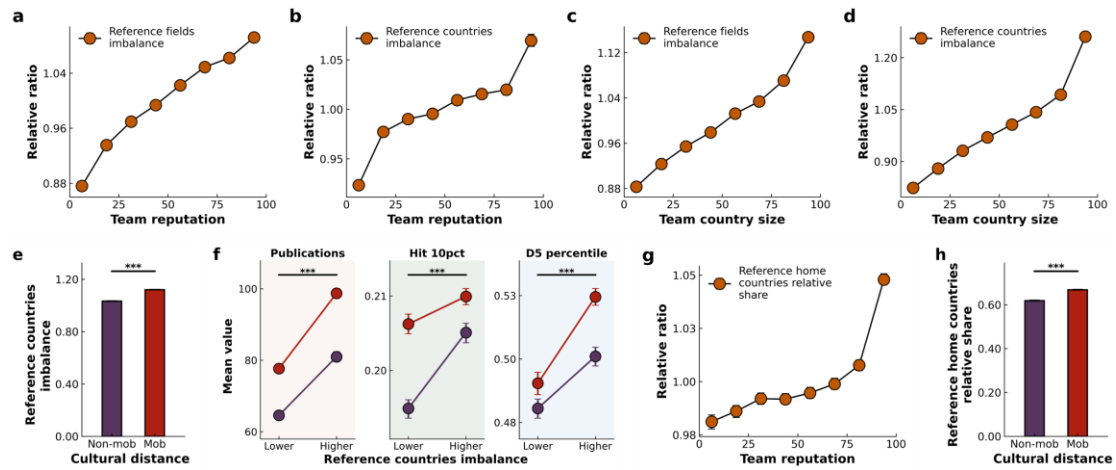


**Fig. S11 Team expertise for non-Mob and Mob scientists.** **a**, Early-career field pivot size as a function of early dense tie ratio. **b**, Early-career field pivot size. **c**, Later-career performance for the lower and higher halves of early field pivot size. **d**, Later-career team expertise distance as a function of early field pivot size. **e**, Later-career betweenness centrality as a function of early field pivot size. **f**, Early coauthor expertise distance as a function of early dense tie ratio. **g**, Early-career coauthor expertise distance. **h**, Later-career performance for the lower and higher halves of early coauthor expertise distance. **i**, Later-career team expertise distance as a function of early coauthor expertise distance. **j**, D5 percentile as a function of team expertise distance. Forming new rather than repeated ties after a move exposes Mob scientists to more expertise-diverse collaborators and larger field pivots in early career, both of which carry forward into more expertise-distant later-career teams and higher disruptiveness. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

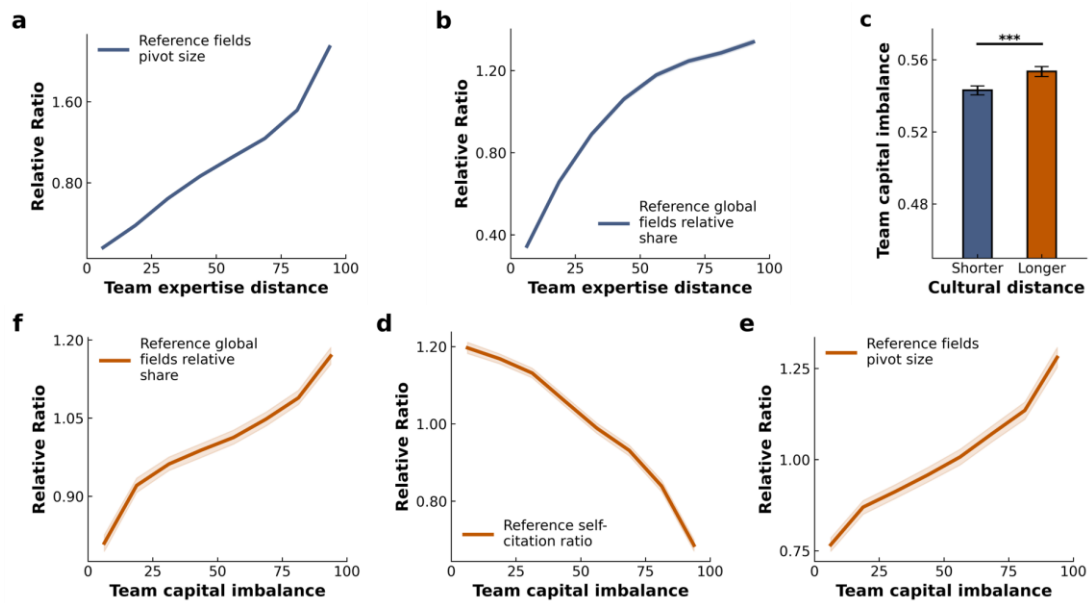




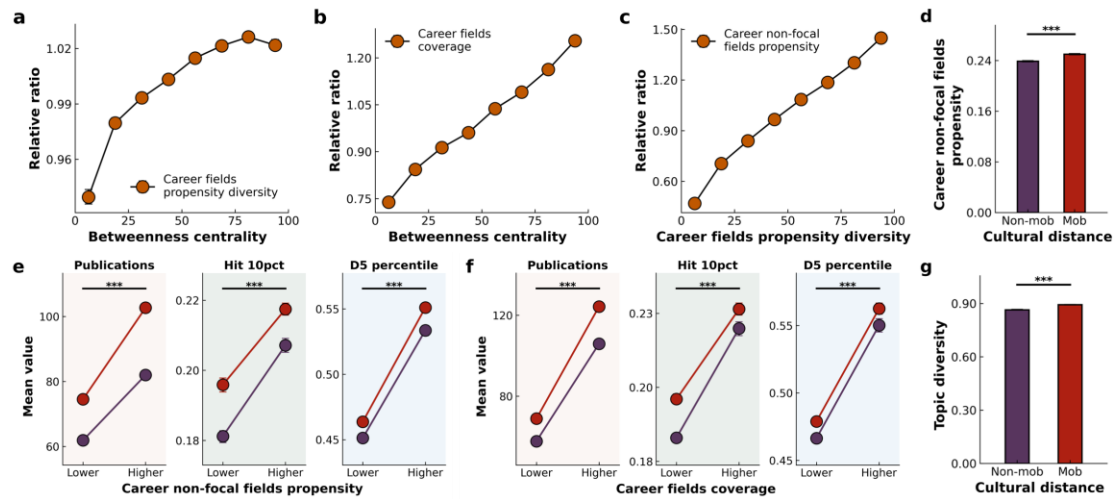
**Fig. S12 Team expertise for shorter- and longer-distance movers.** **a**, Early-career field pivot size. **b**, Later-career team expertise distance as a function of early field pivot size. **c**, Early-career coauthor expertise distance. **d**, Later-career team expertise distance as a function of early coauthor expertise distance. **e**, Later-career team expertise distance. **f**, Early field pivot size as a function of early coauthor capital imbalance. **g**, Early-career coauthor capital imbalance, measured as the coefficient of variation of cumulative coauthor citation impact up to career year five. **h**, Early coauthor expertise distance as a function of early coauthor capital imbalance. **i**, D5 percentile as a function of later-career team expertise distance. **j**, Later-career performance for the lower and higher halves of early coauthor capital imbalance. **k**, Later-career team reputation as a function of early coauthor capital imbalance. Although longer-distance movers rely more on repeated ties (Fig. S9), they enter collaborator pools of more uneven scientific capital, which is associated with larger early field pivots and more expertise-distant later teams, and at the same time with lower later team reputation, matching their higher disruptiveness but lower impact. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



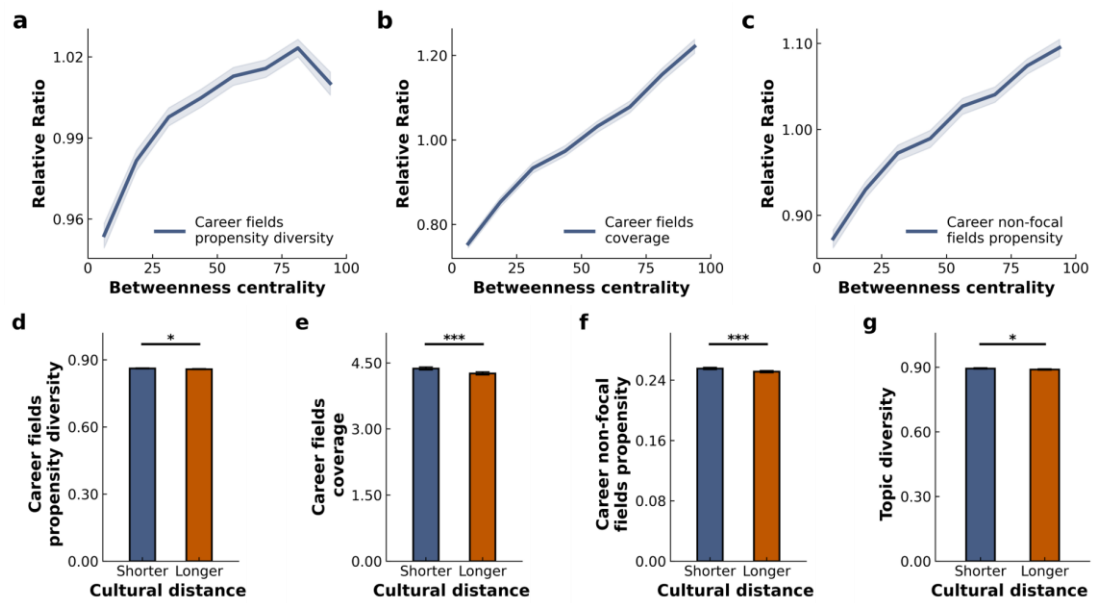
**Fig. S13 Local exploitation and its collaborative origins for non-Mob and Mob scientists.** **a, b,** Later-career reference fields imbalance (**a**) and reference countries imbalance (**b**) as functions of team reputation. **c, d,** The same two measures as functions of team country size. **e,** Reference countries imbalance. **f,** Later-career performance for the lower and higher halves of reference countries imbalance. **g,** Reference home countries relative share as a function of team reputation. **h,** Reference home countries relative share. Higher team reputation and more international teams both accompany a more concentrated use of prior work, and the concentration Mob scientists show at the field level (Fig. 4a, c) recurs at the country level, where they likewise draw more of each paper's references from their own country. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



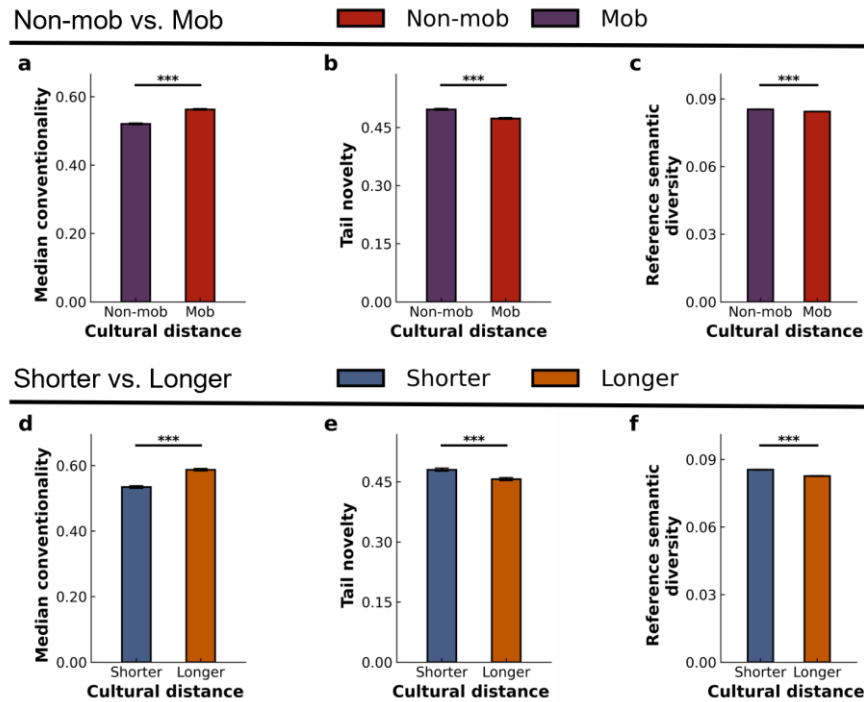
**Fig. S14 Team expertise, team capital imbalance and local knowledge strategy for shorter- and longer-distance movers.** **a, b**, Later-career reference fields pivot size (**a**) and reference global fields relative share (**b**) as functions of team expertise distance. **c**, Later-career team capital imbalance, measured as the per-paper coefficient of variation of coauthors' cumulative citation impact. **d–f**, Reference self-citation ratio (**d**), reference fields pivot size (**e**) and reference global fields relative share (**f**) as functions of team capital imbalance. More expertise-distant teams reference more widely beyond a paper's own fields, and the same holds for teams whose members differ more in scientific capital, both of which are higher among longer-distance movers and align with their more exploration-oriented local knowledge strategy. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



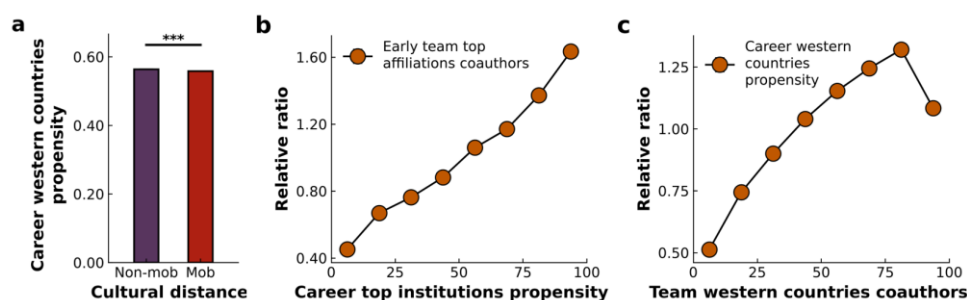
**Fig. S15 Career-level knowledge exploration for non-Mob and Mob scientists.** **a–c**, Career fields propensity diversity (**a**) and career fields coverage (**b**) as functions of later-career betweenness centrality, and career non-focal fields propensity as a function of career fields propensity diversity (**c**). **d**, Career non-focal fields propensity. **e, f**, Later-career performance for the lower and higher halves of career non-focal fields propensity (**e**) and career fields coverage (**f**). **g**, Topic diversity, computed as the normalized entropy of topics detected by community detection on the co-citation network of later-career references. Brokerage positions accompany broader career-level knowledge integration on every measure, and Mob scientists are higher on all of them, so that their local concentration and exploitation in Fig. 4a coexists with wider exploration across the career. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



**Fig. S16 Career-level knowledge exploration for shorter- and longer-distance movers.** **a–c**, Career fields propensity diversity (**a**), career fields coverage (**b**) and career non-focal fields propensity (**c**) as functions of later-career betweenness centrality. **d–g**, Career fields propensity diversity (**d**), career fields coverage (**e**), career non-focal fields propensity (**f**) and topic diversity (**g**). The association with brokerage recurs within the mobile sample, and shorter-distance movers, who occupy the more brokerage-oriented positions (Fig. S10a), explore more broadly at the career level on all four measures, reversing the ordering seen at the paper level in Fig. 4k. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

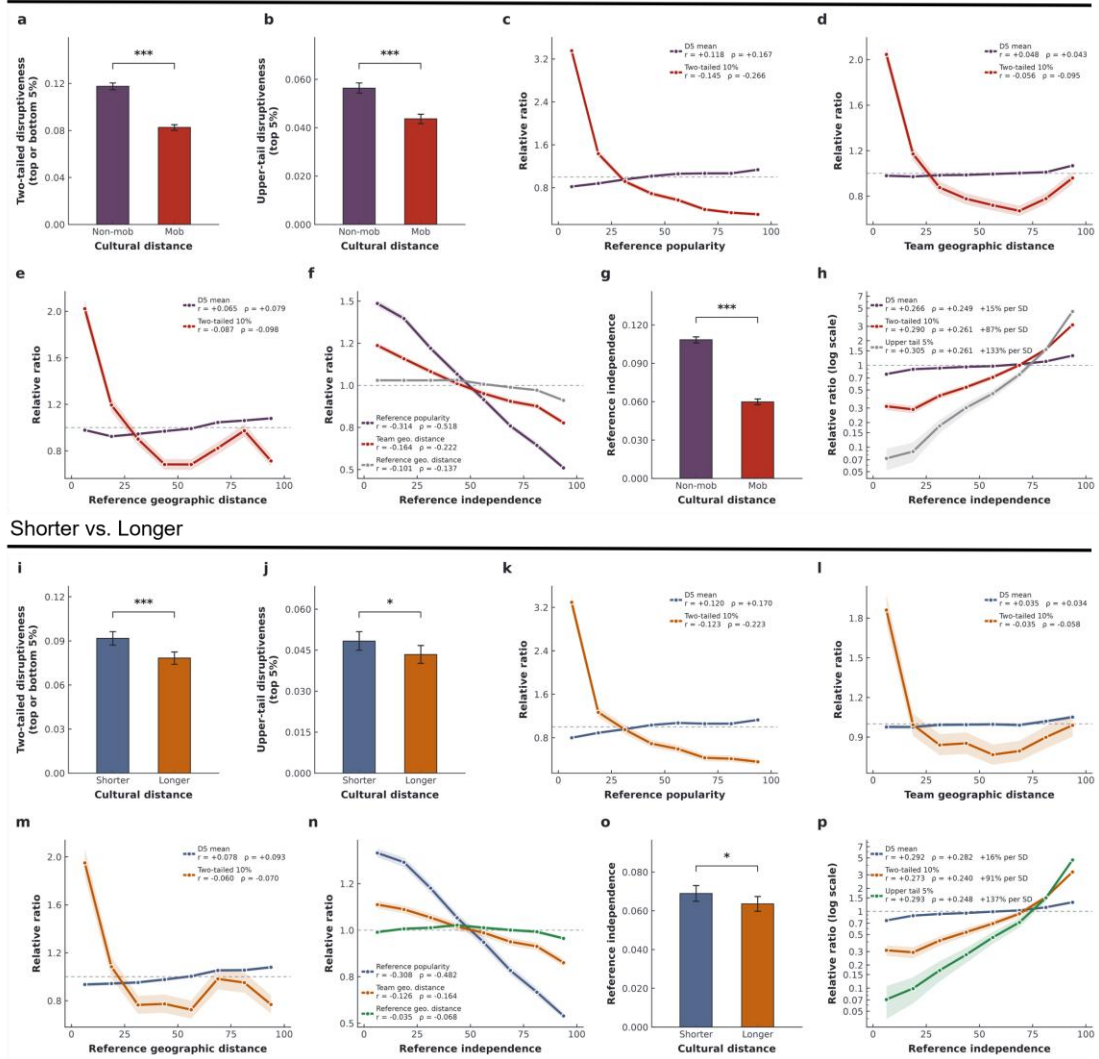


**Fig. S17 Conventionality, novelty and semantic dispersion of recombination.** **a–c**, Median conventionality (**a**), tail novelty (**b**) and reference semantic diversity (**c**) for non-Mob and Mob scientists. **d–f**, The same three measures for shorter- and longer-distance movers. A paper counts as high in median conventionality when its 50th-percentile journal-pair  $z$ -score falls in the upper half of all papers, and as high in tail novelty when its 10th-percentile  $z$ -score falls below zero; reference semantic diversity is the coefficient of variation of pairwise cosine similarities among the SciBERT embeddings of referenced titles. Mob scientists and longer-distance movers ground their work in more conventional combinations yet produce fewer atypical ones and mix semantically near and far references less unevenly, a separation between the centre and the tail of recombination that recurs for disruptiveness in Fig. S19. Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



**Fig. S18 Western-country knowledge sourcing among Mob scientists and their peers.** **a**, Career western countries propensity. **b**, Career top institutions propensity as a function of the mean number of early-career coauthors from top-100 institutions per paper. **c**, Career western countries propensity as a function of the mean number of later-career coauthors from Western countries per paper. Unlike their orientation toward top-institution knowledge (Fig. 5a), Mob scientists draw less on Western-country knowledge overall, in keeping with their fewer later-career Western collaborators; institution-level sourcing instead traces back to early-career elite exposure and persists even after scientists leave elite environments (Table S23). Conventions are as in Fig. S6. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

## Non-mob vs. Mob



**Fig. S19 Center–tail separation in disruptiveness.** Panels **a–h** compare Mob with non-Mob scientists; panels **i–p** compare longer- with shorter-distance movers. **a, i**, Two-tailed 10% extreme disruptiveness, defined as an author-level D5 value in either the top 5% or the bottom 5% of the full distribution. **b, j**, Upper-tail disruptiveness, defined as an author-level D5 value in the top 5%. Bars show group means with bootstrap 95% confidence intervals; asterisks denote one-sided Welch's  $t$ -tests (\* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ ). **c–e, k–m**, Relative ratios of mean D5 percentile and of two-tailed 10% extreme disruptiveness across octiles of reference popularity (**c, k**), team geographic distance (**d, l**) and reference geographic distance (**e, m**). Points are observed octile means divided by the overall mean; shaded bands are bootstrap percentile 95% confidence intervals. Pearson  $r$  and Spearman  $\rho$  are given in the legends. **f, n**, Relative ratios of reference popularity, team geographic distance and reference geographic distance across octiles of reference independence. **g, o**, Reference independence by group. After adjusting for reference count, reference independence was significantly lower



among longer-distance movers than among shorter-distance movers (unstandardized coefficient  $B = -0.020$ ,  $P < 0.001$ ). **h, p**, Relative ratios of mean D5 percentile, two-tailed 10% extreme disruptiveness and upper-tail 5% disruptiveness across octiles of reference independence, on a logarithmic  $y$  axis so that equal vertical distances represent equal proportional changes. Legends additionally report the change per one-standard-deviation increase in reference independence, expressed relative to each outcome's baseline mean. As a robustness check we additionally control for later-career productivity, since averaging over more papers compresses the dispersion of career-level disruptiveness, together with the controls applied to the main outcomes in Supplementary Note 3.1.7 (later-career reference count, field and year fixed effects). The conclusions are unchanged.

## Supplementary Tables

**Table S1 Balance check for the CEM subsample.**

| Covariate<br>(Early career) | Mean     |        | Diff.  | SMD     | P-value |
|-----------------------------|----------|--------|--------|---------|---------|
|                             | Non-mobs | Mobs   |        |         |         |
| <i><b>Pre-match</b></i>     |          |        |        |         |         |
| Publication count           | 6.70     | 10.16  | 3.46   | 0.58    | <0.001  |
| Hit 10pct                   | 0.18     | 0.22   | 0.05   | 0.19    | <0.001  |
| D5 percentile               | 0.50     | 0.48   | -0.02  | -0.08   | <0.001  |
| Coauthor publications       | 17.44    | 23.41  | 5.97   | 0.29    | <0.001  |
| Institution h-index         | 434.48   | 461.45 | 26.97  | 0.11    | <0.001  |
| <i><b>Post-match</b></i>    |          |        |        |         |         |
| Publication count           | 9.88     | 9.98   | 0.10   | 0.01    | 0.0535  |
| Hit 10pct                   | 0.2155   | 0.2177 | 0.0022 | 0.0100  | 0.1291  |
| D5 percentile               | 0.4778   | 0.4780 | 0.0002 | 0.0007  | 0.9132  |
| Coauthor publications       | 23.21    | 23.37  | 0.15   | 0.0075  | 0.2545  |
| Institution h-index         | 464.44   | 462.99 | -1.45  | -0.0069 | 0.2931  |

Covariate means, mean differences (Diff.), standardized mean differences (SMD), and two-sample t-test p-values are shown for non-mobile (“Non-mob”) and mobile (“Mob”) scientists before and after CEM matching. All covariate imbalances were substantially reduced after CEM matching, with all SMDs <0.02. Mean = sample mean. Diff. = mean difference (Mobile - Non-mobile). SMD = standardized mean difference. P-value from two-sample t-test for difference in means.

**Table S2 Correlations between embedding-based and Hofstede cultural distances after CEM matching.**

| Model             | Pearson   |           | Spearman  |           |
|-------------------|-----------|-----------|-----------|-----------|
|                   | Overall   | Mobs      | Overall   | Mobs      |
| GCL (Unweighted)  | 0.76±0.03 | 0.32±0.07 | 0.91±0.01 | 0.35±0.07 |
| GMAE (Unweighted) | 0.73±0.01 | 0.35±0.03 | 0.91±0.01 | 0.38±0.05 |
| GCL (Weighted)    | 0.63±0.02 | 0.23±0.01 | 0.90±0.00 | 0.28±0.01 |
| GMAE (Weighted)   | 0.57±0.01 | 0.28±0.04 | 0.91±0.01 | 0.36±0.08 |

Pearson and Spearman correlations between each embedding-based measure of early-career cultural distance and the Hofstede-based measure, reported for the full matched sample (Overall) and for mobile scientists only (Mobs). Weighted models use graph convolutional networks, which support edge weights during message passing, and unweighted models use GraphSAGE. Each model was run over five random seeds under identical settings; values are reported as mean ± standard deviation across the five runs.

**Table S3 Robustness of the association between early-career cultural distance and long-term academic performance to alternative early-career observation windows.**

| Time Window                   | (1)<br>1-3 years       | (2)<br>1-4 years       | (3)<br>1-6 years       | (4)<br>1-7 years       | (5)<br>1-3 years       | (6)<br>1-4 years       | (7)<br>1-6 years       | (8)<br>1-7 years       |
|-------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| <b>Panel a: Publications</b>  |                        |                        |                        |                        |                        |                        |                        |                        |
| <i>Overall</i>                |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.1176***<br>(0.0089)  | 0.1694***<br>(0.0076)  | 0.2512***<br>(0.0074)  | 0.2705***<br>(0.0074)  | 0.0965***<br>(0.0103)  | 0.1558***<br>(0.0088)  | 0.2543***<br>(0.0089)  | 0.2708***<br>(0.0094)  |
| Controls                      |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 92,914                 | 92,914                 | 92,914                 | 92,914                 | 92,914                 | 92,914                 | 92,914                 | 92,914                 |
| Pseudo-R <sup>2</sup>         | 0.003                  | 0.008                  | 0.018                  | 0.020                  | 0.123                  | 0.128                  | 0.140                  | 0.142                  |
| <i>Mob</i>                    |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0954***<br>(0.0257)  | 0.0964***<br>(0.0200)  | 0.1273***<br>(0.0164)  | 0.1616***<br>(0.0163)  | 0.0963***<br>(0.0264)  | 0.1013***<br>(0.0207)  | 0.1388***<br>(0.0184)  | 0.1567***<br>(0.0166)  |
| Controls                      |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 20,025                 | 32,460                 | 48,076                 | 49,479                 | 20,025                 | 32,460                 | 48,076                 | 49,479                 |
| Pseudo-R <sup>2</sup>         | 0.001                  | 0.001                  | 0.002                  | 0.003                  | 0.179                  | 0.169                  | 0.152                  | 0.152                  |
| <b>Panel b: Hit 10pct</b>     |                        |                        |                        |                        |                        |                        |                        |                        |
| <i>Overall</i>                |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0079***<br>(0.0011)  | 0.0104***<br>(0.0010)  | 0.0156***<br>(0.0009)  | 0.0166***<br>(0.0009)  | 0.0065***<br>(0.0010)  | 0.0083***<br>(0.0009)  | 0.0147***<br>(0.0008)  | 0.0154***<br>(0.0008)  |
| Controls                      |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 92,896                 | 92,896                 | 92,896                 | 92,896                 | 92,896                 | 92,896                 | 92,896                 | 92,896                 |
| R <sup>2</sup>                | 0.001                  | 0.001                  | 0.003                  | 0.003                  | 0.184                  | 0.184                  | 0.186                  | 0.186                  |
| <i>Mob</i>                    |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | -0.0607***<br>(0.0035) | -0.0586***<br>(0.0027) | -0.0631***<br>(0.0022) | -0.0621***<br>(0.0022) | -0.0299***<br>(0.0033) | -0.0274***<br>(0.0025) | -0.0329***<br>(0.0021) | -0.0323***<br>(0.0021) |
| Controls                      |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 20,020                 | 32,454                 | 48,067                 | 49,470                 | 20,020                 | 32,454                 | 48,067                 | 49,470                 |
| R <sup>2</sup>                | 0.015                  | 0.014                  | 0.017                  | 0.016                  | 0.178                  | 0.178                  | 0.168                  | 0.168                  |
| <b>Panel c: D5 percentile</b> |                        |                        |                        |                        |                        |                        |                        |                        |
| <i>Overall</i>                |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0106***<br>(0.0023)  | 0.0147***<br>(0.0020)  | 0.0196***<br>(0.0019)  | 0.0192***<br>(0.0019)  | 0.0100***<br>(0.0023)  | 0.0147***<br>(0.0020)  | 0.0193***<br>(0.0019)  | 0.0189***<br>(0.0019)  |
| Controls                      |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 90,136                 | 90,136                 | 90,136                 | 90,136                 | 90,136                 | 90,136                 | 90,136                 | 90,136                 |
| R <sup>2</sup>                | 0.000                  | 0.001                  | 0.001                  | 0.001                  | 0.051                  | 0.051                  | 0.052                  | 0.052                  |
| <i>Mob</i>                    |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0744***<br>(0.0070)  | 0.0763***<br>(0.0055)  | 0.0729***<br>(0.0045)  | 0.0720***<br>(0.0044)  | 0.0487***<br>(0.0070)  | 0.0513***<br>(0.0055)  | 0.0485***<br>(0.0045)  | 0.0483***<br>(0.0044)  |
| Controls                      |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 19,432                 | 31,561                 | 46,702                 | 48,086                 | 19,432                 | 31,561                 | 46,702                 | 48,086                 |

|                |       |       |       |       |       |       |       |       |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| R <sup>2</sup> | 0.006 | 0.006 | 0.006 | 0.005 | 0.061 | 0.063 | 0.061 | 0.060 |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|

Panel **a** present result for long-term productivity (publications), Panel **b** for impact (Hit 10pct), and Panel **c** for disruptiveness (D5 percentile). Each outcome is reported for the full sample (“Overall”) and for mobile scientists (“Mob”), both without controls (columns 1–4; raw mean differences) and with controls (columns 5–8; i.e., the five CEM-matching covariates). Across all panels, the patterns of long-term academic performance are robust to the definition of the early-career observation window and model specification. Columns 1–4: no controls, early-career window = years 1–3, 1–4, 1–6, 1–7; Columns 5–8: with controls, early-career window = years 1–3, 1–4, 1–6, 1–7; Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

**Table S4 Robustness of early-career cultural distance effects across alternative matching methods.**

| Matching Method               | No-matching            |                        | m: n CEM               |                        | 1:1 CEM                |                        | 1:1 PSM                |                        |
|-------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                               | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    | (8)                    |
| <b>Panel a: Publications</b>  |                        |                        |                        |                        |                        |                        |                        |                        |
| <i>Overall</i>                |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.4830***<br>(0.0052)  | 0.4005***<br>(0.0091)  | 0.2227***<br>(0.0062)  | 0.2273***<br>(0.0062)  | 0.2257***<br>(0.0074)  | 0.2310***<br>(0.0086)  | 0.2562***<br>(0.0073)  | 0.2517***<br>(0.0083)  |
| Controls                      |                        | Yes                    |                        | Yes                    |                        | Yes                    |                        | Yes                    |
| Observations                  | 584,712                | 584,712                | 422,434                | 422,434                | 92,914                 | 92,914                 | 95,676                 | 95,676                 |
| Pseudo-R <sup>2</sup>         | 0.025                  | 0.102                  | 0.014                  | 0.130                  | 0.014                  | 0.137                  | 0.018                  | 0.165                  |
| <i>Mob</i>                    |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0995***<br>(0.0166)  | 0.1094***<br>(0.0184)  | 0.0999***<br>(0.0167)  | 0.1111***<br>(0.0185)  | 0.0986***<br>(0.0167)  | 0.1110***<br>(0.0187)  | 0.0995***<br>(0.0166)  | 0.1093***<br>(0.0184)  |
| Controls                      |                        | Yes                    |                        | Yes                    |                        | Yes                    |                        | Yes                    |
| Observations                  | 47,838                 | 47,838                 | 47,014                 | 47,014                 | 46,457                 | 46,457                 | 47,838                 | 47,838                 |
| Pseudo-R <sup>2</sup>         | 0.001                  | 0.153                  | 0.001                  | 0.153                  | 0.001                  | 0.150                  | 0.001                  | 0.154                  |
| <b>Panel b: Hit 10pct</b>     |                        |                        |                        |                        |                        |                        |                        |                        |
| <i>Overall</i>                |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0339***<br>(0.0007)  | 0.0192***<br>(0.0006)  | 0.0133***<br>(0.0004)  | 0.0130***<br>(0.0004)  | 0.0139***<br>(0.0009)  | 0.0134***<br>(0.0004)  | 0.0135***<br>(0.0009)  | 0.0179***<br>(0.0008)  |
| Controls                      |                        | Yes                    |                        | Yes                    |                        | Yes                    |                        | Yes                    |
| Observations                  | 582,095                | 582,095                | 422,175                | 422,175                | 92,896                 | 92,896                 | 95,676                 | 95,676                 |
| R <sup>2</sup>                | 0.004                  | 0.190                  | 0.002                  | 0.184                  | 0.002                  | 0.185                  | 0.002                  | 0.194                  |
| <i>Mob</i>                    |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | -0.0617***<br>(0.0022) | -0.0310***<br>(0.0021) | -0.0621***<br>(0.0022) | -0.0315***<br>(0.0021) | -0.0627***<br>(0.0022) | -0.0317***<br>(0.0021) | -0.0618***<br>(0.0022) | -0.0310***<br>(0.0021) |
| Controls                      |                        | Yes                    |                        | Yes                    |                        | Yes                    |                        | Yes                    |
| Observations                  | 47,823                 | 47,823                 | 47,005                 | 47,005                 | 46,448                 | 46,448                 | 47,838                 | 47,838                 |
| R <sup>2</sup>                | 0.016                  | 0.171                  | 0.016                  | 0.169                  | 0.017                  | 0.169                  | 0.016                  | 0.171                  |
| <b>Panel c: D5 percentile</b> |                        |                        |                        |                        |                        |                        |                        |                        |
| <i>Overall</i>                |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0088***<br>(0.0014)  | 0.0147***<br>(0.0014)  | 0.0179***<br>(0.0009)  | 0.0176***<br>(0.0009)  | 0.0200***<br>(0.0019)  | 0.0197***<br>(0.0019)  | 0.0085***<br>(0.0019)  | 0.0106***<br>(0.0018)  |
| Controls                      |                        | Yes                    |                        | Yes                    |                        | Yes                    |                        | Yes                    |
| Observations                  | 505,385                | 505,385                | 375,229                | 375,229                | 90,136                 | 90,136                 | 95,200                 | 95,200                 |
| R <sup>2</sup>                | 0.000                  | 0.043                  | 0.001                  | 0.053                  | 0.001                  | 0.052                  | 0.000                  | 0.041                  |
| <i>Mob</i>                    |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0669***<br>(0.0043)  | 0.0440***<br>(0.0043)  | 0.0693***<br>(0.0044)  | 0.0453***<br>(0.0044)  | 0.0733***<br>(0.0046)  | 0.0479***<br>(0.0046)  | 0.0698***<br>(0.0044)  | 0.0471***<br>(0.0044)  |
| Controls                      |                        | Yes                    |                        | Yes                    |                        | Yes                    |                        | Yes                    |
| Observations                  | 46,399                 | 46,399                 | 45,658                 | 45,658                 | 45,108                 | 45,108                 | 47,739                 | 47,739                 |

|                |       |       |       |       |       |       |       |       |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|
| R <sup>2</sup> | 0.005 | 0.062 | 0.005 | 0.062 | 0.006 | 0.061 | 0.005 | 0.051 |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|

We report the estimated effect of early-career cultural distance on long-term academic performance (Panel **a-c**) under different matching approaches: no matching (columns 1–2), m: n CEM (columns 3–4), 1:1 CEM (columns 5–6), and 1:1 PSM (columns 7–8). Across all specifications, the patterns of long-term academic performance are consistent, indicating the robustness of our main findings to the choice of matching algorithm and adjustment strategy. In m: n CEM, weights are assigned as 1.0 for treated observations and as (number of treated / number of controls) within each matched stratum for control observations. Regression coefficients are then estimated via weighted least squares (WLS) using these weights. Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

**Table S5 Robustness of early-career cultural distance effects to alternative estimation strategies methods.**

| Matching Method               | No-matching |            | m: n CEM   |            | 1:1 CEM    |            | 1:1 PSM    |            |
|-------------------------------|-------------|------------|------------|------------|------------|------------|------------|------------|
|                               | (1)         | (2)        | (3)        | (4)        | (5)        | (6)        | (7)        | (8)        |
| <b>Panel a: Publications</b>  |             |            |            |            |            |            |            |            |
| <i>Overall</i>                |             |            |            |            |            |            |            |            |
| Cultural Distance             | 14.7539***  | 16.1493*** | 15.3539*** | 15.5375*** | 16.9070*** | 16.9533*** | 16.8115*** | 17.4150*** |
|                               | (0.4325)    | (0.7374)   | (0.5709)   | (0.6164)   | (0.5746)   | (0.6314)   | (0.5861)   | (0.6501)   |
| Model                         | DML         | DRL        | DML        | DRL        | DML        | DRL        | DML        | DRL        |
| Observations                  | 584,712     | 584,712    | 422,434    | 422,434    | 92,914     | 92,914     | 95,676     | 95,676     |
| <i>Mob</i>                    |             |            |            |            |            |            |            |            |
| Cultural Distance             | 6.5142***   | 4.7548*    | 6.3670***  | 3.7912*    | 6.7688***  | 4.8212**   | 7.8316***  | 3.6709*    |
|                               | (1.5109)    | (2.2039)   | (1.5089)   | (1.6061)   | (1.4866)   | (1.6653)   | (1.5066)   | (1.6555)   |
| Model                         | DML         | DRL        | DML        | DRL        | DML        | DRL        | DML        | DRL        |
| Observations                  | 47,838      | 47,838     | 47,014     | 47,014     | 46,457     | 46,457     | 47,838     | 47,838     |
| <b>Panel b: Hit 10pct</b>     |             |            |            |            |            |            |            |            |
| <i>Overall</i>                |             |            |            |            |            |            |            |            |
| Cultural Distance             | 0.0100***   | 0.0155***  | 0.0107***  | 0.0113***  | 0.0093***  | 0.0099***  | 0.0088***  | 0.0098***  |
|                               | (0.0007)    | (0.0015)   | (0.0008)   | (0.0008)   | (0.0009)   | (0.0009)   | (0.0009)   | (0.0010)   |
| Model                         | DML         | DRL        | DML        | DRL        | DML        | DRL        | DML        | DRL        |
| Observations                  | 582,095     | 582,095    | 422,175    | 422,175    | 92,896     | 92,896     | 95,676     | 95,676     |
| <i>Mob</i>                    |             |            |            |            |            |            |            |            |
| Cultural Distance             | -0.0285***  | -0.0169*** | -0.0307*** | -0.0166*** | -0.0306*** | -0.0175*** | -0.0295*** | -0.0192*** |
|                               | (0.0021)    | (0.0021)   | (0.0021)   | (0.0021)   | (0.0021)   | (0.0021)   | (0.0021)   | (0.0021)   |
| Model                         | DML         | DRL        | DML        | DRL        | DML        | DRL        | DML        | DRL        |
| Observations                  | 47,823      | 47,823     | 47,005     | 47,005     | 46,448     | 46,448     | 47,838     | 47,838     |
| <b>Panel c: D5 percentile</b> |             |            |            |            |            |            |            |            |
| <i>Overall</i>                |             |            |            |            |            |            |            |            |
| Cultural Distance             | 0.0142***   | 0.0121***  | 0.0166***  | 0.0165***  | 0.0183***  | 0.0190***  | 0.0137***  | 0.0129***  |
|                               | (0.0014)    | (0.0034)   | (0.0016)   | (0.0017)   | (0.0020)   | (0.0021)   | (0.0020)   | (0.0022)   |
| Model                         | DML         | DRL        | DML        | DRL        | DML        | DRL        | DML        | DRL        |
| Observations                  | 505,385     | 505,385    | 375,229    | 375,229    | 90,136     | 90,136     | 95,200     | 95,200     |
| <i>Mob</i>                    |             |            |            |            |            |            |            |            |
| Cultural Distance             | 0.0427***   | 0.0332***  | 0.0429***  | 0.0314***  | 0.0457***  | 0.0374***  | 0.0452***  | 0.0283***  |
|                               | (0.0043)    | (0.0041)   | (0.0044)   | (0.0046)   | (0.0046)   | (0.0045)   | (0.0045)   | (0.0045)   |
| Model                         | DML         | DRL        | DML        | DRL        | DML        | DRL        | DML        | DRL        |
| Observations                  | 46,399      | 46,399     | 45,658     | 45,658     | 45,108     | 45,108     | 47,739     | 47,739     |

Results from Double Machine Learning (DML) and Doubly Robust Learner (DRL) estimators across the four matching specifications (columns 1–8: no-matching, m: n CEM, 1:1 CEM, 1:1 PSM). For each long-term performance—Publications (Panel a), Hit 10 pct (Panel b) and D5 percentile (Panel c)—average treatment effects of early-career cultural distance are reported for the full sample (“Overall”) and for mobile scientists only (“Mob”). For the m: n CEM



specification, control observations are weighted by the treated-to-control ratio within each stratum and all estimators are fit by weighted least squares. Across all matching  $\times$  estimation combinations, the direction, magnitude and statistical significance of the cultural-distance effect are highly consistent. In the Overall sample the treatment is a binary mobility indicator; in the Mob sample it is the early-career cultural distance percentile for DML and a three-level categorical variable contrasting Longer with Shorter for DRL. Standard errors are reported in parentheses.  $*P < 0.05$ ;  $**P < 0.01$ ;  $***P < 0.001$ .

**Table S6 Robustness of the association between early-career cultural distance and long-term academic performance to alternative distance functions.**

| Cultural Functions            | (1)<br>L1 raw          | (2)<br>L1 pct          | (3)<br>L1 tri          | (4)<br>L2              | (5)<br>L1 raw          | (6)<br>L1 pct          | (7)<br>L1 tri          | (8)<br>L2              |
|-------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| <b>Panel a: Publications</b>  |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.1004***<br>(0.0167)  | 0.0622***<br>(0.0165)  | 0.0343*<br>(0.0161)    | 0.0927***<br>(0.0167)  | 0.1126***<br>(0.0188)  | 0.0839***<br>(0.0184)  | 0.0775***<br>(0.0167)  | 0.1003***<br>(0.0188)  |
| Controls                      |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 46457                  | 46457                  | 46457                  | 46457                  | 46457                  | 46457                  | 46457                  | 46457                  |
| Pseudo-R <sup>2</sup>         | 0.001                  | 0.000                  | 0.000                  | 0.001                  | 0.151                  | 0.150                  | 0.150                  | 0.150                  |
| <b>Panel b: Hit 10pct</b>     |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | -0.0628***<br>(0.0022) | -0.0608***<br>(0.0022) | -0.0661***<br>(0.0022) | -0.0585***<br>(0.0022) | -0.0317***<br>(0.0021) | -0.0290***<br>(0.0021) | -0.0337***<br>(0.0021) | -0.0285***<br>(0.0021) |
| Controls                      |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 46448                  | 46448                  | 46448                  | 46448                  | 46448                  | 46448                  | 46448                  | 46448                  |
| R <sup>2</sup>                | 0.017                  | 0.016                  | 0.018                  | 0.014                  | 0.169                  | 0.169                  | 0.170                  | 0.169                  |
| <b>Panel c: D5 percentile</b> |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0738***<br>(0.0046)  | 0.0766***<br>(0.0046)  | 0.0728***<br>(0.0046)  | 0.0697***<br>(0.0046)  | 0.0484***<br>(0.0046)  | 0.0514***<br>(0.0046)  | 0.0474***<br>(0.0046)  | 0.0447***<br>(0.0046)  |
| Controls                      |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 45108                  | 45108                  | 45108                  | 45108                  | 45108                  | 45108                  | 45108                  | 45108                  |
| R <sup>2</sup>                | 0.006                  | 0.006                  | 0.006                  | 0.005                  | 0.061                  | 0.062                  | 0.061                  | 0.061                  |

Panel **a** present result for long-term productivity, Panel **b** for impact, and Panel **c** for disruptiveness. Columns 1–4 use no additional controls; columns 5–8 include early-career controls. Columns (1, 5) L1 on raw Hofstede scores (L1 raw); (2, 6) L1 on percentile-transformed Hofstede scores (L1 pct); (3, 7) normalized L1 using only unique country–country pairs (L1 tri); (4, 8) normalized Euclidean (L2); Across all specifications, the patterns of long-term academic performance within Mob scientists are consistent. Estimates are reported for mobile scientists only, because non-mobile scientists are assigned a distance of zero under every metric and the Mob versus non-Mob contrast is therefore identical across specifications. Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

**Table S7 Robustness of the association between early-career cultural distance and long-term academic performance to embedding-based measures of cultural distance.**

|                               | (1)                        | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (7)                    | (8)                    | (9)                    | (10)                   |
|-------------------------------|----------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| Models                        | Word2<br>vec               | GCL<br>(uw)            | GMAE<br>(uw)           | GCL<br>(w)             | GMAE<br>(w)            | Word2<br>vec           | GCL<br>(uw)            | GMAE<br>(uw)           | GCL<br>(w)             | GMAE<br>(w)            |
| <b>Panel a: Publications</b>  |                            |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| <i>Overall</i>                |                            |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.2262***<br>(0.0074)      | 0.2262***<br>(0.0074)  | 0.2262***<br>(0.0074)  | 0.2262***<br>(0.0074)  | 0.2262***<br>(0.0074)  | 0.2311***<br>(0.0086)  | 0.2311***<br>(0.0086)  | 0.2311***<br>(0.0086)  | 0.2311***<br>(0.0086)  | 0.2311***<br>(0.0086)  |
| Controls                      |                            |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 92914                      | 92914                  | 92914                  | 92914                  | 92914                  | 92914                  | 92914                  | 92914                  | 92914                  | 92914                  |
| Pseudo-R <sup>2</sup>         | 0.014                      | 0.014                  | 0.014                  | 0.014                  | 0.014                  | 0.137                  | 0.137                  | 0.137                  | 0.137                  | 0.137                  |
| <i>Mob</i>                    |                            |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0186<br>(0.0166)         | 0.0730***<br>(0.0172)  | 0.0360*<br>(0.0168)    | 0.0431*<br>(0.0168)    | 0.0923***<br>(0.0168)  | 0.0761***<br>(0.0193)  | 0.1422***<br>(0.0171)  | 0.1050***<br>(0.0168)  | 0.0783***<br>(0.0169)  | 0.1186***<br>(0.0187)  |
| Controls                      |                            |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 46523                      | 46523                  | 46523                  | 46523                  | 46523                  | 46523                  | 46523                  | 46523                  | 46523                  | 46523                  |
| Pseudo-R <sup>2</sup>         | 0.000                      | 0.001                  | 0.000                  | 0.000                  | 0.001                  | 0.150                  | 0.152                  | 0.151                  | 0.150                  | 0.151                  |
| <b>Panel b: Hit 10pct</b>     |                            |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| <i>Overall</i>                |                            |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0140***<br>(0.0009)      | 0.0140***<br>(0.0009)  | 0.0140***<br>(0.0009)  | 0.0140***<br>(0.0009)  | 0.0140***<br>(0.0009)  | 0.0134***<br>(0.0008)  | 0.0134***<br>(0.0008)  | 0.0134***<br>(0.0008)  | 0.0134***<br>(0.0008)  | 0.0134***<br>(0.0008)  |
| Controls                      |                            |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 92896                      | 92896                  | 92896                  | 92896                  | 92896                  | 92896                  | 92896                  | 92896                  | 92896                  | 92896                  |
| R <sup>2</sup>                | 0.002                      | 0.002                  | 0.002                  | 0.002                  | 0.002                  | 0.185                  | 0.185                  | 0.185                  | 0.185                  | 0.185                  |
| <i>Mob</i>                    |                            |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | -<br>0.0668***<br>(0.0022) | -0.0476***<br>(0.0022) | -0.0504***<br>(0.0022) | -0.0290***<br>(0.0023) | -0.0436***<br>(0.0023) | -0.0311***<br>(0.0021) | -0.0237***<br>(0.0021) | -0.0190***<br>(0.0021) | -0.0184***<br>(0.0021) | -0.0179***<br>(0.0021) |
| Controls                      |                            |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 46514                      | 46514                  | 46514                  | 46514                  | 46514                  | 46514                  | 46514                  | 46514                  | 46514                  | 46514                  |
| R <sup>2</sup>                | 0.019                      | 0.010                  | 0.011                  | 0.004                  | 0.008                  | 0.169                  | 0.168                  | 0.167                  | 0.167                  | 0.167                  |
| <b>Panel c: D5 percentile</b> |                            |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| <i>Overall</i>                |                            |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0201***<br>(0.0019)      | 0.0201***<br>(0.0019)  | 0.0201***<br>(0.0019)  | 0.0201***<br>(0.0019)  | 0.0201***<br>(0.0019)  | 0.0199***<br>(0.0019)  | 0.0199***<br>(0.0019)  | 0.0199***<br>(0.0019)  | 0.0199***<br>(0.0019)  | 0.0199***<br>(0.0019)  |
| Controls                      |                            |                        |                        |                        |                        | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                  | 90136                      | 90136                  | 90136                  | 90136                  | 90136                  | 90136                  | 90136                  | 90136                  | 90136                  | 90136                  |
| R <sup>2</sup>                | 0.001                      | 0.001                  | 0.001                  | 0.001                  | 0.001                  | 0.052                  | 0.052                  | 0.052                  | 0.052                  | 0.052                  |
| <i>Mob</i>                    |                            |                        |                        |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance             | 0.0582***<br>(0.0046)      | 0.0493***<br>(0.0046)  | 0.0575***<br>(0.0046)  | 0.0193***<br>(0.0046)  | 0.0525***<br>(0.0046)  | 0.0345***<br>(0.0046)  | 0.0322***<br>(0.0045)  | 0.0368***<br>(0.0046)  | 0.0116**<br>(0.0045)   | 0.0346***<br>(0.0045)  |

|                |       |       |       |       |       |       |       |       |       |       |
|----------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| Controls       |       |       |       |       |       | Yes   | Yes   | Yes   | Yes   | Yes   |
| Observations   | 45173 | 45173 | 45173 | 45173 | 45173 | 45173 | 45173 | 45173 | 45173 | 45173 |
| R <sup>2</sup> | 0.004 | 0.003 | 0.003 | 0.000 | 0.003 | 0.060 | 0.060 | 0.061 | 0.059 | 0.060 |

Panel **a** present result for long-term productivity, Panel **b** for impact, and Panel **c** for disruptiveness. We report regression results using five alternative embedding-based measures of early-career cultural distance: Word2vec, GCL (unweighted), GMAE (unweighted), GCL (weighted), and GMAE (weighted). Columns (1)– (5) report results without controls; columns (6)–(10) report results with controls included. Compared to the Hofstede-based analysis, the Mob sample size is larger with embedding-based cultural distances because the embeddings cover all countries represented in the data—including those not in the Hofstede database—allowing us to compute distances for every mobility trajectory. As a result, mobile scientists involving countries lacking Hofstede scores can now be included. For each outcome, coefficients for the “Overall” group are identical across all five embedding-based measures. This is because all embedding approaches assign a distance of zero to non-mobile scientists, and—since the embeddings provide complete coverage—the same set of scientists receives a non-zero distance under every method. Thus, the Overall group regressions are based on an identical dichotomy (zero vs. non-zero) across all embeddings, regardless of the specific method used. Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

**Table S8 Robustness of the association between cultural distance and long-term scientific impact to alternative impact definitions and field classification granularity.**

| Impact indicators     | Field LV0              |                        |                        |                        | Field LV1              |                        |                        |
|-----------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                       | Hit 1pct               | Hit 5pct               | Hit 10pct              | FY C                   | Hit 1pct               | Hit 5pct               | FY C                   |
| <b><i>Overall</i></b> |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance     | 0.0007**<br>(0.0003)   | 0.0027***<br>(0.0006)  | 0.0053***<br>(0.0008)  | 0.0446**<br>(0.0153)   | 0.0005*<br>(0.0003)    | 0.0029***<br>(0.0006)  | 0.0097***<br>(0.0011)  |
| Reference Count       | 0.0006***<br>(0.0000)  | 0.0020***<br>(0.0000)  | 0.0032***<br>(0.0000)  | 0.0337***<br>(0.0006)  | 0.0006***<br>(0.0000)  | 0.0018***<br>(0.0000)  | 0.0043***<br>(0.0000)  |
| Controls              | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations          | 92896                  | 92896                  | 92896                  | 92896                  | 92893                  | 92893                  | 92893                  |
| R <sup>2</sup>        | 0.123                  | 0.221                  | 0.262                  | 0.113                  | 0.123                  | 0.214                  | 0.248                  |
| <b><i>Mob</i></b>     |                        |                        |                        |                        |                        |                        |                        |
| Cultural Distance     | -0.0035***<br>(0.0007) | -0.0153***<br>(0.0016) | -0.0281***<br>(0.0021) | -0.2412***<br>(0.0378) | -0.0038***<br>(0.0007) | -0.0168***<br>(0.0015) | -0.0397***<br>(0.0029) |
| Reference Count       | 0.0006***<br>(0.0000)  | 0.0020***<br>(0.0000)  | 0.0031***<br>(0.0000)  | 0.0339***<br>(0.0008)  | 0.0006***<br>(0.0000)  | 0.0018***<br>(0.0000)  | 0.0041***<br>(0.0001)  |
| Controls              | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations          | 46448                  | 46448                  | 46448                  | 46448                  | 46448                  | 46448                  | 46448                  |
| R <sup>2</sup>        | 0.119                  | 0.210                  | 0.250                  | 0.109                  | 0.117                  | 0.200                  | 0.228                  |

Regression results using seven alternative impact measures, defined by different high-impact thresholds (top 1%, 5%, 10%) and a continuous normalized citation metric (FY C), calculated at both the MAG Field LV1 and LV0 levels. Each metric is computed as either the proportion of publications ranking among the top cited within field and year (Hits) or the mean citation count normalized by the mean in the same field-year group (FY C). All models include reference count and five early-career controls. For FY C, values represent the average multiplicative factor of citations relative to the field-year mean, therefore, for regression analyses, FY C is log-transformed. Consistent with this approach, all subsequent field- and year-normalized measures expressed as a ratio to the group mean and citation indicators are log-transformed for regression analyses, as these measures typically exhibit substantial right skew. Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

**Table S9 Robustness of the association between cultural distance and long-term impact to minimum reference list length.**

|                       | (1)        | (2)         | (3)         | (4)         |
|-----------------------|------------|-------------|-------------|-------------|
| <b><i>Overall</i></b> |            |             |             |             |
| Cultural              | 0.0024*    | 0.0031**    | 0.0046***   | 0.0074***   |
| Distance              | (0.0010)   | (0.0011)    | (0.0012)    | (0.0013)    |
| Reference             | 0.0024***  | 0.0018***   | 0.0013***   | 0.0007***   |
| Count                 | (0.0000)   | (0.0000)    | (0.0000)    | (0.0001)    |
| Reference             | At-least-5 | At-least-10 | At-least-15 | At-least-20 |
| thresholds            | references | references  | references  | references  |
| Controls              | Yes        | Yes         | Yes         | Yes         |
| Observations          | 92651      | 92307       | 91616       | 90422       |
| R <sup>2</sup>        | 0.195      | 0.164       | 0.132       | 0.109       |
| <b><i>Mob</i></b>     |            |             |             |             |
| Cultural              | -0.0409*** | -0.0432***  | -0.0449***  | -0.0461***  |
| Distance              | (0.0025)   | (0.0027)    | (0.0029)    | (0.0032)    |
| Reference             | 0.0024***  | 0.0019***   | 0.0013***   | 0.0007***   |
| Count                 | (0.0001)   | (0.0001)    | (0.0001)    | (0.0001)    |
| Reference             | At-least-5 | At-least-10 | At-least-15 | At-least-20 |
| thresholds            | references | references  | references  | references  |
| Controls              | Yes        | Yes         | Yes         | Yes         |
| Observations          | 46388      | 46288       | 46066       | 45629       |
| R <sup>2</sup>        | 0.193      | 0.164       | 0.131       | 0.106       |

Regression results using different minimum thresholds for the number of cited references per paper ( $\geq 5$ ,  $\geq 10$ ,  $\geq 15$ ,  $\geq 20$ ). All models including reference count and early-career controls. The advantage of Mob scientists in scientific impact and the association of shorter early-career cultural distance with higher long-term impact—remain robust across all sample restrictions. Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

**Table S10 Robustness of the association between cultural distance and long-term impact using citations and normalized citation indicators.**

|                   | (1)<br>C5              | (2)<br>FW C5           | (3)<br>C10             | (4)<br>MF C5           | (5)<br>BC C5           |
|-------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| <i>Overall</i>    |                        |                        |                        |                        |                        |
| Cultural Distance | 0.0099***<br>(0.0016)  | 0.0091***<br>(0.0016)  | 0.0104***<br>(0.0018)  | 0.0051**<br>(0.0017)   | 0.0054***<br>(0.0007)  |
| Reference Count   | 0.0151***<br>(0.0001)  | 0.0149***<br>(0.0001)  | 0.0155***<br>(0.0001)  | 0.0142***<br>(0.0001)  | 0.0028***<br>(0.0000)  |
| Controls          | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations      | 92896                  | 92896                  | 92896                  | 92896                  | 92895                  |
| R <sup>2</sup>    | 0.482                  | 0.477                  | 0.461                  | 0.418                  | 0.242                  |
| <i>Mob</i>        |                        |                        |                        |                        |                        |
| Cultural Distance | -0.0312***<br>(0.0040) | -0.0321***<br>(0.0039) | -0.0438***<br>(0.0043) | -0.0338***<br>(0.0043) | -0.0195***<br>(0.0016) |
| Reference Count   | 0.0148***<br>(0.0001)  | 0.0145***<br>(0.0001)  | 0.0150***<br>(0.0001)  | 0.0142***<br>(0.0001)  | 0.0027***<br>(0.0000)  |
| Controls          | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations      | 46448                  | 46448                  | 46448                  | 46448                  | 46448                  |
| R <sup>2</sup>    | 0.487                  | 0.481                  | 0.460                  | 0.429                  | 0.233                  |

Regression results using five alternative citation-based indicators of scientific impact: five-year citation count (C5), full-window citation count up to 2022 (FW C5), ten-year citation count (C10), main field-normalized C5 (MF C5), and network-based normalized C5 (BC C5). Each model includes reference count and five early-career controls. Results are reported for both the full sample (Overall) and the Mob scientist subsample (Mob). Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

**Table S11 Robustness of the association between cultural distance and long-term disruptiveness to alternative DI-variants.**

|                | (1)<br>D5 values | (2)<br>D10 values | (3)<br>D10 | (4)<br>D5 pct | (5)<br>D5>0 | (6)<br>MF D5 |
|----------------|------------------|-------------------|------------|---------------|-------------|--------------|
| <i>Overall</i> |                  |                   |            |               |             |              |
| Cultural       | 0.0017***        | 0.0016***         | 0.0200***  | 0.0031***     | 0.0024      | 0.0193***    |
| Distance       | (0.0002)         | (0.0002)          | (0.0019)   | (0.0007)      | (0.0017)    | (0.0019)     |
| Reference      | 0.0000           | -0.0001***        | -0.0024*** | -0.0012***    | -0.0100***  | -0.0004***   |
| Count          | (0.0000)         | (0.0000)          | (0.0001)   | (0.0000)      | (0.0001)    | (0.0001)     |
| Controls       | Yes              | Yes               | Yes        | Yes           | Yes         | Yes          |
| Observations   | 90136            | 90103             | 90103      | 90133         | 90133       | 90013        |
| R <sup>2</sup> | 0.015            | 0.017             | 0.062      | 0.099         | 0.247       | 0.044        |
| <i>Mob</i>     |                  |                   |            |               |             |              |
| Cultural       | 0.0014***        | 0.0012**          | 0.0425***  | 0.0226***     | 0.0269***   | 0.0467***    |
| Distance       | (0.0004)         | (0.0004)          | (0.0046)   | (0.0017)      | (0.0041)    | (0.0046)     |
| Reference      | -0.0000***       | -0.0001***        | -0.0027*** | -0.0012***    | -0.0100***  | -0.0005***   |
| Count          | (0.0000)         | (0.0000)          | (0.0001)   | (0.0000)      | (0.0001)    | (0.0001)     |
| Controls       | Yes              | Yes               | Yes        | Yes           | Yes         | Yes          |
| Observations   | 45108            | 45097             | 45097      | 45108         | 45108       | 45072        |
| R <sup>2</sup> | 0.023            | 0.028             | 0.075      | 0.124         | 0.275       | 0.053        |

Regression results using multiple variants of the disruption index. Columns show (1) raw five-year disruption index (D5 values), (2) raw ten-year disruption index (D10 values), (3) D10 percentile, (4) D5 pct, (5) proportion of papers with D5 > 0, and (6) main field D5 calculated using MAG LV0. Aside from the raw D5, D10, and already percentile-scaled D5 pct, all other disruption indicators are converted to percentiles to account for skewed distributions. For the D5 > 0 indicator, we also fit a fractional-logit model on the raw proportion of disruptive papers, and the estimated effects remain robust and consistent with those based on percentile-transformed values. All models include reference count and five early-career controls. Except for column (1), where D5 values are computed using disruption scores supplemented with OpenAlex data after 2017 (consistent with the main text), all other D5 columns are based only on disruption values available up to 2017. Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .



**Table S12 Robustness of the association between cultural distance and disruptiveness to reference and citation count thresholds.**

|                       | (1)                      | (2)                       | (3)                       | (4)                       | (5)                      | (6)                       | (7)                       | (8)                       |
|-----------------------|--------------------------|---------------------------|---------------------------|---------------------------|--------------------------|---------------------------|---------------------------|---------------------------|
| <b>Overall</b>        |                          |                           |                           |                           |                          |                           |                           |                           |
| Cultural Distance     | 0.0083***<br>(0.0019)    | 0.0096***<br>(0.0019)     | 0.0094***<br>(0.0019)     | 0.0071***<br>(0.0019)     | 0.0049**<br>(0.0019)     | 0.0035<br>(0.0019)        | 0.0033<br>(0.0020)        | 0.0035<br>(0.0021)        |
| Reference Count       | 0.0015***<br>(0.0001)    | 0.0018***<br>(0.0001)     | 0.0012***<br>(0.0001)     | 0.0004***<br>(0.0001)     | 0.0031***<br>(0.0001)    | 0.0037***<br>(0.0001)     | 0.0031***<br>(0.0001)     | 0.0025***<br>(0.0001)     |
| Reference thresholds  | At-least-5<br>references | At-least-10<br>references | At-least-15<br>references | At-least-20<br>references | At-least-5<br>references | At-least-10<br>references | At-least-15<br>references | At-least-20<br>references |
| Citation thresholds   |                          |                           |                           |                           | At-least-5<br>citations  | At-least-10<br>citations  | At-least-15<br>citations  | At-least-20<br>citations  |
| Controls              | Yes                      | Yes                       | Yes                       | Yes                       | Yes                      | Yes                       | Yes                       | Yes                       |
| Observations          | 89993                    | 89630                     | 88922                     | 87729                     | 88250                    | 83250                     | 76785                     | 69853                     |
| Pseudo-R <sup>2</sup> | 0.068                    | 0.075                     | 0.074                     | 0.071                     | 0.077                    | 0.081                     | 0.070                     | 0.058                     |
| <b>Mob</b>            |                          |                           |                           |                           |                          |                           |                           |                           |
| Cultural Distance     | 0.0585***<br>(0.0045)    | 0.0650***<br>(0.0045)     | 0.0693***<br>(0.0045)     | 0.0705***<br>(0.0046)     | 0.0533***<br>(0.0045)    | 0.0533***<br>(0.0047)     | 0.0542***<br>(0.0049)     | 0.0576***<br>(0.0051)     |
| Reference Count       | 0.0013***<br>(0.0001)    | 0.0014***<br>(0.0001)     | 0.0008***<br>(0.0001)     | 0.0001<br>(0.0001)        | 0.0028***<br>(0.0001)    | 0.0035***<br>(0.0001)     | 0.0029***<br>(0.0001)     | 0.0022***<br>(0.0001)     |
| Reference thresholds  | At-least-5<br>references | At-least-10<br>references | At-least-15<br>references | At-least-20<br>references | At-least-5<br>references | At-least-10<br>references | At-least-15<br>references | At-least-20<br>references |
| Citation thresholds   |                          |                           |                           |                           | At-least-5<br>citations  | At-least-10<br>citations  | At-least-15<br>citations  | At-least-20<br>citations  |
| Controls              | Yes                      | Yes                       | Yes                       | Yes                       | Yes                      | Yes                       | Yes                       | Yes                       |
| Observations          | 45070                    | 44952                     | 44730                     | 44284                     | 44477                    | 42469                     | 39715                     | 36525                     |
| R <sup>2</sup>        | 0.081                    | 0.086                     | 0.088                     | 0.086                     | 0.086                    | 0.088                     | 0.077                     | 0.063                     |

Regression results for D5 percentile using different minimum thresholds for the number of cited references and citing papers per publication. Columns 1–4 restrict the sample to papers with at least 5, 10, 15, or 20 references. Columns 5–8 further restrict the sample to papers meeting both minimum reference and citation thresholds (e.g., at least 5, 10, 15, or 20 for both references and citations). All models include reference count and early-career controls. Across all thresholds, the positive association between early-career cultural distance and long-term disruptiveness—both overall and among Mob scientists—remains robust. Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

**Table S13 Robustness of the association between early-career cultural distance and long-term disruptiveness to field-year and bibliographic coupling normalization.**

| DIs            | Overall   |            | Mob      |            |
|----------------|-----------|------------|----------|------------|
|                | FY DI     | BC DI      | FY DI    | BC DI      |
| Cultural       | 0.0002*** | 0.0001**   | 0.0002** | 0.0002**   |
| Distance       | (0.0000)  | (0.0000)   | (0.0001) | (0.0001)   |
| Reference      | 0.0000    | -0.0000*** | -0.0000* | -0.0000*** |
| Count          | (0.0000)  | (0.0000)   | (0.0000) | (0.0000)   |
| Controls       | Yes       | Yes        | Yes      | Yes        |
| Observations   | 90124     | 90098      | 45106    | 45098      |
| R <sup>2</sup> | 0.005     | 0.013      | 0.006    | 0.022      |

Regression results estimating the effect of early-career cultural distance on long-term disruptiveness index (DI) using two alternative normalization strategies: field-year normalized D5 (FY DI) and bibliographic coupling normalized D5 (BC DI). Columns report results for the full sample (Overall) and for mobile scientists only (Mob). Standard errors are reported in parentheses. \* $P < 0.05$ ; \*\* $P < 0.01$ ; \*\*\* $P < 0.001$ .

**Table S14 Robustness of early-career cultural distance effect on long-term academic performance to stepwise inclusion of controls (Country, Academic distance, Reference, Field, and Year).**

|                              | (1)                   | (2)                    | (3)                    | (4)                    | (5)                    |
|------------------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|
| <b>Panel a: Publications</b> |                       |                        |                        |                        |                        |
| <i>Overall</i>               |                       |                        |                        |                        |                        |
| Cultural Distance            | 0.1904***<br>(0.0138) | 0.1639***<br>(0.0145)  | 0.1512***<br>(0.0146)  | 0.1234***<br>(0.0147)  | 0.1132***<br>(0.0146)  |
| Controls                     | Yes                   | Yes                    | Yes                    | Yes                    | Yes                    |
| Country FE                   | Yes                   | Yes                    | Yes                    | Yes                    | Yes                    |
| Academic Distance            |                       | 0.1630***<br>(0.0408)  | 0.1797***<br>(0.0410)  | 0.1914***<br>(0.0419)  | 0.1673***<br>(0.0406)  |
| Reference count              |                       |                        | 0.0047***<br>(0.0003)  | 0.0060***<br>(0.0003)  | 0.0029***<br>(0.0003)  |
| Fields FE                    |                       |                        |                        | Yes                    | Yes                    |
| Year FE                      |                       |                        |                        |                        | Yes                    |
| Observations                 | 92914                 | 92914                  | 92914                  | 92914                  | 92914                  |
| Pseudo-R <sup>2</sup>        | 0.1399                | 0.1401                 | 0.1446                 | 0.1653                 | 0.1782                 |
| <i>Mob</i>                   |                       |                        |                        |                        |                        |
| Cultural Distance            | 0.1214***<br>(0.0194) | 0.1695***<br>(0.0227)  | 0.1653***<br>(0.0226)  | 0.1086***<br>(0.0223)  | 0.0603**<br>(0.0217)   |
| Controls                     | Yes                   | Yes                    | Yes                    | Yes                    | Yes                    |
| Country FE                   | Yes                   | Yes                    | Yes                    | Yes                    | Yes                    |
| Academic Distance            |                       | -0.5028***<br>(0.1103) | -0.4352***<br>(0.1093) | -0.3531***<br>(0.1066) | -0.3222**<br>(0.1037)  |
| Reference count              |                       |                        | 0.0035***<br>(0.0004)  | 0.0054***<br>(0.0004)  | 0.0025***<br>(0.0004)  |
| Fields FE                    |                       |                        |                        | Yes                    | Yes                    |
| Year FE                      |                       |                        |                        |                        | Yes                    |
| Observations                 | 46457                 | 46457                  | 46457                  | 46457                  | 46457                  |
| Pseudo-R <sup>2</sup>        | 0.1521                | 0.1528                 | 0.1555                 | 0.1745                 | 0.1869                 |
| <b>Panel b: Hit 10pct</b>    |                       |                        |                        |                        |                        |
| <i>Overall</i>               |                       |                        |                        |                        |                        |
| Cultural Distance            | 0.0105***<br>(0.0016) | 0.0171***<br>(0.0017)  | 0.0100***<br>(0.0016)  | 0.0076***<br>(0.0016)  | 0.0083***<br>(0.0016)  |
| Controls                     | Yes                   | Yes                    | Yes                    | Yes                    | Yes                    |
| Country FE                   | Yes                   | Yes                    | Yes                    | Yes                    | Yes                    |
| Academic Distance            |                       | -0.0441***<br>(0.0047) | -0.0353***<br>(0.0045) | -0.0348***<br>(0.0044) | -0.0330***<br>(0.0044) |
| Reference count              |                       |                        | 0.0028***<br>(0.0000)  | 0.0032***<br>(0.0000)  | 0.0034***<br>(0.0000)  |
| Fields FE                    |                       |                        |                        | Yes                    | Yes                    |

|                               |            |            |            |            |            |
|-------------------------------|------------|------------|------------|------------|------------|
| Year FE                       |            |            |            |            | Yes        |
| Observations                  | 92896      | 92896      | 92896      | 92896      | 92896      |
| R <sup>2</sup>                | 0.2001     | 0.2008     | 0.2662     | 0.2849     | 0.2879     |
| <b><i>Mob</i></b>             |            |            |            |            |            |
| Cultural                      | -0.0209*** | -0.0111*** | -0.0145*** | -0.0169*** | -0.0132*** |
| Distance                      | (0.0023)   | (0.0026)   | (0.0025)   | (0.0025)   | (0.0025)   |
| Controls                      | Yes        | Yes        | Yes        | Yes        | Yes        |
| Country FE                    | Yes        | Yes        | Yes        | Yes        | Yes        |
| Academic                      |            | -0.0991*** | -0.0487*** | -0.0596*** | -0.0589*** |
| Distance                      |            | (0.0124)   | (0.0119)   | (0.0117)   | (0.0117)   |
| Reference                     |            |            | 0.0028***  | 0.0032***  | 0.0036***  |
| count                         |            |            | (0.0000)   | (0.0000)   | (0.0000)   |
| Fields FE                     |            |            |            | Yes        | Yes        |
| Year FE                       |            |            |            |            | Yes        |
| Observations                  | 46448      | 46448      | 46448      | 46448      | 46448      |
| R <sup>2</sup>                | 0.1729     | 0.1740     | 0.2428     | 0.2677     | 0.2731     |
| <b>Panel c: D5 Percentile</b> |            |            |            |            |            |
| <b><i>Overall</i></b>         |            |            |            |            |            |
| Cultural                      | 0.0226***  | 0.0239***  | 0.0259***  | 0.0179***  | 0.0133***  |
| Distance                      | (0.0035)   | (0.0038)   | (0.0038)   | (0.0037)   | (0.0037)   |
| Controls                      | Yes        | Yes        | Yes        | Yes        | Yes        |
| Country FE                    | Yes        | Yes        | Yes        | Yes        | Yes        |
| Academic                      |            | -0.0087    | -0.0113    | -0.0118    | -0.0248*   |
| Distance                      |            | (0.0105)   | (0.0105)   | (0.0102)   | (0.0101)   |
| Reference                     |            |            | -0.0008*** | 0.0003***  | -0.0012*** |
| count                         |            |            | (0.0001)   | (0.0001)   | (0.0001)   |
| Fields FE                     |            |            |            | Yes        | Yes        |
| Year FE                       |            |            |            |            | Yes        |
| Observations                  | 90136      | 90136      | 90136      | 90136      | 90136      |
| R <sup>2</sup>                | 0.0521     | 0.0521     | 0.0534     | 0.0983     | 0.1240     |
| <b><i>Mob</i></b>             |            |            |            |            |            |
| Cultural                      | 0.0518***  | 0.0403***  | 0.0415***  | 0.0347***  | 0.0165**   |
| Distance                      | (0.0051)   | (0.0057)   | (0.0057)   | (0.0056)   | (0.0056)   |
| Controls                      | Yes        | Yes        | Yes        | Yes        | Yes        |
| Country FE                    | Yes        | Yes        | Yes        | Yes        | Yes        |
| Academic                      |            | 0.1174***  | 0.0986***  | 0.0640*    | 0.0620*    |
| Distance                      |            | (0.0269)   | (0.0269)   | (0.0263)   | (0.0260)   |
| Reference                     |            |            | -0.0010*** | 0.0001     | -0.0013*** |
| count                         |            |            | (0.0001)   | (0.0001)   | (0.0001)   |
| Fields FE                     |            |            |            | Yes        | Yes        |
| Year FE                       |            |            |            |            | Yes        |
| Observations                  | 45108      | 45108      | 45108      | 45108      | 45108      |

| R <sup>2</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.0622 | 0.0626 | 0.0649 | 0.1086 | 0.1318 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|--------|
| <p>We report the estimated effects of early-career cultural distance on long-term academic performance, with sequential inclusion of reference count (6-25), field, year, country, and academic distance controls. Each outcome is shown for three performance indicators: productivity (Panel <b>a</b>, Publications), impact (Panel <b>b</b>, Hit 10pct), and disruptiveness (Panel <b>c</b>, D5 percentile). Columns (1)– (5) correspond to the stepwise addition of controls as follows: (1) Country fixed effects (Country FE); (2) + Academic distance (AD); (3) + Reference count; (4) + Field fixed effects (Fields FE); (5) + Year (career-start year) fixed effects (Year FE). Estimates are based on the CEM-matched sample. The results demonstrate that the association between early-career cultural distance and long-term academic performance remains robust and largely significant after stepwise adjustment for reference, field, year, country, and academic distance. Standard errors are reported in parentheses. *<math>P &lt; 0.05</math>; **<math>P &lt; 0.01</math>; ***<math>P &lt; 0.001</math>.</p> |        |        |        |        |        |

**Table S15 Labour advantages and team reputation as potential channels of the long-term advantages of Mob scientists.**

**Panel a: Means**

| Variables       | Group   | Publications |        |         | Hit 10pct |        |         |
|-----------------|---------|--------------|--------|---------|-----------|--------|---------|
|                 |         | Lower        | Higher | P-value | Lower     | Higher | P-value |
| Coauthor Count  | non-Mob | 35.20        | 112.63 | <0.001  |           |        |         |
|                 | Mob     | 43.79        | 129.65 | <0.001  |           |        |         |
| Team Reputation | non-Mob |              |        |         | 0.1244    | 0.1333 | <0.001  |
|                 | Mob     |              |        |         | 0.2704    | 0.2737 | <0.001  |

**Panel b: Publications**

|                                           | (1)                   | (2)                   | (3)                    | (4)                    |
|-------------------------------------------|-----------------------|-----------------------|------------------------|------------------------|
| Team Country Size                         | 0.0493***<br>(0.0023) |                       | -0.0288***<br>(0.0046) | 0.0044<br>(0.0041)     |
| Coauthor Count                            |                       | 0.0001***<br>(0.0000) | 0.0001***<br>(0.0000)  | 0.0002***<br>(0.0000)  |
| Team Country Size $\times$ Coauthor Count |                       |                       |                        | -0.0000***<br>(0.0000) |
| Cultural Distance                         | 0.2036***<br>(0.0079) | 0.2083***<br>(0.0078) | 0.2052***<br>(0.0085)  | 0.1914***<br>(0.0071)  |
| Controls                                  | Yes                   | Yes                   | Yes                    | Yes                    |
| Observations                              | 90162                 | 90304                 | 90162                  | 90162                  |
| Pseudo-R <sup>2</sup>                     | 0.214                 | 0.234                 | 0.242                  | 0.277                  |

**Panel c Hit 10pct**

|                                            | (1)                   | (2)                   | (3)                   | (4)                    |
|--------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------|
| Team Country Size                          | 0.0129***<br>(0.0002) |                       | 0.0076***<br>(0.0002) | -0.0076***<br>(0.0012) |
| Team Reputation                            |                       | 0.2766***<br>(0.0014) | 0.2681***<br>(0.0014) | 0.2482***<br>(0.0021)  |
| Team Country Size $\times$ Team Reputation |                       |                       |                       | 0.0163***<br>(0.0012)  |
| Cultural Distance                          | 0.0109***<br>(0.0008) | 0.0033***<br>(0.0007) | 0.0024***<br>(0.0007) | 0.0033***<br>(0.0007)  |
| Controls                                   | Yes                   | Yes                   | Yes                   | Yes                    |
| Observations                               | 90162                 | 90304                 | 90162                 | 90162                  |
| R <sup>2</sup>                             | 0.214                 | 0.422                 | 0.432                 | 0.433                  |

Panel a reports group means of long-term performance for scientists in the lower and higher halves of each collaboration measure, split at the median of the full matched sample, with two-sided Mann–Whitney *U* tests. Panel b reports Poisson regressions of Publications with robust standard errors, and Panel c ordinary-least-squares regressions of Hit 10pct; both include the

five early-career covariates used for matching. Team country size is the mean number of countries represented per paper in years 6–25, and team reputation is the mean cumulative number of top-1% cited papers among a paper’s coauthors up to the previous year, rank-transformed because of its right-skewed distribution. Super-team papers with more than 29 authors are excluded. Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

**Table S16 Labour advantages and team reputation as potential channels of the divergence between shorter- and longer-distance movers.**

**Panel a: Means**

| Variables       | Group   | Publications |        |         | Hit 10pct |        |         |
|-----------------|---------|--------------|--------|---------|-----------|--------|---------|
|                 |         | Lower        | Higher | P-value | Lower     | Higher | P-value |
| Team Size       | Shorter | 72.48        | 106.76 | <0.001  |           |        |         |
|                 | Longer  | 78.68        | 106.65 | <0.001  |           |        |         |
| Team Reputation | Shorter |              |        |         | 0.2649    | 0.2854 | <0.001  |
|                 | Longer  |              |        |         | 0.1208    | 0.1504 | <0.001  |

**Panel b: Publications**

|                                            | (1)                   | (2)                   | (3)                   | (4)                   |
|--------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Repeat Col. Interval (Inverse)             | 2.3846***<br>(0.0412) |                       | 2.4216***<br>(0.0410) | 2.3126***<br>(0.1143) |
| Team Size                                  |                       | 0.0538***<br>(0.0032) | 0.0578***<br>(0.0030) | 0.0460***<br>(0.0127) |
| Repeat Col. Interval (Inverse) × Team Size |                       |                       |                       | 0.0202<br>(0.0234)    |
| Cultural Distance                          | 0.0294<br>(0.0168)    | 0.0726***<br>(0.0180) | -0.0022<br>(0.0159)   | -0.0023<br>(0.0160)   |
| Controls                                   | Yes                   | Yes                   | Yes                   | Yes                   |
| Observations                               | 44769                 | 45081                 | 44769                 | 44769                 |
| Pseudo-R <sup>2</sup>                      | 0.330                 | 0.194                 | 0.361                 | 0.361                 |

**Panel c Hit 10pct**

|                                          | (1)                    | (2)                    | (3)                    | (4)                    |
|------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| Team New Collaborators                   | 0.0092***<br>(0.0006)  |                        | -0.0009<br>(0.0005)    | -0.0135***<br>(0.0012) |
| Team Reputation                          |                        | 0.2822***<br>(0.0020)  | 0.2826***<br>(0.0020)  | 0.2493***<br>(0.0035)  |
| Team New Collaborators × Team Reputation |                        |                        |                        | 0.0210***<br>(0.0018)  |
| Cultural Distance                        | -0.0311***<br>(0.0022) | -0.0191***<br>(0.0018) | -0.0191***<br>(0.0018) | -0.0188***<br>(0.0018) |
| Controls                                 | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                             | 45026                  | 45152                  | 45026                  | 45026                  |
| R <sup>2</sup>                           | 0.173                  | 0.421                  | 0.422                  | 0.423                  |

As in Table S15, but comparing shorter- with longer-distance movers within the mobile sample. Labour advantage is measured by per-paper team size rather than by coauthor count, because the two groups differ more sharply on the former; conclusions are unchanged when coauthor count is used instead. Repeat Col. Interval (Inverse) denotes coordination efficiency, the inverse



of the mean interval between repeated collaborations with the same coauthor. Team new collaborators is the mean number of first-time coauthors per paper in years 6–25. Super-team papers with more than 29 authors are excluded. Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

**Table S17 Network position, coordination efficiency and team expertise distance as potential channels of the long-term advantages of Mob scientists.**

**Panel a: Means**

| Variables          | Group   | Publications |        |         | D5 percentile |        |         |
|--------------------|---------|--------------|--------|---------|---------------|--------|---------|
|                    |         | Lower        | Higher | P-value | Lower         | Higher | P-value |
| Repeat Col.        | non-Mob | 47.93        | 98.24  | <0.001  |               |        |         |
| Interval (Inverse) | Mob     | 57.51        | 118.52 | <0.001  |               |        |         |
| Team Expertise     | non-Mob |              |        |         | 0.4553        | 0.5264 | <0.001  |
| Distance           | Mob     |              |        |         | 0.5264        | 0.5433 | <0.001  |

**Panel b: Publications**

|                                                            | (1)                   | (2)                   | (3)                   | (4)                   |
|------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Betweenness Centrality                                     | 1.0387***<br>(0.0434) |                       | 1.3524***<br>(0.0392) | 0.5835**<br>(0.1952)  |
| Repeat Col. Interval (Inverse)                             |                       | 2.4065***<br>(0.0420) | 2.4905***<br>(0.0423) | 1.4235***<br>(0.2572) |
| Betweenness Centrality × Repeat Col.<br>Interval (Inverse) |                       |                       |                       | 1.3287***<br>(0.3116) |
| Cultural Distance                                          | 0.1837***<br>(0.0092) | 0.1842***<br>(0.0075) | 0.1208***<br>(0.0078) | 0.1203***<br>(0.0078) |
| Controls                                                   | Yes                   | Yes                   | Yes                   | Yes                   |
| Observations                                               | 90022                 | 89363                 | 89363                 | 89363                 |
| Pseudo-R <sup>2</sup>                                      | 0.176                 | 0.322                 | 0.365                 | 0.366                 |

**Panel c Hit 10pct**

|                                             | (1)                   | (2)                   | (3)                   | (4)                    |
|---------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------|
| Betweenness Centrality                      | 0.0339***<br>(0.0028) |                       | 0.0300***<br>(0.0024) | 0.0471***<br>(0.0043)  |
| Team Reputation                             |                       | 0.2766***<br>(0.0014) | 0.2765***<br>(0.0014) | 0.3044***<br>(0.0060)  |
| Betweenness Centrality × Team<br>Reputation |                       |                       |                       | -0.0360***<br>(0.0074) |
| Cultural Distance                           | 0.0115***<br>(0.0009) | 0.0033***<br>(0.0007) | 0.0019*<br>(0.0007)   | 0.0018*<br>(0.0007)    |
| Controls                                    | Yes                   | Yes                   | Yes                   | Yes                    |
| Observations                                | 90022                 | 90304                 | 90022                 | 90022                  |
| R <sup>2</sup>                              | 0.185                 | 0.422                 | 0.423                 | 0.423                  |

**Panel d D5 percentile**

|                        | (1)       | (2) | (3)    | (4)        |
|------------------------|-----------|-----|--------|------------|
| Betweenness Centrality | 0.0388*** |     | 0.0084 | -0.1148*** |

|                                                  |           |           |           |           |
|--------------------------------------------------|-----------|-----------|-----------|-----------|
|                                                  | (0.0061)  |           | (0.0061)  | (0.0117)  |
| Team Expertise Distance                          |           | 0.1903*** | 0.1896*** | -0.0519** |
|                                                  |           | (0.0043)  | (0.0043)  | (0.0200)  |
| Betweenness Centrality × Team Expertise Distance |           |           |           | 0.3066*** |
|                                                  |           |           |           | (0.0248)  |
| Cultural Distance                                | 0.0180*** | 0.0183*** | 0.0179*** | 0.0176*** |
|                                                  | (0.0019)  | (0.0018)  | (0.0019)  | (0.0019)  |
| Controls                                         | Yes       | Yes       | Yes       | Yes       |
| Observations                                     | 89863     | 89863     | 89863     | 89863     |
| R <sup>2</sup>                                   | 0.052     | 0.072     | 0.072     | 0.074     |

Panel a reports group means of long-term performance for scientists in the lower and higher halves of each measure, split at the median of the full matched sample, with one-sided Mann–Whitney  $U$  tests. Panels b–d report Poisson regressions of Publications with robust standard errors and ordinary-least-squares regressions of Hit 10pct and D5 percentile, all including the five early-career covariates used for matching. Betweenness centrality, component size diversity, the relative giant component and mean component density are computed on the unweighted later-career collaboration network, excluding super-team papers with more than 29 authors. Substituting mean component density for coordination efficiency in Panel b yields the same positive main effect on Publications and the same positive interaction with betweenness centrality. Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

**Table S18 Team expertise distance and early coauthor capital imbalance as potential channels of the disruptiveness advantage of longer-distance movers.**

**Panel a: Means**

| Variables               | D5 percentile |                   |         |        |                  |         |
|-------------------------|---------------|-------------------|---------|--------|------------------|---------|
|                         | Lower         | Shorter<br>Higher | P-value | Lower  | Longer<br>Higher | P-value |
| Team Expertise distance | 0.4560        | 0.5223            | <0.001  | 0.4992 | 0.5716           | <0.001  |

**Panel b: Early field pivot size**

|                                                      | (1)                | (2)                   | (3)                   | (4)                   |
|------------------------------------------------------|--------------------|-----------------------|-----------------------|-----------------------|
| Cultural Distance                                    | 0.0055<br>(0.0029) |                       | 0.0033<br>(0.0029)    | -0.0109<br>(0.0061)   |
| Early Coauthor Capital Imbalance                     |                    | 0.0816***<br>(0.0034) | 0.0814***<br>(0.0034) | 0.0668***<br>(0.0065) |
| Cultural Distance × Early Coauthor Capital Imbalance |                    |                       |                       | 0.0279**<br>(0.0105)  |
| Controls                                             | Yes                | Yes                   | Yes                   | Yes                   |
| Observations                                         | 43130              | 41467                 | 41467                 | 41467                 |
| R <sup>2</sup>                                       | 0.030              | 0.044                 | 0.044                 | 0.044                 |

**Panel c Early coauthor expertise distance**

|                                                      | (1)                   | (2)                   | (3)                   | (4)                   |
|------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Cultural Distance                                    | 0.0428***<br>(0.0054) |                       | 0.0237***<br>(0.0054) | -0.0022<br>(0.0111)   |
| Early Coauthor Capital Imbalance                     |                       | 0.1347***<br>(0.0061) | 0.1330***<br>(0.0061) | 0.1066***<br>(0.0117) |
| Cultural Distance × Early Coauthor Capital Imbalance |                       |                       |                       | 0.0506**<br>(0.0190)  |
| Controls                                             | Yes                   | Yes                   | Yes                   | Yes                   |
| Observations                                         | 45152                 | 43310                 | 43310                 | 43310                 |
| R <sup>2</sup>                                       | 0.007                 | 0.020                 | 0.021                 | 0.021                 |

**Panel d D5 percentile**

|                                       | (1)                   | (2)                   | (3)                   | (4)                    |
|---------------------------------------|-----------------------|-----------------------|-----------------------|------------------------|
| Cultural Distance                     | 0.0487***<br>(0.0046) | 0.0458***<br>(0.0045) | 0.0460***<br>(0.0045) | 0.0455***<br>(0.0045)  |
| Team Expertise Distance               |                       | 0.1749***<br>(0.0059) | 0.1743***<br>(0.0059) | -0.1591***<br>(0.0310) |
| Betweenness Centrality                | 0.0389***<br>(0.0094) |                       | 0.0093<br>(0.0093)    | -0.1566***<br>(0.0178) |
| Team Expertise Distance × Betweenness |                       |                       |                       | 0.4106***              |

|                |       |       |       |          |
|----------------|-------|-------|-------|----------|
| Centrality     |       |       |       | (0.0375) |
| Controls       | Yes   | Yes   | Yes   | Yes      |
| Observations   | 45007 | 45007 | 45007 | 45007    |
| R <sup>2</sup> | 0.062 | 0.079 | 0.079 | 0.082    |

Panel a reports group means of D5 percentile for shorter- and longer-distance movers in the lower and higher halves of team expertise distance, split at the median of the mobile sample, with one-sided Mann–Whitney  $U$  tests. Panels b and c take early-career field pivot size and early coauthor expertise distance as outcomes, and Panel d takes D5 percentile; all are ordinary-least-squares regressions including the five early-career covariates used for matching. Early coauthor capital imbalance is the coefficient of variation of coauthors' cumulative citation impact up to career year five, and results are robust to using the Gini coefficient instead. Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

**Table S19 Early-career exploration followed by later-career exploitation as a potential channel of the impact advantage of Mob scientists.**

**Panel a Hit 10pct (Early-Career Field Pivot Size)**

|                                                            | (1)                   | (2)                   | (3)                   | (4)                    |
|------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------|
| Early-Career Field Pivot Size                              | 0.0178***<br>(0.0027) |                       | 0.0342***<br>(0.0026) | -0.0343***<br>(0.0102) |
| Reference Fields Imbalance                                 |                       | 0.1421***<br>(0.0021) | 0.1439***<br>(0.0022) | 0.1327***<br>(0.0027)  |
| Early-Career Field Pivot Size × Reference Fields Imbalance |                       |                       |                       | 0.0894***<br>(0.0129)  |
| Cultural Distance                                          | 0.0124***<br>(0.0009) | 0.0076***<br>(0.0008) | 0.0071***<br>(0.0009) | 0.0072***<br>(0.0009)  |
| Controls                                                   | Yes                   | Yes                   | Yes                   | Yes                    |
| Observations                                               | 84907                 |                       | 84559                 | 84559                  |
| R <sup>2</sup>                                             | 0.189                 | 0.222                 | 0.229                 | 0.229                  |

**Panel b Hit 10pct (Team Reputation)**

|                                              | (1)                   | (2)                   | (3)                   | (4)                   |
|----------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Team Reputation                              | 0.2766***<br>(0.0014) |                       | 0.2650***<br>(0.0015) | 0.1724***<br>(0.0049) |
| Reference Fields Imbalance                   |                       | 0.1421***<br>(0.0021) | 0.0707***<br>(0.0018) | 0.0086*<br>(0.0036)   |
| Team Reputation × Reference Fields Imbalance |                       |                       |                       | 0.1205***<br>(0.0061) |
| Cultural Distance                            | 0.0033***<br>(0.0007) | 0.0076***<br>(0.0008) | 0.0010<br>(0.0007)    | 0.0011<br>(0.0007)    |
| Controls                                     | Yes                   | Yes                   | Yes                   | Yes                   |
| Observations                                 | 90304                 | 89881                 | 89881                 | 89881                 |
| R <sup>2</sup>                               | 0.422                 | 0.222                 | 0.431                 | 0.433                 |

Ordinary-least-squares regressions of Hit 10pct among matched Mob and non-Mob scientists, including the five early-career covariates used for matching. Reference fields imbalance is the coefficient of variation of the field-size-normalized distribution of referenced fields in years 6–25, and early-career field pivot size is one minus the cosine similarity between the field distribution of a paper's references and that of the author's cumulative prior publications. Replacing reference fields imbalance with reference home fields relative share yields coefficients of the same sign throughout (Table S22). Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

**Table S20 Local knowledge strategy, team reputation and team expertise distance as potential channels of the divergence between shorter- and longer-distance movers.**

**Panel a Hit 10pct**

|                                                           | (1)                    | (2)                    | (3)                    | (4)                    |
|-----------------------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| Reference Home Fields Relative Share                      | 0.1234***<br>(0.0053)  | 0.0629***<br>(0.0045)  | 0.0725***<br>(0.0046)  | -0.0412***<br>(0.0085) |
| Team Reputation                                           |                        | 0.2795***<br>(0.0020)  | 0.2768***<br>(0.0020)  | 0.1070***<br>(0.0121)  |
| Career Fields Propensity Diversity                        |                        |                        | 0.0354***<br>(0.0043)  |                        |
| Reference Home Fields Relative Share ×<br>Team Reputation |                        |                        |                        | 0.2166***<br>(0.0150)  |
| Cultural Distance                                         | -0.0296***<br>(0.0021) | -0.0184***<br>(0.0018) | -0.0181***<br>(0.0018) | -0.0189***<br>(0.0018) |
| Controls                                                  | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                                              | 45144                  | 45144                  | 45144                  | 45144                  |
| R <sup>2</sup>                                            | 0.178                  | 0.424                  | 0.425                  | 0.426                  |

**Panel b D5 Percentile**

|                                                                     | (1)                   | (2)                    | (3)                    | (4)                    |
|---------------------------------------------------------------------|-----------------------|------------------------|------------------------|------------------------|
| Reference Global Fields Relative Share                              | 0.1538***<br>(0.0113) | -0.0482***<br>(0.0135) | -0.0556***<br>(0.0134) | -0.1631***<br>(0.0269) |
| Team Expertise Distance                                             |                       | 0.1891***<br>(0.0071)  | 0.1205***<br>(0.0074)  | 0.1440***<br>(0.0115)  |
| Career Fields Propensity Diversity                                  |                       |                        | 0.3262***<br>(0.0113)  |                        |
| Reference Global Fields Relative Share ×<br>Team Expertise Distance |                       |                        |                        | 0.2557***<br>(0.0517)  |
| Cultural Distance                                                   | 0.0457***<br>(0.0046) | 0.0462***<br>(0.0045)  | 0.0501***<br>(0.0045)  | 0.0462***<br>(0.0045)  |
| Controls                                                            | Yes                   | Yes                    | Yes                    | Yes                    |
| Observations                                                        | 45108                 | 45007                  | 45007                  | 45007                  |
| R <sup>2</sup>                                                      | 0.065                 | 0.080                  | 0.096                  | 0.080                  |

Ordinary-least-squares regressions of Hit 10pct (Panel a) and D5 percentile (Panel b) within the mobile sample, including the five early-career covariates used for matching. Reference home fields relative share is the share of field-size-normalized referencing frequencies drawn from a paper's own fields, and reference global fields relative share is its complement. Career fields propensity diversity is the normalized Shannon entropy of a scientist's AUC-based propensities across all fields referenced in years 6–25. Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

**Table S21 Career-level knowledge exploration as a potential channel of the long-term advantages of Mob scientists.**

**Panel a Publication (Coauthor Count)**

|                                                     | (1)                   | (2)                   | (3)                   | (4)                    |
|-----------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------|
| Coauthor Count                                      | 0.0001***<br>(0.0000) |                       | 0.0001***<br>(0.0000) | 0.0002***<br>(0.0000)  |
| Career Fields Propensity Diversity                  |                       | 2.2205***<br>(0.1508) | 1.9857***<br>(0.0538) | 2.0668***<br>(0.0516)  |
| Coauthor Count × Career Fields Propensity Diversity |                       |                       |                       | -0.0002***<br>(0.0000) |
| Cultural Distance                                   | 0.2083***<br>(0.0078) | 0.1972***<br>(0.0079) | 0.1874***<br>(0.0081) | 0.1866***<br>(0.0078)  |
| Controls                                            | Yes                   | Yes                   | Yes                   | Yes                    |
| Observations                                        | 90,304                | 90,304                | 90,304                | 90,304                 |
| R <sup>2</sup>                                      | 0.234                 | 0.212                 | 0.283                 | 0.284                  |

**Panel b Publication (Repeat Col. Interval (Inverse))**

|                                                                     | (1)                   | (2)                   | (3)                   | (4)                   |
|---------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Repeat Col. Interval (Inverse)                                      | 2.4065***<br>(0.0420) |                       | 2.2951***<br>(0.0321) | 1.1998*<br>(0.5927)   |
| Career Fields Propensity Diversity                                  |                       | 2.2205***<br>(0.1508) | 1.9147***<br>(0.0933) | 1.2628**<br>(0.4367)  |
| Repeat Col. Interval (Inverse) × Career Fields Propensity Diversity |                       |                       |                       | 1.2602<br>(0.6710)    |
| Cultural Distance                                                   | 0.1842***<br>(0.0075) | 0.1972***<br>(0.0079) | 0.1611***<br>(0.0069) | 0.1611***<br>(0.0069) |
| Controls                                                            | Yes                   | Yes                   | Yes                   | Yes                   |
| Observations                                                        | 89,363                | 90,304                | 89,363                | 89,363                |
| R <sup>2</sup>                                                      | 0.322                 | 0.212                 | 0.366                 | 0.367                 |

**Panel c D5 Percentile**

|                                                              | (1)                   | (2)                   | (3)                   | (4)                    |
|--------------------------------------------------------------|-----------------------|-----------------------|-----------------------|------------------------|
| Team Expertise Distance                                      | 0.1903***<br>(0.0043) |                       | 0.1310***<br>(0.0046) | -0.4191***<br>(0.0290) |
| Career Fields Propensity Diversity                           |                       | 0.3558***<br>(0.0073) | 0.2756***<br>(0.0079) | 0.1390***<br>(0.0106)  |
| Team Expertise Distance × Career Fields Propensity Diversity |                       |                       |                       | 0.6258***<br>(0.0326)  |
| Cultural Distance                                            | 0.0183***<br>(0.0018) | 0.0159***<br>(0.0018) | 0.0160***<br>(0.0018) | 0.0145***<br>(0.0018)  |
| Controls                                                     | Yes                   | Yes                   | Yes                   | Yes                    |



|                |        |        |        |        |
|----------------|--------|--------|--------|--------|
| Observations   | 89,863 | 90,136 | 89,863 | 89,863 |
| R <sup>2</sup> | 0.072  | 0.076  | 0.084  | 0.088  |

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Poisson regressions of Publications with robust standard errors (Panels a and b) and ordinary-least-squares regressions of D5 percentile (Panel c) among matched Mob and non-Mob scientists, including the five early-career covariates used for matching. Career fields propensity diversity is the normalized Shannon entropy of a scientist's AUC-based propensities across all fields referenced in years 6–25. Replacing it with career fields coverage, the number of distinct fields engaged over the same period, yields the same conclusions for Publications. The interaction with coauthor count in Panel a is negative, consistent with larger teams facing higher costs of field switching, whereas the interaction with team expertise distance in Panel c is positive, indicating that carrying breadth personally across the career and assembling expertise-diverse teams reinforce each other in predicting disruptiveness. Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

**Table S22 Local exploitation versus exploration (reference home/global fields relative share) among Mob versus non-Mob scientists and associations with long-term performance.**

**Panel a Hit 10pct**

|                                                           | (1)                   | (2)                   | (3)                   | (4)                   |
|-----------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Reference Home Fields Relative Share                      | 0.1175***<br>(0.0037) | 0.0588***<br>(0.0031) | 0.0737***<br>(0.0032) | -0.0140*<br>(0.0058)  |
| Team Reputation                                           |                       | 0.2743***<br>(0.0014) | 0.2702***<br>(0.0015) | 0.1208***<br>(0.0083) |
| Career Fields Propensity Diversity                        |                       |                       | 0.0540***<br>(0.0030) | 0.0474***<br>(0.0030) |
| Reference Home Fields Relative Share ×<br>Team Reputation |                       |                       |                       | 0.1893***<br>(0.0104) |
| Cultural Distance                                         | 0.0115***<br>(0.0009) | 0.0026***<br>(0.0007) | 0.0020**<br>(0.0007)  | 0.0021**<br>(0.0007)  |
| Controls                                                  | Yes                   | Yes                   | Yes                   | Yes                   |
| Observations                                              | 90249                 | 90249                 | 90249                 | 90249                 |
| R <sup>2</sup>                                            | 0.192                 | 0.424                 | 0.426                 | 0.428                 |

**Panel b D5 Percentile**

|                                                                     | (1)                   | (2)                   | (3)                    | (4)                    |
|---------------------------------------------------------------------|-----------------------|-----------------------|------------------------|------------------------|
| Reference Global Fields Relative Share                              | 0.1720***<br>(0.0080) | -0.0201*<br>(0.0094)  | -0.0349***<br>(0.0094) | -0.1903***<br>(0.0189) |
| Team Expertise Distance                                             |                       | 0.1960***<br>(0.0050) | 0.1406***<br>(0.0053)  | 0.0730***<br>(0.0089)  |
| Career Fields Propensity Diversity                                  |                       |                       | 0.2772***<br>(0.0079)  | 0.2915***<br>(0.0081)  |
| Reference Global Fields Relative Share ×<br>Team Expertise Distance |                       |                       |                        | 0.3495***<br>(0.0369)  |
| Cultural Distance                                                   | 0.0216***<br>(0.0019) | 0.0180***<br>(0.0019) | 0.0155***<br>(0.0018)  | 0.0149***<br>(0.0018)  |
| Controls                                                            | Yes                   | Yes                   | Yes                    | Yes                    |
| Observations                                                        | 90135                 | 89862                 | 89862                  | 89862                  |
| R <sup>2</sup>                                                      | 0.056                 | 0.072                 | 0.084                  | 0.085                  |

Ordinary-least-squares regressions of Hit 10pct (Panel a) and D5 percentile (Panel b) among matched Mob and non-Mob scientists, including the five early-career covariates used for matching. These specifications replace reference fields imbalance, used in Tables S19 and S21, with the reference home and global fields relative shares, which separate within-field concentration from cross-field referencing and therefore yield directionally interpretable coefficients for exploitation and exploration. Coefficients on cultural distance and on team reputation, team expertise distance and career fields

propensity diversity retain the same signs and significance as in Tables S19 and S21, so the conclusions do not depend on which paper-level measure is used. Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

**Table S23 Knowledge core sourcing and collaborator exposure by later-career location.**

| Later-career<br>Location | Metrics                             | Mean     |        | Diff.   | N        |        | P-value |
|--------------------------|-------------------------------------|----------|--------|---------|----------|--------|---------|
|                          |                                     | Non-mobs | Mobs   |         | Non-mobs | Mobs   |         |
| Panel a Institutions     |                                     |          |        |         |          |        |         |
| Non-top institution      | Top institution coauthors per paper | 0.3688   | 0.4318 | 0.0630  | 32,739   | 34,554 | <0.001  |
|                          | Career top institutions propensity  | 0.5189   | 0.5223 | 0.0033  | 32,713   | 34,572 | <0.001  |
| Top-100 institution      | Top institution coauthors per paper | 2.4548   | 2.3116 | -0.1433 | 13,594   | 11,832 | <0.001  |
|                          | Career top institutions propensity  | 0.5287   | 0.5308 | 0.0021  | 13,562   | 11,831 | <0.001  |
| Panel b Country          |                                     |          |        |         |          |        |         |
| Non-western country      | Western country coauthors per paper | 0.4144   | 0.6670 | 0.2526  | 10,022   | 13,449 | <0.001  |
|                          | Career western countries propensity | 0.5307   | 0.5384 | 0.0078  | 10,014   | 13,456 | <0.001  |
| Western country          | Western country coauthors per paper | 3.1614   | 3.1097 | -0.0517 | 36,311   | 32,937 | <0.001  |
|                          | Career western countries propensity | 0.5731   | 0.5668 | -0.0063 | 36,151   | 32,921 | <0.001  |

Diff. is calculated as the Mob mean minus the non-Mob mean. P values are from two-sided Welch's t-tests. Sample sizes (N) vary slightly across metrics because of missing information in collaborator or reference-source measures. Scientists are assigned to strata by the modal institution or country of their later-career affiliations.

**Table S24 Core-oriented knowledge sourcing and long-term performance.**

|                                        | (1)<br>Hit 10pct      | (2)<br>Hit 10pct      | (3)<br>Hit 10pct       | (4)<br>Hit 10pct       | (5)<br>D5 Percentile   | (6)<br>D5 Percentile   |
|----------------------------------------|-----------------------|-----------------------|------------------------|------------------------|------------------------|------------------------|
| <b>Group</b>                           | <b><i>Overall</i></b> | <b><i>Overall</i></b> | <b><i>Mob</i></b>      | <b><i>Mob</i></b>      | <b><i>Mob</i></b>      | <b><i>Mob</i></b>      |
| Career Top Institutions Propensity     | 0.8876***<br>(0.0155) | 0.1042***<br>(0.0147) |                        |                        |                        |                        |
| Career Western Countries<br>Propensity |                       |                       | 0.2336***<br>(0.0136)  | 0.1250***<br>(0.0114)  | -0.1383***<br>(0.0292) | -0.1513***<br>(0.0291) |
| Team Reputation                        |                       | 0.2626***<br>(0.0015) |                        | 0.2783***<br>(0.0020)  |                        |                        |
| Paper-level local exploitation         |                       | 0.0683***<br>(0.0019) |                        | 0.0617***<br>(0.0045)  |                        |                        |
| Team Expertise Distance                |                       |                       |                        |                        |                        | 0.1747***<br>(0.0059)  |
| Cultural Distance                      | 0.0107***<br>(0.0008) | 0.0010<br>(0.0007)    | -0.0270***<br>(0.0022) | -0.0164***<br>(0.0018) | 0.0450***<br>(0.0046)  | 0.0429***<br>(0.0046)  |
| Controls                               | Yes                   | Yes                   | Yes                    | Yes                    | Yes                    | Yes                    |
| Observations                           | 90181                 | 89843                 | 45113                  | 45112                  | 45084                  | 44984                  |
| R <sup>2</sup>                         | 0.212                 | 0.431                 | 0.173                  | 0.426                  | 0.062                  | 0.080                  |

Ordinary-least-squares regressions including the five early-career covariates used for matching. Columns 1 and 2 use the full matched sample; columns 3–6 use the mobile sample only. Career top institutions propensity and career western countries propensity are AUC-based measures of the relative propensity to reference work from top-100 institutions and from Western countries, respectively, over years 6–25. Paper-level local exploitation is measured by reference fields imbalance in columns 1 and 2, and by reference home fields relative share in columns 3 and 4. Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

**Table S25 Team capital, deepening expertise, and national bias as potential channels of the impact advantage of shorter-distance movers.**

|                                          | (1)                      | (2)                    | (3)                    | (4)                      |
|------------------------------------------|--------------------------|------------------------|------------------------|--------------------------|
| Reference Geographic Distance            | -4.00e-6***<br>(3.11e-7) |                        |                        | -2.00e-6***<br>(2.65e-7) |
| Team Reputation                          |                          | 0.2795***<br>(0.0020)  | 0.2783***<br>(0.0020)  | 0.2777***<br>(0.0020)    |
| Reference Home Fields Relative Share     |                          | 0.0629***<br>(0.0045)  | 0.0617***<br>(0.0045)  | 0.0626***<br>(0.0045)    |
| Career Western Countries Propensity      |                          |                        | 0.1250***<br>(0.0114)  | 0.1081***<br>(0.0118)    |
| Cultural Distance                        | -0.0289***<br>(0.0022)   | -0.0184***<br>(0.0018) | -0.0164***<br>(0.0018) | -0.0153***<br>(0.0018)   |
| Controls                                 | Yes                      | Yes                    | Yes                    | Yes                      |
| Observations                             | 45078                    | 45144                  | 45112                  | 45061                    |
| R <sup>2</sup>                           | 0.171                    | 0.424                  | 0.426                  | 0.427                    |
| Reduction in effect of Cultural Distance | 8.80%                    | 41.96%                 | 48.34%                 | 51.71%                   |

Standard errors are reported in parentheses. \*P < 0.05; \*\*P < 0.01; \*\*\*P < 0.001. Reference Geographic Distance (km) is reported in scientific notation because of its small scale. Reference Home Fields Relative Share is the paper-level local exploitation measure. The baseline cultural-distance coefficient is -0.0317 from the baseline specification in Table S4; reduction is calculated as (|baseline coefficient| - |model coefficient|) / |baseline coefficient|. In addition, when added to team reputation and paper-level local exploitation, reference geographic distance attenuates the cultural-distance coefficient to 47.00%, whereas adding career western countries propensity attenuates it to 48.34%. After team reputation, paper-level local exploitation, and career western countries propensity are already included, further adding reference geographic distance increases the attenuation only modestly to 51.71%; therefore, we focus on team reputation, paper-level local exploitation, and career Western-country propensity as the primary channels.

**Table S26 Associations between search portfolio and long-term disruptiveness.**

**Panel a Mob vs. non-Mob**

|                                      | (1)                       | (2)                   | (3)                       | (4)                       | (5)                       |
|--------------------------------------|---------------------------|-----------------------|---------------------------|---------------------------|---------------------------|
| Reference Geographic Distance        | 7.00e-06***<br>(5.10e-07) |                       |                           |                           |                           |
| Reference Age                        |                           | 0.0048***<br>(0.0003) |                           | 0.0074***<br>(0.0003)     | 0.0043***<br>(0.0004)     |
| Reference Popularity                 |                           |                       | 3.40e-04***<br>(9.95e-06) | 3.87e-04***<br>(1.02e-05) | -1.00e-05<br>(3.01e-05)   |
| Reference Age × Reference Popularity |                           |                       |                           |                           | 4.90e-05***<br>(3.32e-06) |
| Cultural Distance                    | 0.0152***<br>(0.0019)     | 0.0187***<br>(0.0019) | 0.0163***<br>(0.0019)     | 0.0144***<br>(0.0019)     | 0.0144***<br>(0.0018)     |
| Controls                             | Yes                       | Yes                   | Yes                       | Yes                       | Yes                       |
| Observations                         | 89912                     | 90136                 | 90136                     | 90136                     | 90136                     |
| R <sup>2</sup>                       | 0.054                     | 0.054                 | 0.064                     | 0.069                     | 0.071                     |

**Panel b Shorter vs. Longer**

|                                      | (1)                       | (2)                   | (3)                       | (4)                       | (5)                       |
|--------------------------------------|---------------------------|-----------------------|---------------------------|---------------------------|---------------------------|
| Reference Geographic Distance        | 8.00e-06***<br>(6.60e-07) |                       |                           |                           |                           |
| Reference Age                        |                           | 0.0028***<br>(0.0004) |                           | 0.0057***<br>(0.0005)     | 0.0020***<br>(0.0006)     |
| Reference Popularity                 |                           |                       | 3.32e-04***<br>(1.38e-05) | 3.73e-04***<br>(1.42e-05) | -6.90e-05<br>(4.51e-05)   |
| Reference Age × Reference Popularity |                           |                       |                           |                           | 5.20e-05***<br>(5.07e-06) |
| Cultural Distance                    | 0.0427***<br>(0.0046)     | 0.0464***<br>(0.0046) | 0.0450***<br>(0.0045)     | 0.0421***<br>(0.0045)     | 0.0424***<br>(0.0045)     |
| Controls                             | Yes                       | Yes                   | Yes                       | Yes                       | Yes                       |
| Observations                         | 45054                     | 45108                 | 45108                     | 45108                     | 45108                     |
| R <sup>2</sup>                       | 0.064                     | 0.062                 | 0.073                     | 0.076                     | 0.079                     |

Ordinary-least-squares regressions of D5 percentile including the five early-career covariates used for matching; Panel a uses the full matched sample and Panel b the mobile sample only. Reference geographic distance is the mean distance between a focal paper's affiliations and those of its cited references, reference age the mean age of cited work, and reference popularity the mean five-year citation count (C5) of cited work, all measured over years 6–25. Using ten-year or full-window citation counts instead of C5 yields the same conclusions. Because references of focal papers published after 2016 lack a complete five-year window, we also recompute the measure excluding them rather than counting them at their truncated value, again with unchanged

conclusions (including Fig. S19). Standard errors are reported in parentheses. Reference Geographic Distance and Reference Popularity are reported in scientific notation because of their small scales. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .



**Table S27 Team expertise, peripheral-oriented knowledge sourcing, and broader search combined with standing on the shoulders of giants as potential channels of the disruptiveness advantage of longer-distance movers.**

|                                          | (1)                   | (2)                    | (3)                       | (4)                       | (5)                       | (6)                       |
|------------------------------------------|-----------------------|------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| Team Expertise Distance                  | 0.1749***<br>(0.0059) | 0.1747***<br>(0.0059)  | 0.1734***<br>(0.0059)     | 0.1721***<br>(0.0059)     | 0.1804***<br>(0.0058)     | 0.1787***<br>(0.0058)     |
| Career Western Countries Propensity      |                       | -0.1513***<br>(0.0291) | -0.0856**<br>(0.0301)     | -0.0823**<br>(0.0301)     | -0.1317***<br>(0.0299)    | -0.1270***<br>(0.0299)    |
| Reference Geographic Distance            |                       |                        | 7.00e-06***<br>(6.70e-07) | 7.00e-06***<br>(6.71e-07) | 6.00e-06***<br>(6.67e-07) | 6.00e-06***<br>(6.67e-07) |
| Reference Age                            |                       |                        |                           | 0.0018***<br>(0.0004)     | 0.0048***<br>(0.0005)     | 0.0014*<br>(0.0006)       |
| Reference Popularity                     |                       |                        |                           |                           | 3.88e-04***<br>(1.41e-05) | -2.50e-05<br>(4.52e-05)   |
| Reference Age × Reference Popularity     |                       |                        |                           |                           |                           | 4.90e-05***<br>(5.07e-06) |
| Cultural Distance                        | 0.0458***<br>(0.0045) | 0.0429***<br>(0.0046)  | 0.0395***<br>(0.0046)     | 0.0389***<br>(0.0046)     | 0.0344***<br>(0.0045)     | 0.0348***<br>(0.0045)     |
| Controls                                 | Yes                   | Yes                    | Yes                       | Yes                       | Yes                       | Yes                       |
| Observations                             | 45007                 | 44984                  | 44943                     | 44943                     | 44943                     | 44943                     |
| R <sup>2</sup>                           | 0.079                 | 0.080                  | 0.082                     | 0.083                     | 0.098                     | 0.100                     |
| Reduction in effect of Cultural Distance | 4.38%                 | 10.44%                 | 17.54%                    | 18.79%                    | 28.18%                    | 27.35%                    |

Standard errors are reported in parentheses. \*P < 0.05; \*\*P < 0.01; \*\*\*P < 0.001. Reference Geographic Distance and Reference Popularity are reported in scientific notation because of their small scales. The baseline cultural-distance coefficient is 0.0479 from the baseline specification in Table S4.

**Table S28 Team expertise, career exploration, and broader search combined with standing on the shoulders of giants as potential channels of the disruptiveness advantage of movers.**

|                                             | (1)                   | (2)                   | (3)                       | (4)                       | (5)                       | (6)                       |
|---------------------------------------------|-----------------------|-----------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| Team Expertise Distance                     | 0.1903***<br>(0.0043) | 0.1310***<br>(0.0046) | 0.1280***<br>(0.0046)     | 0.1263***<br>(0.0046)     | 0.1330***<br>(0.0046)     | 0.1341***<br>(0.0046)     |
| Career Fields Propensity Diversity          |                       | 0.2756***<br>(0.0079) | 0.2849***<br>(0.0080)     | 0.2811***<br>(0.0080)     | 0.2704***<br>(0.0080)     | 0.2625***<br>(0.0080)     |
| Reference Geographic Distance               |                       |                       | 6.00e-06***<br>(5.10e-07) | 6.00e-06***<br>(5.10e-07) | 4.00e-06***<br>(5.10e-07) | 4.00e-06***<br>(5.10e-07) |
| Reference Age                               |                       |                       |                           | 0.0030***<br>(0.0003)     | 0.0057***<br>(0.0003)     | 0.0034***<br>(0.0004)     |
| Reference Popularity                        |                       |                       |                           |                           | 3.78e-04***<br>(9.95e-06) | 9.00e-05**<br>(2.98e-05)  |
| Reference Age $\times$ Reference Popularity |                       |                       |                           |                           |                           | 3.50e-05***<br>(3.32e-06) |
| Cultural Distance                           | 0.0183***<br>(0.0018) | 0.0160***<br>(0.0018) | 0.0119***<br>(0.0019)     | 0.0115***<br>(0.0019)     | 0.0086***<br>(0.0018)     | 0.0087***<br>(0.0018)     |
| Controls                                    | Yes                   | Yes                   | Yes                       | Yes                       | Yes                       | Yes                       |
| Observations                                | 89863                 | 89863                 | 89647                     | 89647                     | 89647                     | 89647                     |
| R <sup>2</sup>                              | 0.072                 | 0.084                 | 0.086                     | 0.087                     | 0.101                     | 0.102                     |
| Reduction in effect of Cultural Distance    | 7.11%                 | 18.80%                | 39.59%                    | 41.62%                    | 56.38%                    | 55.84%                    |

Standard errors are reported in parentheses. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ . Reference Geographic Distance and Reference Popularity are reported in scientific notation because of their small scales. The baseline cultural-distance coefficient is 0.0197 from the baseline specification in Table S4.

**Table S29 Robustness of evidence on potential channels to embedding-based measures of cultural distance.**

| Measure                                                | Hofstede | GMAE (uw)  | GCL (uw)   | Word2vec   | GCL (w)    | GMAE (w)   | Robust |
|--------------------------------------------------------|----------|------------|------------|------------|------------|------------|--------|
| <b>Panel a: Mob versus non-Mob</b>                     |          |            |            |            |            |            |        |
| Team country size                                      | +        | +11.03%*** | +11.03%*** | +11.03%*** | +11.03%*** | +11.03%*** | 5/5    |
| Team reputation                                        | +        | +7.89%***  | +7.89%***  | +7.89%***  | +7.89%***  | +7.89%***  | 5/5    |
| Betweenness centrality                                 | +        | +6.93%***  | +6.93%***  | +6.93%***  | +6.93%***  | +6.93%***  | 5/5    |
| Team expertise distance                                | +        | +2.16%***  | +2.16%***  | +2.16%***  | +2.16%***  | +2.16%***  | 5/5    |
| Reference fields imbalance                             | +        | +4.81%***  | +4.81%***  | +4.81%***  | +4.81%***  | +4.81%***  | 5/5    |
| Reference home fields relative share                   | +        | +1.56%***  | +1.56%***  | +1.56%***  | +1.56%***  | +1.56%***  | 5/5    |
| Career fields propensity diversity                     | +        | +1.55%***  | +1.55%***  | +1.55%***  | +1.55%***  | +1.55%***  | 5/5    |
| Career fields coverage                                 | +        | +12.38%*** | +12.38%*** | +12.38%*** | +12.38%*** | +12.38%*** | 5/5    |
| Career top institutions propensity                     | +        | +0.51%***  | +0.51%***  | +0.51%***  | +0.51%***  | +0.51%***  | 5/5    |
| Citation geographic dispersion                         | +        | +5.58%***  | +5.58%***  | +5.58%***  | +5.58%***  | +5.58%***  | 5/5    |
| <b>Panel b: Shorter- versus longer-distance movers</b> |          |            |            |            |            |            |        |
| Team new collaborators                                 | –        | –6.47%***  | –7.47%***  | –8.94%***  | –7.60%***  | –5.87%***  | 5/5    |
| Team reputation                                        | –        | –11.77%*** | –11.38%*** | –14.50%*** | –5.52%***  | –9.38%***  | 5/5    |
| Coordination efficiency                                | +        | +1.71%***  | +2.58%***  | +1.97%***  | +2.37%***  | +1.07%***  | 5/5    |
| Reference self-citation ratio                          | –        | –7.27%***  | –9.57%***  | –4.83%***  | –8.13%***  | +1.20%     | 4/5    |
| Reference fields pivot size                            | +        | +8.22%***  | +11.21%*** | +8.49%***  | +12.63%*** | +5.48%***  | 5/5    |
| Reference global fields relative share                 | +        | +5.83%***  | +6.18%***  | +7.95%***  | +6.33%***  | +5.08%***  | 5/5    |
| Career western countries propensity                    | –        | –2.18%***  | –3.66%***  | –2.64%***  | –3.11%***  | –1.85%***  | 5/5    |
| Reference geographic distance                          | +        | +17.25%*** | +40.00%*** | +20.51%*** | +31.74%*** | +4.72%***  | 5/5    |
| Career western countries citation propensity           | –        | –2.41%***  | –3.51%***  | –2.97%***  | –2.74%***  | –1.88%***  | 5/5    |

The Hofstede column reports the direction of the original difference shown in the corresponding main-text figure. In Panel a, “+” indicates Mob > non-Mob and “–” indicates Mob < non-Mob. In Panel b, “+” indicates Longer > Shorter and “–” indicates Longer < Shorter. The five embedding-based columns report the relative difference in group means, expressed as a percentage of the reference group. The reference group is non-Mob in Panel a and Shorter in Panel b. P values are from one-sided Welch t-tests in the direction reported in the main text. “Robust” counts how many of the five alternative measures reproduce the direction reported in

the main text at  $P < 0.05$ . Across all 19 variables, 94 of 95 tests (99%) are robust. \*\*\* $P < 0.001$ , \*\* $P < 0.01$ , \* $P < 0.05$ .

## Supplementary References

1. Lin, Z., Yin, Y., Liu, L. & Wang, D. SciSciNet: A large-scale open data lake for the science of science research. *Sci. Data* **10**, 315 (2023).
2. Funk, R. J. & Owen-Smith, J. A Dynamic Network Measure of Technological Change. *Manag. Sci.* **63**, 791–817 (2017).
3. Holst, V., Algaba, A., Tori, F., Wenmackers, S. & Ginis, V. Dataset Artefacts are the Hidden Drivers of the Declining Disruptiveness in Science. Preprint at <https://doi.org/10.48550/arXiv.2402.14583> (2024).
4. Park, M., Leahey, E. & Funk, R. J. Robust Evidence for Declining Disruptiveness: Assessing the Role of Zero-Backward-Citation Works. Preprint at <https://doi.org/10.48550/arXiv.2503.00184> (2026).
5. Murray, D. *et al.* Unsupervised embedding of trajectories captures the latent structure of scientific migration. *Proc. Natl. Acad. Sci.* **120**, e2305414120 (2023).
6. Mikolov, T., Sutskever, I., Chen, K., Corrado, G. S. & Dean, J. Distributed representations of words and phrases and their compositionality. *Adv. Neural Inf. Process. Syst.* **26**, (2013).
7. You, Y. *et al.* Graph contrastive learning with augmentations. *Adv. Neural Inf. Process. Syst.* **33**, 5812–5823 (2020).
8. Hou, Z. *et al.* GraphMAE: Self-Supervised Masked Graph Autoencoders. in

*Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* 594–604 (ACM, Washington DC USA, 2022).  
doi:10.1145/3534678.3539321.

9. Hamilton, W., Ying, Z. & Leskovec, J. Inductive representation learning on large graphs. *Adv. Neural Inf. Process. Syst.* **30**, (2017).
10. Kipf, T. N. & Welling, M. Semi-supervised classification with graph convolutional networks. *ArXiv Prepr. ArXiv160902907* <https://arxiv.org/abs/1609.02907> (2016).
11. Ying, Z., Bourgeois, D., You, J., Zitnik, M. & Leskovec, J. Gnnexplainer: Generating explanations for graph neural networks. *Adv. Neural Inf. Process. Syst.* **32**, (2019).
12. Rosenbaum, P. R. & Rubin, D. B. The central role of the propensity score in observational studies for causal effects. *Biometrika* **70**, 41–55 (1983).
13. Chernozhukov, V. *et al.* Double/debiased machine learning for treatment and structural parameters. (2018).
14. Nie, X. & Wager, S. Quasi-oracle estimation of heterogeneous treatment effects. *Biometrika* **108**, 299–319 (2021).
15. Radicchi, F., Fortunato, S. & Castellano, C. Universality of citation distributions: Toward an objective measure of scientific impact. *Proc. Natl. Acad. Sci.* **105**, 17268–17272 (2008).

16. Petersen, A. M., Arroyave, F. & Pammolli, F. The disruption index is biased by citation inflation. Preprint at <http://arxiv.org/abs/2306.01949> (2023).
17. Waltman, L. A review of the literature on citation impact indicators. *J. Informetr.* **10**, 365–391 (2016).
18. Ke, Q., Gates, A. J. & Barabási, A.-L. A network-based normalized impact measure reveals successful periods of scientific discovery across disciplines. *Proc. Natl. Acad. Sci.* **120**, e2309378120 (2023).
19. Bornmann, L., Devarakonda, S., Tekles, A. & Chacko, G. Are disruption index indicators convergently valid? The comparison of several indicator variants with assessments by peers. *Quant. Sci. Stud.* **1**, 1242–1259 (2020).
20. Leydesdorff, L., Tekles, A. & Bornmann, L. A proposal to revise the disruption indicator. *Prof. Inf.* **30**, (2021).
21. Wang, J., Frietsch, R., Neuhäusler, P. & Hooi, R. International collaboration leading to high citations: Global impact or home country effect? *J. Informetr.* **18**, 101565 (2024).
22. Gomez, C. J., Herman, A. C. & Parigi, P. Leading countries in global science increasingly receive more citations than other countries doing similar research. *Nat. Hum. Behav.* **6**, 919–929 (2022).
23. Hao, Q., Xu, F., Li, Y. & Evans, J. AI Expands Scientists' Impact but Contracts Science's Focus. Preprint at <https://doi.org/10.48550/arXiv.2412.07727> (2024).

24. All Nobel Prizes. *NobelPrize.org* <https://www.nobelprize.org/prizes/lists/all-nobel-prizes/>.
25. Zhang, S., Wapman, K. H., Larremore, D. B. & Clauset, A. Labor advantages drive the greater productivity of faculty at elite universities. *Sci. Adv.* **8**, (2022).
26. Shi, D., Liu, W. & Wang, Y. Has China's Young Thousand Talents program been successful in recruiting and nurturing top-caliber scientists? *Science* **379**, 62–65 (2023).
27. Wagner, C. S., Whetsell, T. A. & Mukherjee, S. International research collaboration: Novelty, conventionality, and atypicality in knowledge recombination. *Res. Policy* **48**, 1260–1270 (2019).
28. Li, W., Zhang, S., Zheng, Z., Cranmer, S. J. & Clauset, A. Untangling the network effects of productivity and prominence among scientists. *Nat. Commun.* **13**, 4907 (2022).
29. Petersen, A. M. Quantifying the impact of weak, strong, and super ties in scientific careers. *Proc. Natl. Acad. Sci.* **112**, E4671–E4680 (2015).
30. Burt, R. S. Structural Holes and Good Ideas. *Am. J. Sociol.* **110**, 349–399 (2004).
31. Yan, E. & Ding, Y. Applying centrality measures to impact analysis: A coauthorship network analysis. *J. Am. Soc. Inf. Sci. Technol.* **60**, 2107–2118 (2009).
32. Guimerà, R., Uzzi, B., Spiro, J. & Amaral, L. A. N. Team Assembly Mechanisms



- Determine Collaboration Network Structure and Team Performance. *Science* **308**, 697–702 (2005).
33. Cummings, J. N. & Kiesler, S. Coordination costs and project outcomes in multi-university collaborations. *Res. Policy* **36**, 1620–1634 (2007).
34. Argyres, N., Rios, L. A. & Silverman, B. S. On the heels of giants: Internal network structure and the race to build on prior innovation. *Strateg. Manag. J.* **46**, 1728–1761 (2025).
35. Liu, L. *et al.* Hot streaks in artistic, cultural, and scientific careers. *Nature* **559**, 396–399 (2018).
36. Liu, L., Dehmamy, N., Chown, J., Giles, C. L. & Wang, D. Understanding the onset of hot streaks across artistic, cultural, and scientific careers. *Nat. Commun.* **12**, 5392 (2021).
37. Schich, M. *et al.* A network framework of cultural history. *Science* **345**, 558–562 (2014).
38. Hill, R. *et al.* The pivot penalty in research. *Nature* 1–8 (2025).
39. Chowdhary, S., Gallo, L., Musciotto, F. & Battiston, F. Team careers in science: formation, composition and success of persistent collaborations. Preprint at <https://doi.org/10.48550/arXiv.2407.09326> (2024).
40. Venturini, S., Sikdar, S., Rinaldi, F., Tudisco, F. & Fortunato, S. Collaboration and

- topic switches in science. *Sci. Rep.* **14**, 1258 (2024).
41. Zeng, A., Fan, Y., Di, Z., Wang, Y. & Havlin, S. Impactful scientists have higher tendency to involve collaborators in new topics. *Proc. Natl. Acad. Sci.* **119**, (2022).
  42. Aghion, P. & Howitt, P. A model of growth through creative destruction. (1990).
  43. Fleming, L. Recombinant Uncertainty in Technological Search. *Manag. Sci.* **47**, 117–132 (2001).
  44. Uzzi, B., Mukherjee, S., Stringer, M. & Jones, B. Atypical Combinations and Scientific Impact. *Science* **342**, 468–472 (2013).
  45. March, J. G. Exploration and Exploitation in Organizational Learning. *Organ. Sci.* **2**, 71–87 (1991).
  46. Li, L., Lin, Y. & Wu, L. Can Recombination Displace Dominant Scientific Ideas. Preprint at <https://doi.org/10.48550/arXiv.2506.15959> (2025).
  47. Sekara, V. *et al.* The chaperone effect in scientific publishing. *Proc. Natl. Acad. Sci.* **115**, 12603–12607 (2018).
  48. Nguyen, K.-T. *Trust and Innovation within the Firm: Evidence from Matched CEO-Firm Data*. <https://thedocs.worldbank.org/en/doc/438741576867823506-0090022019/render/Paper13NguyenKieuTrangTrustInnovationEMCWB.pdf> (2018).
  49. Zeng, A. *et al.* Increasing trend of scientists to switch between topics. *Nat. Commun.*

- 10**, 3439 (2019).
50. Park, M., Leahey, E. & Funk, R. J. Papers and patents are becoming less disruptive over time. *Nature* **613**, 138–144 (2023).
  51. Beltagy, I., Lo, K. & Cohan, A. SciBERT: A Pretrained Language Model for Scientific Text. Preprint at <http://arxiv.org/abs/1903.10676> (2019).
  52. Foster, J. G., Rzhetsky, A. & Evans, J. A. Tradition and Innovation in Scientists' Research Strategies. *Am. Sociol. Rev.* **80**, 875–908 (2015).
  53. Howells, J. R. L. Tacit Knowledge, Innovation and Economic Geography. *Urban Stud.* **39**, 871–884 (2002).
  54. Boschma, R. Proximity and Innovation: A Critical Assessment. *Reg. Stud.* **39**, 61–74 (2005).
  55. Li, W., Aste, T., Caccioli, F. & Livan, G. Early coauthorship with top scientists predicts success in academic careers. *Nat. Commun.* **10**, 5170 (2019).
  56. Laursen, K. & Salter, A. Open for innovation: the role of openness in explaining innovation performance among U.K. manufacturing firms. *Strateg. Manag. J.* **27**, 131–150 (2006).
  57. Katila, R. & Ahuja, G. SOMETHING OLD, SOMETHING NEW: A LONGITUDINAL STUDY OF SEARCH BEHAVIOR AND NEW PRODUCT INTRODUCTION. *Acad. Manage. J.* **45**, 1183–1194 (2002).

58. Rosenkopf, L. & Almeida, P. Overcoming Local Search Through Alliances and Mobility. *Manag. Sci.* **49**, 751–766 (2003).
59. Wu, L., Wang, D. & Evans, J. A. Large teams develop and small teams disrupt science and technology. *Nature* **566**, 378–382 (2019).
60. Storper, M. & Venables, A. J. Buzz: face-to-face contact and the urban economy. *J. Econ. Geogr.* **4**, 351–370 (2004).
61. Yu, X., Rahwan, T. & Jia, T. Knowledge Independence Breeds Disruption but Limits Recognition. *ArXiv Prepr. ArXiv250409589* <https://arxiv.org/abs/2504.09589> (2025).
62. Hsieh, C.-T., Hurst, E., Jones, C. I. & Klenow, P. J. The Allocation of Talent and U.S. Economic Growth. *Econometrica* **87**, 1439–1474 (2019).
63. Kerr, S. P., Kerr, W., Özden, Ç. & Parsons, C. High-Skilled Migration and Agglomeration. *Annu. Rev. Econ.* **9**, 201–234 (2017).
64. Sugimoto, C. R. *et al.* Scientists have most impact when they're free to move. *Nature* **550**, 29–31 (2017).
65. Kahn, S. & MacGarvie, M. New evidence on international postdocs in the US: Less pay, different experiences. *Res. Policy* **53**, 105077 (2024).
66. Franzoni, C., Scellato, G. & Stephan, P. The mover's advantage: The superior performance of migrant scientists. *Econ. Lett.* **122**, 89–93 (2014).

67. Holding, B. C., Acciai, C., Schneider, J. W. & Nielsen, M. W. Quantifying the mover's advantage: transatlantic migration, employment prestige, and scientific performance. *High. Educ.* **87**, 1749–1767 (2024).
68. Netz, N., Hampel, S. & Aman, V. What effects does international mobility have on scientists' careers? A systematic review. *Res. Eval.* **29**, 327–351 (2020).
69. Hunt, J. & Gauthier-Loiselle, M. How much does immigration boost innovation? *Am. Econ. J. Macroecon.* **2**, 31–56 (2010).
70. Kerr, W. R. & Lincoln, W. F. The Supply Side of Innovation: H-1B Visa Reforms and U.S. Ethnic Invention. *J. Labor Econ.* **28**, 473–508 (2010).
71. Moser, P., Voena, A. & Waldinger, F. German Jewish émigrés and US invention. *Am. Econ. Rev.* **104**, 3222–3255 (2014).
72. Zhao, X., Akbaritabar, A., Kashyap, R. & Zagheni, E. A gender perspective on the global migration of scholars. *Proc. Natl. Acad. Sci.* **120**, e2214664120 (2023).
73. Czaika, M. & De Haas, H. The Globalization of Migration: Has the World Become More Migratory? *Int. Migr. Rev.* **48**, 283–323 (2014).
74. Han, F., Zhang, R., Zhang, S. & Yuan, J. International mobility characteristics, effects of, and effects on elite scientists. *J. Informetr.* **18**, 101485 (2024).
75. Liu, T. *et al.* Spatiotemporal mobility network of global scientists, 1970–2020. *Int. J. Geogr. Inf. Sci.* **38**, 1991–2018 (2024).

76. Gui, Q., Liu, C. & Du, D. Globalization of science and international scientific collaboration: A network perspective. *Geoforum* **105**, 1–12 (2019).
77. Czaika, M. & Orazbayev, S. The globalisation of scientific mobility, 1970–2014. *Appl. Geogr.* **96**, 1–10 (2018).
78. Weitzman, M. L. Recombinant growth. *Q. J. Econ.* **113**, 331–360 (1998).
79. Arthur, W. B. *The Nature of Technology: What It Is and How It Evolves*. (Simon and Schuster, 2009).
80. Almeida, P. & Kogut, B. Localization of Knowledge and the Mobility of Engineers in Regional Networks. *Manag. Sci.* **45**, 905–917 (1999).
81. Joel, M. *A Culture of Growth: The Origins of the Modern Economy*. (Princeton University Press, 2016).
82. Hornung, E. Immigration and the diffusion of technology: The Huguenot diaspora in Prussia. *Am. Econ. Rev.* **104**, 84–122 (2014).
83. Muthukrishna, M. & Henrich, J. Innovation in the collective brain. *Philos. Trans. R. Soc. B Biol. Sci.* **371**, 20150192 (2016).
84. Posch, M., Schulz, J. & Henrich, J. How Cultural Diversity Drives Innovation: Surnames and Patents in US History. *J. Polit. Econ.* **134**, 1245–1305 (2026).
85. Kim, K., Askin, N. & Evans, J. A. Disrupted routines anticipate musical exploration. *Proc. Natl. Acad. Sci.* **121**, e2306549121 (2024).

86. Godart, F. C., Maddux, W. W., Shipilov, A. V. & Galinsky, A. D. Fashion with a Foreign Flair: Professional Experiences Abroad Facilitate the Creative Innovations of Organizations. *Acad. Manage. J.* **58**, 195–220 (2015).
87. Wu, X., Wu, L., Park, M., Leahey, E. & Funk, R. J. Is Innovation Becoming Less Disruptive? An Inventory of the Literature. Preprint at <https://doi.org/10.48550/arXiv.2602.05140> (2026).
88. Decker, R. A., Haltiwanger, J., Jarmin, R. S. & Miranda, J. Where has all the skewness gone? The decline in high-growth (young) firms in the US. *Eur. Econ. Rev.* **86**, 4–23 (2016).
89. Akcigit, U. & Ates, S. T. Ten facts on declining business dynamism and lessons from endogenous growth theory. *Am. Econ. J. Macroecon.* **13**, 257–298 (2021).
90. Bloom, N., Jones, C. I., Van Reenen, J. & Webb, M. Are ideas getting harder to find? *Am. Econ. Rev.* **110**, 1104–1144 (2020).
91. Schimmelpfennig, R., Razek, L., Schnell, E. & Muthukrishna, M. Paradox of diversity in the collective brain. *Philos. Trans. R. Soc. B Biol. Sci.* **377**, 20200316 (2021).
92. Jones, B. F. The burden of knowledge and the “death of the renaissance man”: Is innovation getting harder? *Rev. Econ. Stud.* **76**, 283–317 (2009).
93. Gordon, R. J. *The Rise and Fall of American Growth: The U.S. Standard of Living since the Civil War.* (Princeton University Press, 2016).

doi:10.1515/9781400873302.

94. Jones, C. I. Recipes and Economic Growth: A Combinatorial March Down an Exponential Tail. *J. Polit. Econ.* **131**, 1994–2031 (2023).
95. Akcigit, U. & Ates, S. T. What Happened to US Business Dynamism? *J. Polit. Econ.* **131**, 2059–2124 (2023).
96. Kang, H. & Eklund, J. C. Fast Friends: The Impact of Short-Term Visits on Firms' Invention Outcomes. *Organ. Sci.* **36**, 1154–1176 (2025).
97. Glennon, B. How Do Restrictions on High-Skilled Immigration Affect Offshoring? Evidence from the H-1B Program. *Manag. Sci.* **70**, 907–930 (2024).
98. Bell, A., Chetty, R., Jaravel, X., Petkova, N. & Van Reenen, J. Who becomes an inventor in America? The importance of exposure to innovation. *Q. J. Econ.* **134**, 647–713 (2019).
99. Huang, J., Cao, C. & Liu, H. Indigenization and inclusion in Chinese academia. *Nat. Hum. Behav.* **10**, 492–504 (2026).
100. Xie, Y., Lin, X., Li, J., He, Q. & Huang, J. Caught in the crossfire: Fears of Chinese–American scientists. *Proc. Natl. Acad. Sci.* **120**, e2216248120 (2023).
101. Jia, R., Roberts, M. E., Wang, Y. & Yang, E. The impact of US–China tensions on US science: Evidence from the NIH investigations. *Proc. Natl. Acad. Sci.* **121**, e2301436121 (2024).



102. Chu, J. S. G. & Evans, J. A. Slowed canonical progress in large fields of science. *Proc. Natl. Acad. Sci.* **118**, e2021636118 (2021).
103. Iaria, A., Schwarz, C. & Waldinger, F. Frontier knowledge and scientific production: Evidence from the collapse of international science. *Q. J. Econ.* **133**, 927–991 (2018).
104. Gomez, C. J. The growing concentration of national influence in global science. *Nat. Hum. Behav.* 1–17 (2026).